

Fundamental Theorem of Matrix Product States

A Formalization Blueprint

Authors

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Chapter 1

Introduction

We study the fundamental theorem for periodic, translation-invariant matrix product vectors. The basic object is a family of matrices $\{A^i\}$, viewed as a tensor that generates vectors $|V^{(N)}(A)\rangle$ on chains of length N .

Chapters 2 and 4 describe the algebraic core. Chapter 2 introduces word evaluation, MPV coefficients, gauge equivalence, injectivity, blocking, overlap, and the block-injective canonical-form data. Chapter 4 proves the single-block fundamental theorem by a purely algebraic argument through trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether.

The later chapters develop the spectral and canonical-form machinery needed for the full multi-block theorem. Chapter 5 studies positive maps and quantum channels. Chapters 8 and 9 supply Perron–Frobenius theory and overlap decay. Chapters 10 and 11 connect blocking, primitivity, irreducibility, and the normal/canonical-form reductions. Chapters 12 and 13 carry out block permutation and BNT theory; Chapter 14 proves the Fundamental Theorem.

Chapter 2

Matrix Product Vectors

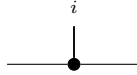
Fix a physical dimension d and a bond dimension D . This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor). A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \quad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

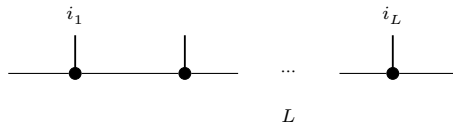


The black node denotes the tensor A , the horizontal legs are virtual, and the upper leg is the physical index i .

Definition 2.2 (Word evaluation). Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \dots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^\emptyset = \mathbb{1}_D$. In tensor-network notation,



in which each black node denotes the same local tensor A , the virtual legs remain open, and the physical legs record the word $(i_1, \dots, i_L)$.

Lemma 2.3 (Multiplicativity of word evaluation). *For any words w_1, w_2 , $A^{w_1 w_2} = A^{w_1} A^{w_2}$.*

Proof. Immediate from the definition of A^w . $\square$

Definition 2.4 (Matrix product vector). The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

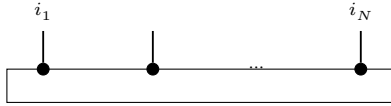
Equivalently, for a configuration $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}).$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components; the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w , we also write

$$c_w(A) := \text{tr}(A^w),$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction

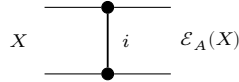


of N copies of the local tensor A , with the outer virtual legs closed by the trace.

Definition 2.5 (Transfer map). The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes A , the lower node denotes $A^\dagger$, and the physical index is summed over between them.

Remark 2.6. The transfer map is automatically positive: if $X \geq 0$ then $\mathcal{E}_A(X) \geq 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 5.

2.2 Periodic-chain aliases

Definition 2.7 (Periodic-chain tensor alias). For a fixed period m , a *periodic-chain tensor* is a site-dependent tensor family indexed by $\{0, \dots, m-1\}$, with local physical dimension d and bond dimension D .

Definition 2.8 (Periodic same-state and gauge-equivalence relations). For periodic-chain tensors A, B at fixed period m :

- equality of the induced periodic-chain states;
- cyclic gauge equivalence.

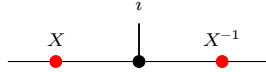
Lemma 2.9 (Equivalence structures for periodic relations). *Both periodic relations are equivalence relations (reflexive, symmetric, transitive).*

2.3 Gauge equivalence and same MPV

Definition 2.10 (Gauge equivalence). Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that

$$B^i = X A^i X^{-1} \quad \text{for all } i \in \{0, \dots, d-1\}.$$

In tensor-network notation,



the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index i , and the two red side nodes denote the gauge matrices acting on the virtual legs.

Definition 2.11 (Same MPV). Two tensors A and B (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size N .

Definition 2.12 (Same MPV — different bond dimensions). For tensors A and B of possibly different bond dimensions D_1 and D_2 , we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every N .

Remark 2.13. Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

Definition 2.14 (Proportional MPV). Two tensors A and B (of possibly different bond dimensions) *generate proportional MPV families* if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$.

Lemma 2.15 (Scaling of word evaluation). *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for any word w ,*

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where $|w|$ is the length of w .

Proof. Induction on w : the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ . $\square$

Definition 2.16 (Gauge-phase equivalence). Two tensors A and B are *gauge-phase equivalent* if there exist $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that

$$B^i = \zeta X A^i X^{-1} \quad \text{for all } i.$$

Remark 2.17. After spectral-radius normalization one may restrict ζ to a unit-modulus phase $e^{i\phi}$. We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

Lemma 2.18 (Gauge covariance of word evaluation). *If $B^i = X A^i X^{-1}$ for all i , then for every word w ,*

$$B^w = X A^w X^{-1}.$$

Proof. Induction on the word length, using $XX^{-1} = \mathbb{1}$. $\square$

Theorem 2.19 (Gauge invariance of MPVs). *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. Let $B^i = X A^i X^{-1}$. By Lemma 2.18, $B^w = X A^w X^{-1}$ for every word w . Then $\text{tr}(B^w) = \text{tr}(X A^w X^{-1}) = \text{tr}(A^w)$ by cyclicity of the trace. $\square$

Theorem 2.20 (Gauge-phase scaling of MPVs). *If*

$$B^i = \zeta X A^i X^{-1} \quad \text{for all } i,$$

then for every system size N and configuration σ one has

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. Combine gauge covariance of word evaluation with scalar scaling, then take traces. $\square$

2.4 Injectivity and normality

Definition 2.21 (Injectivity). A tensor A is *injective* if its matrices span the full matrix algebra: $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.22 (L -block injectivity). A tensor A is *L -block injective* if $\text{span}_{\mathbb{C}}\{A^{i_1} \dots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Definition 2.23 (Normal tensor). A tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0 -block injective.

Remark 2.24. The algebraic characterization of normality in Definition 2.23 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map $\mathcal{E}_A$ is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Theorem 10.50, Theorem 10.60, and Theorem 11.30. In particular, every subsequent use of “normal” here refers to this single notion; Definition 11.38 employs the spectral formulation, but it describes the same class of tensors.

2.5 Canonical form

Definition 2.25 (Block-injective canonical form). Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,

4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

This is the block-injective version of the canonical-form data; see [PGVWC07] and [CPGSV21].

Remark 2.26. Here we use the block-injective canonical form: each block is injective in the sense of Definition 2.21. This is stronger than the NT-block canonical form of [CPGSV21], where the blocks are only normal in the spectral sense. Chapter 11 treats the normal and periodic canonical forms needed for the general multi-block theory, whereas the present section isolates the stronger block-injective form used in the injective and block-injective arguments.

Definition 2.27 (Block-diagonal tensor from blocks). Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

Theorem 2.28 (MPV decomposition). *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(A_k)_\sigma.$$

Proof. For each physical index i , the block-diagonal tensor $A^i = \bigoplus_k \mu_k A_k^i$ is a block-diagonal matrix. By Lemma 2.15, the word evaluation of each block picks up a factor $\mu_k^{|w|}$, and the full word evaluation is block-diagonal with blocks $\mu_k^N A_k^w$. Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\text{tr}(A^w) = \sum_k \mu_k^N \text{tr}(A_k^w)$. $\square$

2.6 Blocking

Definition 2.29 (Blocked tensor). Given a tensor A and a blocking length L , the *L -blocked tensor* is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D , whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1, \dots, i_L)} = A^{i_1} A^{i_2} \dots A^{i_L}.$$

Lemma 2.30 (Blocked word evaluation). *If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L -tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\tilde{w}$ of length mL in the original alphabet and one has*

$$(A^{[L]})^w = A^{\tilde{w}}.$$

Proof. Induct on the blocked word w and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations. $\square$

Lemma 2.31 (Blocking preserves MPV coefficients). *For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L -tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that*

$$V^{(N)}(A^{[L]})_{\sigma} = V^{(NL)}(A)_{\tilde{\sigma}}.$$

Proof. Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.30). $\square$

Theorem 2.32 (Blocking preserves same MPV). *If A and B generate the same MPV family, then for any blocking length L , the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

Proof. By Lemma 2.31, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. $\square$

2.7 MPV overlap

Definition 2.33 (MPV as Hilbert-space vector). We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ -component $V^{(N)}(A)_{\sigma}$ as in Definition 2.4. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.34 (MPV inner product). The *MPV inner product* between two tensors A and B at system size N is

$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0, \dots, d-1\}^N} \overline{V^{(N)}(A)_{\sigma}} V^{(N)}(B)_{\sigma},$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.35 (MPV overlap). The *MPV overlap* between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_{\sigma} \overline{V^{(N)}(B)_{\sigma}}.$$

This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.36, it is the complex conjugate of the Hilbert-space inner product from Definition 2.34.

Lemma 2.36 (Overlap equals conjugate inner product). $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}$.

Proof. The overlap sums $V_{\sigma} \overline{W_{\sigma}}$ while the inner product sums $\overline{V_{\sigma}} W_{\sigma}$. Complex conjugation swaps these. $\square$

Lemma 2.37 (Overlap determined by positive-length MPV). *If $V^{(N)}(A)_{\sigma} = V^{(N)}(A')_{\sigma}$ and $V^{(N)}(B)_{\sigma} = V^{(N)}(B')_{\sigma}$ for all $N > 0$ and all σ , then $O_{AB}(N) = O_{A'B'}(N)$ for all positive N .*

Proof. Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical. $\square$

Chapter 3

Matrix Product Operators and Density Operators

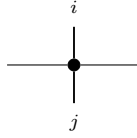
Fix a physical dimension d and a bond dimension D . An MPO tensor carries two physical indices, one ket index and one bra index, together with the virtual left and right indices.

3.1 MPO tensors

Definition 3.1 (MPO tensor). An *MPO tensor* is a family of matrices

$$\{M^{ij}\}_{i,j=0}^{d-1}, \quad M^{ij} \in M_D(\mathbb{C}),$$

indexed by a ket index i and a bra index j . Diagrammatically,



The black node denotes the tensor M , the horizontal legs are virtual, the upper leg is the ket index i , and the lower leg is the bra index j .

Definition 3.2 (Doubled-index MPS view). By combining the pair (i, j) into a single physical index in $\{0, \dots, d^2 - 1\}$, one obtains an MPS tensor with physical dimension d^2 and bond dimension D .

Definition 3.3 (Word evaluation). For words $u = (i_1, \dots, i_N)$ and $v = (j_1, \dots, j_N)$ in $\{0, \dots, d-1\}^N$, the *word evaluation* is

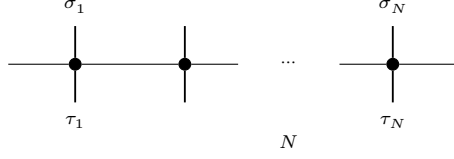
$$M^{u,v} := M^{i_1 j_1} M^{i_2 j_2} \dots M^{i_N j_N} \in M_D(\mathbb{C}).$$

The empty pair of words evaluates to $\mathbb{1}_D$, and word pairs of different lengths evaluate to 0.

Definition 3.4 (MPO operator family). The operator generated by M on N sites is the matrix $\rho^{(N)}(M)$ indexed by configurations $\sigma, \tau \in \{0, \dots, d-1\}^N$ and given by

$$\rho^{(N)}(M)_{\sigma, \tau} = \text{tr}(M^{\sigma, \tau}).$$

Diagrammatically,



each black node denotes the same local tensor M , and the matrix entry $\rho^{(N)}(M)_{\sigma,\tau}$ is obtained by closing the outer virtual legs cyclically and taking the resulting trace.

Definition 3.5 (Hermitian MPO tensor). An MPO tensor is *Hermitian* if

$$M^{ij} = (M^{ji})^\dagger \quad \text{for all } i, j \in \{0, \dots, d-1\}.$$

3.2 Transfer maps

Definition 3.6 (MPO transfer map). The *transfer map* associated to M is the linear map $\mathcal{E}_M : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_M(X) = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger.$$

Lemma 3.7 (Identification with the doubled-index MPS transfer map). *The transfer map of an MPO tensor agrees with the transfer map of its doubled-index MPS view.*

Proof. After identifying the single physical index of the doubled tensor with a pair (i, j) , the single sum defining the MPS transfer map is exactly the double sum defining $\mathcal{E}_M$. $\square$

Theorem 3.8 (Positivity of the MPO transfer map). *If $X \geq 0$, then $\mathcal{E}_M(X) \geq 0$.*

Proof. By Lemma 3.7, $\mathcal{E}_M$ is the transfer map of the doubled-index MPS tensor. The corresponding MPS transfer map is positive, so $\mathcal{E}_M(X) \geq 0$ whenever $X \geq 0$. $\square$

3.3 Zero correlation length

Definition 3.9 (Zero correlation length for MPO tensors). An MPO tensor M has *zero correlation length* when its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

This is the mixed-state analogue of the renormalization fixed-point condition for MPS tensors; see [CPGSV21, Definition 4.2].

Theorem 3.10 (MPO ZCL iff doubled-index MPS RFP). *An MPO tensor has zero correlation length if and only if its doubled-index MPS view is a renormalization fixed point.*

Proof. By Lemma 3.7, the transfer map of M agrees with the transfer map of its doubled-index MPS view. Thus $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$ holds exactly when the transfer map of M .toMPSTensor is idempotent, which is the RFP condition. $\square$

3.4 MPDOs and LPDOs

Definition 3.11 (MPDO). An MPO tensor is an *MPDO* if

$$\rho^{(N)}(M) \geq 0 \quad \text{for every } N \in \mathbb{N}.$$

This is the global positivity condition of [CPGSV17a, §4.1].

Definition 3.12 (LPDO). An MPO tensor is an *LPDO* if there exist an ancilla dimension d_K , an inner bond dimension D' , an identification of bond indices $e : \{0, \dots, D-1\} \simeq \{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ (equivalently, D and $(D')^2$ have the same cardinality), and matrices $A^{(i,k)} \in M_{D'}(\mathbb{C})$ such that

$$M^{ij} = \left(\sum_{k=0}^{d_K-1} A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e} \quad \text{for all } i, j,$$

where $\otimes$ denotes the Kronecker product, $\overline{(\cdot)}$ denotes entrywise complex conjugation, and $(\cdot)_{e,e}$ denotes reindexing along e from $\{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ back to $\{0, \dots, D-1\}$. This is the local purification form of [CPGSV17a, §4.3].

Theorem 3.13 (LPDOs are MPDOs). *Every LPDO tensor is an MPDO.*

Proof. By the mixed-product property $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, the N -site product decomposes as a sum of Kronecker products after transporting along the bond-space identification e :

$$M^{\sigma_1 \tau_1} \dots M^{\sigma_N \tau_N} = \left(\sum_{\kappa} P_{\kappa} \otimes \overline{Q_{\kappa}} \right) \Big|_e,$$

where $P_{\kappa} = A^{(\sigma_1, \kappa_1)} \dots A^{(\sigma_N, \kappa_N)}$ and $Q_{\kappa} = A^{(\tau_1, \kappa_1)} \dots A^{(\tau_N, \kappa_N)}$. Since the trace is invariant under the reindexing by e , we apply $\text{tr}(X \otimes Y) = \text{tr}(X) \text{tr}(Y)$ to obtain

$$\rho^{(N)}(M)_{\sigma, \tau} = \sum_{\kappa} \text{tr}(P_{\kappa}) \overline{\text{tr}(Q_{\kappa})} = \sum_{\kappa} \psi_{\kappa}(\sigma) \overline{\psi_{\kappa}(\tau)},$$

where $\psi_{\kappa}(\sigma) = \text{tr}(P_{\kappa})$. Hence $\rho^{(N)}(M) = \sum_{\kappa} |\psi_{\kappa}\rangle \langle \psi_{\kappa}| \geq 0$. $\square$

3.5 Purification RFP

Definition 3.14 (Purifying MPS tensor). Given a purifying family $A^{(i,k)} \in M_{D'}(\mathbb{C})$ for $i \in \{0, \dots, d-1\}$ and $k \in \{0, \dots, d_K-1\}$, the *purifying MPS tensor* is the MPS tensor with physical dimension $d \cdot d_K$ and bond dimension D' obtained by bundling the pair (i, k) into a single index.

Definition 3.15 (Purification RFP). An MPO tensor M has a *purification RFP* if it admits an LPDO witness whose purifying MPS tensor is a pure-state renormalization fixed point. This is [CPGSV21, Definition 4.3].

Remark 3.16 (PRFP does not imply MPO ZCL). The current purification-RFP condition is strictly weaker than MPO zero correlation length: there exists an MPO tensor with purification RFP whose transfer map is not idempotent. Equivalently, the implication from the Lean predicate `IsPRFP` to the predicate `IsZCL` fails.

Theorem 3.17 (Purification RFP implies LPDO). *A purification-RFP tensor is, in particular, an LPDO.*

Proof. The PRFP condition provides an LPDO witness by construction. $\square$

3.6 Pure-state recovery inside the MPO formalism

Definition 3.18 (MPDO renormalization fixed point). An MPO tensor M is an *MPDO renormalization fixed point* if its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

Theorem 3.19 (MPDO RFP iff MPO ZCL). *The provisional MPDO renormalization fixed-point condition is equivalent to the zero-correlation-length condition for MPO tensors.*

Proof. Both conditions assert $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$; they are definitionally identical. $\square$

Theorem 3.20 (MPDO RFP iff doubled-index MPS RFP). *The provisional MPDO RFP condition is equivalent to the pure-state RFP condition for the doubled-index MPS tensor.*

Proof. Unfold the MPDO RFP condition to its definitional equivalent, MPO ZCL, then apply Theorem 3.10. $\square$

Definition 3.21 (Diagonal pure-state embedding as an MPO). An MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ determines an MPO tensor by placing A^i on the diagonal in the bra and ket indices:

$$M^{ij} = \delta_{ij} A^i.$$

Equivalently, $M^{ii} = A^i$ and $M^{ij} = 0$ for $i \neq j$.

Theorem 3.22 (Transfer map of the diagonal pure-state embedding). *The transfer map of the diagonal MPO associated to A agrees with the original MPS transfer map:*

$$\mathcal{E}_{A^{\text{MPO}}} = \mathcal{E}_A.$$

Proof. In the double sum defining $\mathcal{E}_{A^{\text{MPO}}}$, all off-diagonal terms vanish because $M^{ij} = 0$ for $i \neq j$, while the diagonal terms are exactly $A^i X(A^i)^\dagger$. Thus only the sum over i remains, which is the defining formula for $\mathcal{E}_A$. $\square$

Theorem 3.23 (Pure-state recovery of the RFP condition). *For an MPS tensor A , its diagonal MPO embedding is MPDO-RFP if and only if A is an RFP:*

$$A^{\text{MPO}} \text{ is MPDO-RFP} \iff A \text{ is RFP}.$$

Proof. Both conditions assert idempotence of the corresponding transfer map. Theorem 3.22 identifies those transfer maps, so the two predicates are equivalent. $\square$

Theorem 3.24 (Pure-state recovery of zero correlation length). *For an MPS tensor A , the diagonal MPO embedding is MPDO-RFP if and only if A has zero correlation length.*

Proof. Combine Theorem 3.23 with the characterization of zero correlation length by idempotence of the MPS transfer map (Theorem 20.26). $\square$

3.7 Horizontal and Vertical Canonical Forms

Definition 3.25 (Horizontal canonical form for an MPO). An MPO tensor is in *horizontal canonical form* if its doubled-index MPS tensor generates the same MPV family as a block-diagonal tensor $\bigoplus_k \mu_k A_k$ whose blocks are injective, left-canonical, and have nonzero weights μ_k .

Definition 3.26 (Vertical canonical form for an MPO). An MPO tensor is in *vertical canonical form* if its diagonal MPS tensor $\{M^{ii}\}_{i=0}^{d-1}$ generates the same MPV family as a block-diagonal tensor obtained from a basis of normal tensors by repeating each block according to a multiplicity and weighting each copy by a positive scalar. This is the flattened form of the decomposition $\bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ from [CPGSV17a, §4.4].

Remark 3.27 (Finite-length span and selector criteria for the abstract biCF hypothesis). Let (A_k) be a family of block tensors, and fix a length L . Write A_k^w for the length- L word evaluation of A_k along a word w . The predicate `WordTupleSpanTop` says that the tuple-valued evaluations

$$w \mapsto (A_k^w)_k$$

span the full product algebra of block matrices. The predicate `HasBiCF` records the corresponding finite-length separation property: there exists L such that if

$$\sum_k \text{tr}(\Delta_k A_k^w) = 0 \quad \text{for every word } w \text{ of length } L,$$

then every block coefficient Δ_k vanishes. The theorem `hasBiCF_of_wordTupleSpanTop` proves that the spanning condition implies this separation property, by nondegeneracy of the product trace pairing. The packaging theorem `horizontalCFData_of_wordTupleSpanTop` then inserts the resulting biCF witness into `HorizontalCFData` once injectivity, left-canonicity, and nonzero weights are given.

The predicate `PropBlockInjective` isolates the finite-length content of Proposition IV.3: after some common blocking length, each block is block-injective, and after a second finite blocking length one can form word combinations which equal the identity on one chosen block and vanish on all others. Concatenating an injective prefix with such a selector suffix yields the product-algebra span condition above. This is formalized by `wordTupleSpanTop_of_propBlockInjective`, and the resulting biCF witness is packaged into horizontal canonical-form data by `horizontalCFData_of_propBlockI`.

Lemma 3.28 (Blockwise equality from agreement on the MPV family). *Let $\bigoplus_k \mu_k A_k$ be a block-diagonal tensor whose blocks are injective, left-canonical, weighted by nonzero scalars, and jointly block-injective in the sense that for some blocking length L the tuple of trace pairings against length- L block words is faithful across all blocks simultaneously (the block-injective canonical form, biCF). If two physical operators induce the same first-site action on every MPV generated by this tensor, then their first-site insertions agree on each block A_k .*

Proof. This is Lemma L from [CPGSV17a, Appendix A]. Expanding the hypothesis over a word of length $L + 1$ beginning with the fixed site and folding each block contribution into a trace pairing against insertedTensor $W A_k$ yields, upon taking the Y - Z difference, a tuple of block matrices whose trace pairings against every length- L block word vanish. The biCF hypothesis then forces each block difference to be zero, and dividing by the nonzero weight μ_k^{L+1} concludes the blockwise equality of the two first-site insertions. $\square$

Theorem 3.29 (Horizontal canonical form implies vertical canonical form). *Every MPDO in horizontal canonical form is also in vertical canonical form.*

Proof. Starting from the horizontal canonical-form decomposition, one uses the MPDO positivity condition to compare the first-site action of the doubled-index tensor on the diagonal sector. Lemma 3.28 then shows that the diagonal tensor splits along the same normal blocks, with the coefficients rearranged into positive scalar weights. Flattening those positive diagonal coefficients produces the required vertical canonical-form decomposition. $\square$

3.8 Fusion isometries

Following [CPGSV17a, §4.5], the current Lean development formalizes the transfer-map content of the fusion-isometry picture. For each blocked size $n \geq 1$, the doubled-index blocked tensor of an MPO has a blocked transfer map E_n acting on bond-space matrices, and the support algebra is modeled by a subspace through which E_n factors as a retract.

Definition 3.30 (Transfer-map fusion-isometry datum). A *transfer-map fusion-isometry datum* at blocked size n for an MPO tensor consists of a subspace $\mathcal{A}_n \subseteq M_D(\mathbb{C})$ together with linear maps

$$T_n : M_D(\mathbb{C}) \rightarrow \mathcal{A}_n, \quad S_n : \mathcal{A}_n \rightarrow M_D(\mathbb{C})$$

such that

$$T_n \circ S_n = \text{id}_{\mathcal{A}_n}, \quad S_n \circ T_n = E_n,$$

where E_n denotes the blocked transfer map.

Definition 3.31 (Fusion-isometry RFP predicate). An MPO tensor M satisfies the *fusion-isometry formulation* of the renormalization fixed-point condition when for every blocked size $n \geq 1$ there exists transfer-map fusion-isometry data for the blocked transfer map E_n .

Theorem 3.32 (Fusion-isometry data imply the provisional MPDO RFP condition). *The fusion-isometry formulation implies the current predicate.*

Proof. Apply the definition at blocked size $n = 1$. If $E_1 = S_1 \circ T_1$ and $T_1 \circ S_1 = \text{id}$, then

$$E_1^2 = S_1 T_1 S_1 T_1 = S_1 (T_1 S_1) T_1 = S_1 T_1 = E_1.$$

Since E_1 is the original transfer map, this is exactly . $\square$

Theorem 3.33 (Transfer-map fusion criterion for MPDO RFP). *The fusion-isometry formulation is equivalent to .*

Proof. The forward implication is Theorem 3.32. Conversely, if the transfer map E is idempotent, then every positive blocked transfer map equals E and therefore factors through its range. This range-factorization gives the required fusion-isometry data. $\square$

Corollary 3.34 (Pure-state recovery). *For the diagonal MPO associated to a pure MPS tensor, the fusion-isometry formulation reduces to the usual pure-state RFP condition.*

Proof. Combine Theorem 3.33 with the diagonal pure-state recovery theorem for the provisional MPO predicate. $\square$

3.9 Algebra structure

Definition 3.35 (Algebra-structure data for an MPDO). A family of support algebras $\{\mathcal{A}_n\}_{n \geq 0}$, one at every blocking size, equipped with blocking maps $m_n : \mathcal{A}_n \otimes \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$ arising from horizontal concatenation of blocks and unital inclusion maps $\iota_n : \mathcal{A}_n \rightarrow \mathcal{A}_{n+1}$ arising from appending a single site, together with a coherence condition relating m_n and ι_n .

Definition 3.36 (Algebra-structure RFP predicate (provisional)). An MPO tensor M satisfies the *algebra-structure formulation* of the renormalization fixed-point condition when there exist algebra-structure data compatible with M , in the sense of [CPGSV17a, Theorem IV.13 (ii)]. The compatibility relation between algebra-structure data and M is, in this provisional formulation, the trivial relation that holds for every pair; consequently the predicate above holds vacuously of every MPO tensor. The exact compatibility condition of [CPGSV17a, Theorem IV.13 (ii)] is to be substituted in place of the trivial relation in subsequent work.

Chapter 4

The Single-Block Fundamental Theorem

This chapter proves that if an injective MPS tensor A generates the same MPV family as another tensor B of the same bond dimension, then A and B are gauge equivalent. The proof is purely algebraic and does not use the later overlap or normal-form reductions. The trace-pairing maps introduced below are auxiliary devices attached to the chosen generating families $\{A^i\}$ and $\{B^i\}$; they serve only to construct the linear map that is later shown to be a gauge transform.

4.1 Trace pairing

Theorem 4.1 (Nondegeneracy of the trace pairing). *For $M \in M_D(\mathbb{C})$, $\text{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. Testing against the matrix units E_{ji} (the matrix with 1 in position (j, i) and 0 elsewhere) gives $\text{tr}(E_{ji}M) = M_{ij}$. Hence if $\text{tr}(MN) = 0$ for all N , then every entry of M vanishes. $\square$

Lemma 4.2 (Same MPV implies trace agreement). *If A and B generate the same MPV family, then $\text{tr}(A^w) = \text{tr}(B^w)$ for every word w .*

Proof. Viewing the word w as a configuration of length $|w|$, Definition 2.4 gives the corresponding MPV coefficient as the trace coefficient $c_w(A) = \text{tr}(A^w)$. The same-MPV hypothesis gives equality at every system size. $\square$

Definition 4.3 (Trace pairing map). For a tensor A , attach to the chosen family $\{A^i\}$ the auxiliary linear map $\Phi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ defined by

$$\Phi_A(M)_i = \text{tr}(M \cdot A^i).$$

Theorem 4.4 (Injectivity of the trace pairing map). *If A is injective, then $\ker \Phi_A = \{0\}$.*

Proof. If $\Phi_A(M) = 0$, then $\text{tr}(M \cdot A^i) = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, this gives $\text{tr}(MN) = 0$ for all N , so $M = 0$ by Theorem 4.1. $\square$

Theorem 4.5 (Nonvanishing trace on injective tensors). *If $D \geq 1$, A is injective, A and B generate the same MPV family, and $B^i = 0$ for all i , then we have a contradiction.*

Proof. If $B^i = 0$ for all i , then Lemma 4.2 gives $\text{tr}(A^w) = 0$ for every word w , in particular for all length-1 words. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, the trace functional vanishes identically, contradicting $\text{tr}(\mathbb{1}_D) = D \neq 0$. $\square$

4.2 Linear extension

Theorem 4.6 (Existence and uniqueness of the linear extension). *If A is injective and A, B generate the same MPV family, then there exists a unique linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ satisfying $T(A^i) = B^i$ for all i .*

Proof. Write $\ell_A : \mathbb{C}^d \rightarrow M_D(\mathbb{C})$ for the map that sends a coefficient vector c to $\sum_i c_i A^i$ (linear combination with the A^i). Applying Lemma 4.2 to all words of length 2 shows $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$. Since A is injective, ℓ_A is surjective, giving $\text{range}(\Phi_A) \subseteq \text{range}(\Phi_B)$. By Theorem 4.4, $\ker \Phi_A = \{0\}$, so rank-nullity gives $\dim(\text{range}(\Phi_A)) = \dim(M_D(\mathbb{C}))$. Hence $\dim(\text{range}(\Phi_B)) \geq \dim(M_D(\mathbb{C}))$, and rank-nullity again gives $\ker \Phi_B = \{0\}$. A left inverse g of Φ_B gives $T := g \circ \Phi_A$, and uniqueness follows because the $\{A^i\}$ span $M_D(\mathbb{C})$. $\square$

Theorem 4.7 (Multiplicativity of the linear extension). *If A is injective and A, B generate the same MPV family, and $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is the unique linear map satisfying $T(A^i) = B^i$ for all i , then $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. **Step 1** ($\Phi_B \circ T = \Phi_A$). Applying Lemma 4.2 to words of length 2 gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\Phi_B \circ T = \Phi_A$.

Step 2 ($T(A^i A^j) = B^i B^j$). Lemma 4.2 applied to the word (i, j, k) gives $\text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k)$ for all i, j, k . So $\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k$. The same rank-nullity argument used in the proof of Theorem 4.6 gives $\ker \Phi_B = \{0\}$. Hence $T(A^i A^j) = B^i B^j$.

Step 3 (bilinear extension). For fixed j , the maps $M \mapsto T(M \cdot A^j)$ and $M \mapsto T(M) \cdot B^j$ agree on the spanning set $\{A^i\}$, hence on all of $M_D(\mathbb{C})$. Applying this once more with j replaced by a general matrix gives full multiplicativity. $\square$

4.3 Simplicity and Skolem–Noether

Theorem 4.8 (Simplicity of $M_D(\mathbb{C})$). *A nonzero multiplicative $\mathbb{C}$ -linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. The kernel of a multiplicative linear map is a two-sided ideal. Since $M_D(\mathbb{C})$ is a simple ring (it has no nontrivial two-sided ideals), the kernel is either $\{0\}$ or $M_D(\mathbb{C})$. The latter would force the map to be zero, contradicting the hypothesis. Hence the map is injective, and surjectivity follows from finite-dimensionality. $\square$

Theorem 4.9 (Skolem–Noether). *Every $\mathbb{C}$ -algebra automorphism of $M_D(\mathbb{C})$ is inner: for any $f \in \text{Aut}_{\mathbb{C}\text{-alg}}(M_D(\mathbb{C}))$, there exists $X \in \text{GL}_D(\mathbb{C})$ such that $f(M) = XMX^{-1}$ for all M .*

Proof. Via the canonical identification $M_D(\mathbb{C}) \cong \text{End}_{\mathbb{C}}(\mathbb{C}^D)$, the automorphism f induces an automorphism of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. Every algebra automorphism of $\text{End}(V)$ is conjugation by an invertible linear map (this is a standard result in the theory of central simple algebras). Translating back to matrices gives the invertible X . $\square$

Lemma 4.10 (Promotion to algebra homomorphism). *If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a multiplicative surjective $\mathbb{C}$ -linear map, then $T(\mathbb{1}) = \mathbb{1}$. Equivalently, the same underlying function defines a $\mathbb{C}$ -algebra homomorphism.*

Proof. By surjectivity, there exists X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X) \cdot T(\mathbb{1}) = T(X \cdot \mathbb{1}) = T(X) = \mathbb{1}$, using multiplicativity and $T(X) = \mathbb{1}$. With $T(\mathbb{1}) = \mathbb{1}$ established, T preserves the unit and hence is a $\mathbb{C}$ -algebra homomorphism. $\square$

4.4 Proof of the single-block theorem

Theorem 4.11 (Single-block Fundamental Theorem). *Let A be an injective MPV tensor and let B be an MPV tensor of the same bond dimension. If A and B generate the same MPV family, then they are gauge equivalent: there exists $X \in \text{GL}_D(\mathbb{C})$ such that $B^i = XA^iX^{-1}$ for all i .*

Proof. By Theorem 4.6, there is a unique linear T with $T(A^i) = B^i$. By Theorem 4.7, T is multiplicative. T is nonzero: if $T = 0$ then $B^i = 0$ for all i , contradicting Theorem 4.5. By Theorem 4.8, T is bijective. By Lemma 4.10, T is a $\mathbb{C}$ -algebra automorphism. Theorem 4.9 gives an invertible X with $T(M) = XMX^{-1}$, and evaluating at $M = A^i$ yields $B^i = XA^iX^{-1}$. $\square$

4.5 Alternative fixed-length non-TI setup

The proof above treats a single translation-invariant tensor and equality of the full MPV family for all system sizes. The alternative algebraic proof of [MGRPG⁺18] instead keeps a fixed periodic chain length and allows the local tensors to vary from site to site. The first step is therefore a separate setup for finite non-translation-invariant chains.

Definition 4.12 (Periodic non-TI chain data). Fix a chain length n . A *non-translation-invariant periodic MPS chain* consists of sitewise tensors

$$A_k = \{A_k^i\}_{i=0}^{d-1}, \quad k = 0, \dots, n-1,$$

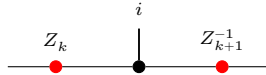
where each $A_k^i \in M_D(\mathbb{C})$. For a basis configuration $\sigma = (\sigma_0, \dots, \sigma_{n-1})$, its coefficient is

$$c_{A,\sigma} := \text{tr}(A_0^{\sigma_0} A_1^{\sigma_1} \dots A_{n-1}^{\sigma_{n-1}}).$$

Two such chains generate the *same chain state* if these coefficients agree for every σ . The chain is *injective* if each local tensor A_k is injective in the usual single-site sense. Finally, two chains are *cyclically gauge equivalent* if there are invertible bond matrices $Z_0, \dots, Z_{n-1}$ such that

$$B_k^i = Z_k A_k^i Z_{k+1}^{-1}$$

for every site k and physical index i , with indices understood modulo n so that $Z_n = Z_0$. Locally, this gauge relation is represented by



where the black node denotes the site tensor named in the surrounding formula and the two red side nodes denote the bond gauges on the neighboring virtual legs.

Lemma 4.13 (Cyclic gauge equivalence is an equivalence relation). *On fixed-length periodic chains, cyclic gauge equivalence is reflexive, symmetric, and transitive.*

Proof. Reflexivity uses the identity gauge on every bond. Symmetry inverts each bond gauge, and transitivity multiplies the gauges pointwise along the chain. $\square$

Chapter 5

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

5.1 Positive maps and channels

Definition 5.1 (Positive map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Remark 5.2. We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Definition 5.3 (Trace-preserving map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *trace-preserving* if $\text{tr}(E(X)) = \text{tr}(X)$ for all X .

Definition 5.4 (Completely positive map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger \quad \text{for all } X \in M_D(\mathbb{C}).$$

We use the Kraus characterisation because it matches the later Kadison–Schwarz arguments. By Choi’s theorem, this is equivalent to $E \otimes \text{id}_n$ being positive for all n .

Theorem 5.5 (CP maps are positive). *Every completely positive map is positive.*

Proof. If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. $\square$

Theorem 5.6 (Channels are positive). *Every quantum channel is positive.*

Proof. A quantum channel is completely positive by definition, so Theorem 5.5 applies. $\square$

Theorem 5.7 (Positive maps preserve Hermiticity). *If E is positive and $X = X^\dagger$, then $E(X) = E(X)^\dagger$.*

Proof. Decompose $X = X_+ - X_-$ into its positive and negative parts. Positivity makes $E(X_+)$ and $E(X_-)$ positive semidefinite, hence Hermitian, so their difference $E(X)$ is Hermitian. $\square$

Definition 5.8 (Quantum channel). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Definition 5.9 (Density matrices). The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \text{tr}(\rho) = 1\}.$$

Theorem 5.10 (Density matrices are compact). $\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).

Proof. The PSD cone is closed (preimage of the closed nonnegative cone under continuous quadratic forms), and if $\rho \geq 0$ with $\text{tr}(\rho) = 1$ then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are nonnegative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. $\square$

Theorem 5.11 (Density matrices are convex). $\mathcal{D}_D$ is convex.

Proof. Convex combination of PSD matrices is PSD, and trace is linear. $\square$

Theorem 5.12 (Density matrices are nonempty). For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.

Proof. $\frac{1}{D}\mathbb{1}_D$ is a density matrix. $\square$

Theorem 5.13 (Channels preserve density matrices). If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.

Proof. Complete positivity implies positivity (Theorem 5.5), so $E(\rho) \geq 0$; trace preservation gives $\text{tr}(E(\rho)) = \text{tr}(\rho) = 1$. $\square$

5.2 Irreducibility

Definition 5.14 (Orthogonal projection). A matrix $P \in M_D(\mathbb{C})$ is an *orthogonal projection* if $P = P^\dagger$ and $P^2 = P$.

Definition 5.15 (Irreducible map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Thm. 6.2, condition (1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 5.16 (Injectivity implies irreducibility). If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.

Proof. If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^i P = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M , so $P = 0$ or $P = \mathbb{1}$. $\square$

Remark 5.17 (Transfer map is CP). The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is completely positive: it is already written in Kraus form, with Kraus operators $\{A^i\}$.

Theorem 5.18 (Transfer map is a channel). If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).

Proof. Complete positivity is the observation of Remark 5.17. Trace preservation: $\text{tr}(\mathcal{E}_A(X)) = \sum_i \text{tr}(A^i X (A^i)^\dagger) = \text{tr}((\sum_i (A^i)^\dagger A^i)X) = \text{tr}(X)$ by cyclicity and the normalization hypothesis. $\square$

5.3 Cesàro fixed-point theorem

Definition 5.19 (Cesàro mean). The *Cesàro mean* of a linear map E at order N , applied to a starting matrix ρ_0 , is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 5.20 (Cesàro telescope). $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

Proof. Telescoping: $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^N E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. $\square$

Theorem 5.21 (Hermitian fixed-point decomposition). *Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_{\pm} \geq 0$, $\text{tr}(P_+ P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Prop. 6.8].*

Proof. Let Q be the support projection of P_+ . Trace preservation and positivity give $\text{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_{\pm}) = P_{\pm}$. $\square$

Theorem 5.22 (Cesàro fixed-point theorem). *Every quantum channel $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.*

Proof. Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 5.12). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ (by convexity and channel preservation). By compactness (Theorem 5.10), extract a convergent subsequence $\sigma_{N_k} \rightarrow \rho$. The telescope identity (Theorem 5.20) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N} \|E^N(\rho_0) - \rho_0\| \rightarrow 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Thm. 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. $\square$

5.4 Fixed-point projection

Definition 5.23 (Fixed-point projection). Given a linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho.$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 5.24 (Power decomposition). *If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,*

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. P is idempotent ($P^2 = P$), the fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. $\square$

5.5 Peripheral spectrum and primitivity

Definition 5.25 (Peripheral spectrum). The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T .

Definition 5.26 (Peripheral eigenvalues). The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Prop. 6.1], so these coincide with the eigenvalues of maximal modulus.

Note: The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1. For a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Theorem 5.27 (1 is a peripheral eigenvalue). *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then 1 is a peripheral eigenvalue of E .*

Proof. ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. □

Remark 5.28 (Peripheral eigenvalues lie on the unit circle). Every peripheral eigenvalue λ satisfies $|\lambda| = 1$. This is simply the defining condition in Definition 5.26.

Theorem 5.29 (Finiteness of peripheral eigenvalues). *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

Theorem 5.30 (Power-stable peripheral eigenvalues are roots of unity). *Let E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.*

Proof. Since μ^n is an eigenvalue for every n , the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. □

Theorem 5.31 (Peripheral powers imply root of unity). *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is closed under powers ($\mu \in \text{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \text{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. The closure hypothesis provides $\mu^n \in \text{peripheral}(E)$ for all n , which in particular means μ^n is an eigenvalue. Theorem 5.30 then gives $\mu^p = 1$ for some $p > 0$. □

5.5.1 Periodicity removal by powering

Remark 5.32. The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p -blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 5.33 (Common exponent for a finite set of roots of unity). *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. Take p to be the least common multiple of the exponents p_μ . □

Lemma 5.34 (Powering collapses peripheral eigenvalues). *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then $\text{peripheral}(E^p) = \{1\}$.*

Proof. The fixed point ρ gives $1 \in \text{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \text{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \text{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. $\square$

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 5.35 (Bi-canonical special case). When a Kraus map happens to be both unital and trace-preserving (the “bi-canonical” case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 5.41) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 5.31. This is the simplest argument, but it requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

5.5.2 Kadison–Schwarz input for peripheral and fixed-point arguments

Definition 5.36 (Kraus maps, adjoints, and normalizations). For operators $\{K_i\}_{i=0}^{d-1} \subseteq M_D(\mathbb{C})$, the associated *Kraus map* and *adjoint Kraus map* are

$$E(X) = \sum_i K_i X K_i^\dagger, \quad E^*(X) = \sum_i K_i^\dagger X K_i.$$

The Kraus family is *unital* when $\sum_i K_i K_i^\dagger = \mathbb{1}$ and *trace-preserving* when $\sum_i K_i^\dagger K_i = \mathbb{1}$.

Theorem 5.37 (Kadison–Schwarz inequality). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map. Then for every $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Eq. (5.2)].

Proof. Apply the unital completely positive map entrywise to the positive 2×2 block matrix $\begin{pmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{pmatrix}$ and take the Schur complement of the lower-right block. $\square$

Theorem 5.38 (Hilbert–Schmidt contraction). *If a Kraus map is both unital and trace-preserving, then*

$$\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. The Kadison–Schwarz gap is positive semidefinite by Theorem 5.37. Trace preservation then identifies the traces of its two terms. $\square$

Theorem 5.39 (KS gap decomposition). *For a unital Kraus map,*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_i R_i^\dagger R_i, \quad R_i := X K_i^\dagger - K_i^\dagger E(X).$$

Proof. Expand the right-hand side, distribute the products, and use $\sum_i K_i K_i^\dagger = \mathbb{1}$ to recover the Kadison–Schwarz gap. $\square$

Theorem 5.40 (KS equality implies Kraus commutation). *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then*

$$X K_i^\dagger = K_i^\dagger E(X) \quad \text{for all } i.$$

Proof. The Kadison–Schwarz gap is a sum of positive semidefinite terms $\sum_i R_i^\dagger R_i$. If the gap vanishes, each R_i must vanish. $\square$

Theorem 5.41 (KS equality for peripheral eigenvectors). *Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

Proof. The Kadison–Schwarz gap is positive semidefinite by Theorem 5.37. Trace preservation and the relation $E(X) = \mu X$ with $|\mu| = 1$ show that its trace is zero, hence the gap vanishes. $\square$

Theorem 5.42 (Left multiplicative identity from KS equality). *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then for every $Y \in M_D(\mathbb{C})$,*

$$E(X^\dagger Y) = E(X)^\dagger E(Y).$$

Proof. The commutation relation of Theorem 5.40 lets one move $X^\dagger$ past each Kraus operator inside the sum. $\square$

Definition 5.43 (Multiplicative domain). The *multiplicative domain* of a Kraus map E is the set of matrices that are simultaneously left- and right-multiplicative for E .

Theorem 5.44 (Multiplicative domain is a $*$ -subalgebra). *For a unital Kraus map, the multiplicative domain is a $*$ -subalgebra of $M_D(\mathbb{C})$.*

Proof. The multiplicative identities are stable under addition, multiplication, and adjoint, and they contain the identity matrix. $\square$

5.5.3 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV17a, Appendix A].

Lemma 5.45 (Peripheral closure via adjoint fixed point). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:*

1. *the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and*
2. *E is irreducible.*

Then $\text{peripheral}(E)$ is closed under powers: if $\mu \in \text{peripheral}(E)$ then $\mu^n \in \text{peripheral}(E)$ for every $n \in \mathbb{N}$.

Proof. Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By Kadison–Schwarz, $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\text{tr}(\rho G) = \text{tr}(\rho E(X^\dagger X)) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(E^*(\rho) X^\dagger X) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(\rho X^\dagger X) - |\mu|^2 \text{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 8.3). So X is invertible. By Theorem 5.40 (KS equality implies Kraus commutation), X commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 5.42) gives $E(X^n) = \mu^n X^n$, so μ^n is peripheral. $\square$

Theorem 5.46 (Roots of unity via adjoint fixed point). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital, and assume that the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point and that E is irreducible. Then every peripheral eigenvalue of E is a root of unity.*

Proof. Combine the closure-under-powers result (Lemma 5.45) with Theorem 5.31. $\square$

Remark 5.47. Lemma 5.45 and Theorem 5.46 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV17a, Appendix A], which uses a faithful invariant state rather than bi-canonicity.

Definition 5.48 (Channel period). The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\text{peripheral}(E)|$.

Definition 5.49 (Primitive map). A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\text{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Thm. 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 5.50 (Primitivity equals period one). *A linear map with a nonzero fixed point is primitive if and only if its period is 1.*

Proof. If $\text{peripheral}(E) = \{1\}$, then $|\text{peripheral}(E)| = 1$. Conversely, if $|\text{peripheral}(E)| = 1$, then since $1 \in \text{peripheral}(E)$ (Theorem 5.27), the only element is 1. $\square$

Theorem 5.51 (Primitive maps have a spectral gap on the complement). *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$. Let*

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the fixed-point projection associated to ρ . Assume that:

1. *E is primitive;*
2. *every eigenvalue μ of E satisfies $|\mu| \leq 1$;*
3. *whenever $E(X) = X$ and $\text{tr}(X) = 0$, one has $X = 0$.*

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then

$$E(X) = \nu X + P(X).$$

If $\text{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E ; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\text{tr}(X) \neq 0$, trace preservation and $\text{tr}(P(X)) = \text{tr}(X)$ give

$$\text{tr}(X) = \text{tr}(E(X)) = \nu \text{tr}(X) + \text{tr}(P(X)) = \nu \text{tr}(X) + \text{tr}(X),$$

so $\nu \text{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. $\square$

5.6 Representations

The following definitions support the Choi–Jamiołkowski isomorphism (Theorem 5.57) and the representation theorems of [Wol12, Ch. 2].

Definition 5.52 (Partial traces). Let $X \in M_{d,d'}(\mathbb{C})$ be a bipartite matrix indexed by $(\text{Fin } d \times \text{Fin } d')^2$. The *left partial trace* $\text{tr}_A(X)$ and *right partial trace* $\text{tr}_B(X)$ are the $d' \times d'$ and $d \times d$ matrices defined by

$$(\text{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \quad (\text{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

Definition 5.53 (Maximally entangled state). The *maximally entangled state* on $\mathbb{C}^d \otimes \mathbb{C}^d$ is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

Concretely, $(|\Omega\rangle\langle\Omega|)_{(i_1,i_2),(j_1,j_2)} = \frac{1}{d} \delta_{i_1 j_1} \delta_{i_2 j_2}$ and $\text{tr}(|\Omega\rangle\langle\Omega|) = 1$.

Definition 5.54 (Tensor product of a map with identity). Given a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the *tensor extension* $T \otimes \text{id}$ acts on bipartite matrices $X \in M_{D \times D}(\mathbb{C})$ by applying T to each “slice”:

$$(T \otimes \text{id})(X)_{(i_1,i_2),(j_1,j_2)} = (T(X^{(i_2,j_2)}))_{i_1 j_1},$$

where $X_{ab}^{(i_2,j_2)} = X_{(a,i_2),(b,j_2)}$ is the bipartite slice.

5.7 Choi–Jamiołkowski isomorphism

Definition 5.55 (Choi matrix). The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is

$$\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|) \in M_{D \times D}(\mathbb{C}).$$

Concretely, $\tau_{(i_1,i_2),(j_1,j_2)} = \frac{1}{D} (T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiołkowski convention of [Wol12, Prop. 2.1].

Theorem 5.56 (Choi matrix of the identity map). *The identity channel has Choi matrix $|\Omega\rangle\langle\Omega|$.*

Proof. Apply the identity map in the definition of the Choi matrix. $\square$

Theorem 5.57 (Prop. 2.1: CP iff Choi matrix is PSD). *A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is completely positive (in the Kraus sense) if and only if its Choi matrix $\tau \geq 0$.*

Proof. ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a,b)} = c K_i(a,b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a,b) = v_m(a,b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. $\square$

Theorem 5.58 (Prop. 2.1: TP implies partial trace condition). *If T is trace-preserving, then $\text{tr}_A(\tau) = \frac{1}{D} \mathbb{1}_D$.*

Proof. $(\text{tr}_A(\tau))_{ij} = \text{tr}(T(\Omega^{(ij)})) = \text{tr}(\Omega^{(ij)})$ by trace preservation, where $\Omega^{(ij)} = \frac{1}{D} E_{ij}$ is the (i,j) -slice of $|\Omega\rangle\langle\Omega|$. Taking the trace gives $\frac{1}{D} \delta_{ij}$. $\square$

Theorem 5.59 (Prop. 2.1: Hermiticity correspondence). *The Choi matrix τ is Hermitian if and only if $T(B^\dagger) = T(B)^\dagger$ for all $B \in M_D(\mathbb{C})$.*

Proof. The Choi matrix entries satisfy $\tau_{(a,i),(b,j)} = \frac{1}{D}(T(E_{ij}))_{ab}$. Hermiticity of τ says $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$, which unravels to $T(E_{ji})_{ba} = \overline{T(E_{ij})_{ab}}$, i.e., $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$. By linearity this extends to all B . $\square$

Theorem 5.60 (Prop. 2.1: Trace of Choi matrix for TP map). *If T is trace-preserving, then $\text{tr}(\tau) = 1$.*

Proof. $\text{tr}(\tau) = \text{tr}(\text{tr}_A(\tau)) = \text{tr}(\frac{1}{D}\mathbb{1}_D) = 1$. $\square$

5.8 Kraus representation theorem

Theorem 5.61 (Thm. 2.1: TP implies Kraus normalization). *If $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving, then $\sum_i K_i^\dagger K_i = \mathbb{1}$.*

Proof. For any N , $\text{tr}((\sum_i K_i^\dagger K_i) N) = \sum_i \text{tr}(K_i^\dagger K_i N) = \sum_i \text{tr}(K_i N K_i^\dagger) = \text{tr}(T(N)) = \text{tr}(N)$. Non-degeneracy of the trace pairing forces $\sum_i K_i^\dagger K_i = \mathbb{1}$. $\square$

Theorem 5.62 (Thm. 2.1: Kraus normalization implies TP). *If $\sum_i K_i^\dagger K_i = \mathbb{1}$, then $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving.*

Proof. $\text{tr}(T(X)) = \sum_i \text{tr}(K_i X K_i^\dagger) = \sum_i \text{tr}(K_i^\dagger K_i X) = \text{tr}((\sum_i K_i^\dagger K_i) X) = \text{tr}(X)$. $\square$

Theorem 5.63 (Thm. 2.1: Unitality implies Kraus unital normalization). *If $T(X) = \sum_i K_i X K_i^\dagger$ satisfies $T(\mathbb{1}) = \mathbb{1}$, then*

$$\sum_i K_i K_i^\dagger = \mathbb{1}.$$

Proof. Evaluate the Kraus formula at the identity matrix. $\square$

Theorem 5.64 (Thm. 2.1: Unitary freedom in Kraus operators). *Let U be a unitary $r \times r$ matrix ($U^\dagger U = \mathbb{1}$) and suppose $K_j = \sum_\ell U_{j\ell} \tilde{K}_\ell$. Then $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same Kraus map:*

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger \quad \forall X \in M_D(\mathbb{C}).$$

Proof. Substitute the combination formula, expand the triple sum over j, ℓ, ℓ' , swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. $\square$

Theorem 5.65 (Isometric mixing preserves the Kraus map). *Let W be an isometry between two Kraus index spaces, so $W^\dagger W = \mathbb{1}$. If*

$$K_j = \sum_\ell W_{j\ell} \tilde{K}_\ell,$$

then the Kraus families $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same completely positive map.

Proof. Expand the Kraus sums, interchange the finite summations, and use $W^\dagger W = \mathbb{1}$ to collapse the coefficient matrix. $\square$

Theorem 5.66 (Thm. 2.1: Orthonormal Kraus transition is unitary). *Let $\{K_j\}_{j=0}^{r-1}$ and $\{K'_j\}_{j=0}^{r-1}$ be two Hilbert–Schmidt orthonormal Kraus families, and suppose*

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell} K'_\ell.$$

Then the transition matrix U is unitary: $U^\dagger U = \mathbb{1}_r$.

Proof. Expand $\text{tr}(K_j^\dagger K_i)$ using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity $UU^\dagger = \mathbb{1}_r$, hence also $U^\dagger U = \mathbb{1}_r$. $\square$

Theorem 5.67 (Thm. 2.1: Dual map equality from primal map equality). *If two Kraus families satisfy $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X , then $\sum_\alpha B_\alpha^\dagger Y B_\alpha = \sum_j A_j^\dagger Y A_j$ for all Y .*

Proof. Use the trace pairing: for all X , $\text{tr}(X^\dagger (\sum B_\alpha^\dagger Y B_\alpha)) = \text{tr}((\sum B_\alpha X^\dagger B_\alpha^\dagger) Y) = \text{tr}((\sum A_j X^\dagger A_j^\dagger) Y) = \text{tr}(X^\dagger (\sum A_j^\dagger Y A_j))$. Nondegeneracy of the trace pairing gives the result. $\square$

Theorem 5.68 (Thm. 2.1: Equal Stinespring Gramians). *If two Kraus families define the same CPM, then $\sum_\alpha B_\alpha^\dagger B_\alpha = \sum_j A_j^\dagger A_j$.*

Proof. Specialise Theorem 5.67 at $Y = \mathbb{1}$. $\square$

Theorem 5.69 (Thm. 2.1 item 4: Rectangular Kraus freedom (necessary direction)). *If $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X and $r_2 \leq r_1$ (where $\{B_\alpha\}_{\alpha=0}^{r_1-1}$ is the larger family and $\{A_j\}_{j=0}^{r_2-1}$ the smaller), then there exists a rectangular isometry V ($r_1 \times r_2$, $V^\dagger V = \mathbb{1}_{r_2}$) such that $B_\alpha = \sum_j V_{\alpha j} A_j$.*

Proof. Pad the smaller family to size r_1 , establish Gram matrix equality $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$ from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry V with $V^\dagger V = \mathbb{1}_{r_2}$. $\square$

Theorem 5.70 (Thm. 2.1 item 4: Rectangular Kraus freedom (general index variant)). *Variant of Theorem 5.69 with general finite index types ι_1, ι_2 in place of $\text{Fin } r_1, \text{Fin } r_2$.*

Proof. Reindex via equivalences to Fin and apply Theorem 5.69. $\square$

Theorem 5.71 (Wolf Thm. 2.18: Isometric freedom of Kraus representations). *Let $\{B_\alpha\}_{\alpha \in \iota_1}$ and $\{A_j\}_{j \in \iota_2}$ be finite Kraus families with $|\iota_2| \leq |\iota_1|$. Then the following are equivalent:*

1. $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.
2. There exists a matrix $V = (V_{\alpha j})$ with $V^\dagger V = \mathbb{1}$ such that $B_\alpha = \sum_j V_{\alpha j} A_j$ for every α .

Proof. The implication (1) $\Rightarrow$ (2) is Theorem 5.70. Conversely, expand $\sum_\alpha B_\alpha X B_\alpha^\dagger$ using $B_\alpha = \sum_j V_{\alpha j} A_j$, interchange the finite sums, and use $V^\dagger V = \mathbb{1}$ to collapse the coefficient matrix to the diagonal. $\square$

Theorem 5.72 (Wolf Thm. 2.18: Unitary freedom of Kraus representations). *Let $\{B_\alpha\}_{\alpha \in \iota}$ and $\{A_j\}_{j \in \iota}$ be two Kraus families with the same finite index set. Then the following are equivalent:*

1. $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.

2. There exists a unitary matrix $U = (U_{\alpha j})$ such that $B_\alpha = \sum_j U_{\alpha j} A_j$ for every α .

Proof. Apply Theorem 5.71 with $|\iota| = |\iota|$. In the square case an isometry matrix is unitary, and the converse is immediate. $\square$

5.9 Stinespring dilation

Definition 5.73 (Stinespring isometry). Given Kraus operators $\{K_j\}_{j=0}^{r-1}$ with $K_j \in M_D(\mathbb{C})$, the *Stinespring isometry* $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$ is the $(D \cdot r) \times D$ matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements $V = \sum_j K_j \otimes |j\rangle$.

Lemma 5.74 (Entry formula for the Stinespring isometry). For indices $i, k \in \{0, \dots, D-1\}$ and $j \in \{0, \dots, r-1\}$,

$$V_{(i,j),k} = (K_j)_{ik}.$$

Theorem 5.75 (Stinespring Gram identity). The Stinespring isometry satisfies

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

Theorem 5.76 (Thm. 2.2: Stinespring dual representation). For Kraus operators $\{K_j\}$ and $A \in M_D(\mathbb{C})$,

$$V^\dagger (A \otimes \mathbb{1}_r) V = \sum_j K_j^\dagger A K_j = T^*(A).$$

This is the Stinespring representation of the dual (Heisenberg picture) map T^* .

Proof. Compute $(V^\dagger (A \otimes \mathbb{1}_r) V)_{ab}$ by summing over the product index (i, j) ; the δ_{jl} from $\mathbb{1}_r$ collapses the sum over l , giving $\sum_j K_j^\dagger A K_j$. $\square$

Theorem 5.77 (Thm. 2.2: Isometry condition iff TP). $V^\dagger V = \mathbb{1}_D$ if and only if $\sum_j K_j^\dagger K_j = \mathbb{1}_D$. In particular, V is an isometry precisely when the Kraus map is trace-preserving.

Proof. $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}} V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}} (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$. $\square$

Theorem 5.78 (Thm. 2.2: Stinespring Schrödinger representation). The Kraus map $T(\rho) = \sum_j K_j \rho K_j^\dagger$ equals the partial trace over the dilation space:

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger).$$

This is the Schrödinger-picture Stinespring representation.

Proof. Expand $(V \rho V^\dagger)_{(i,k),(j,k)}$ and sum over k to recover $\sum_k K_k \rho K_k^\dagger$. $\square$

Theorem 5.79 (Existence of a Stinespring dilation). Every completely positive map admits an ancilla dimension r , Kraus operators $\{K_j\}_{j=0}^{r-1}$, and the concrete representation $\pi(A) = A \otimes \mathbb{1}_r$ such that

$$E(A) = V^\dagger \pi(A) V.$$

Proof. Choose a Kraus representation of E and apply the explicit Heisenberg-picture Stinespring formula. $\square$

Theorem 5.80 (CPTP maps admit isometric Stinespring dilations). *Every quantum channel admits a Stinespring dilation whose matrix V is an isometry, equivalently $V^\dagger V = \mathbb{1}$.*

Proof. Choose Kraus operators for the channel and use trace preservation to obtain the Stinespring isometry condition $V^\dagger V = \mathbb{1}$. $\square$

5.10 POVMs and Naimark dilation

Definition 5.81 (Positive operator-valued measure). A *positive operator-valued measure* with n outcomes on $\mathbb{C}^D$ is a family $\{E_i\}_{i=1}^n$ of positive semidefinite operators on $\mathbb{C}^D$ satisfying the resolution of identity

$$\sum_{i=1}^n E_i = \mathbb{1}_D.$$

Definition 5.82 (Naimark Kraus square roots). For a POVM $\{E_i\}$, each effect $E_i \geq 0$ admits a square-root factorisation $E_i = M_i^\dagger M_i$ with $M_i \in M_D(\mathbb{C})$. The operators M_i are the *Naimark Kraus square roots*.

Theorem 5.83 (Naimark square roots satisfy the Kraus normalization). *For a POVM $\{E_i\}$ with square roots $E_i = M_i^\dagger M_i$, one has*

$$\sum_i M_i^\dagger M_i = \mathbb{1}.$$

Proof. Sum the identities $E_i = M_i^\dagger M_i$ and use the defining resolution of the identity for the POVM. $\square$

Definition 5.84 (Naimark isometry). The *Naimark isometry* $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$ of a POVM is the Stinespring-style assembly

$$V = \sum_{i=1}^n M_i \otimes |i\rangle, \quad V_{(a,i),k} = (M_i)_{a,k}.$$

Definition 5.85 (Naimark projective measurement). The *Naimark projectors* on $\mathbb{C}^D \otimes \mathbb{C}^n$ are

$$P_i = \mathbb{1}_D \otimes |i\rangle\langle i|, \quad (P_i)_{(a,b),(c,d)} = \delta_{a,c} \delta_{b,i} \delta_{d,i}.$$

Theorem 5.86 (Naimark isometry condition). *The Naimark isometry satisfies $V^\dagger V = \mathbb{1}_D$.*

Proof. The Stinespring Gram identity gives $V^\dagger V = \sum_i M_i^\dagger M_i = \sum_i E_i = \mathbb{1}_D$ by the resolution of identity. $\square$

Theorem 5.87 (Naimark projectors form a projective measurement). *The Naimark projectors satisfy $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_i P_j = 0$ for $i \neq j$, and $\sum_i P_i = \mathbb{1}_{\mathbb{C}^D \otimes \mathbb{C}^n}$.*

Proof. Each identity follows from direct entrywise computation using the delta-function form of $(P_i)_{(a,b),(c,d)}$. $\square$

Theorem 5.88 (Thm. 2.6: Naimark dilation). *Every POVM $\{E_i\}_{i=1}^n$ arises as a projective measurement on a dilation: for the isometry V and projectors P_i above,*

$$E_i = V^\dagger P_i V.$$

Proof. Computing entrywise, $(V^\dagger P_i V)_{k,\ell} = \sum_a \overline{(M_i)_{a,k}} (M_i)_{a,\ell} = (M_i^\dagger M_i)_{k,\ell} = (E_i)_{k,\ell}$. $\square$

Theorem 5.89 (Existential Naimark dilation). *For every POVM $\{E_i\}$ on $\mathbb{C}^D$ there exist a dilation dimension r , an isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$, and a projective measurement $\{P_i\}$ on the dilation satisfying $E_i = V^\dagger P_i V$.*

Proof. Take $r = n$ and the explicit witnesses $V = \sum_i M_i \otimes |i\rangle$ and $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. $\square$

Definition 5.90 (Packaged Naimark dilation). A *Naimark dilation* of a POVM $\{E_i\}$ consists of an isometry V into a larger Hilbert space together with a projective measurement $\{P_i\}$ there such that $E_i = V^\dagger P_i V$ for all i .

Theorem 5.91 (Canonical dilation as a packaged Naimark dilation). *The explicit isometry $V = \sum_i M_i \otimes |i\rangle$ and projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ satisfy the packaged axioms of a Naimark dilation.*

Proof. Combine the previously proved identities $V^\dagger V = \mathbb{1}_D$, $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_i P_j = 0$ for $i \neq j$, $\sum_i P_i = \mathbb{1}$, and $V^\dagger P_i V = E_i$. $\square$

Theorem 5.92 (Concrete uniqueness for canonical projectors). *Let $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$ satisfy $V^\dagger P_i V = E_i$ for the canonical projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. Then there exists an isometry W on the dilated space such that $V = W V_0$, where V_0 is the canonical Naimark isometry of $\{E_i\}$.*

Proof. For each outcome i , the i -th block of V has Gram matrix E_i . Comparing with the canonical square root M_i gives $V_i^\dagger V_i = M_i^\dagger M_i$, hence $V_i = U_i M_i$ for a unitary U_i on $\mathbb{C}^D$. Assemble the U_i block-diagonally to obtain an isometry $W = \bigoplus_i U_i$ on $\mathbb{C}^D \otimes \mathbb{C}^n$, and then $V = W \sum_i M_i \otimes |i\rangle = W V_0$. $\square$

Definition 5.93 (POVM from a PSD resolution of identity on a dilation). Given an isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^{d'}$ with $V^\dagger V = \mathbb{1}_D$ and a family $\{P_i\}_{i=1}^n$ of positive semidefinite operators on $\mathbb{C}^{d'}$ summing to the identity, the pulled-back operators

$$E_i := V^\dagger P_i V$$

form a POVM. In particular, any projective measurement on the dilation (a special case of a PSD resolution of identity) pulls back to a POVM.

Definition 5.94 (Quantum instrument). A *quantum instrument* with n outcomes is a family $\{\Phi_i\}_{i=1}^n$ of completely positive maps on $M_D(\mathbb{C})$ whose sum $\sum_i \Phi_i$ is trace-preserving.

Definition 5.95 (Instrument operations). Associated to an instrument are the total channel $\sum_i \Phi_i$, the unnormalized update $\rho \mapsto \Phi_i(\rho)$ for each outcome i , the outcome probability $p_i(\rho) = \text{tr}(\Phi_i(\rho))$, and the normalized posterior state $\Phi_i(\rho)/p_i(\rho)$ whenever $p_i(\rho) \neq 0$.

Theorem 5.96 (Instrument total map is a channel). *The total map $\sum_i \Phi_i$ of an instrument is a quantum channel.*

Proof. Complete positivity of the sum follows from closure of CP maps under finite addition; trace preservation is built into the definition. $\square$

Theorem 5.97 (Conservation of probability for instruments). *For every state $\rho \in M_D(\mathbb{C})$, the outcome probabilities $p_i(\rho) := \text{tr}(\Phi_i(\rho))$ sum to $\text{tr}(\rho)$.*

Proof. Linearity of trace and trace preservation of $\sum_i \Phi_i$ give $\sum_i \text{tr}(\Phi_i(\rho)) = \text{tr}((\sum_i \Phi_i)(\rho)) = \text{tr}(\rho)$. $\square$

Theorem 5.98 (Non-negativity of instrument probabilities). *If $\rho \in M_D(\mathbb{C})$ is positive semidefinite, then each outcome probability $p_i(\rho) = \text{tr}(\Phi_i(\rho))$ is a non-negative real number.*

Proof. Each Φ_i is completely positive, hence positive, so $\Phi_i(\rho)$ is positive semidefinite and its trace is a non-negative real. $\square$

5.10.1 Trace-pairing expansion in transfer-matrix form

Definition 5.99 (Trace-self-dual basis). A basis $\{\sigma_i\}_i$ of $M_D(\mathbb{C})$ is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \text{tr}(\sigma_i X) \sigma_i.$$

Definition 5.100 (Trace-pairing coefficients). For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a trace-self-dual basis $\{\sigma_i\}$, define coefficients

$$t_{ij} := \text{tr}(\sigma_i T(\sigma_j)).$$

Theorem 5.101 (Trace-pairing expansion of a linear map). *If $\{\sigma_i\}$ is trace-self-dual, then every linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits the expansion*

$$T(\rho) = \sum_i \sum_j t_{ij} \text{tr}(\sigma_j \rho) \sigma_i, \quad t_{ij} = \text{tr}(\sigma_i T(\sigma_j)).$$

Proof. Expand ρ and each $T(\sigma_j)$ in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula. $\square$

5.11 Determinant of a quantum channel

Definition 5.102 (Channel determinant). For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the *channel determinant* $\det T$ is the determinant of the matrix of T with respect to the standard matrix-unit basis of $M_D(\mathbb{C})$. This is the quantity studied in [Wol12, §6.1.1].

Remark 5.103 (Channel determinant as an operator determinant). The channel determinant agrees with the ordinary determinant of the linear endomorphism $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$.

Definition 5.104 (Unitary channel). For a unitary matrix $U \in \mathcal{U}(D)$, the *unitary channel* is $T(\rho) = U\rho U^\dagger$. It is automatically a quantum channel.

Theorem 5.105 (Unitary channels have determinant one). *If $T(\rho) = U\rho U^\dagger$ is a unitary channel, then $\det T = 1$ and hence $|\det T| = 1$.*

Proof. Vectorization identifies T with a Kronecker product $\bar{U} \otimes U$, whose determinant is $\det \bar{U}^D (\det U)^D = 1$. $\square$

Theorem 5.106 (Thm. 6.1(1): Determinant bound for positive TP maps). *For any positive trace-preserving map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, $|\det T| \leq 1$. This is [Wol12, Thm. 6.1(1)].*

Proof. Every eigenvalue μ of T satisfies $|\mu| \leq 1$ (positivity and trace preservation force spectral radius ≤ 1). Since $\det T$ is the product of the eigenvalues (counted with algebraic multiplicity), the bound $|\det T| = \prod_i |\mu_i| \leq 1$ follows by induction. $\square$

Theorem 5.107 (Thm. 6.1(2): Determinant one iff unitary channel). *For a CPTP map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$,*

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

This is [Wol12, Thm. 6.1(2)].

Proof. Reverse direction is Theorem 5.105. Forward direction: $|\det T| = 1$ together with $|\mu| \leq 1$ for every eigenvalue μ (from trace preservation and positivity) forces every eigenvalue to satisfy $|\mu| = 1$. Hence T is bijective. A bijective CPTP map is a $*$ -automorphism: the Heisenberg dual $T^*(Y) = \sum_i K_i^\dagger Y K_i$ is unital (since T is TP), so the Kadison–Schwarz inequality $T^*(Y^\dagger Y) \geq T^*(Y)^\dagger T^*(Y)$ becomes an equality for every Y . Hence T^* , and therefore T , preserves products. By the Skolem–Noether theorem, every $*$ -automorphism of $M_D(\mathbb{C})$ has the form $T(\rho) = U\rho U^\dagger$ for some unitary U . Kraus freedom then gives $K_i = c_i U$ with $\sum_i |c_i|^2 = 1$. $\square$

5.12 Fixed-point algebra

Definition 5.108 (Kraus fixed points). The *fixed-point set* of a Kraus map $E(X) = \sum_i K_i X K_i^\dagger$ is $\text{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

Theorem 5.109 (Fixed points lie in the multiplicative domain). *Let E be a trace-preserving Kraus map whose Schrödinger-picture map has a positive definite fixed point $\rho > 0$. Then every adjoint fixed point belongs to the multiplicative domain of the adjoint Kraus family.*

Proof. The Kadison–Schwarz equality for fixed points forces both one-sided multiplicative identities, so the fixed point lies in the full multiplicative domain. $\square$

Theorem 5.110 (Thm. 6.12: Fixed points form a $*$ -subalgebra). *Let E be a unital Kraus map whose adjoint map E^* has a positive definite fixed point $\rho > 0$. Then $\text{Fix}(E)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$. This is [Wol12, Thm. 6.12].*

Proof. Closure under addition and scalar multiplication is immediate from linearity. Closure under $\dagger$: if $E(X) = X$ then $E(X^\dagger) = X^\dagger$ because fixed points are stable under adjoint. Closure under multiplication: if $E(X) = X$ and $E(Y) = Y$, the weighted Kadison–Schwarz argument gives $E(X^\dagger X) = E(X)^\dagger E(X)$ and, applied to $X^\dagger$, also $E(X X^\dagger) = E(X) E(X)^\dagger$. Thus X lies in the multiplicative domain, and Theorem 5.42 yields $E(XY) = E(X)E(Y) = XY$. $\square$

Remark 5.111. Theorem 5.109 records the trace-preserving / Heisenberg-picture counterpart of the same multiplicative-domain argument.

Theorem 5.112 (Projected elements lie in the multiplicative domain). *Let E be a unital Kraus map whose adjoint map has a positive definite fixed point, and suppose $E^2 = E$. Then $E(X)$ lies in the multiplicative domain of E for every $X \in M_D(\mathbb{C})$.*

Proof. Idempotence gives $E(E(X)) = E(X)$, so $E(X)$ is a fixed point. By the argument in Theorem 5.110, the Kadison–Schwarz gap for the fixed point $E(X)$ vanishes, hence

$$E(E(X)^\dagger E(X)) = E(X)^\dagger E(X).$$

Fixed points are closed under $\dagger$, so the same argument applied to $E(X)^\dagger$ gives

$$E(E(X)E(X)^\dagger) = E(X)E(X)^\dagger.$$

Therefore Theorem 7.28 places $E(X)$ in the multiplicative domain. $\square$

Theorem 5.113 (Choi–Effros left absorption). *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(E(X)E(Y)).$$

Proof. The previous theorem places $E(X)$ in the multiplicative domain, so $E(E(X)Y) = E(E(X))E(Y) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. $\square$

Theorem 5.114 (Choi–Effros right absorption). *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(XE(Y)) = E(E(X)E(Y)).$$

Proof. The previous theorem places $E(Y)$ in the multiplicative domain, so $E(XE(Y)) = E(X)E(E(Y)) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. $\square$

Theorem 5.115 (Choi–Effros symmetric absorption). *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(XE(Y)).$$

Proof. Both sides equal $E(E(X)E(Y))$ by the two preceding theorems. $\square$

Theorem 5.116 (Choi–Effros projected product). *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)E(Y)) = E(X)E(Y).$$

Proof. The previous theorem places $E(X)$ and $E(Y)$ in the multiplicative domain. Therefore multiplicativity gives

$$E(E(X)E(Y)) = E(E(X))E(E(Y)).$$

Since E is idempotent under the standing hypotheses, $E(E(X)) = E(X)$ and $E(E(Y)) = E(Y)$, so

$$E(E(X)E(Y)) = E(X)E(Y).$$

$\square$

Definition 5.117 (Adjoint fixed points). The *adjoint fixed-point set* $\text{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$, where $E^*(X) = \sum_i K_i^\dagger X K_i$.

Theorem 5.118 (Thm. 6.12 in the Heisenberg picture). *If the Kraus map is trace-preserving and has a positive definite fixed point $\rho > 0$ with $E(\rho) = \rho$, then $\text{Fix}(E^*)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$.*

Proof. Apply Theorem 5.110 to the adjoint family $\{K_i^\dagger\}$. $\square$

Definition 5.119 (Kraus commutant). The *Kraus commutant* is $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0 \ \forall i\}$. It is a $*$ -subalgebra of $M_D(\mathbb{C})$.

Theorem 5.120 (Thm. 6.13: Adjoint fixed points = Kraus commutant). *Under the hypotheses of Theorem 5.118, the adjoint fixed-point *-subalgebra equals the Kraus commutant:*

$$\text{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

This is [Wol12, Thm. 6.13].

Proof. ($\supseteq$): if X commutes with all K_i and $K_i^\dagger$, then $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$ by trace preservation. ($\subseteq$): any *-subalgebra inside $\text{Fix}(E^*)$ lies in the Kraus commutant, because for $X \in \text{Fix}(E^*)$ with $X^\dagger X \in \text{Fix}(E^*)$ the Kadison–Schwarz gap vanishes, which forces $XK_i = K_iX$ and $XK_i^\dagger = K_i^\dagger X$. $\square$

Theorem 5.121 (The Kraus commutant is the largest *-subalgebra inside the adjoint fixed points). *Under the hypotheses above, the Kraus commutant is the largest *-subalgebra contained in the adjoint fixed-point set.*

Proof. By Theorem 5.120, every *-subalgebra inside the adjoint fixed points is automatically contained in the Kraus commutant, while the Kraus commutant itself is such a *-subalgebra. $\square$

5.13 Quantum dynamical semigroups

The one-parameter semigroup theory of Wolf Chapter 7 is treated in Chapter 16. That later chapter records the exponential representation of norm-continuous semigroups, perturbation bounds, and the GKSL/Lindblad description of generators. We do not duplicate that material here.

Chapter 6

Quantum Entropy

This chapter develops the von Neumann entropy and its basic properties for finite-dimensional quantum systems.

6.1 Von Neumann entropy

Definition 6.1 (Von Neumann entropy). For a Hermitian matrix $\rho \in M_D(\mathbb{C})$ with eigenvalues $\lambda_0, \dots, \lambda_{D-1}$, the *von Neumann entropy* is

$$S(\rho) = - \sum_{i=0}^{D-1} \lambda_i \log \lambda_i,$$

where $0 \log 0 := 0$.

Theorem 6.2 (Entropy is non-negative). *For any density matrix ρ , $S(\rho) \geq 0$.*

Proof. Each eigenvalue λ_i of a density matrix satisfies $0 \leq \lambda_i \leq 1$, and $-x \log x \geq 0$ on $[0, 1]$. $\square$

Lemma 6.3 (Eigenvalue sum). *The eigenvalues of a density matrix sum to 1.*

Proof. Follows from $\text{tr}(\rho) = \sum_i \lambda_i = 1$. $\square$

Theorem 6.4 (Eigenvalue bound). *Each eigenvalue of a density matrix lies in $[0, 1]$.*

Proof. Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1. $\square$

Theorem 6.5 (Entropy upper bound). *For a density matrix $\rho \in M_D(\mathbb{C})$ with $D \geq 1$,*

$$S(\rho) \leq \log D.$$

Proof. By Jensen's inequality applied to the concave function $-x \log x$, the entropy is maximized when all eigenvalues equal $1/D$. $\square$

6.2 Tripartite partial traces

Definition 6.6 (Partial trace over A). For a tripartite matrix ρ_{ABC} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$, the partial trace over A is

$$(\mathrm{tr}_A \rho_{ABC})_{(b_1, c_1)(b_2, c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a, b_1, c_1)(a, b_2, c_2)}.$$

Definition 6.7 (Partial trace over C). The partial trace over C :

$$(\mathrm{tr}_C \rho_{ABC})_{(a_1, b_1)(a_2, b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1, b_1, c)(a_2, b_2, c)}.$$

Definition 6.8 (Partial trace over AC). The partial trace over A and C :

$$(\mathrm{tr}_{AC} \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a, b_1, c)(a, b_2, c)}.$$

Lemma 6.9 (Partial trace preserves Hermiticity). *If ρ_{ABC} is Hermitian, then $\mathrm{tr}_A(\rho_{ABC})$, $\mathrm{tr}_C(\rho_{ABC})$, and $\mathrm{tr}_{AC}(\rho_{ABC})$ are all Hermitian. The same holds for bipartite partial traces $\mathrm{tr}_A(\rho_{AB})$ and $\mathrm{tr}_B(\rho_{AB})$.*

Proof. Follows from $\overline{\rho_{ji}} = \rho_{ij}$ applied entry-wise inside the summation defining each partial trace. $\square$

6.3 Strong subadditivity

Theorem 6.10 (Strong subadditivity). *For a tripartite density matrix ρ_{ABC} ,*

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 6.11 (SSA equality). A tripartite density matrix ρ_{ABC} satisfies *SSA equality* if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 6.12 (Quantum Markov decomposition). A *Hayashi Markov decomposition* of a tripartite state ρ_{ABC} consists of a finite direct-sum decomposition

$$H_B \cong \bigoplus_j H_{B_j^L} \otimes H_{B_j^R},$$

together with a unitary change of basis on B , a probability vector $(p_j)_j$, and density matrices $\rho_{AB_j^L}$ and $\rho_{B_j^R C}$ such that, in the adapted basis, the state becomes

$$\bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}.$$

Theorem 6.13 (Hayashi SSA-equality characterization). *For a tripartite density matrix ρ_{ABC} ,*

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$$

holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B .

Status: Recorded in Lean as a sanctioned external-input axiom in `TNLean/Axioms/Entropy.lean`. This is the equality criterion needed by Appendix C of [arXiv:1606.00608](#). We cite Hayashi, J. Phys. A 37 (2004) L205–L208, and Ruskai, J. Math. Phys. 43 (2002) 4358; the structural block-decomposition formulation used in the MPDO paper also appears in Hayden–Jozsa–Petz–Winter, Commun. Math. Phys. 246 (2004) 359–374.

6.4 Mutual information

Definition 6.14 (Mutual information). The *quantum mutual information* of a bipartite state ρ_{AB} is

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

Theorem 6.15 (Mutual information is non-negative). *For any bipartite density matrix ρ_{AB} , $I(A:B) \geq 0$.*

Proof. Apply strong subadditivity (Theorem 6.10) with trivial B (one-dimensional middle system). The SSA inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ reduces to subadditivity $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$, which gives $I(A:C) \geq 0$. $\square$

6.5 `TNLean.Entropy` namespace

The modules in `TNLean.Entropy` (issue #613) place the von Neumann entropy and mutual information in the `Entropy` namespace, restate the sanctioned strong-subadditivity axiom there, and introduce the name `QuantumMarkovDecomposition` for the Hayashi SSA-equality decomposition. The external inputs are accepted as given; their full proofs are tracked separately in the entropy umbrella issues.

Definition 6.16 (Von Neumann entropy, namespaced). The namespaced wrapper `Entropy.vonNeumannEntropy`(ρ, h) unfolds to $S(\rho) = -\sum_i \lambda_i \log \lambda_i$ as in Definition 6.1.

Theorem 6.17 (Strong subadditivity, namespaced wrapper). *For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,*

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

In the Lean development, `Entropy.strongSubadditivity` is a theorem wrapper around the single sanctioned SSA axiom `strong_subadditivity` of `TNLean/Axioms/Entropy.lean`; no new axiom is introduced by the `TNLean.Entropy.*` declarations. The underlying axiom’s proof (Lieb–Ruskai 1973) is deferred to the umbrella entropy issue.

Definition 6.18 (Quantum Markov decomposition, namespaced). The namespaced alias `Entropy.QuantumMarkovDecomposition` is the name used in the `Entropy` namespace for the decomposition record of Definition 6.12.

Theorem 6.19 (SSA equality iff quantum Markov decomposition). *For any tripartite density matrix ρ_{ABC} , equality in strong subadditivity holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B .*

In the Lean development, this is a theorem wrapper around the sanctioned axiom `hayashi_ssa_equality_characterization` of `TNLean/Axioms/Entropy.lean`; no new axiom is introduced by `TNLean/Entropy/MarkovChain.lean`.

Definition 6.20 (Mutual information, namespaced). The namespaced wrapper `Entropy.mutualInformation`(ρ_{AB}, h) unfolds to $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ as in Definition 6.14.

Theorem 6.21 (Subadditivity from trivial-middle SSA). *For a tripartite density matrix ρ_{ABC} with $\dim B = 1$,*

$$S(\rho_{ABC}) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

This follows from SSA because the 1-dimensional middle subsystem contributes zero entropy. The statement carries an implicit dependence on the axiom of Theorem 6.17 and is therefore not marked.

Chapter 7

Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory from Wolf's Chapter 5 [Wol12]. We begin with the Kadison–Schwarz inequality for Kraus maps, then describe the multiplicative domain and its algebraic structure. We also establish the extension to normal operators for positive subunital maps, the resulting order and spectral contractivity statements, and Wolf's standard example of a Schwarz map that is not completely positive.

7.1 Kadison–Schwarz inequality

A Kraus map is the concrete realisation of a completely positive map (Definition 5.4). The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map is a Kraus map by construction (Remark 5.17), so these results apply directly to transfer maps.

Definitions 7.1–7.4 below restate the Kraus-map notions from Chapter 5 for the reader's convenience.

Definition 7.1 (Kraus map). Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 5.4) if and only if it can be written in this form.

Definition 7.2 (Adjoint Kraus map). The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A , this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 7.3 (Unital Kraus map). A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$. In the later gauge language this is the right-canonical condition.

Definition 7.4 (Trace-preserving Kraus map). A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 7.5 (Kadison–Schwarz inequality). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Eq. (5.2)].

Proof. The 2×2 block matrix $\begin{bmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{bmatrix}$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves positive semidefiniteness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2, 2)$ -block maps to $\mathbb{1}$). The Schur complement of the $(2, 2)$ -block $\mathbb{1}$ gives the result. $\square$

Theorem 7.6 (Kadison–Schwarz for the adjoint channel). *If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.*

Proof. The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 7.5 to $\{K_i^\dagger\}$. $\square$

7.1.1 Two-positive maps and the generalized Schwarz inequality

Definition 7.7 (k -positive map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is k -positive if $E \otimes \mathbb{1}_k$ is positive on $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$.

Definition 7.8 (Two-positive map). A linear map is *two-positive* if it is 2-positive.

Theorem 7.9 (Completely positive maps are k -positive). *Every completely positive map is k -positive for all $k \geq 1$.*

Theorem 7.10 ($(k+1)$ -positive implies k -positive). *Positivity under amplification is monotone in k .*

Theorem 7.11 (Completely positive implies two-positive). *Every completely positive map is two-positive.*

Theorem 7.12 (Two-positive implies positive). *Every two-positive map is positive.*

Definition 7.13 (Unital linear map). A linear map E is unital if $E(\mathbb{1}) = \mathbb{1}$.

Theorem 7.14 (Kadison–Schwarz for unital two-positive maps). *If E is unital and two-positive, then*

$$E(X^\dagger X) \geq E(X)^\dagger E(X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. Form the 2×2 block matrix $\begin{pmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{pmatrix} \geq 0$. Apply $E \otimes \text{id}_2$ (positive by 2-positivity) and read off the $(2, 2)$ -block Schur complement: $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$. $\square$

Theorem 7.15 (Kraus-form Kadison–Schwarz as a special case). *The Kraus-map Kadison–Schwarz inequality is recovered from the two-positive formulation.*

7.1.2 Douglas-type factorization lemmas

Theorem 7.16 (Range inclusion implies right factorization). *If $\text{ran}(A) \subseteq \text{ran}(B)$, then there exists C such that $A = BC$.*

Proof. In finite dimensions, $\text{ran}(A) \subseteq \text{ran}(B)$ implies $A = B(B^\dagger B)^\dagger B^\dagger A$, where $(B^\dagger B)^\dagger$ denotes the Moore–Penrose pseudoinverse. Set $C = (B^\dagger B)^\dagger B^\dagger A$. $\square$

Theorem 7.17 (Vectorwise range criterion implies factorization). *If $Av \in \text{ran}(B)$ for every vector v , then there exists C with $A = BC$.*

Theorem 7.18 (Hilbert–Schmidt contraction). *If a Kraus map is both unital and trace-preserving, then $\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$ for all $X \in M_D(\mathbb{C})$.*

Proof. By Theorem 7.5, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\text{tr}(E(X^\dagger X)) \geq \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. $\square$

Lemma 7.19 (Trace pairing for Kraus map and adjoint map). *For all $X, Y \in M_D(\mathbb{C})$,*

$$\text{tr}(X E(Y)) = \text{tr}(E^*(X) Y).$$

Lemma 7.20 (Positive-definite trace test). *If $A > 0$, $X \geq 0$, and $\text{tr}(AX) = 0$, then $X = 0$.*

Theorem 7.21 (Fixed-point peripheral eigenvectors satisfy KS equality). *Let E be unital and let E^* be trace-preserving with a positive definite fixed point $\rho > 0$. If $E(X) = \mu X$ with $|\mu| = 1$, then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

7.2 Multiplicative domain

Theorem 7.22 (KS gap decomposition). *For a unital Kraus map E , the Kadison–Schwarz gap decomposes as*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := XK_i^\dagger - K_i^\dagger E(X)$.

Proof. Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. $\square$

Theorem 7.23 (KS equality implies Kraus commutation). *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$ for some X , then $XK_i^\dagger = K_i^\dagger E(X)$ for all i .*

Proof. By Theorem 7.22, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. $\square$

Theorem 7.24 (KS equality for peripheral eigenvectors). *Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then the KS gap vanishes: $E(X^\dagger X) = E(X)^\dagger E(X)$.*

Proof. The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 7.5. Its trace satisfies $\text{tr}(G) = \text{tr}(E(X^\dagger X)) - \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\text{tr}(E(X)^\dagger E(X)) = |\mu|^2 \text{tr}(X^\dagger X) = \text{tr}(X^\dagger X)$. So $\text{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. $\square$

Theorem 7.25 (Left multiplicative identity from KS equality). *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.*

Proof. From Theorem 7.23, $XK_i^\dagger = K_i^\dagger E(X)$ for all i . Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

□

7.2.1 Full multiplicative-domain structure

Definition 7.26 (Right and left multiplicative domains). For a Kraus map E , define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : E(XY) = E(X)E(Y) \text{ for all } Y \in M_D(\mathbb{C})\}$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : E(YX) = E(Y)E(X) \text{ for all } Y \in M_D(\mathbb{C})\}.$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Eqs. (5.11)–(5.12)].

Definition 7.27 (Full multiplicative domain). The *multiplicative domain* of E is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

Theorem 7.28 (Multiplicative-domain characterization). *Let E be a unital Kraus map. Then*

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger$$

and

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

Together these are the two one-sided characterizations from [Wol12, Thm. 5.7].

Proof. If X lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 7.25, and the right-domain implication follows by applying Theorem 7.25 to $X^\dagger$. □

Theorem 7.29 (One-sided multiplicative domains are subalgebras). *For a unital Kraus map E , both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.*

Proof. Closure under addition follows from linearity of E . Closure under multiplication: if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all Z , using Theorem 7.28 iteratively, and the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of E . □

Theorem 7.30 (Multiplicative domain is a $*$ -subalgebra). *For a unital Kraus map E , the full multiplicative domain $\mathcal{A}(E)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Thm. 5.7].*

Proof. If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then $E(X^\dagger X) = E(X)^\dagger E(X)$ and $E(XX^\dagger) = E(X)E(X)^\dagger$. The characterization of Theorem 7.28 shows $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 7.29, this gives a $*$ -subalgebra. $\square$

Theorem 7.31 (Restriction to multiplicative domain is a $*$ -homomorphism). *If E is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \rightarrow M_D(\mathbb{C})$ is a $*$ -algebra homomorphism. This is the homomorphism conclusion of [Wol12, Thm. 5.7].*

Proof. Multiplicativity on $\mathcal{A}(E)$ is immediate from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation from Theorem 7.23 gives $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. $\square$

7.2.2 Schwarz inequality for normal operators

Theorem 7.32 (CP Schwarz inequality for normal operators). *Let $E^*(X) = \sum_i K_i^\dagger X K_i$ be the adjoint Kraus map of a trace-preserving Kraus family, and let $A \in M_D(\mathbb{C})$ be normal. Then the Schwarz gap*

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A)$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \leq E^*(A^\dagger A).$$

This is the completely positive case of [Wol12, Prop. 5.1].

Proof. The positivity statement is exactly Theorem 7.6. The second formulation is just the corresponding Loewner-order reformulation. $\square$

Theorem 7.33 (Prop. 5.1: Schwarz inequality for normal operators). *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is normal, then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Prop. 5.1].

Proof. Since A is normal, diagonalize $A = UDU^\dagger$ with D diagonal and U unitary. Express $A^\dagger A = U|D|^2U^\dagger$ as a sum $\sum_{i,j} \bar{d}_i d_j P_i P_j$ where $P_i = U e_i e_i^\dagger U^\dagger$. Each P_i is rank-one PSD, and the images $T(P_i)$ pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown nonneg by a simultaneous-diagonalization argument: positivity of T ensures the scalar PSD matrix $(\langle v_k, T(P_i P_j) v_k \rangle)_{i,j}$ dominates the product $(\langle v_k, T(P_i) v_k \rangle \langle v_k, T(P_j) v_k \rangle)_{i,j}$, and combining over a common eigenbasis of the $T(P_i)$ yields the result. $\square$

Remark 7.34. The proof follows the positivity-on-abelian argument of [Wol12, Prop. 1.6]: a positive map restricted to the commutative $*$ -subalgebra generated by the spectral projectors of A satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

7.2.3 Schwarz inequality for subnormal and commuting-dominant operators

Theorem 7.35 (Thm. 5.5: Schwarz inequality for subnormal operators). *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is subnormal (i.e. there exists a normal operator on a larger space whose compression to $\mathbb{C}^D$ is A), then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Thm. 5.5].

Proof. Embed A as the northwest corner of a normal operator N on $\mathbb{C}^D \oplus \mathbb{C}^E$. Extend T to a positive subunitary map $\tilde{T}$ on the larger space (acting as T on the northwest block, zero elsewhere). Apply Theorem 7.33 to $\tilde{T}$ and N , then compress the resulting inequality back to $\mathbb{C}^D$. $\square$

Theorem 7.36 (Thm. 5.6: Schwarz inequality for commuting-dominant operators). *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant $B \geq 0$ with $A^\dagger A \leq B$ and B commuting with A (in the sense $BA = AB$, $BA^\dagger = A^\dagger B$), then*

$$T(A^\dagger)T(A) \leq T(B).$$

This is [Wol12, Thm. 5.6].

Proof. Let $B^{1/2}$ be the positive square root of B . Since B commutes with A and $A^\dagger$, the contraction $C = B^{-1/2}A$ (defined on the range of $B^{1/2}$) is subnormal. An ε -regularisation $B_\varepsilon = B + \varepsilon\mathbb{1}$ makes $B_\varepsilon > 0$ and the contraction C_ε globally defined. Apply Theorem 7.35 to C_ε , multiply both sides by $T(B_\varepsilon)$, use positivity and monotonicity, and take $\varepsilon \rightarrow 0$. $\square$

Theorem 7.37 (CP variant: Kadison–Schwarz for commuting-dominant inputs). *Let E^* be the adjoint of a trace-preserving Kraus map. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant B commuting with A , $A^\dagger$, and satisfying $A^\dagger A \leq B$, then $E^*(A^\dagger)E^*(A) \leq E^*(B)$.*

Proof. Apply Theorem 7.36 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis). $\square$

7.2.4 Positive maps preserve order and spectral intervals

Theorem 7.38 (Positive maps are monotone). *If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive and $A \leq B$, then $T(A) \leq T(B)$.*

Proof. Since $B - A \geq 0$, positivity gives $T(B - A) \geq 0$. By linearity, this is $T(B) - T(A) \geq 0$. $\square$

Theorem 7.39 (Positive maps preserve adjoints). *If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_D(\mathbb{C})$.*

Proof. Write $A = B + iC$ with B and C Hermitian. Positive maps send Hermitian matrices to Hermitian matrices, so they preserve this decomposition and commute with taking adjoints. $\square$

Theorem 7.40 (Order intervals are preserved). *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$. If $a \leq 0 \leq b$ and $a\mathbb{1} \leq A \leq b\mathbb{1}$, then*

$$a\mathbb{1} \leq T(A) \leq b\mathbb{1}.$$

This is the matrix form of [Wol12, Eq. (5.21)].

Proof. By monotonicity, $T(a\mathbb{1}) \leq T(A) \leq T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = aT(\mathbb{1})$. Since $a \leq 0$ and $T(\mathbb{1}) \leq \mathbb{1}$, we have $aT(\mathbb{1}) \geq a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = bT(\mathbb{1}) \leq b\mathbb{1}$ since $b \geq 0$. Therefore $a\mathbb{1} \leq T(A) \leq b\mathbb{1}$. $\square$

Theorem 7.41 (Spectral interval contractivity). *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If $\text{spec}(A) \subseteq [a, b]$ with $a \leq 0 \leq b$, then*

$$\text{spec}(T(A)) \subseteq [a, b].$$

This is [Wol12, Eq. (5.21)].

Proof. For Hermitian matrices, the inclusion $\text{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \leq A \leq b\mathbb{1}$. Apply Theorem 7.40 and translate the resulting inequalities back into spectral bounds. $\square$

7.2.5 A positive Schwarz map that is not completely positive

Example 7.42 (Wolf Example 5.3 map). Consider the linear map $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\text{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Ex. 5.3].

Theorem 7.43 (Wolf Example 5.3 is positive). *The map from Example 7.42 is positive.*

Proof. It is a nonnegative linear combination of the transpose map and the map $A \mapsto \text{tr}(A)\mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. $\square$

Theorem 7.44 (Wolf Example 5.3 satisfies the Schwarz inequality). *For every $A \in M_2(\mathbb{C})$,*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0,$$

where T is the map from Example 7.42.

Proof. Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda\mathbb{1}) = F(A)$ for every scalar λ , so it is enough to treat the case $\text{tr}(A) = 0$. In that case a direct 2×2 computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. $\square$

Theorem 7.45 (Wolf Example 5.3 is not completely positive). *The map from Example 7.42 is not completely positive.*

Proof. Compute the Choi matrix of T and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. $\square$

7.3 Operator convexity and Jensen inequalities

This section collects the operator convexity/concavity results, trace inequalities, the operator Jensen inequality for positive maps, and the Lieb concavity theorem. These are standard results in matrix analysis (Bhatia Ch. V, Wolf Thm. 5.1, Hansen–Pedersen 2003, Lieb 1973). They are currently taken as assumptions pending upstream formalization of operator concavity and convexity for real powers $x \mapsto x^p$ on the Loewner order, together with operator concavity of the matrix logarithm.

7.3.1 Diagonal Jensen inequality

Lemma 7.46 (Diagonal Jensen inequality). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex on $[0, \infty)$, let A be a positive semidefinite matrix indexed by a finite type n , and let $v \in \mathbb{C}^n$ satisfy $\langle v, v \rangle = 1$. Then*

$$f(\Re \langle v, Av \rangle) \leq \Re \langle v, f(A)v \rangle,$$

where $f(A)$ is defined by the Hermitian continuous functional calculus.

Proof. Write $A = U \operatorname{diag}(\mu) U^*$ by the spectral theorem and set $w = U^*v$. Since U is unitary, $p_i := |w_i|^2$ satisfies $\sum_i p_i = \|v\|^2 = 1$, so (p_i) is a probability distribution on n . The eigenvalues μ_i lie in $[0, \infty)$ because A is positive semidefinite. A direct computation yields $\Re \langle v, Av \rangle = \sum_i p_i \mu_i$ and $\Re \langle v, f(A)v \rangle = \sum_i p_i f(\mu_i)$, so the scalar Jensen inequality applied to the weights (p_i) and points (μ_i) gives the conclusion. See Bhatia, *Matrix Analysis*, Ch. V. $\square$

7.3.2 Trace concavity and convexity of matrix powers

Theorem 7.47 (Trace concavity of real powers). *For $p \in [0, 1]$, PSD matrices A_1, A_2 , and $t \in [0, 1]$:*

$$t \Re \operatorname{tr}(A_1^p) + (1-t) \Re \operatorname{tr}(A_2^p) \leq \Re \operatorname{tr}((tA_1 + (1-t)A_2)^p).$$

Follows from operator concavity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. Let $A = tA_1 + (1-t)A_2$ and let $\{\psi_j\}$ with eigenvalues $\mu_j \geq 0$ be an orthonormal eigenbasis of A . A direct computation gives $\Re \operatorname{tr}(A^p) = \sum_j \mu_j^p$ and $\mu_j = t a_j + (1-t) b_j$ where $a_j = \Re \langle \psi_j, A_1 \psi_j \rangle$, $b_j = \Re \langle \psi_j, A_2 \psi_j \rangle$. Scalar concavity of $x \mapsto x^p$ on $[0, \infty)$ yields $\mu_j^p \geq t a_j^p + (1-t) b_j^p$, and the diagonal Jensen inequality (Lemma 7.46, concave form) gives $a_j^p \leq \Re \langle \psi_j, A_1^p \psi_j \rangle$ (and similarly for b_j). Summing over j and using $\sum_j \Re \langle \psi_j, M \psi_j \rangle = \Re \operatorname{tr} M$ yields the conclusion. See Bhatia, *Matrix Analysis*, Ch. V. $\square$

Theorem 7.48 (Trace convexity of real powers). *For $p \in [1, 2]$, PSD matrices A_1, A_2 , and $t \in [0, 1]$:*

$$\Re \operatorname{tr}((tA_1 + (1-t)A_2)^p) \leq t \Re \operatorname{tr}(A_1^p) + (1-t) \Re \operatorname{tr}(A_2^p).$$

Follows from operator convexity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. Identical eigenbasis reduction as in Theorem 7.47, with scalar convexity (rather than concavity) of $x \mapsto x^p$ on $[0, \infty)$ for $p \geq 1$ and the convex form of the diagonal Jensen inequality. $\square$

7.3.3 Operator Jensen inequality for positive maps

Theorem 7.49 (Jensen for concave real powers). *For a positive subunital map T and $p \in [0, 1]$:*

$$T(A^p) \leq (TA)^p.$$

This is [Wol12, Thm. 5.1] for concave $f(x) = x^p$.

Theorem 7.50 (Jensen for concave real powers — restatement). *Restates Theorem 7.49 for downstream use as a property of a positive map. Axiom-dependent: not fully formalized until Theorem 7.49 is proved.*

Proof. Direct application of the axiom. □

Theorem 7.51 (Jensen for convex real powers). *For a positive subunital map T and $p \in [1, 2]$:*

$$(TA)^p \leq T(A^p).$$

This is [Wol12, Thm. 5.1] for convex $f(x) = x^p$.

Theorem 7.52 (Jensen for convex real powers — restatement). *Restates Theorem 7.51 for downstream use as a property of a positive map. Axiom-dependent: not fully formalized until Theorem 7.51 is proved.*

Proof. Direct application of the axiom. □

Theorem 7.53 (Jensen for concave log). *For a positive unital map T and positive-definite A :*

$$T(\log A) \leq \log(TA).$$

This is [Wol12, Thm. 5.1] for $f = \log$. Requires unitality ($T(\mathbb{1}) = \mathbb{1}$), not merely subunitality.

Theorem 7.54 (Jensen for concave log — restatement). *Restates Theorem 7.53 for downstream use as a property of a positive map. Axiom-dependent: not fully formalized until Theorem 7.53 is proved.*

Proof. Direct application of the axiom. □

7.3.4 Lieb concavity theorem

Theorem 7.55 (Lieb concavity). *For $s \in [0, 1]$, any matrix K , and positive-definite matrices A_1, A_2, B_1, B_2 , the map $(A, B) \mapsto \text{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave:*

$$t \Re \text{tr}(K^\dagger A_1^s K B_1^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A_2^s K B_2^{1-s}) \leq \Re \text{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K (tB_1 + (1-t)B_2)^{1-s}).$$

See Lieb (1973) and Ando (1979).

Theorem 7.56 (Lieb concavity — restatement). *Restates Theorem 7.55 for downstream use. Axiom-dependent: not fully formalized until Theorem 7.55 is proved.*

Proof. Direct application of the axiom. □

Chapter 8

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory used in subsequent chapters: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf’s spectral characterization of irreducibility. The conceptual source is Wolf’s treatment of irreducible positive maps [Wol12, Ch. 6, especially Thms. 6.2–6.5 and Cor. 6.3] and [EHK78].

8.1 Positive definiteness

Remark 8.1. Wolf’s general theory treats irreducible positive maps first ([Wol12, Thm. 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Thm. 6.3]). The injective-tensor version (Theorem 8.2) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 8.3) follows via the support-projection argument of [Wol12, Thm. 6.2].

Theorem 8.2 (PSD fixed point is positive definite). *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is nonneg; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all i : the kernel of ρ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, we have $\rho M v = 0$ for every matrix M . Given any vector w , choose M with $M v = w$. Then $\rho w = 0$ for every w , so $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Thm. 6.3, item 2], the same conclusion is reached via the growth condition $(\mathbb{1} + T)^{d-1}(X) > 0$ of [Wol12, Thm. 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. $\square$

Theorem 8.3 (PSD fixed point is positive definite — irreducible version). *If $\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then ρ is positive definite.*

Proof. Suppose ρ is not positive definite. Then ρ has a nontrivial kernel; let Q be the orthogonal projection onto the support of ρ (the complement of the kernel). A direct computation using the fixed-point equation $\mathcal{E}_A(\rho) = \rho$ shows that Q is an invariant projection for $\mathcal{E}_A$: it satisfies $Q \mathcal{E}_A(Q X Q) Q = \mathcal{E}_A(Q X Q)$ for all X . By irreducibility of $\mathcal{E}_A$ (Definition 5.15), every invariant projection is 0 or $\mathbb{1}$. If $Q = 0$ then $\rho = 0$, contradicting $\rho \neq 0$. If $Q = \mathbb{1}$ then ρ is positive definite,

contradicting our assumption. Hence ρ must be positive definite. This is the support-projection direction of [Wol12, Thm. 6.2, condition (1)]. $\square$

Theorem 8.4 (Orthogonal trace condition). *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\text{tr}(BA) = 0$. Then there exists t with $1 \leq t \leq D - 1$ such that*

$$\text{tr}(B \cdot E^t(A)) > 0.$$

Proof. The growth theorem for irreducible CP maps ([Wol12, Thm. 6.2, item 2]) gives $(\mathbb{1} + E)^{D-1}(A) > 0$ (positive definite). Since $B \geq 0$ is nonzero, $\text{tr}(B \cdot (\mathbb{1} + E)^{D-1}(A)) > 0$. Expanding by the binomial theorem:

$$\text{tr}\left(B \cdot \sum_{k=0}^{D-1} \binom{D-1}{k} E^k(A)\right) > 0.$$

Each term $\text{tr}(B \cdot E^k(A)) \geq 0$ since both factors are PSD. The $k = 0$ term vanishes by hypothesis ($\text{tr}(BA) = 0$). Hence at least one $k \in \{1, \dots, D - 1\}$ contributes a strictly positive term. This is [Wol12, Thm. 6.2, item 4]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument. $\square$

8.2 Uniqueness

Theorem 8.5 (Uniqueness of PSD fixed point). *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. Both ρ and σ are positive definite by Theorem 8.2. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1}\sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H . Then $\tau := \sigma - c_0\rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of c_0). By Theorem 8.2, τ must be zero, giving $\sigma = c_0\rho$.

Relation to Wolf: In [Wol12, Thm. 6.3, item 2], uniqueness is proved via the growth condition of [Wol12, Thm. 6.2]. The same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 8.2) in place of the growth condition. $\square$

Remark 8.6. The proof produces a positive real scalar $c_0 > 0$: it is the minimal eigenvalue of the positive definite Hermitian matrix $H = (S^\dagger)^{-1}\sigma S^{-1}$.

Theorem 8.7 (Uniqueness of positive eigenvalue for irreducible CP maps). *Let $D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. Extract Kraus operators K for E . Irreducibility of E passes to the adjoint transfer map $E^\dagger(\cdot) = \sum_i K_i^\dagger(\cdot)K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 8.17) together with the irreducible positive-definiteness upgrade (Theorem 8.3) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\tau$. For any nonzero PSD X with $E(X) = sX$, the trace pairing identity yields

$$s \cdot \text{tr}(\tau X) = \text{tr}(\tau \cdot E(X)) = \text{tr}(E^\dagger(\tau) \cdot X) = t \cdot \text{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\text{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. **Relation to Wolf:** This is [Wol12, Thm. 6.3, item 3], establishing non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps. The proof follows Wolf’s dual-map trace argument. $\square$

8.3 Existence and the Perron–Frobenius theorem

Theorem 8.8 (Existence of PSD fixed point). *Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

Proof. Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 5.18). The Cesàro fixed-point theorem (Theorem 5.22) then provides the fixed point; injectivity is not needed for this step. $\square$

Theorem 8.9 (Quantum Perron–Frobenius). *Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.*

Proof. Combine Theorems 8.8, 8.2, and 8.5. $\square$

8.4 Right- and left-canonical gauges

Theorem 8.10 (Right-canonical (unital) gauge). *Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization used in subsequent chapters.*

Proof. Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_\rho (A^i)^\dagger (S^\dagger)^{-1}$. Summing over i and using $\sum_i A^i (A^i)^\dagger = \rho$ gives $S^{-1}\rho(S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. $\square$

Theorem 8.11 (Left-canonical (trace-preserving) gauge). *Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization used in subsequent chapters.*

Proof. Expand each summand: $(SA^iS^{-1})^\dagger (SA^iS^{-1}) = (S^\dagger)^{-1} (A^i)^\dagger \underbrace{S^\dagger S}_\sigma A^i S^{-1}$. Summing over i and using $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ gives $(S^\dagger)^{-1} \sigma S^{-1} = (S^\dagger)^{-1} S^\dagger S S^{-1} = \mathbb{1}$. $\square$

Remark 8.12. The right-canonical gauge of Theorem 8.10 and the left-canonical gauge of Theorem 8.11 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

8.5 Similarity preserves irreducibility

Lemma 8.13 (Similarity transform preserves irreducibility). *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible ($\det C \neq 0$), and let $c > 0$. Define the similarity-transformed map*

$$E'(X) = c \cdot C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. Suppose Q is an invariant projection for E' : $Q \neq 0$, $Q \neq \mathbb{1}$, and $(\mathbb{1} - Q)E'(QXQ)Q = 0$ for all X . Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E . Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the full similarity version of [Wol12, Prop. 6.6]; the scalar case ($C = \mathbb{1}$) gives the scaling result stated next. $\square$

Lemma 8.14 (Similarity is an involution). *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transforms by C and C^{-1} compose to the identity:*

$$X \mapsto C(C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}) C^\dagger = E(X).$$

Proof. Direct computation: the inner conjugation $C \cdot (C^{-1} X (C^\dagger)^{-1}) \cdot C^\dagger$ collapses to X using $CC^{-1} = \mathbb{1}$ and $(C^\dagger)^{-1}C^\dagger = \mathbb{1}$, and the outer wrapping by $C(\cdot)C^\dagger$ then cancels the inner $C^{-1}(\cdot)(C^\dagger)^{-1}$ on $E(X)$ for the same reason. $\square$

Lemma 8.15 (Irreducibility is invariant under similarity). *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transform $X \mapsto C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}$ is irreducible if and only if E is.*

Proof. The forward direction follows from Lemma 8.13 applied with the inverse C^{-1} (whose determinant is also nonzero): if the similarity transform of E by C is irreducible, then so is its similarity transform by C^{-1} , which equals E by Lemma 8.14. The reverse direction is Lemma 8.13 itself (with $c = 1$). $\square$

Theorem 8.16 (Scaling preserves irreducibility). *If E is an irreducible map and $c \neq 0$, then cE is also irreducible. This is the scalar special case ($C = \mathbb{1}$) of [Wol12, Prop. 6.6].*

Proof. An invariant projection for cE is an invariant projection for E (scale out the nonzero constant from the commutation relation). $\square$

8.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Thm. 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Thm. 6.3]. The TP-gauge application follows [CPGSV17a, Appendix A].

Theorem 8.17 (PSD eigenvector for positive maps). *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero PSD matrix σ . Then there exist a nonzero PSD matrix ρ and a positive real $r > 0$ such that $E(\rho) = r\rho$.*

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Thm. 6.5]; see also [EHK78]). The proof applies Brouwer’s fixed-point theorem to the normalization map $\rho \mapsto E(\rho)/\text{tr}(E(\rho))$ on the compact convex set of density matrices.

Theorem 8.18 (Adjoint transfer map is nonzero). *If some Kraus operator $A^i \neq 0$, then the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is nonzero.*

Proof. If $\mathcal{E}_A^\dagger = 0$ then $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$, so $\sum_i (A^i)^\dagger A^i = 0$. Since each summand $(A^i)^\dagger A^i$ is PSD, each $(A^i)^\dagger = 0$ and hence each $A^i = 0$. $\square$

Theorem 8.19 (Positive-definite adjoint eigenvector for irreducible tensors). *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.*

This combines the general PF existence ([Wol12, Thm. 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Thm. 6.3, item 2]); the application to the adjoint transfer map follows [CPGSV17a, Appendix A].

Proof. By Theorem 8.18, $\mathcal{E}_A^\dagger \neq 0$. Since $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is a Kraus map, it is positive. Irreducibility of A transfers to the adjoint transfer map. More precisely: if P is invariant for $\mathcal{E}_A$ (i.e. $(\mathbb{1} - P)K_i P = 0$ for all Kraus operators K_i), then taking adjoints gives $P K_i^\dagger (\mathbb{1} - P) = 0$, which means $\mathbb{1} - P$ is invariant for $\mathcal{E}_A^\dagger$. Hence $\mathcal{E}_A$ irreducible $\iff \mathcal{E}_A^\dagger$ irreducible, and together with $\mathcal{E}_A^\dagger \neq 0$ this implies $\mathcal{E}_A^\dagger(\tau) \neq 0$ for every nonzero PSD matrix τ . Hence Theorem 8.17 applies to $\mathcal{E}_A^\dagger$, giving $\sigma_0 \geq 0$ (PSD, nonzero) and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$. Rescale: set $T^i = r^{-1/2}(A^i)^\dagger$. Then $\mathcal{E}_T(\sigma_0) = \sigma_0$, and $\mathcal{E}_T$ is irreducible (by Theorem 8.16, since irreducibility of $\mathcal{E}_A$ transfers to the adjoint and is preserved under scaling). Since $\mathcal{E}_T$ is irreducible with a PSD fixed point, the positive-definiteness upgrade (Theorem 8.3) gives σ_0 positive definite. $\square$

Theorem 8.20 (TP-gauge existence for irreducible tensors). *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r , a positive definite matrix σ , and a gauge-equivalent tensor B such that*

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This is the “spectral rescaling + TP gauge” step of [CPGSV17a, Appendix A]. The similarity is generally non-unitary.

Proof. By Theorem 8.19, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the TP gauge construction (Theorem 8.11), $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$ satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. $\square$

8.7 Exponential positivity for irreducible CP maps

Theorem 8.21 (Exponential truncation is positive definite). *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then the finite exponential truncation*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is the finite-sum core of [Wol12, Thm. 6.2, item (3)].

Proof. If a nonzero vector v annihilated this quadratic form, then every term $E^k(A)$ with $0 \leq k \leq D-1$ would annihilate v . The growth theorem for irreducible CP maps (that is, $(\mathbb{1} + E)^{D-1}(A) > 0$ for irreducible E and nonzero PSD A , as established in the proof of Theorem 8.4) then forces $(\mathbb{1} + E)^{D-1}(A)$ to annihilate v , contradicting its positive definiteness. $\square$

Theorem 8.22 (Exponential positivity for irreducible CP maps). *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then*

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is [Wol12, Thm. 6.2, item (3)].

Proof. By Theorem 8.21, the first D terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. $\square$

8.8 Ergodicity of irreducible channels

Theorem 8.23 (Unique density-matrix fixed point for an irreducible channel). *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix σ such that $E(\sigma) = \sigma$. Moreover σ is positive definite. This is the qualitative form of [Wol12, Cor. 6.3].*

Proof. Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1. $\square$

Theorem 8.24 (Cesàro convergence for irreducible channels). *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let ρ be any density matrix. Then the Cesàro means*

$$\frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho)$$

converge to the unique positive definite density-matrix fixed point of E . This is [Wol12, Cor. 6.3].

Proof. Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 8.23, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. $\square$

8.9 Spectral radius at the Perron eigenvalue

Theorem 8.25 (Spectral radius is the Perron eigenvalue). *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho > 0$ and $E(\rho) = r\rho$ with $r > 0$. Then the spectral radius of E is r . This is [Wol12, Thm. 6.3, item (4)].*

Proof. Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by r^{-1} . The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving map F satisfies $\rho(F) \leq 1$, since $\text{tr}(F^n(X)) = \text{tr}(X)$ for all n bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of E . $\square$

Theorem 8.26 (Real spectral-radius identity). *Under the hypotheses of Theorem 8.25, the real-valued spectral radius of E is also equal to r .*

Proof. This is the real-valued reformulation of Theorem 8.25. $\square$

8.10 Spectral characterization of irreducibility

Definition 8.27 (Wolf spectral properties). A completely positive map E on $M_D(\mathbb{C})$ has the *spectral properties* of [Wol12, Thm. 6.4] if there exist a positive real number r , a positive definite right eigenvector ρ , and a positive definite left eigenvector σ for the adjoint map such that $E(\rho) = r\rho$, $E^\dagger(\sigma) = r\sigma$, every positive semidefinite right eigenvector for eigenvalue r is a scalar multiple of ρ , and the spectral radius of E is equal to r .

Theorem 8.28 (Irreducibility gives the spectral properties). *Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 8.27.*

Proof. Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 8.25. $\square$

Theorem 8.29 (Spectral properties imply irreducibility). *If a completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 8.27, then it is irreducible.*

Proof. Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 8.27. $\square$

Theorem 8.30 (Irreducibility iff Wolf’s spectral properties). *Let E be a nonzero completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if it has the spectral properties of Definition 8.27. This is [Wol12, Thm. 6.4].*

Proof. Combine Theorems 8.28 and 8.29. $\square$

Chapter 9

Spectral Gap and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary spectral-gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

9.1 Mixed transfer operator

Definition 9.1 (Mixed transfer operator). The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Definition 9.2 (Rectangular mixed transfer operator). For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F_{AB}^{\text{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \rightarrow M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\text{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Remark 9.3 (Self-transfer). Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.

Theorem 9.4 (Iterated mixed transfer). For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,

$$F_{AB}^N(X) = \sum_{\sigma \in \{0, \dots, d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

where $A^\sigma = A^{\sigma_1} \dots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).

Proof. Induction on N : the base case is $F^0(X) = X$; the inductive step applies F_{AB} to each summand and re-indexes via $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$. $\square$

Theorem 9.5 (Matrix trace of iterated mixed transfer at the identity). *For $N \geq 0$,*

$$\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^{\sigma}(B^{\sigma})^{\dagger}).$$

This is the matrix trace of F_{AB}^N evaluated at the identity matrix. It should not be confused with the operator trace $\text{Tr}(F_{AB}^N)$, which equals the MPV overlap (Theorem 9.7).

Proof. Apply the matrix trace to each summand in Theorem 9.4 with $X = \mathbb{1}$: $\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^{\sigma} \cdot \mathbb{1} \cdot (B^{\sigma})^{\dagger}) = \sum_{\sigma} \text{tr}(A^{\sigma}(B^{\sigma})^{\dagger})$. $\square$

9.2 MPV overlap as transfer trace

Recall from Definition 2.35 that

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_{\sigma} \overline{V^{(N)}(B)_{\sigma}}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

Lemma 9.6 (Trace expansion over matrix units). *For any linear endomorphism T on $M_D(\mathbb{C})$, the operator trace can be expanded as*

$$\text{Tr}(T) = \sum_{p,q=0}^{D-1} (T(E_{pq}))_{pq},$$

where E_{pq} is the matrix with 1 in position (p, q) and 0 elsewhere.

Proof. The operator trace $\text{Tr}(T)$ equals the sum of $\text{tr}(T(E_{pq}) \cdot E_{qp})$ over all (p, q) , and $\text{tr}(M \cdot E_{qp}) = M_{pq}$. $\square$

Theorem 9.7 (Overlap equals transfer trace). *For tensors A, B of bond dimension $D \geq 1$,*

$$O_{AB}(N) = \text{Tr}(F_{AB}^N).$$

Proof. Expand the operator trace using Lemma 9.6 as a sum over matrix units E_{pq} . Apply Theorem 9.4 to evaluate $F_{AB}^N(E_{pq})$, extract the (p, q) -entry, and reassemble. The resulting double sum over (p, q) and σ equals $\sum_{\sigma} V^{(N)}(A)_{\sigma} \overline{V^{(N)}(B)_{\sigma}} = O_{AB}(N)$. $\square$

Theorem 9.8 (Rectangular overlap as transfer trace). *For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,*

$$O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N).$$

Proof. Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Theorem 9.7. $\square$

Lemma 9.9 (Word trace as entrywise inner product). *For a word $\sigma \in \{0, \dots, d-1\}^N$,*

$$\text{tr}(A^{\sigma}(B^{\sigma})^{\dagger}) = \sum_{j,k=0}^{D-1} (A^{\sigma})_{jk} \overline{(B^{\sigma})_{jk}}.$$

Proof. Expand the matrix trace and the (j, j) -entry of $A^{\sigma}(B^{\sigma})^{\dagger}$. $\square$

9.3 Frobenius norm estimates

The spectral-gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write $\|\cdot\|_F$ for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm $\|\cdot\|_{\text{HS}}$ on finite-dimensional spaces. We record the squared version and an isometric embedding into Euclidean space.

Definition 9.10 (Frobenius norm squared). For a (possibly rectangular) matrix $X \in M_{m \times n}(\mathbb{C})$, the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |X_{ij}|^2.$$

Lemma 9.11 (Trace formula for Frobenius norm). $\|X\|_F^2 = \text{Re tr}(X^\dagger X)$.

Proof. Expand the matrix product and trace entry by entry. $\square$

Definition 9.12 (Euclidean-space embedding). The map $\text{vec} : M_{m \times n}(\mathbb{C}) \rightarrow \mathbb{C}^{mn}$ that flattens matrix entries is an isometric embedding with respect to the Frobenius norm: $\|\text{vec}(X)\|^2 = \|X\|_F^2$.

Lemma 9.13 (Norm of embedded matrix). $\|\text{vec}(X)\|^2 = \|X\|_F^2$.

Proof. Expand both sides as sums over entries and use $|z|^2 = \bar{z}z$. $\square$

9.4 Eigenvalue bound and spectral gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization. For the rigidity theorem, one gauges A and B separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument to a block embedding of a mixed-transfer eigenvector.

Theorem 9.14 (Eigenvalue bound). *Let A, B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \leq 1$.*

Proof. Expand $F_{AB}^n(X)$ as a sum over words and use Cauchy–Schwarz, together with $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$, to obtain a uniform Frobenius-norm bound $\|F_{AB}^n(X)\|_{\text{HS}} \leq C_D \|X\|_{\text{HS}}$ for all n , where C_D depends only on D . If $F_{AB}(X) = \mu X$ and $X \neq 0$, then $|\mu|^n \|X\|_{\text{HS}} = \|F_{AB}^n(X)\|_{\text{HS}} \leq C_D \|X\|_{\text{HS}}$ for every n . Hence $|\mu| \leq 1$.

The argument uses only the trace-preserving normalization and does not appeal to Perron–Frobenius theory. $\square$

Theorem 9.15 (Spectral radius bound). $\rho(F_{AB}) \leq 1$ for normalized tensors.

Proof. The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \leq 1$ by Theorem 9.14. $\square$

Theorem 9.16 (Spectral radius ≥ 1 implies gauge-phase equivalence). *Let A, B be injective normalized MPS tensors. If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. Since $\rho(F_{AB}) \leq 1$ (Theorem 9.15) and $\rho(F_{AB}) \geq 1$ by hypothesis, there exists an eigenvalue μ with $|\mu| = 1$ and eigenvector $X \neq 0$: $F_{AB}(X) = \mu X$. The proof proceeds in five steps.

Step 1 (Separate gauging). Choose positive definite fixed points ρ_A, ρ_B of the transfer maps $\mathcal{E}_A, \mathcal{E}_B$ (Theorem 8.9) and gauge each tensor separately to a unital Kraus family (Theorem 8.10):

set $A'^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$ and similarly for B . Let $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$ denote the correspondingly gauged eigenvector.

Step 2 (Block Kraus family and off-diagonal embedding). Form the block Kraus operators $C^i = \begin{pmatrix} A'^i & 0 \\ 0 & B'^i \end{pmatrix}$. These constitute a unital Kraus family on $M_{2D}(\mathbb{C})$. The off-diagonal matrix $Y = \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}$ satisfies $E_C(Y) = \mu Y$.

Step 3 (KS equality and Kraus commutation). Because $|\mu| = 1$ and E_C is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 7.24) gives $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$. By the Kraus commutation relation (Theorem 7.23), Y commutes with every block Kraus operator: $C^i Y = Y C^i$ for all i .

Step 4 (Intertwining identity and invertibility). Expanding the commutation relation block by block gives $A'^i X' = X' B'^i$ for every i . Since $\{A'^i\}$ span $M_D(\mathbb{C})$ (injectivity of A), X' is a nonzero intertwiner from $M_D(\mathbb{C})$ to $M_D(\mathbb{C})$. The left multiplicative identity (Theorem 7.25) gives $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$, so $X'^\dagger X'$ is a positive semi-definite fixed point of $\mathcal{E}_{B'}$, positive definite by injectivity. Hence X' is invertible.

Step 5 (Skolem–Noether conclusion). The map $M \mapsto X' M (X')^{-1}$ is a $\mathbb{C}$ -algebra automorphism of $M_D(\mathbb{C})$ sending B'^i to A'^i . By Skolem–Noether (Theorem 4.9) this is an inner automorphism. Undoing the separate gauges gives $B^i = \bar{\mu} X A^i X^{-1}$ for all i , establishing gauge-phase equivalence (Definition 2.16).

The crucial point is that one gauges the individual transfer maps to unital form; one does not require the mixed transfer map itself to be simultaneously unital and trace-preserving. $\square$

Theorem 9.17 (Gauged peripheral eigenvector yields Kraus intertwining). *Let A and B be normalized tensors, and let $X \neq 0$ satisfy $F_{AB}(X) = \mu X$ with $|\mu| = 1$. Choose positive-definite fixed points ρ_A, ρ_B of the transfer maps and gauge to unital families $A'^i = S_A^{-1} A^i S_A$ and $B'^i = S_B^{-1} B^i S_B$ with $S_A S_A^\dagger = \rho_A$ and $S_B S_B^\dagger = \rho_B$. Then the transported matrix $X' = S_A^{-1} X (S_B^\dagger)^{-1}$ is nonzero, the families A' and B' are unital, and for every i one has*

$$X' (B'^i)^\dagger = \mu (A'^i)^\dagger X', \quad A'^i X' = \mu X' B'^i.$$

Proof. Block-embed the gauged families and transported eigenvector into a single unital Kraus map, apply Kadison–Schwarz equality for the modulus-one eigenvalue, then extract the intertwining identities from the block structure. $\square$

Theorem 9.18 (Intertwining yields gauge-phase equivalence). *Let A and B be tensors of the same bond dimension. Suppose there are invertible gauges taking them to families A' and B' , an invertible matrix X' , and a scalar μ with $|\mu| = 1$ such that*

$$A'^i X' = \mu X' B'^i \quad \text{for all } i.$$

Then A and B are gauge-phase equivalent.

Proof. Compose the two gauge matrices with the intertwiner to obtain a single invertible matrix conjugating A to a scalar multiple of B . $\square$

Remark 9.19 (Two gauge conventions). The right-canonical gauge used here, $A'^i = \rho^{-1/2} A^i \rho^{1/2}$, makes $\{A'^i\}$ a *unital* Kraus family ($\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$). A different convention, $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$, makes the family *trace-preserving* ($\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$). Both conventions appear in practice: the unital version via direct matrix square-root inversion (used in the spectral-gap proof above), and the trace-preserving version (used in the canonical-form reduction of Chapter 11).

Theorem 9.20 (Strict spectral gap). *Let A, B be injective normalized MPS tensors that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. $\rho(F_{AB}) \leq 1$ by Theorem 9.15. If $\rho(F_{AB}) = 1$, Theorem 9.16 gives gauge-phase equivalence, contradicting the hypothesis. $\square$

9.5 Spectral gap — rectangular case

Theorem 9.21 (Rectangular intertwining forces equal bond dimension). *Let A and B be injective tensors of bond dimensions D_1 and D_2 . If there exist $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ and $\mu \in \mathbb{C}$ such that*

$$X(B^i)^\dagger = \mu (A^i)^\dagger X, \quad A^i X = \mu X B^i \quad \text{for all } i,$$

then $D_1 = D_2$.

Proof. The intertwining relation shows that $\ker(X)$ is invariant under all generators of B . Since B is injective, its generators span the full matrix algebra, so $\ker(X)$ is trivial and $D_2 \leq D_1$. Applying the same argument to $X^\dagger$ gives $D_1 \leq D_2$. $\square$

Theorem 9.22 (Rectangular spectral gap). *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2 , with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\text{rect}}) < 1$.*

Proof. The same word-expansion argument as in Theorem 9.14 gives $|\mu| \leq 1$ for every eigenvalue of F_{AB}^{rect} . If $\rho(F_{AB}^{\text{rect}}) = 1$, choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge A and B separately to unital Kraus families and apply the block-embedding Kadison–Schwarz argument as in the square case. The resulting intertwining relations satisfy the hypotheses of Theorem 9.21, so $D_1 = D_2$, contradicting the hypothesis. $\square$

Theorem 9.23 (Rectangular overlap decay). *If $D_1 \neq D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. The spectral gap $\rho(F_{AB}^{\text{rect}}) < 1$ (Theorem 9.22) implies $(F_{AB}^{\text{rect}})^N \rightarrow 0$, so the operator trace converges to 0. By Theorem 9.8, $O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N) \rightarrow 0$. $\square$

9.6 MPV overlap decay

Theorem 9.24 (Transfer powers converge to zero). *Let A, B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \rightarrow 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. The spectral gap $\rho(F_{AB}) < 1$ (Theorem 9.20) implies $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$ by the Gelfand formula, so $F_{AB}^n(X) \rightarrow 0$ for every X . $\square$

Theorem 9.25 (Overlap decay). *Let A, B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. By Theorem 9.7, $O_{AB}(N) = \text{Tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 9.6): $\text{Tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}$. By Theorem 9.24, each $F_{AB}^N(E_{pq}) \rightarrow 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \rightarrow 0$. $\square$

9.7 Block separation

These asymptotic facts are the ingredients used later in the canonical-form separation arguments.

Remark 9.26 (Cross-correlation decay). If A and B are injective normalized tensors that are not gauge-phase equivalent, then for every $X \in M_D(\mathbb{C})$ one has $\text{tr}(F_{AB}^N(X)) \rightarrow 0$ as $N \rightarrow \infty$. This is the scalar trace consequence of the operator convergence $F_{AB}^N(X) \rightarrow 0$ from Theorem 9.24.

Remark 9.27 (Self-correlation persists). If ρ is a fixed point of $\mathcal{E}_A$, then $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$ for all $N \geq 0$. In particular this applies to the distinguished QPF fixed point from Theorem 8.9.

9.8 Spectral gap under irreducible-TP hypotheses

The preceding spectral-gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with trace-preserving normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), following Cirac et al. [CPGSV17b]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

Theorem 9.28 (Gauge-equivalence from irreducible-TP spectral radius). *Let A, B be irreducible trace-preserving normalized MPS tensors of bond dimension D . If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. Choose a modulus-one eigenvector as in Theorem 9.16. The proof follows the five-step structure of Theorem 9.16, with one key adaptation. Steps 1–3 (separate gauging, block Kraus family, KS equality and Kraus commutation) proceed identically: the gauging uses the unique positive definite fixed points guaranteed by irreducibility (Theorem 8.9) and the right-canonical gauge (Theorem 8.10). The change is in Step 4: the Kadison–Schwarz equality gives $X'^\dagger X'$ as a nonzero positive semidefinite fixed point of $\mathcal{E}_{B'}$. Under injectivity this is immediately positive definite (the fixed point is unique up to scaling). Under the weaker hypothesis of irreducibility, Theorem 8.3 upgrades it to positive definite, giving invertibility of X' . Formally, Theorem 9.17 combines the separately gauged eigenvector into the required intertwining relations. Once X' is invertible, Theorem 9.18 upgrades the intertwining to gauge-phase equivalence. $\square$

Theorem 9.29 (Strict spectral gap (irreducible-TP)). *Let A, B be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. Contrapositive of Theorem 9.28, combined with $\rho(F_{AB}) \leq 1$ (Theorem 9.15). $\square$

Theorem 9.30 (Transfer power decay (irreducible-TP)). *Under the hypotheses of Theorem 9.29, $F_{AB}^n(X) \rightarrow 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. By the Gelfand formula, $\rho(F_{AB}) < 1$ implies $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$. $\square$

Theorem 9.31 (Overlap decay (irreducible-TP)). *Let A, B be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. Combine the spectral gap (Theorem 9.29) with the overlap–trace identity (Theorem 9.7). $\square$

Theorem 9.32 (Rectangular spectral gap (irreducible-TP)). *Let A (bond dimension D_1) and B (bond dimension D_2) be irreducible trace-preserving normalized tensors with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\text{rect}}) < 1$.*

Proof. Assume for contradiction that $\rho = 1$. Choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C})$. The Cauchy–Schwarz rigidity argument produces an intertwiner X' with $A'^i X' = X' B'^i$ for all i . Since $\ker(X')$ is invariant under all B'^i , irreducibility forces $\ker(X') = \{0\}$; similarly for $X'^\dagger$. This gives $D_1 = D_2$, contradicting the hypothesis. $\square$

Theorem 9.33 (Rectangular overlap decay (irreducible-TP)). *If $D_1 \neq D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. Combine Theorem 9.32 with Theorem 9.8. $\square$

9.9 Conditional expectation from a faithful fixed point

Definition 9.34 (Conditional expectation). Given a $*$ -subalgebra $S \subseteq M_D(\mathbb{C})$, a linear map $P : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a conditional expectation onto S if it is idempotent, unital, has range contained in S , and satisfies $P(X) = X$ for all $X \in S$.

Definition 9.35 (Scalar conditional expectation). For $\sigma \geq 0$ with $\text{tr}(\sigma) \neq 0$, define

$$E_\sigma(X) := \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}.$$

Lemma 9.36 (Evaluation formula). $E_\sigma(X) = \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}$.

Theorem 9.37 (Unitality). $E_\sigma(\mathbb{1}) = \mathbb{1}$.

Theorem 9.38 (Idempotence). $E_\sigma(E_\sigma(X)) = E_\sigma(X)$ for all X .

Lemma 9.39 (Range is scalar). For every X , there exists $c \in \mathbb{C}$ with $E_\sigma(X) = c \mathbb{1}$.

Lemma 9.40 (Fixes scalar operators). For $c \in \mathbb{C}$, $E_\sigma(c \mathbb{1}) = c \mathbb{1}$.

Theorem 9.41 (Absorbs the adjoint map). If $T(\sigma) = \sigma$, then $E_\sigma \circ T^* = E_\sigma$.

Theorem 9.42 (Adjoint map absorbs scalar projection). If T is trace-preserving, then $T^* \circ E_\sigma = E_\sigma$.

Lemma 9.43 (Image lies in adjoint fixed points). For all X , $E_\sigma(X)$ is fixed by T^* .

Theorem 9.44 (Scalar map is a conditional expectation). Let K be trace-preserving, let $\rho > 0$ be a positive-definite fixed point of the associated channel $T(\rho) = \sum_i K_i \rho K_i^\dagger$ (so $T(\rho) = \rho$), and assume every adjoint fixed point of T^* is a scalar multiple of $\mathbb{1}$. Then E_ρ is a conditional expectation onto the adjoint fixed-point $*$ -subalgebra.

Theorem 9.45 (Wolf Cor. 6.6 — complete positivity of E_σ). For any positive semidefinite σ , the scalar conditional expectation E_σ is a completely positive map.

Proof. Write $S = \sqrt{\sigma}$ and define the Kraus family $K_{j,i} = |j\rangle\langle i| S$ indexed by $(j, i) \in \{1, \dots, D\}^2$. Using $|j\rangle\langle i| M |i\rangle\langle j| = M_{ii} |j\rangle\langle j|$ and summing first over i and then over j ,

$$\sum_{j,i} K_{j,i} X K_{j,i}^\dagger = \sum_j |j\rangle\langle j| \text{tr}(SXS) = \text{tr}(\sigma X) \mathbb{1},$$

where we used cyclicity of the trace and $S^2 = \sigma$. Hence $E_\sigma = (\text{tr} \sigma)^{-1} \cdot (X \mapsto \sum_{j,i} K_{j,i} X K_{j,i}^\dagger)$ is the Kraus map scaled by the nonnegative real number $(\text{tr} \sigma)^{-1}$, and therefore completely positive. $\square$

Theorem 9.46 (Wolf Cor. 6.6 — σ -trace preservation of E_σ). *If $\text{tr } \sigma \neq 0$, then for every X*

$$\text{tr}(\sigma E_\sigma(X)) = \text{tr}(\sigma X).$$

Proof. Direct computation: $\text{tr}(\sigma \cdot \frac{\text{tr}(\sigma X)}{\text{tr } \sigma} \mathbb{1}) = \frac{\text{tr}(\sigma X)}{\text{tr } \sigma} \text{tr } \sigma = \text{tr}(\sigma X)$. $\square$

9.10 Stationary support

This section records the support-projection theory behind stationary states, following [Wol12, §6.4].

Lemma 9.47 (Lower-triangular Kraus condition implies corner invariance). *Let A be an MPS tensor and let P be an orthogonal projection. If*

$$(\mathbb{1} - P)A^i P = 0 \quad \text{for all } i,$$

then

$$P \mathcal{E}_A(PXP) P = \mathcal{E}_A(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. Expand the transfer map and use $(\mathbb{1} - P)A^i P = 0$ to reduce each Kraus term to the P -corner. $\square$

Theorem 9.48 (Support projection invariance (Wolf Lemma 6.4)). *Let E be a quantum channel and $\rho \geq 0$ a positive semidefinite fixed point ($E(\rho) = \rho$). Let P denote the support projection of ρ . Then*

$$P E(PXP) P = E(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. Write E in Kraus form $E(X) = \sum_i K_i X K_i^\dagger$. From $E(\rho) = \rho$ and $\rho \geq 0$ one deduces $(\mathbb{1} - P)K_i P = 0$ for all i . Apply Lemma 9.47. $\square$

Definition 9.49 (Stationary state). For an irreducible channel E on $M_D(\mathbb{C})$ (with $D \geq 1$), the *stationary state* is the unique density-matrix fixed point ρ_∞ satisfying $E(\rho_\infty) = \rho_\infty$, $\rho_\infty > 0$, and $\text{tr}(\rho_\infty) = 1$. Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 8.9).

Definition 9.50 (Stationary support). The *stationary support* of an irreducible channel E is the support projection of the stationary state (Definition 9.49).

Theorem 9.51 (Prop. 6.9: stationary support is full). *For an irreducible channel E , the stationary support equals the identity: $\text{supp}(\rho_\infty) = \mathbb{1}$.*

Proof. The support projection P of ρ_∞ is an orthogonal projection satisfying $PE(PXP)P = E(PXP)$ for all X (Theorem 9.48). By irreducibility, $P = 0$ or $P = \mathbb{1}$. Since $\rho_\infty \neq 0$, we have $P \neq 0$, so $P = \mathbb{1}$. $\square$

Remark 9.52 (Props. 6.10–6.11: non-trivial formulations). The non-trivial formulations needed here are:

- **Prop. 6.10** (converse of Prop. 6.9): if the support projection of *every* PSD fixed point is full, then the channel is irreducible.
- **Prop. 6.11**: minimality of the stationary support among invariant projections, without assuming irreducibility.

These statements are deferred in the present chapter.

9.11 Wedderburn decomposition of the fixed-point algebra

Theorem 9.53 (Fixed-point algebra is semisimple). *The adjoint-fixed-point $*$ -subalgebra of a trace-preserving Kraus map on $M_D(\mathbb{C})$ is a semisimple ring.*

Proof. The subalgebra is finite-dimensional (as a subalgebra of $M_D(\mathbb{C})$), hence Artinian. By [Wol12, Thm. 6.14], it suffices to show the Jacobson radical J vanishes. If $x \in J$, then $x^*x \in J$ (left-ideal closure). The radical of an Artinian ring is nilpotent, so x^*x is nilpotent. A self-adjoint nilpotent matrix satisfies $\|x^*x\|^{2^k} = \|(x^*x)^{2^k}\| = 0$, hence $x^*x = 0$, and the C^* -identity gives $x = 0$. $\square$

Theorem 9.54 (Abstract Wedderburn–Artin decomposition (Wolf Thm. 6.14)). *There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that the adjoint-fixed-point $*$ -subalgebra is $\mathbb{C}$ -algebra isomorphic to $\prod_{k=1}^n M_{d_k}(\mathbb{C})$.*

Proof. Combine semisimplicity (Theorem 9.53) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field. $\square$

Theorem 9.55 (Wedderburn data with ambient dimension bound). *There exist integers n , block sizes $d_1, \dots, d_n \geq 1$, multiplicities $m_1, \dots, m_n \geq 1$, and a $\mathbb{C}$ -algebra isomorphism from the adjoint-fixed-point $*$ -subalgebra to*

$$\prod_{k=1}^n M_{d_k}(\mathbb{C})$$

such that

$$\sum_{k=1}^n d_k m_k \leq D.$$

Proof. Apply the abstract Wedderburn decomposition (Theorem 9.54). Since the adjoint-fixed-point algebra is realized as a subalgebra of the ambient matrix algebra $M_D(\mathbb{C})$, this realization also yields multiplicities m_k and the bound $\sum_k d_k m_k \leq D$. $\square$

Remark 9.56 (Pending part of Wolf Thm. 6.14). The stronger ambient statement that the adjoint-fixed-point algebra is unitarily conjugate to

$$0 \oplus \bigoplus_k M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$$

together with the density-block form $M_{d_k}(\mathbb{C}) \otimes \rho_k$ is not yet formalized.

Theorem 9.57 (Wolf Cor. 6.7). *Let $T(X) = \sum_i K_i X K_i^\dagger$ be trace-preserving, and let $\rho > 0$ satisfy $T(\rho) = \rho$. If*

$$F_T := \{X \in M_D(\mathbb{C}) \mid T(X) = X\},$$

then $\rho^{-1/2} F_T \rho^{-1/2}$ is a $$ -subalgebra of $M_D(\mathbb{C})$.*

Proof. Set $B_i = \rho^{-1/2} K_i \rho^{1/2}$. The equation $T(\rho) = \rho$ implies that the family $\{B_i\}$ is unital, and trace preservation of T implies that ρ is a positive-definite fixed point of the adjoint map of the gauged family. Apply Theorem 5.110 to B . The identity

$$\sum_i B_i X B_i^\dagger = X \iff T(\rho^{1/2} X \rho^{1/2}) = \rho^{1/2} X \rho^{1/2}$$

identifies the fixed points of the gauged map with $\rho^{-1/2} F_T \rho^{-1/2}$. $\square$

9.12 Further irreducibility, primitivity, and fixed-point criteria

This section records several criteria and equivalences from Wolf's Chapter 6 that are used later.

Theorem 9.58 (Wolf Thm. 6.2(3): exponential positivity condition). *Let E be an irreducible completely positive map, $A \geq 0$ with $A \neq 0$, and $t > 0$. Then $\exp(tE)(A)$ is positive definite. Conversely, if $\exp(tE)(A)$ is positive definite for every such t, A , then E is irreducible (full equivalence).*

Theorem 9.59 (Wolf Thm. 6.8: Kraus-span equivalence). *Under normalization, primitivity in Wolf's sense is equivalent to eventual full Kraus rank.*

Theorem 9.60 (Wolf Thm. 6.8: combined equivalence). *Under normalization, primitivity in Wolf's sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.*

Remark 9.61 (Wolf Thm. 6.8: pairwise equivalences). These are the pairwise equivalences appearing in Wolf's Theorem 6.8.

Theorem 9.62 (Wolf Thm. 6.15 (scalar fixed-point algebra case)). *In the scalar fixed-point algebra setting, the weighted map $X \mapsto \frac{\text{tr}(\rho X)}{\text{tr}(\rho)} \mathbb{1}$ is a conditional expectation onto the adjoint fixed-point $*$ -subalgebra.*

9.13 Multi-cycle block-permutation data

Definition 9.63 (Block-permutation data). Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$. A *block-permutation structure* for T consists of a finite index set ι , a family of orthogonal projections $P_k \in M_D(\mathbb{C})$ for $k \in \iota$, and a permutation $\sigma \in \text{Sym}(\iota)$ (not necessarily a single cycle — in general σ may have multiple disjoint cycles), such that

$$T(P_{\sigma(k)}) = P_k, \quad T(P_k X) = T(P_k) T(X), \quad T(X P_k) = T(X) T(P_k)$$

for every $k \in \iota$ and $X \in M_D(\mathbb{C})$.

Definition 9.64 (Multi-cycle block-permutation data). A *multi-cycle decomposition* of $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ records a finite cycle index ι , a per-cycle period $m : \iota \rightarrow \mathbb{N}_{>0}$, a projection family $P_{c,k} \in M_D(\mathbb{C})$ for $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$ consisting of orthogonal projections, together with the per-cycle cyclic action

$$T(P_{c,k+1}) = P_{c,k} \quad (k \in \{0, \dots, m(c)-1\}, c \in \iota)$$

and the multiplicative-domain factorisations $T(P_{c,k} X) = T(P_{c,k}) T(X)$ and $T(X P_{c,k}) = T(X) T(P_{c,k})$.

Theorem 9.65 (Per-cycle corner preservation). *For every cycle $c \in \iota$ and index $k \in \{0, \dots, m(c)-1\}$, the iterate $T^{m(c)}$ preserves the corner $P_{c,k} M_D(\mathbb{C}) P_{c,k}$.*

Proof. By induction on n , the n -th iterate T^n sends $P_{c,k+n} X P_{c,k+n}$ to $P_{c,k} T^n(X) P_{c,k}$, using the multiplicative-domain factorisations and the cyclic action $T(P_{c,j+1}) = P_{c,j}$ at each induction step. Specialising to $n = m(c)$ and using that the indices $k + m(c) = k$ in $\{0, \dots, m(c)-1\}$ yields corner preservation of $T^{m(c)}$. $\square$

Theorem 9.66 (Common-period corner preservation). *If $N \in \mathbb{N}$ is divisible by every per-cycle period $m(c)$, then T^N preserves every corner $P_{c,k} M_D(\mathbb{C}) P_{c,k}$ of the decomposition.*

Proof. Write $N = m(c)q$. Corner preservation is stable under taking powers, so $T^N = (T^{m(c)})^q$ preserves each corner by Theorem 9.65. $\square$

Definition 9.67 (Flattening to single-index block-permutation data). A multi-cycle decomposition gives a single-index block-permutation structure on the sigma index $\Sigma_{c \in \iota} \{0, \dots, m(c)-1\}$: the permutation is the sigma-product of per-cycle cyclic shifts $k \mapsto k+1$ (whose cycle decomposition has one cycle per $c \in \iota$), the projections are $(c, k) \mapsto P_{c,k}$, and the multiplicative-domain factorisations descend componentwise. The resulting map forgets the explicit cycle-indexing.

9.14 Peripheral eigenvalue group structure

Lemma 9.68 (Peripheral eigenvectors are invertible). *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If $X \neq 0$ satisfies $E(X) = \mu X$ with $|\mu| = 1$, then X is invertible.*

Proof. The KS equality gives $E(X^\dagger X) = X^\dagger X$, so $X^\dagger X$ is a nonzero PSD fixed point. Irreducibility upgrades it to positive definite, hence invertible, so $\det(X) \neq 0$. $\square$

Lemma 9.69 (Peripheral eigenvalues: product closure). *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If μ, ν are peripheral eigenvalues of E , then so is $\mu\nu$.*

Proof. Take eigenvectors X, Y for μ, ν . The KS equality at X puts X in the multiplicative domain, so $E(YX) = E(Y)E(X) = (\nu\mu)(YX)$. Since X, Y are units (Lemma 9.68), $YX \neq 0$ and $|\mu\nu| = 1$. $\square$

Theorem 9.70 (Peripheral eigenvalues: cyclic structure). *Let E be an irreducible unital Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is not required). Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. The peripheral eigenvalues form a finite subset of $\mathbb{C}$, contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of $\mathbb{C}^\times$. Any finite subgroup of $\mathbb{C}^\times$ is cyclic. The generator γ has order $m = |\text{peripheral spectrum}|$. $\square$

Theorem 9.71 (Peripheral eigenvalues form a cyclic group). *Let E be an irreducible unital trace-preserving Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that $m \mid D$ and the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. Apply Theorem 9.70 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: m mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace (by trace preservation), so $m \cdot \text{tr}(P_0) = D$ with $\text{tr}(P_0) \in \mathbb{N}$. $\square$

Theorem 9.72 (Period divides bond dimension). *Let E be as in Theorem 9.71, and let γ be a primitive m -th root of unity generating the peripheral eigenvalues. Then $m \mid D$.*

Proof. Follows from the cyclic projection decomposition and trace preservation, as in the final step of Theorem 9.71. $\square$

9.14.1 Cyclic decomposition of irreducible Schwarz maps

Definition 9.73 (Corner preservation). For an orthogonal projection P and linear map E , the corner $PM_D(\mathbb{C})P$ is preserved if $E(PXP) = PE(PXP)P$ for all X .

Definition 9.74 (Corner subspace). The corner subspace associated with P is $\{PXP : X \in M_D(\mathbb{C})\}$.

Definition 9.75 (Restricted corner map). The restriction of E to an invariant corner $PM_D(\mathbb{C})P$.

Definition 9.76 (Irreducible restriction on a corner). Irreducibility of the corner-restricted map.

Definition 9.77 (Rank of an orthogonal projection). The rank of an orthogonal projection $P \in M_D(\mathbb{C})$, i.e. the dimension of the corner algebra $PM_D(\mathbb{C})P$.

Theorem 9.78 (Compression isometry for an orthogonal projection). *Let $P \in M_D(\mathbb{C})$ be an orthogonal projection of rank n . The corner algebra $PM_D(\mathbb{C})P$ is linearly isomorphic to the full matrix algebra $M_n(\mathbb{C})$. The isomorphism is built from the spectral diagonalisation of P by conjugating the top-left $n \times n$ block by the unitary diagonalising P .*

Theorem 9.79 (Rank equals trace for an orthogonal projection). *The rank of an orthogonal projection P equals its trace: $n = \text{tr } P$.*

Theorem 9.80 (Peripheral unitary eigenvector). *For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.*

Theorem 9.81 (Powers on a peripheral-unitary orbit). *If $E(U) = \mu U$ with $|\mu| = 1$ and U unitary, then $E(U^n) = \mu^n U^n$ for all n .*

Theorem 9.82 (Normalized peripheral unitary). *Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.*

Theorem 9.83 (Cyclic projections from a peripheral unitary). *A primitive m -th peripheral root yields m orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by the channel.*

Theorem 9.84 (Cyclic decomposition for irreducible Schwarz maps). *Under irreducibility and faithful fixed point hypotheses, there is a full cyclic projection decomposition realizing the peripheral period.*

Theorem 9.85 (Power maps preserve each cyclic sector). *Appropriate powers of the channel preserve each sector corner.*

Theorem 9.86 (Irreducibility of restricted sector dynamics). *Each sector restriction is irreducible.*

Theorem 9.87 (Primitivity of restricted sector dynamics). *Each sector restriction is primitive.*

Theorem 9.88 (Corner invariance at the period). *The channel raised to the period preserves every cyclic corner.*

9.14.2 Group structure of the peripheral spectrum

Theorem 9.89 (Peripheral product closure). *The product of two peripheral eigenvalues is peripheral.*

Theorem 9.90 (Peripheral inverse closure). *The inverse of a peripheral eigenvalue is peripheral.*

Theorem 9.91 (Peripheral cyclic group with period divisibility). *The peripheral eigenvalues form a finite cyclic group whose order divides the dimension.*

Theorem 9.92 (Peripheral eigenvalue multiplicity one). *Each peripheral eigenvalue has algebraic multiplicity one.*

Proof. Given a peripheral unitary eigenvector U for γ , the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ -eigenvector X . By the scalar fixed-point lemma for irreducible unital maps, $XU^\dagger = c \cdot I$, hence $X = c \cdot U$ and the eigenspace is one-dimensional. $\square$

9.15 Primitive overlap convergence (spectral-gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument. For primitive normalized transfer maps, Chapter 5 supplies this spectral-gap input under explicit hypotheses.

Theorem 9.93 (Complementary spectral gap implies trace convergence to 1). *Let E be trace-preserving with fixed point ρ ($\text{tr}(\rho) \neq 0$) and let P be the fixed-point projection (Definition 5.23). If $\rho_{\text{spec}}(E - P) < 1$, then $\text{Tr}(E^n) \rightarrow 1$ as $n \rightarrow \infty$.*

Proof. By the power decomposition (Theorem 5.24), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\text{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \rightarrow 0$, so $\text{Tr}(E^n) \rightarrow \text{Tr}(P)$. The operator trace of P equals 1: since $P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$, we have $\text{Tr}(P) = \frac{\text{tr}(\rho)}{\text{tr}(\rho)} = 1$. $\square$

Theorem 9.94 (Self-overlap convergence from a complementary spectral gap). *Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection), then $O_{AA}(N) \rightarrow 1$ as $N \rightarrow \infty$.*

Proof. By Theorem 9.7 (with $A = B$), $O_{AA}(N) = \text{Tr}(\mathcal{E}_A^N)$. By Theorem 9.93, $\text{Tr}(\mathcal{E}_A^N) \rightarrow 1$. $\square$

Chapter 10

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D , the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

10.1 Cumulative span

Definition 10.1 (Word span). The *word span* at length n is the subspace

$$S_n(A) = \text{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, §II].

Definition 10.2 (Cumulative span). The *cumulative span* at level n is

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A).$$

Lemma 10.3 (Monotonicity of cumulative span). $T_n(A) \leq T_{n+1}(A)$ for all n .

Proof. T_{n+1} includes all words of length $\leq n+1$, which contains all words of length $\leq n$. $\square$

Lemma 10.4 (Dimension bound). $\dim T_n(A) \leq D^2$ for all n .

Proof. $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. $\square$

Theorem 10.5 (Cumulative span stabilisation). If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \geq n$.

Proof. If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2} , so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. $\square$

Theorem 10.6 (Dimension growth). If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.

Proof. $T_n \leq T_{n+1}$ (Lemma 10.3) and $T_n \neq T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. $\square$

Lemma 10.7 (Word span characterises block injectivity). $S_L(A) = M_D(\mathbb{C})$ if and only if A is L -block injective.

Proof. Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. $\square$

10.2 Fitting decomposition

Definition 10.8 (Fitting decomposition). A *Fitting decomposition* of a linear endomorphism $f : V \rightarrow V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ -eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V ,
4. the generalized eigenspaces are linearly independent.

Theorem 10.9 (Fitting decomposition exists). *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. $\square$

Theorem 10.10 (Nilpotent power bound). *If f is nilpotent on a space of dimension n , then $f^n = 0$.*

Proof. A nilpotent endomorphism on an n -dimensional space has nilpotency index $\leq n$. $\square$

10.3 Nonzero trace product

Theorem 10.11 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound). *If A is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.*

Proof. By Theorem 10.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 10.4), after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0 , hence $T_{L_0} = M_D(\mathbb{C})$). $\square$

Theorem 10.12 (Cumulative span reaches $M_D(\mathbb{C})$). *If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. Take n from Theorem 10.11; since $n \leq D^2$ and T is monotone, $T_{D^2} = M_D(\mathbb{C})$. $\square$

Theorem 10.13 (Nonzero trace product). *If A is normal, then there exists a word w of length $\leq D^2$ such that $\text{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is $D \geq 1$; when $D = 0$ the normality hypothesis is vacuous.*

Proof. By Theorem 10.12, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. In the nontrivial case $D \geq 1$, at least one has nonzero trace (since $\text{tr}(\mathbb{1}) = D \neq 0$). $\square$

Theorem 10.14 (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound). *If A is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \text{kr}(A)$. This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches $M_D(\mathbb{C})$ by index $D^2 - d' + 1 \leq D^2$.*

Proof. The key observation is that if $T_n(A) \neq T_{n+1}(A)$, the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, we have $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. $\square$

Theorem 10.15 (Sharp nonzero trace product). *If A is normal, then there exists a word w of length $\leq D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1] with the sharp length bound.*

Proof. By Theorem 10.14, the identity $\mathbf{1}$ lies in $T_{D^2-d'+1}(A)$, hence is a linear combination of word products of length $\leq D^2 - d' + 1$. At least one such product has nonzero trace. $\square$

Theorem 10.16 (Sharp nonzero trace product — positive length). *If A is normal and $D \geq 2$, then there exists a word w of positive length $1 \leq |w| \leq D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This strengthens Theorem 10.15 by requiring $|w| \geq 1$, which is needed for the blocking argument in Theorem 10.28.*

Proof. Define the *positive-level* cumulative span $V_n = \text{span}\{A^w : 1 \leq |w| \leq n\}$. For $D \geq 2$, the span $S_0(A) = \text{span}\{\mathbf{1}\}$ has dimension 1, so $V_1 \supseteq S_1(A)$ has dimension $\geq d' \geq 1$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if V_n stabilises and is not $M_D(\mathbb{C})$, then V_n is closed under left-multiplication by each A^i , so $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$ for all $N \geq 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbf{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since $\text{tr}(\mathbf{1}) = D \geq 2 \neq 0$). $\square$

10.4 Eigenvector extraction

Theorem 10.17 (Nonzero trace implies eigenvector). *If $M \in M_D(\mathbb{C})$ has $\text{tr}(M) \neq 0$, then M has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ with $M\varphi = \mu\varphi$.*

Proof. A matrix with nonzero trace has a nonzero eigenvalue (since trace = sum of eigenvalues counted with multiplicity). The Fitting decomposition (Theorem 10.9) ensures that the corresponding generalized eigenspace is nontrivial. On that generalized eigenspace, $M - \mu\mathbf{1}$ is nilpotent, so its kernel is nontrivial. Any nonzero vector in $\ker(M - \mu\mathbf{1})$ is therefore a genuine eigenvector with eigenvalue μ . $\square$

Theorem 10.18 (Eigenvalue extraction). *If A is normal, then there exist a word w_0 with $|w_0| \leq D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.*

Proof. By Theorem 10.13, there is a word w_0 with $\text{tr}(A^{w_0}) \neq 0$ and $|w_0| \leq D^2$. By Theorem 10.17, A^{w_0} has a nonzero eigenvector. $\square$

10.5 Eigenvector spreading

Definition 10.19 (Cumulative vector span). Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w\varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 10.20 (Vector span stabilisation). *If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for all $m \geq n$.*

Proof. Same argument as Theorem 10.5: left-multiplication by A^i cannot escape the stabilised span. $\square$

Lemma 10.21 (Vector span from matrix span). *If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.*

Proof. Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v is in T_N . Applying such matrices to φ recovers all of $\mathbb{C}^D$. $\square$

Theorem 10.22 (Eigenvector spreading). *Let A be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].*

Proof. The vector span K_n is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises (Theorem 10.20). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 10.12, so Lemma 10.21 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D - 1$. $\square$

10.6 Proof of the cumulative Wielandt bound

Definition 10.23 (Wielandt analysis). A *Wielandt analysis* for a normal MPS tensor A consists of: a word w_0 of length $\leq D^2$, a nonzero eigenvalue μ , and a nonzero eigenvector φ of the matrix A^{w_0} , such that $A^{w_0}\varphi = \mu\varphi$.

Theorem 10.24 (Wielandt analysis exists). *Every normal MPS tensor admits a Wielandt analysis.*

Proof. Immediate from Theorem 10.18. $\square$

Theorem 10.25 (Wielandt chain). *Let A be a normal MPS tensor with bond dimension $D \geq 1$. Then the following hold simultaneously:*

1. $T_{D^2}(A) = M_D(\mathbb{C})$ (cumulative span stabilises);
2. there exists a word w_0 with $|w_0| \leq D^2$ and $\text{tr}(A^{w_0}) \neq 0$;
3. there exist $w_0, \mu \neq 0, \varphi \neq 0$ with $|w_0| \leq D^2$ and $A^{w_0}\varphi = \mu\varphi$;
4. for every $\varphi \neq 0$, $K_{D^2}(A, \varphi) = \mathbb{C}^D$.

Proof. Items (1)–(3) follow from earlier results (Theorems 10.12, 10.24). Item (4) combines these with the eigenvector spreading bound (Theorem 10.22). $\square$

Theorem 10.26 (Wielandt bound). *If A is a normal MPS tensor with bond dimension D , then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most D^2 . This is the cumulative-span part of [SPGWC10, Theorem 1]. The paper’s theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion $S_N(A) = M_D(\mathbb{C})$, proved below in Theorem 10.41.*

Proof. Immediate from Theorem 10.12. $\square$

Remark 10.27 (Explicit blocking-length estimates). Theorem 10.26 gives the D^2 cumulative-span bound. The full-rank index bound $\iota(A) \leq (D^2 - d + 1) \cdot D^2$ (Theorem 1 case (1) of [SPGWC10]) is proved in Theorem 10.28 below. The sharp blocking-length estimates for injectivity (D^4) and bi-canonical form ($3D^5$) from [SPGWC10] are not proved here.

Theorem 10.28 (Wielandt’s inequality — general bound). *Let A be a normalized MPS tensor ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) that is primitive in the paper sense. Then*

$$\iota(A) \leq (D^2 - \text{kr}(A) + 1) \cdot D^2,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

Proof.

1. $D = 1$: In this case $\iota(A) = 0$, so the bound holds trivially.
2. $D \geq 2$: By Theorem 10.16, there exists a *positive-length* word w with $|w| \leq D^2 - d' + 1$ and $\text{tr}(A^w) \neq 0$, where $d' = \text{kr}(A)$. The matrix A^w has nonzero trace, hence a nonzero eigenvalue μ and eigenvector φ (Theorem 10.17). Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then B is normal (Theorem 10.36), and the encoding of w as a single block index gives $B^{i_0} = A^w$.
 - (a) If A^w is invertible (invertible case), $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.
 - (b) If A^w is non-invertible (non-invertible case), the eigenvector φ spans a proper subspace, and the Wielandt chain argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

In the invertible case, the invertible Kraus operator B^{i_0} allows direct dimension growth, giving $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$ where $k_B = \text{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; Theorem 10.41 only gives the qualitative conclusion $\exists N : S_N(B) = M_D(\mathbb{C})$ without this explicit bound. In both cases $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$. □

Remark 10.29 (Quantitative vs. qualitative content of Theorem 10.28). The quantitative bound $\iota(A) \leq (D^2 - d' + 1) \cdot D^2$ is cited from [SPGWC10, Theorem 1, case (1)]. The blueprint’s own lemmas prove the qualitative conclusion used on the fundamental-theorem route, namely Theorem 10.41: there exists N with $S_N(A) = M_D(\mathbb{C})$. The quantitative bound is not on the fundamental-theorem critical path.

10.7 Fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices).

Definition 10.30 (Fixed-length vector span). The *fixed-length vector span* at length n is

$$H_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

Lemma 10.31 (Word span products). $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$: the product of word spans at lengths m and n is contained in the word span at length $m + n$.

Proof. Every product $A^{w_1}A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1w_2}$ with $|w_1w_2| = m + n$. $\square$

Lemma 10.32 (Eigenvector padding). Suppose $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$. Then for all n ,

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ then already $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. Any word w with $|w| \leq n$ can be padded to exact length n by appending $n - |w|$ copies of the index i_0 . The eigenvector relation gives $A^{w \cdot i_0^k}\varphi = \mu^k \cdot A^w\varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n . The reverse inclusion $H_n \leq K_n$ is immediate. $\square$

Theorem 10.33 (Fixed-length vector spanning implies matrix spanning). Assume:

1. $H_n(A, \varphi) = \mathbb{C}^D$ (length- n word products applied to φ span all of $\mathbb{C}^D$), and
2. for each standard basis vector e_j , the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. Fix a matrix unit E_{ij} . By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 10.31. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. $\square$

Lemma 10.34 (Blocking transfer). If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$.

Proof. Each length- n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length- nL word in the original alphabet $\{0, \dots, d - 1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. $\square$

Remark 10.35 (Relation with Lemma 2(b)). In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with a product argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 10.33 is the second step. Theorem 10.40 supplies the rank-one input in blocked form, and Theorem 10.41 then gives a blocked, fixed-length word-span saturation result.

10.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) argument to the *blocked tensor* $A^{[L]}$ and develops the “rectangular span” used to produce rank-one elements.

Theorem 10.36 (Blocking preserves normality). If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.

Proof. Choose N with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M \cdot \mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A)$$

by Lemma 10.31. Iterating this inclusion gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$; taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma 10.37, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. $\square$

Lemma 10.37 (Chunking: word span embeds into blocked word span). $S_{nL}(A) \subseteq S_n(A^{[L]})$ for all n, L .

Proof. Every length- nL word in the original alphabet can be split into n consecutive chunks of length L , each of which is a single blocked Kraus operator. By induction on n , using the word span product lemma (Lemma 10.31). $\square$

Theorem 10.38 (Word eigenvector gives blocked eigenvector). *If $A^{w_0}\varphi = \mu\varphi$ for a word w_0 of length L , then $(A^{[L]})^{j_0}\varphi = \mu\varphi$, where j_0 is the encoded blocked index corresponding to w_0 .*

Proof. By definition, $(A^{[L]})^{j_0} = A^{w_0}$. $\square$

Theorem 10.39 (Blocked fixed-length matrix spanning). *Suppose A is normal and $L > 0$, and we have:*

1. *a word w_0 of length L with eigenvector $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0, \varphi \neq 0$),*
2. *a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0, \psi \neq 0$),*
3. *a rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for some m .*

Then there exists N such that $S_N(A) = M_D(\mathbb{C})$.

Remark. The proof gives the explicit bound $N = ((D-1) + m + (D-1)) \cdot L$; the statement above records only the existential conclusion.

Proof. Apply Theorem 10.33 to the blocked tensor $A^{[L]}$, which is normal by Theorem 10.36. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 10.38. Transfer back to the original tensor via Lemma 10.34. $\square$

Theorem 10.40 (Rank-one extraction for blocked tensor). *Suppose A is normal and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0 , nonzero vectors φ, ψ , nonzero scalars μ, ν , and $m \in \mathbb{N}$ such that:*

1. $A^{\sigma_0}\varphi = \mu\varphi$,
2. $(A^{\tau_0})^T\psi = \nu\psi$,
3. $\varphi\psi^T \in S_m(A^{[N_0]})$.

Proof. Set $B = A^{[N_0]}$. Since $S_{N_0}(A) = M_D(\mathbb{C})$, we have $S_1(B) = M_D(\mathbb{C})$ by definition of the blocked tensor. Hence $\mathbb{1} \in S_1(B)$, so in the nontrivial case $D \geq 1$ not all blocked generators can have trace zero. Choose j_0 with $\text{tr}(B^{j_0}) \neq 0$. By Theorem 10.17, B^{j_0} has a nonzero eigenvector φ with eigenvalue $\mu \neq 0$. Writing j_0 as the blocked index corresponding to a word σ_0 of length N_0 gives $A^{\sigma_0}\varphi = \mu\varphi$. Applying the same argument to the transpose blocked tensor gives a word τ_0 of length N_0 , a nonzero vector ψ , and $\nu \neq 0$ with $(A^{\tau_0})^T\psi = \nu\psi$. Since A is normal and $N_0 \geq 1$, the blocked tensor B is normal by Theorem 10.36. Therefore there exists M with $S_M(B) = M_D(\mathbb{C})$. In particular $\varphi\psi^T \in S_M(B)$. Taking $m = M$ proves (3). $\square$

Theorem 10.41 (Fixed-length word-span saturation via blocking). *If A is a normal MPS tensor with bond dimension D , then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].*

Proof. By normality, some N_0 has $S_{N_0}(A) = M_D(\mathbb{C})$. If $N_0 = 0$ we are done. Otherwise, Theorem 10.40 produces eigenvectors and a rank-one element $\varphi\psi^T \in S_m(A^{[N_0]})$. Theorem 10.39 then gives $S_{((D-1)+m+(D-1)) \cdot N_0}(A) = M_D(\mathbb{C})$. $\square$

Remark 10.42 (Blocking argument for fixed-length spanning). Theorem 10.41 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 10.40) and then applying the fixed-length matrix-spanning theorem (Theorem 10.39). Thus the fixed-length span of A is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

10.9 Primitive MPS tensors

Definition 10.43 (Primitive MPS tensor). Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \quad \rho \geq 0, \quad \rho \neq 0, \quad \mathcal{E}_A(\rho) = \rho,$$

and, if P is the fixed-point projection associated to ρ , then $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$. When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor. This is weaker than the paper's notion, which requires $\rho > 0$; see Remark 10.49. The stronger positive-definite hypothesis is imposed separately in Theorems 10.50 and 10.61.

Theorem 10.44 (Primitive overlap convergence). *If A is a primitive MPS tensor, then $O_{AA}(N) \rightarrow 1$.*

Proof. Immediate from the self-overlap convergence under a complementary spectral gap (Theorem 9.94). $\square$

10.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 10.43: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1 .

Theorem 10.45 (Primitive MPS tensors have nonzero word products). *If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length exactly n with $A^w \neq 0$.*

Proof. The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. $\square$

Theorem 10.46 (Iterated transfer does not vanish). *If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. By Theorem 10.45, there exists a word w of length n with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. $\square$

10.10.1 From primitivity to normality

The primitivity condition of Definition 10.43 combines TP normalization, a nonzero PSD fixed point, and a complementary spectral gap. It is connected to normality through the following results.

Theorem 10.47 (Fixed-point uniqueness from spectral gap). *If A is a primitive MPS tensor with PSD fixed point ρ , then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ : $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$.*

Proof. Set $\sigma' = \sigma - \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$. Then $\text{tr}(\sigma') = 0$ and $(E - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $E - P_\rho$, contradicting the spectral gap $\rho_{\text{spec}}(E - P_\rho) < 1$. $\square$

Theorem 10.48 (Complement powers tend to zero). *If A is a primitive MPS tensor with PSD fixed point ρ , then $(\mathcal{E}_A - P_\rho)^n \rightarrow 0$ in operator norm.*

Proof. The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the spectral-gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. $\square$

Remark 10.49 (Primitivity: spectral gap vs. positive definiteness). The primitivity condition used in §10.10.1 has three parts: TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, a nonzero PSD fixed point ρ , and the spectral gap $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. The notion of “primitive” in [SPGWC10, Proposition 3] additionally requires ρ to be positive definite (full rank). Without positive definiteness, the spectral-gap condition alone does *not* imply irreducibility or normality. A counterexample is $D = 2$, $\rho = \text{diag}(1, 0)$.

With the additional hypothesis $\rho > 0$, Theorem 10.50 gives irreducibility. Together with Burnside’s theorem (Chapter 11, §11.4.1) and Theorem 10.59, one obtains

$$\text{primitive} + \rho \text{ positive definite} \Rightarrow \text{irreducible} \Rightarrow \text{alg}(A) = M_D(\mathbb{C}) \Rightarrow \text{normal}.$$

The last step uses the aperiodicity condition $\mathbb{1} \in S_1(A)$ from Remark 10.54. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies the paper’s primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

10.10.2 Primitive implies irreducible

Theorem 10.50 (Primitive with positive-definite fixed point implies irreducible tensor). *If A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV17a, §2.3]).*

Proof. By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq \mathbb{1}$, $(\mathbb{1} - P)A^i P = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(\mathbb{1} - P)A^i P = 0$ implies $(\mathbb{1} - P) \cdot \mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).
2. *Convergence:* By Theorem 10.48, $\mathcal{E}_A^n(\sigma) \rightarrow \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$. Hence $(\mathbb{1} - P) \cdot \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho = 0$.
3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\text{tr}(P\rho P) > 0$, so $\frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \neq 0$. Therefore $(\mathbb{1} - P)\rho = 0$.
4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(\mathbb{1} - P)\rho = 0$ we get $\mathbb{1} - P = 0$, i.e. $P = \mathbb{1}$, contradicting $P \neq \mathbb{1}$.

$\square$

10.10.3 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect the spectral-gap primitivity of Definition 10.43 directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

Theorem 10.51 (Primitive MPS with positive-definite fixed point has irreducible transfer map). *If A is a primitive MPS tensor with positive-definite fixed point ρ , then the transfer map $\mathcal{E}_A$ is irreducible.*

The proof uses fixed-point uniqueness (Theorem 10.47): every PSD fixed point is proportional to ρ , so Wolf's criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.

Proof. By Theorem 10.47, if $\sigma \geq 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\text{tr} \sigma}{\text{tr} \rho} \rho$. Since $\rho > 0$, Wolf's uniqueness criterion for irreducibility applies. $\square$

Theorem 10.52 (Primitive MPS with positive-definite fixed point is strongly irreducible). *If A is a primitive MPS tensor with positive-definite fixed point ρ , then A is strongly irreducible in the sense of [SPGWC10, Proposition 3(c)]: $\mathcal{E}_A$ is irreducible, the fixed point ρ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$.*

Proof. Combine Theorem 10.51 (irreducibility) with the peripheral-spectrum primitivity from the spectral-gap hypothesis (Theorem 10.44). $\square$

Theorem 10.53 (Primitive MPS with positive-definite fixed point is normal). *If A is a primitive MPS tensor with positive-definite fixed point ρ , then A is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.*

This is the key connection between the spectral-gap primitivity of Definition 10.43 and normality, filling the gap that the aperiodicity-based argument in §10.11 leaves open.

Proof. By Theorem 10.52, A is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 9.60 identifies this primitivity condition directly with normality (equivalently, eventual full Kraus rank). $\square$

10.11 Cumulative span to word span

Cumulative spanning ($T_N(A) = M_D(\mathbb{C})$, the union of word spans of all lengths $\leq N$) is weaker than normality ($S_L(A) = M_D(\mathbb{C})$ for a single length L). The gap is closed by an aperiodicity hypothesis.

Remark 10.54 (Aperiodicity condition). We say A satisfies the *aperiodicity condition* if $\mathbb{1} \in S_1(A)$, i.e. the identity matrix lies in the span of the Kraus operators $\{A^i\}$. This is an additional hypothesis. The one-sided TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ does *not* imply $\mathbb{1} \in S_1(A)$; it only gives a quadratic identity among the Kraus operators. The aperiodicity condition is kept explicit in this section for generality. For the paper-style primitive condition (peripheral-spectrum primitivity together with a positive-definite fixed point), normality follows directly from Wolf's primitivity equivalence (Theorem 9.60) without passing through aperiodicity.

Remark 10.55 (Counterexample: cumulative spanning does not imply normality). In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ (the off-diagonal matrix units). Then $\text{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n . Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n , and A is not normal. The aperiodicity condition $\mathbb{1} \in S_1(A)$ fails: $S_1(A) = \text{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem 10.56 (Word span monotonicity from aperiodicity). *If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for all n .*

Proof. For any $M \in S_n(A)$, $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$ by Lemma 10.31. $\square$

Theorem 10.57 (Cumulative span equals word span under aperiodicity). *If $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for all n .*

Proof. By Theorem 10.56, the word spans are monotone: $S_0 \leq S_1 \leq \dots$. The supremum $T_n = S_0 + \dots + S_n$ therefore equals the largest term S_n . $\square$

Theorem 10.58 (Cumulative spanning plus aperiodicity implies normality). *If $T_N(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.*

Proof. By Theorem 10.57, $S_N(A) = T_N(A) = M_D(\mathbb{C})$. $\square$

Theorem 10.59 (Algebra span plus aperiodicity implies normality). *If $\text{alg}(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.*

Proof. Since $\text{alg}(A) = M_D(\mathbb{C})$, there exists N with $T_N(A) = M_D(\mathbb{C})$ (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 10.58. $\square$

Theorem 10.60 (Irreducible tensor plus aperiodicity implies normality). *If A is an irreducible tensor, $D > 0$, and $\mathbb{1} \in S_1(A)$, then A is normal. This assembles the full chain:*

$$\text{irreducible tensor} \xrightarrow{11.21} \text{irreducible action} \xrightarrow{\text{Burnside}} \text{alg}(A) = M_D(\mathbb{C}) \xrightarrow{10.59} \text{normal}.$$

This corresponds to the irreducibility-to-normality direction of [SPGWC10, Proposition 3]. For the Burnside step, we appeal forward to Theorems 11.21 and 11.22 in Chapter 11; this is a presentation-level forward dependency, but the mathematical dependency is acyclic.

Proof. By Theorem 11.21, A acts irreducibly on $\mathbb{C}^D$. By Theorem 11.22, $\text{alg}(A) = M_D(\mathbb{C})$. By Theorem 10.59 with the aperiodicity hypothesis, A is normal. $\square$

Theorem 10.61 (Primitive + positive definite fixed point imply normality). *If A is a primitive MPS tensor with fixed point ρ and ρ is positive definite, then A is normal. No aperiodicity assumption is needed.*

Proof. By Theorem 10.52, A is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 9.60 therefore gives normality directly, hence eventual full Kraus rank. $\square$

Remark 10.62 (Normal canonical form avoids the aperiodicity step). The aperiodicity results in §10.11 are needed only for the block-injective argument through Definition 2.23, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical theory used later in Chapters 11–13 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV17a]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

Chapter 11

Canonical Form Reduction

This chapter reduces a general MPS tensor to block-diagonal data via iterated invariant-subspace decomposition. Two reduction schemes are relevant. The block-injective reduction aims at injective TP-normalized blocks and the canonical form conditions of Chapter 12. The normal-canonical reduction uses a weighted family of irreducible TP-normalized blocks with primitive transfer maps and pairwise distinct nonzero weight moduli, in the sense of [CPGSV17a]. The Perron–Frobenius eigenvector existence and TP-gauge construction are in Chapter 8; this chapter uses them to produce left-canonical normalization data and the normal canonical form data. The reduction draws on [PGVWC07], [CPGSV21], and [Wol12, Ch. 6].

11.1 Block-diagonal construction

Definition 11.1 (Block-diagonal tensor from blocks). Recall the block-diagonal tensor of Definition 2.27. Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block MPS tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, we write $\bigoplus_{k=1}^r \mu_k A_k$ for the tensor of total bond dimension $D = \sum_k D_k$ whose Kraus operators are $A^i = \bigoplus_{k=1}^r \mu_k A_k^i$, obtained by placing the scaled block Kraus operators on the diagonal.

Theorem 11.2 (Per-block single-block FT). *If all block tensors A_k are injective and the per-block same-MPV condition holds (A_k and B_k generate the same MPV family for each k), then each pair (A_k, B_k) is gauge equivalent.*

Proof. Apply the single-block fundamental theorem (Theorem 4.11) to each block independently. $\square$

Theorem 11.3 (Per-block same MPV implies global same MPV). *If A_k and B_k generate the same MPV family for each block k , then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

Proof. By the MPV decomposition (Theorem 2.28), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. $\square$

Theorem 11.4 (Global multi-block fundamental theorem). *If each block A_k is injective and each pair (A_k, B_k) generates the same MPV family, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. Apply the per-block single-block fundamental theorem (Theorem 11.2) to obtain blockwise gauge equivalences from injectivity and equality of MPV families, then combine them into a global gauge for the block-diagonal tensors. $\square$

11.2 Transfer map normalization

Theorem 11.5 (Scaling preserves injectivity). *If A is injective and $\zeta \neq 0$, then ζA is injective.*

Proof. Scaling by a nonzero scalar preserves linear independence and hence spanning. $\square$

Theorem 11.6 (Transfer map under scaling). $\mathcal{E}_{\zeta A}(X) = |\zeta|^2 \mathcal{E}_A(X)$.

Proof. Each summand: $(\zeta A^i)X(\zeta A^i)^\dagger = |\zeta|^2 A^i X(A^i)^\dagger$. $\square$

Theorem 11.7 (Phase-scaling preserves normalization). *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $|\zeta| = 1$, then $\sum_i (\zeta A^i)^\dagger (\zeta A^i) = \mathbb{1}$.*

Proof. $(\zeta A^i)^\dagger (\zeta A^i) = \bar{\zeta} \zeta (A^i)^\dagger A^i = (A^i)^\dagger A^i$, since $\bar{\zeta} \zeta = |\zeta|^2 = 1$. $\square$

Theorem 11.8 (MPV under block normalization). *For any nonzero scaling factors μ_k , set $\eta_k := \mu_k/|\mu_k|$ and $B_k := |\mu_k|A_k$. Then $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \eta_k B_k$ generate the same MPV family. Equivalently, the k -th block contribution factors as*

$$\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma,$$

so the modulus contributes only through $|\mu_k|^N$, while the remaining factor η_k is a unit-modulus phase.

Proof. By the MPV decomposition (Theorem 2.28), the k -th contribution of $\bigoplus_k \mu_k A_k$ is $\mu_k^N V^{(N)}(A_k)_\sigma$. For $B_k = |\mu_k|A_k$, Lemma 2.15 gives $V^{(N)}(B_k)_\sigma = |\mu_k|^N V^{(N)}(A_k)_\sigma$. Hence $\eta_k^N V^{(N)}(B_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma = \mu_k^N V^{(N)}(A_k)_\sigma$ for every block, so the total MPV families agree. $\square$

Remark 11.9 (Why blockwise cancellation cannot be separated without normalization). The implication

$$\mathcal{V}(\bigoplus_{k=1}^r \mu_k A_k) = \mathcal{V}(\bigoplus_{k=1}^r \mu_k B_k) \implies \mathcal{V}(A_k) = \mathcal{V}(B_k) \text{ for all } k$$

is false if one assumes only that the weights μ_k are pairwise distinct and nonzero. Already for $r = 2$, with $\mu = (1, 2)$ and scalar 1×1 blocks,

$$A = (1, 3/2), \quad B = (3, 1/2),$$

one has

$$1^N(1^N - 3^N) + 2^N((3/2)^N - (1/2)^N) = 0$$

for every $N \geq 1$, so the two direct sums generate the same MPV family, even though the first blocks already disagree at $N = 1$. The reason is exactly the rescaling freedom described above: a nonzero scalar may be moved from the weight μ_k into the block tensor A_k , or conversely, without changing the contribution $\mu_k^N V^{(N)}(A_k)_\sigma$. Thus distinct nonzero weights alone do not identify the individual blocks.

The canonical-form hypotheses remove this ambiguity. If each block is first placed in the normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, then further rescaling is restricted to unit-modulus phases, and the strict ordering $|\mu_1| > \dots > |\mu_r|$ used in [PGVWC07, CPGSV21] prevents trading a nontrivial modulus between the weight and the block tensor.

11.3 Invariant subspace decomposition

Theorem 11.10 (Unitary conjugation preserves the MPV family). *If U is a unitary matrix, then A and $U^\dagger A U$ generate the same MPV family.*

Proof. By cyclicity of the trace: $\text{tr}((U^\dagger A^{i_1} U) \dots (U^\dagger A^{i_N} U)) = \text{tr}(A^{i_1} \dots A^{i_N})$. $\square$

Definition 11.11 (Support projection). For a positive semi-definite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ . It is constructed via the spectral decomposition: if $\rho = U \text{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$, then $P = U \text{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger$, where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Definition 11.12 (Invariant projection predicate). An MPS tensor A has an *invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P)A^i P = 0$ for all i . A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 11.15).

Theorem 11.13 (PSD fixed point gives invariant projection). *Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be the support projection of ρ . Then for all i , $(\mathbb{1} - P)A^i P = 0$: the range of P is invariant under all A^i .*

Proof. From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i \rho (A^i)^\dagger$, $(\mathbb{1} - P)A^i \rho (A^i)^\dagger (\mathbb{1} - P) = 0$ for each i . Since ρ is supported on the range of P , this forces $(\mathbb{1} - P)A^i P = 0$.

This is the finite-dimensional version of [Wol12, Prop. 6.10] (“Stationary subspaces”: the support projection of any stationary state is sub-harmonic, i.e. invariant under the channel). The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Thm. 6.2]; see also [CPGSV21, §IV.A.1]. $\square$

Theorem 11.14 (Two-block decomposition). *If the transfer map has a PSD fixed point ρ that is not positive definite (i.e. ρ has a nontrivial kernel), then there exist MPS tensors A_1, A_2 of smaller bond dimensions such that the two-block tensor $A_1 \oplus A_2$ generates the same MPV family as A : $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$.*

Proof. The support projection P of ρ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and ρ is not PD). Theorem 11.13 gives $(\mathbb{1} - P)A^i P = 0$ for all i . A unitary change of basis that block-diagonalizes P puts A^i in upper-triangular block form, with diagonal blocks B_{11}^i and B_{22}^i . Theorem 11.10 ensures the MPV is preserved by the conjugation. The trace of the upper-triangular product equals the sum of the diagonal-block traces, so the block-diagonal tensor obtained by discarding the off-diagonal blocks has the same MPV coefficients as the conjugated tensor, and hence as A . $\square$

11.4 Iterated reduction

Definition 11.15 (Irreducible tensor). An MPS tensor A is *irreducible* if it has no nontrivial invariant projection.

Theorem 11.16 (Irreducible block decomposition). *Every MPS tensor A admits a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ where each A_k is irreducible and $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$.*

Proof. By strong induction on the bond dimension D . If A is already irreducible, take $r = 1$. Otherwise, by Definition 11.15, A has a nontrivial invariant projection P . Diagonalizing P yields the same two-block upper-triangular splitting used in Theorem 11.14; the block-diagonal tensor obtained by discarding the off-diagonal part has the same MPV family as A , and the two diagonal blocks have strictly smaller bond dimensions. Iterating this splitting produces a block-diagonal

tensor $\bigoplus_k A_k$ whose blocks are irreducible. Apply the induction hypothesis to each block and concatenate. (If one prefers a positive-eigenvector argument without assuming TP normalization, the correct replacement for the channel-specific Cesàro theorem is Theorem 8.17; the induction itself needs only the invariant projection.)

Both [PGVWC07] (Theorem 4, “TI canonical form”) and [CPGSV21] (§IV.A.1) obtain the block decomposition by the same invariant-projection splitting used here. An alternative derivation, developed in [Wol12, Thm. 6.14], characterises the fixed-point set of a quantum channel as a $*$ -algebra and then applies the Wedderburn–Artin decomposition ([Wol12, Eq. (1.39)]) to read off the block structure directly. The block decomposition is obtained by iterating the invariant-projection splitting and using strong induction on D to ensure termination. $\square$

Theorem 11.17 (Unitary diagonal positive fixed point in TP gauge). *Let A be an irreducible MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \quad \text{and} \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

Proof. The TP hypothesis makes $\mathcal{E}_A$ a channel, so it has a nonzero PSD fixed point. The irreducible tensor condition implies that $\mathcal{E}_A$ is irreducible as a CP map, and Theorem 8.3 upgrades the fixed point to a positive definite one. The spectral theorem diagonalizes this matrix by a unitary conjugation, and unitary conjugation preserves the TP normalization. $\square$

11.4.1 From irreducibility to full spanning via Burnside’s theorem

The irreducibility condition (Definition 11.15) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under the Kraus operators.

Definition 11.18 (Algebra span). For an MPS tensor A , let $\text{alg}(A)$ be the unital $\mathbb{C}$ -subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}$:

$$\text{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d-1 \rangle \subseteq M_D(\mathbb{C}).$$

Definition 11.19 (Invariant submodule). A subspace $W \leq \mathbb{C}^D$ is *A-invariant* if $A^i W \subseteq W$ for every i .

Definition 11.20 (Irreducible action). The Kraus operators of A act *irreducibly* on $\mathbb{C}^D$ if every A -invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 11.21 (Irreducible tensor implies irreducible action). *If A is irreducible as a tensor (Definition 11.15), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 11.20).*

Proof. Let $W \leq \mathbb{C}^D$ be an A -invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^i P = 0$ for all i . Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. $\square$

Theorem 11.22 (Burnside’s theorem for Kraus algebras). *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra:*

$$\text{alg}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside’s theorem (equivalently, Jacobson’s density theorem); see [Jac09].

Proof. Transport $\text{alg}(A)$ along the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \text{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. $\square$

Remark 11.23. This Burnside/Jacobson-density step is a substantial external algebraic input in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

Theorem 11.24 (Irreducible action implies eventual cumulative spanning). *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.*

Proof. By Theorem 11.22, $\text{alg}(A) = M_D(\mathbb{C})$. Every element of the algebra span is a linear combination of word products, hence belongs to $T_n(A)$ for some n . Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \leq T_1(A) \leq \dots$ stabilizes, so one can choose a uniform N with $T_N(A) = M_D(\mathbb{C})$. $\square$

11.5 Proportional single-block theorem

Theorem 11.25 (Proportional MPV implies gauge-phase equivalence). *Let A and B be injective MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If A and B generate proportional MPV families $(V^{(N)}(A))_\sigma = c_N V^{(N)}(B)_\sigma$ for some $c_N \in \mathbb{C}$ and $O_{AA}(N) \rightarrow 1$, $O_{BB}(N) \rightarrow 1$, then A and B are gauge-phase equivalent.*

Proof. By contradiction. If A and B are not gauge-phase equivalent, then $O_{AB}(N) \rightarrow 0$ (Theorem 9.25). From the proportionality hypothesis, $O_{AB}(N) = c_N O_{BB}(N)$. Since $O_{BB}(N) \rightarrow 1$, it follows that $c_N \rightarrow 0$. But $O_{AA}(N) = |c_N|^2 O_{BB}(N) \rightarrow 0$, contradicting $O_{AA}(N) \rightarrow 1$.

In [PGVWC07], the proportional case is handled as part of the canonical form argument. Thus the proportional case is obtained directly from the spectral gap in Chapter 9. $\square$

Theorem 11.26 (Proportional MPV + primitive implies gauge-phase equivalence). *Let A and B be MPS tensors with primitive transfer maps E_A, E_B and positive definite fixed points ρ_A, ρ_B . If A and B generate proportional MPV families, then A and B are gauge-phase equivalent. This strengthens Theorem 11.25: primitivity supplies left-canonicity, irreducibility, and overlap convergence, reducing the hypothesis count from seven to four.*

Proof. From primitivity, extract a left-canonical normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), overlap convergence ($O_{AA}(N) \rightarrow 1$), and irreducibility (Theorem 10.50). Apply Theorem 11.25. $\square$

11.6 Canonical form from primitivity

The canonical form conditions (Definition 12.16) include an explicit hypothesis that the self-overlap $O_{A_k A_k}(N) \rightarrow 1$ for each block. The first theorem below derives this from blockwise primitive fixed-point data. The second derives the same conclusion from per-block peripheral-spectrum primitivity.

Theorem 11.27 (Canonical form from primitive fixed-point data). *Let $(\mu_k, A_k)_{k=1}^r$ be a collection of block tensors and scaling factors satisfying:*

1. *each A_k is injective,*

2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. $|\mu_1| > \dots > |\mu_r|$ (strictly decreasing),
4. all $\mu_k \neq 0$,
5. each block has positive bond dimension,
6. each block admits a primitive fixed point.

Then (μ_k, A_k) satisfies the canonical form conditions.

Proof. All canonical-form clauses except the overlap condition are already assumptions. For each block, the primitive fixed-point hypothesis gives $O_{A_k A_k}(N) \rightarrow 1$. $\square$

Theorem 11.28 (Canonical form from peripheral primitivity). *Let $(\mu_k, A_k)_{k=1}^r$ be a collection of block tensors and scaling factors satisfying:*

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. $|\mu_1| \geq \dots \geq |\mu_r|$ (non-increasing),
4. all $\mu_k \neq 0$,
5. each transfer map $\mathcal{E}_{A_k}$ is primitive, i.e. has peripheral spectrum $\{1\}$.

Then (μ_k, A_k) satisfies the canonical form conditions (Definition 12.16).

Proof. Fix a block k and write $E := \mathcal{E}_{A_k}$. We check that each primitive block already satisfies the defining properties of the canonical form conditions. By injectivity and TP normalization, Theorem 8.9 provides a nonzero positive definite fixed point for E . To apply the spectral-gap criterion (Theorem 5.51), we check its three hypotheses:

1. E is primitive, since the peripheral spectrum is $\{1\}$ by assumption;
2. every eigenvalue of E has modulus at most 1, because $E = F_{A_k A_k}$ is a normalized mixed transfer operator and Theorem 9.14 applies;
3. any trace-zero fixed point of E is zero; this is the trace-zero uniqueness consequence extracted in Chapter 8 from the uniqueness of the PSD fixed point (Theorem 8.5).

Hence E has a spectral gap on the complement of its fixed-point projection. Theorem 9.94 then gives $O_{A_k A_k}(N) \rightarrow 1$. Since this holds for every block, the overlap clause required in Definition 12.16 follows from the other hypotheses. $\square$

Remark 11.29. Theorem 11.28 shows that the overlap convergence hypothesis in the canonical form conditions is redundant once peripheral-spectrum primitivity is known. The argument factors through the spectral-gap formulation of Chapter 9: peripheral-spectrum primitivity is first converted to that spectral-gap condition, and the overlap limit is then deduced from it. In the paper [CPGSV21], primitivity is obtained from irreducibility plus periodicity removal by blocking to period one. Theorem 11.30 is precisely this periodicity-removal step under irreducibility and TP normalization.

Theorem 11.30 (Blocking yields a peripheral-spectrum primitive transfer map). *Let A be an irreducible MPS tensor with $D > 0$. Assume that A satisfies the TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that the transfer map $\mathcal{E}_{A^{[p]}}$ of the blocked tensor $A^{[p]}$ is primitive, i.e. its peripheral spectrum is $\{1\}$.*

Proof. The period- p blocking step is the diagrammatic move of grouping the chain into consecutive blocks of length p . Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$. The TP normalization of A makes K unital, and irreducibility of A gives irreducibility of the associated Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint map of K , so Theorem 5.46 shows that every peripheral eigenvalue is a root of unity. Since $\text{peripheral}(\mathcal{E}_A)$ is finite (Theorem 5.29), Lemma 5.33 gives a common exponent p such that $\mu^p = 1$ for all $\mu \in \text{peripheral}(\mathcal{E}_A)$. Lemma 5.34 then implies $\text{peripheral}(\mathcal{E}_A^p) = \{1\}$. Finally, blocking corresponds to taking a power of the transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive. $\square$

Remark 11.31. Theorem 11.30 is the “periodicity removal by blocking” step in [CPGSV17a, Appendix A] for an irreducible block already placed in TP gauge. The later existence theorems then reuse this step.

11.6.1 Adjoint transport for blocking

Lemma 11.32 (Characteristic polynomial under conjugate transpose). *For a complex matrix M , the characteristic polynomial of $M^\dagger$ is obtained from the characteristic polynomial of M by coefficientwise complex conjugation.*

Proof. Write $M^\dagger$ as the transpose of the coefficientwise conjugate matrix. The characteristic polynomial is invariant under transpose and commutes with coefficientwise conjugation. $\square$

Theorem 11.33 (Eigenvalues of the adjoint are conjugated). *Let E be a linear endomorphism of a finite-dimensional complex inner-product space, and let $\mu \in \mathbb{C}$. Then μ is an eigenvalue of E if and only if $\bar{\mu}$ is an eigenvalue of the adjoint $E^\dagger$.*

Proof. Express eigenvalues as roots of the characteristic polynomial, convert the adjoint to the conjugate-transposed matrix in an orthonormal basis, and apply Lemma 11.32. $\square$

Theorem 11.34 (Primitivity is invariant under adjoint). *A linear endomorphism is primitive if and only if its adjoint is primitive.*

Proof. The peripheral eigenvalues of the adjoint are the complex conjugates of those of the original map, so both maps have the same peripheral spectrum on the unit circle. $\square$

Lemma 11.35 (Transfer map of the conjugate-transposed Kraus family). *For an MPS tensor A , the transfer map of the conjugate-transposed Kraus family $\{(A^i)^\dagger\}$ is the adjoint of $\mathcal{E}_A$ with respect to the Frobenius inner product.*

Proof. Compare both sides against the Frobenius pairing $\langle X, Y \rangle = \text{tr}(YX^\dagger)$ and rewrite the Kraus sums. $\square$

Theorem 11.36 (Blocking primitivity with a positive-dimension hypothesis). *Let A be an irreducible MPS tensor with positive bond dimension and TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that $\mathcal{E}_{A^{[p]}}$ is primitive.*

Proof. This is Theorem 11.30 with the positive-dimension assumption expressed as a typeclass hypothesis. $\square$

Theorem 11.37 (Blocking primitivity in left-canonical gauge). *Let A be a left-canonical irreducible MPS tensor with positive bond dimension. Then there exists a blocking length $p > 0$ such that $\mathcal{E}_{A^{[p]}}$ is primitive.*

Proof. This is the left-canonical reformulation of Theorem 11.30. $\square$

11.7 Normal canonical form

Definition 11.38 (Normal canonical form conditions). A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *normal canonical form conditions* if:

1. each A_k is irreducible,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),
4. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here “normal” is used in the spectral sense of [CPGSV17a]: irreducible blocks whose TP-normalized transfer maps are primitive. As noted in the remark after Definition 2.23, this is equivalent to the eventual block-injectivity formulation of Definition 2.23; see [SPGWC10, Proposition 3].

Theorem 11.39 (Self-overlap is derived in normal canonical form). *If (μ_k, A_k) satisfies the normal canonical form conditions, then $O_{A_k A_k}(N) \rightarrow 1$ for every block k .*

Proof. Peripheral-spectrum primitivity is part of the data. Together with irreducibility and TP normalization, it yields the spectral-gap primitive theorem needed for overlap convergence, so the self-overlap condition follows from the other hypotheses rather than appearing as a separate assumption. $\square$

Theorem 11.40 (Modulus-one eigenvalue rigidity for irreducible TP blocks). *Let A and B be irreducible MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If the mixed transfer map has spectral radius at least 1 (equivalently, it has a peripheral eigenvalue of modulus 1), then A and B are gauge-phase equivalent.*

Proof. The argument is the same as in Theorem 9.16 (with the gauging formula given by Theorem 8.10 directly), but the fixed-point input comes from irreducibility rather than injectivity. Take a modulus-one eigenvector of the mixed transfer map. Because A and B are TP-normalized, each transfer map has a nonzero PSD fixed point by Theorem 8.8. Irreducibility upgrades these fixed points to positive definite ones by Theorem 8.3. Theorem 8.10 therefore gauges A and B separately to unital Kraus families. In this unital gauge, the embedded modulus-one eigenvector saturates the Kadison–Schwarz inequality, and the same equality / multiplicative-domain argument as in Theorem 9.16 produces an invertible intertwiner between the two Kraus families. That intertwiner is exactly the required gauge-phase equivalence. $\square$

11.8 Steps toward canonical form existence

This section combines the components of the canonical-form construction following [CPGSV17a, §2.3 and Appendix A]. The approach here requires blocking to remove non-trivial periodicity. A separate reduction for periodic blocks without blocking is developed in [DICCSPG17]; see Chapter 14 for a discussion of that result.

Theorem 11.41 (Irreducible block decomposition). *Every MPS tensor A generates the same MPV family as a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ with each block irreducible.*

Proof. Apply Theorem 11.16. □

Theorem 11.42 (Left-canonical normalization for irreducible TP blocks). *Let A be an irreducible MPS tensor already in TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that:*

1. $B^i := U^\dagger A^i U$ is TP,
2. $\mathcal{E}_B(\Lambda) = \Lambda$,
3. Λ is diagonal and positive definite,
4. A and B generate the same MPV family.

This is the left-canonical normalization step (Canonical Form II) of [CPGSV17a, Appendix A]. The unitary conjugation occurs only after TP normalization has already been achieved; it is a second step inside TP gauge, not a unitary normalization of an arbitrary irreducible block.

Proof. Apply Theorem 11.17 to obtain the unitary and the diagonal PD fixed point. Unitary conjugation preserves trace-preservation and the MPV family (Theorem 11.10). □

Theorem 11.43 (Irreducible block decomposition with left-canonical normalization). *For any MPS tensor A , there exist irreducible blocks $(A_k)_{k=1}^r$ that generate the same MPV family as A . Moreover, for each block k , if that block has already been placed in TP gauge and $D_k > 0$, one obtains a left-canonical normalization (unitary, diagonal PD fixed point, MPV preservation). This yields the left-canonical normalization; further reduction to normal canonical form is carried out in Chapter 9.*

Proof. Combine Theorems 11.41 and 11.42. □

Theorem 11.44 (Normal canonical form from a primitive block decomposition). *Suppose A generates the same MPV family as a weighted block-diagonal tensor $\bigoplus_k \mu_k A_k$, and assume that:*

1. each A_k is irreducible,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),
4. the moduli $|\mu_k|$ are pairwise distinct,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Then there exist a blocking length $p > 0$, weights (ν_k) , and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form conditions.

Proof. No new blocking is needed here: the input blocks are already primitive, so one may take $p = 1$. The theorem is stated with a blocking length because earlier reduction steps may require a common blocking period; under these hypotheses that period can be chosen to be 1. Since the moduli $|\mu_k|$ are pairwise distinct, there is a reordering of the blocks for which they become strictly decreasing. After this reindexing, set $\nu_k := \mu_k$ and $B_k := A_k$ in the new order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are unchanged by the reordering, so the reordered family satisfies Definition 11.38 and generates the same MPV family as A . $\square$

11.9 Zero-block separation

In the irreducible block decomposition, some blocks may have all-zero Kraus operators. Such blocks contribute to the MPV only at word length $N = 0$ (where their contribution equals their bond dimension). The following results separate the zero blocks from the “live” (nonzero) blocks.

Definition 11.45 (Zero MPS tensor). The *zero MPS tensor* of physical dimension d and bond dimension D , denoted $\mathbf{0}_{d,D}$, is the tensor whose Kraus operators are all zero.

Theorem 11.46 (MPV of zero tensor). For any spin configuration σ of length N ,

$$V^{(N)}(\mathbf{0}_{d,D})_\sigma = \begin{cases} D & \text{if } N = 0, \\ 0 & \text{if } N \geq 1. \end{cases}$$

Proof. For $N = 0$, $(\mathbf{0}_{d,D})^\emptyset = \mathbb{1}_D$, so the MPV equals $\text{tr}(\mathbb{1}_D) = D$. For $N \geq 1$, every word product of a zero tensor is zero. $\square$

Theorem 11.47 (Zero-block separation). Every MPS tensor A admits a block decomposition into:

1. a trivial block of bond dimension D_0 with all Kraus operators zero (accumulating all all-zero irreducible blocks), and
2. a finite family of nontrivial blocks $(B_k)_{k=1}^r$, each irreducible, each with at least one nonzero Kraus operator, and each with positive bond dimension,

such that

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d,D_0})_\sigma + V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma$$

for every spin configuration σ of length N . At $N = 0$ this reads $D = D_0 + \sum_{k=1}^r D_k$. For $N \geq 1$ the zero-tail term vanishes.

Proof. Apply the irreducible block decomposition (Theorem 11.16) and partition the blocks by whether each has a nonzero Kraus operator. The all-zero blocks satisfy a dimension bound via irreducibility; their total bond dimension is D_0 . The MPV identity follows from the block structure. $\square$

11.10 Arbitrary-input TP-gauge reduction

Theorem 11.48 (Arbitrary-input TP-gauge reduction with zero-block separation). *For any MPS tensor A , there exist:*

1. a zero-tail bond dimension D_0 ,
2. nontrivial blocks $(B_k)_{k=1}^r$ with nonzero weights $(\mu_k)_{k=1}^r$, each block irreducible and left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), with positive bond dimension,

such that

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d,D_0})_\sigma + V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

This is the furthest unconditional arbitrary-input reduction step available before periodicity removal.

Proof. Apply Theorem 11.47 to separate the trivial block and nontrivial blocks. Then apply the blockwise TP-gauge theorem to the nontrivial blocks. The MPV identity chains through both steps. $\square$

11.11 Blocking and primitivity reduction

Theorem 11.49 (Blocking distributes over MPV equivalence). *If A generates the same MPV family as $\bigoplus_k \mu_k B_k$, then for any $p \geq 0$,*

$$A^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]}.$$

Proof. The identity $V^{(N)}(A^{[p]})_\sigma = V^{(Np)}(A)_{\sigma_{\text{flat}}}$ holds for any tensor by definition of blocking. The MPV equivalence of A and $\bigoplus_k \mu_k B_k$ at length Np then gives the equivalence at length N with weights μ_k^p . $\square$

Theorem 11.50 (Primitive channels remain primitive under powers). *If $\mathcal{E}$ has peripheral spectrum $\{1\}$ (is primitive) and $m \geq 1$, then $\mathcal{E}^m$ is also primitive.*

Proof. The peripheral spectrum of $\mathcal{E}^m$ is the set of m -th powers of that of $\mathcal{E}$. Since $\{1\}^m = \{1\}$, the result follows. $\square$

Theorem 11.51 (Blocking-period divisibility preserves primitivity). *If the transfer map of $A^{[p]}$ is primitive, $p \mid q$, and $q \geq 1$, then the transfer map of $A^{[q]}$ is also primitive.*

Proof. Write $q = p \cdot m$. Then $\mathcal{E}_{A^{[q]}} = \mathcal{E}_{A^{[p]}}^m$. Apply Theorem 11.50 with $\mathcal{E} = \mathcal{E}_{A^{[p]}}$. $\square$

Theorem 11.52 (Iterated blocking multiplies periods at the transfer-map level). *For any MPS tensor A and integers $m, n \geq 0$,*

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}.$$

Proof. Rewrite the outer blocked transfer map as the n -th power of $\mathcal{E}_{A^{[m]}}$, then rewrite $\mathcal{E}_{A^{[m]}}$ as $\mathcal{E}_A^m$. This gives $(\mathcal{E}_A^m)^n$, which simplifies to $\mathcal{E}_A^{mn}$. Finally rewrite $\mathcal{E}_A^{mn}$ back as the transfer map of $A^{[mn]}$. $\square$

Theorem 11.53 (Common blocking period for a block family). *Let $(B_k)_{k=1}^r$ be a finite family of MPS tensors with positive bond dimensions, each admitting a blocking period p_k making its transfer map primitive. Then $p = \text{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k .*

Proof. Each $p_k \mid p$, so Theorem 11.51 applies. $\square$

Definition 11.54 (Common-period blocking of a block family). Given a finite family of tensors (B_k) and positive integers (p_k) , the common-period blocking is the family obtained by blocking every B_k at the least common multiple of the periods p_k .

Theorem 11.55 (Primitivity of the common-period blocking). *Suppose each B_k has positive bond dimension and each blocked transfer map $\mathcal{E}_{B_k^{[p_k]}}$ is primitive. Then every member of the common-period blocked family also has primitive transfer map.*

Proof. Each p_k divides the common blocking length, so Theorem 11.51 applies to every block. $\square$

Theorem 11.56 (Formula for common-period blocking). *Each block of the common-period blocking is exactly the corresponding tensor blocked at the common least common multiple.*

Proof. This is the definition of the common-period blocking. $\square$

11.12 Cyclic sector decomposition

Theorem 11.57 (Compression to a supported projection sector). *Let A be an MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = P$ for an orthogonal projection P , and suppose $PA^iP = A^i$ for every i . Then there exist a left-canonical tensor C of bond dimension $n = \text{tr}(P)$ and a $*$ -algebra isomorphism $\varphi : M_n(\mathbb{C}) \xrightarrow{\sim} PM_D(\mathbb{C})P$ (preserving both multiplication and adjoint) with*

$$V^{(N)}(C)_\sigma = \text{tr}(P \cdot A^\sigma), \quad \varphi(\mathcal{E}_{C^\dagger}(X)) = \mathcal{E}_{A^\dagger}(\varphi(X))$$

for every σ and every $X \in M_n(\mathbb{C})$, together with $\varphi(XY) = \varphi(X)\varphi(Y)$ and $\varphi(X^\dagger) = \varphi(X)^\dagger$.

Proof. Diagonalize P via a unitary U (spectral theorem). The eigenvalues are in $\{0, 1\}$. Set $n = \text{tr} P$; let C^i be the top-left $n \times n$ block of $U^\dagger A^i U$ (restricted to eigenvalue-1 indices). TP follows from the projection normalization; the MPV identity follows from trace cyclicity. $\square$

Corollary 11.58 (Compression preserves positive-length MPV coefficients). *Under the hypotheses of Theorem 11.57, the compressed tensor C additionally satisfies $V^{(N)}(C)_\sigma = V^{(N)}(A)_\sigma$ for all positive chain lengths $N > 0$.*

Proof. For $N > 0$ the word $A^{i_1} \dots A^{i_N}$ is nonempty, so P acts as the identity on the left by idempotence and the support hypothesis. The MPV identity then follows from trace cyclicity via Theorem 11.57. $\square$

Theorem 11.59 (Block decomposition from commuting projections). *Let A be a left-canonical MPS tensor, and let $(P_k)_{k=1}^m$ be orthogonal projections summing to $\mathbb{1}$ that commute with every Kraus operator: $P_k A^i = A^i P_k$ for all i, k . Then there exist left-canonical blocks $(C_k)_{k=1}^m$ with unit weights such that A generates the same MPV family as $\bigoplus_{k=1}^m C_k$. Moreover, each sector satisfies the per-sector trace relation $V^{(N)}(C_k)_\sigma = \text{tr}(P_k \cdot A^{i_1} \dots A^{i_N})$ for every word $\sigma = (i_1, \dots, i_N)$, and each block carries a compression linear equivalence $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$ intertwining the compressed adjoint transfer map with the sector adjoint transfer map.*

Proof. For each k , the tensor $P_k A^i$ is supported on P_k (using commutativity and $P_k^2 = P_k$). The TP condition $\sum_i (P_k A^i)^\dagger (P_k A^i) = P_k$ follows from $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Apply Theorem 11.57 to each sector. The MPV identity uses $\sum_k P_k = \mathbb{1}$ and linearity of trace. $\square$

Theorem 11.60 (Block decomposition from adjoint-fixed projections). *Let A be a left-canonical MPS tensor, and let $(P_k)_{k=1}^m$ be orthogonal projections summing to $\mathbb{1}$ that are fixed by the adjoint transfer map: $\mathcal{E}_A^\dagger(P_k) = P_k$ for every k . Then the same block decomposition conclusion holds as in Theorem 11.59, including the per-sector trace relation.*

Proof. The Kadison–Schwarz equality condition for a TP map (Theorem 7.23) implies that a projection fixed by $\mathcal{E}_A^\dagger$ commutes with every Kraus operator. Apply Theorem 11.59. $\square$

Theorem 11.61 (Conjugate-transposed Kraus family for irreducible TP tensors). *Let A be a left-canonical irreducible tensor. Then the conjugate-transposed family $K_i := (A^i)^\dagger$ is unital and irreducible, and there exists a positive-definite matrix ρ fixed by the adjoint map associated to K .*

Proof. Left-canonicity gives $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so the conjugate-transposed family is unital. Irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides the required positive-definite fixed point. $\square$

Theorem 11.62 (One full cyclic turn fixes the sector projections). *If a family of projections $(P_k)_{k=1}^m$ satisfies $\mathcal{E}^\dagger(P_{k+1}) = P_k$, then the m -th iterate of $\mathcal{E}^\dagger$ fixes each projection P_k .*

Proof. Iterate the cyclic relation m times. Because the indices are taken modulo m , the projection returns to its starting position. $\square$

Theorem 11.63 (Blocked adjoint transfer map as an iterate). *For every tensor A , blocking length m , and matrix X ,*

$$\mathcal{E}_{(A^{[m]})^\dagger}(X) = (\mathcal{E}_{A^\dagger})^m(X).$$

Proof. Write both transfer maps as Frobenius adjoints of ordinary blocked transfer maps. Then use Theorem 11.52. $\square$

Theorem 11.64 (Cyclic sector decomposition after blocking). *Let A be a left-canonical irreducible tensor of period m . Then the blocked tensor $A^{[m]}$ admits:*

1. *left-canonical sector tensors $(C_k)_{k=1}^m$,*
2. *orthogonal projections $(P_k)_{k=1}^m$ summing to $\mathbb{1}$ and satisfying $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$,*
3. *compression linear equivalences $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$,*
4. *a direct-sum MPV decomposition $A^{[m]} \sim \bigoplus_{k=1}^m C_k$.*

Proof. The cyclic decomposition of the adjoint transfer map gives projections (P_k) with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Theorem 11.62 upgrades this to $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and Theorem 11.63 identifies $(\mathcal{E}_{A^\dagger})^m$ with the adjoint transfer map of the blocked tensor. Apply Theorem 11.60. $\square$

Theorem 11.65 (Cyclic sector decomposition for irreducible TP tensors). *Let A be a left-canonical irreducible tensor. Then there exist $m \geq 1$ and left-canonical blocks $(C_k)_{k=1}^m$ such that the m -blocked tensor $A^{[m]}$ decomposes as $\bigoplus_{k=1}^m C_k$ (with unit weights).*

Proof. Apply Theorem 11.61 to obtain the conjugate-transposed Kraus family, an irreducible adjoint transfer map, and a positive-definite fixed point. The cyclic peripheral-spectrum theorem then provides a period m and the required cyclic spectral data. Apply Theorem 11.64. $\square$

11.12.1 Sector irreducibility helpers

Lemma 11.66 (Pairwise orthogonality from projection sum). *Let $(P_k)_{k=1}^m$ be orthogonal projections summing to the identity: $\sum_k P_k = \mathbb{1}$. Then for $i \neq j$ we have $P_i P_j = 0$.*

Proof. Sandwich the sum between P_i to get $\sum_k P_i P_k P_i = P_i$. Since the i -th summand equals P_i by idempotence, all other summands vanish. Positivity ($BB^* = 0 \Rightarrow B = 0$) then gives $P_i P_k = 0$ for each $k \neq i$. $\square$

Lemma 11.67 (Corner preservation from adjoint-fixed projection). *Let A be a left-canonical MPS tensor and let P be an orthogonal projection fixed by $\mathcal{E}_A^\dagger$. Then $\mathcal{E}_A^\dagger$ preserves the corner algebra $PM_D(\mathbb{C})P$.*

Proof. The fixed-point condition and the Kadison–Schwarz equality imply $[P, A_i] = 0$ for every Kraus operator. Thread the idempotent relation $P^2 = P$ through the corner sandwich. $\square$

Lemma 11.68 (Orbit-sum fixed point). *Let T be a linear endomorphism and Q a matrix satisfying $T^m(Q) = Q$. Then the orbit sum $\sum_{l=0}^{m-1} T^l(Q)$ is a fixed point of T .*

Proof. Telescoping: T shifts each summand $T^l(Q) \mapsto T^{l+1}(Q)$ and the periodicity condition $T^m(Q) = Q$ wraps the last term $T^{m-1}(Q) \mapsto T^m(Q) = Q$ back to the first. $\square$

Lemma 11.69 (Orbit iterate supported on shifted sector). *Let $(P_k)_{k=1}^m$ be orthogonal projections with cyclic shift $T(P_{k+1}) = P_k$, and suppose T is multiplicative on each sector corner. If Q is supported on sector P_k (i.e. $QP_k = Q = P_k Q$), then for each $0 \leq l < m$ the iterate $T^l(Q)$ is supported on the shifted sector P_{k-l} .*

Proof. Induction on l . The base case is the hypothesis $QP_k = Q = P_k Q$. For the inductive step, the multiplicative-domain identities and the cyclic shift relation $T(P_j) = P_{j-1}$ transfer the support from sector P_j to sector P_{j-1} . $\square$

Lemma 11.70 (Orbit iterate preserves orthogonal projection). *Under the same cyclic shift and multiplicative-domain hypotheses as Lemma 11.69, suppose additionally that T maps every orthogonal projection supported on a single sector to an orthogonal projection. If Q is an orthogonal projection supported on P_k , then $T^l(Q)$ is an orthogonal projection for each $0 \leq l < m$.*

Proof. Induction on l . The base case is the hypothesis on Q . At each step, Lemma 11.69 guarantees that $T^n(Q)$ is supported on a single sector, so the one-step preservation hypothesis applies. $\square$

Lemma 11.71 (Orbit sum of a full sector projection). *Let $(P_k)_{k=1}^m$ be orthogonal projections with $\sum_k P_k = \mathbb{1}$ and cyclic shift $T(P_{k+1}) = P_k$. Then for every sector index k ,*

$$\sum_{l=0}^{m-1} T^l(P_k) = \mathbb{1}.$$

Proof. By induction, $T^l(P_k) = P_{k-l}$ for each l . The orbit sum therefore runs over all distinct sectors, and the reindexing $l \mapsto k-l$ recovers $\sum_j P_j = \mathbb{1}$. $\square$

Theorem 11.72 (Sector irreducibility from orbit-sum lift). *Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be the cyclic sector projections. Suppose the orbit-sum lift hypothesis holds: for every sector k and every subprojection $Q \leq P_k$ fixed by $(\mathcal{E}_A^\dagger)^m$, there exists a global projection R that is invariant under $\mathcal{E}_A^\dagger$ and faithfully reflects triviality ($Q = 0 \Leftrightarrow R = 0$, $Q = P_k \Leftrightarrow R = \mathbb{1}$). Then $(\mathcal{E}_A^\dagger)^m$ is irreducible on each corner P_k .*

Proof. Direct application of the cyclic-decomposition irreducibility theorem from spectral theory. $\square$

Lemma 11.73 (Fixed-point upgrade from irreducibility and trace preservation). *Let A be a left-canonical MPS tensor whose transfer map $\mathcal{E}_A$ is irreducible, and let Q be an orthogonal projection such that the corner $Q\mathcal{M}_DQ$ is preserved by $((\mathcal{E}_A^\dagger)^m)$. Then $((\mathcal{E}_A^\dagger)^m)(Q) = Q$.*

Proof. Let $\rho > 0$ be the distinguished fixed point of $\mathcal{E}_A$. Since $Q\mathcal{M}_DQ$ is preserved, the difference $Q - (\mathcal{E}_A^\dagger)^m(Q)$ can be rewritten as $Q(\mathcal{E}_A^\dagger)^m(\mathbb{1} - Q)Q$, hence it is positive semidefinite. The weighted trace pairing with ρ is invariant under $(\mathcal{E}_A^\dagger)^m$, so this positive semidefinite matrix has weighted trace zero. Faithfulness of the ρ -weighted trace therefore forces it to vanish, giving the claim. $\square$

Lemma 11.74 (Orbit-sum construction of the sector lift). *Let A be a left-canonical MPS tensor, $(P_k)_{k=1}^m$ a cyclic family of orthogonal sector projections with $\sum_k P_k = \mathbb{1}$, and set $T = \mathcal{E}_A^\dagger$. Assume two upstream inputs, both discharged from the block-diagonal canonical-form analysis of [DCCSPG17, Lemma lem:bdcf]:*

1. (Projection step.) *For each sector index k and each orthogonal projection X supported on P_k , the image $T(X)$ is again an orthogonal projection.*
2. (Fixed-point upgrade.) *For each sector index k and each orthogonal subprojection $Q \leq P_k$, if Q is a corner fixed point of T^m then $T^m(Q) = Q$.*

Then for every sector k and every orthogonal subprojection $Q \leq P_k$ which is a corner fixed point of T^m , the orbit sum

$$R := \sum_{l=0}^{m-1} T^l(Q)$$

is an orthogonal projection, preserved by T , with $Q = 0 \Leftrightarrow R = 0$ and $Q = P_k \Leftrightarrow R = \mathbb{1}$. In particular R witnesses the orbit-sum lift hypothesis of Theorem 11.72.

Proof. The iterates $T^l(Q)$ are each orthogonal projections (repeatedly using the projection-step hypothesis), supported on the shifted sector P_{k-l} , and are therefore pairwise orthogonal because distinct sectors are. This gives both the Hermitian and idempotent conjuncts for R . Fixedness under T follows from $T^m(Q) = Q$ and Lemma 11.68, and corner preservation from the adjoint trace-preserving relation. The zero equivalence uses $R \cdot Q = Q$ (the diagonal term $l = 0$ is Q , off-diagonal terms vanish by pairwise orthogonality); the full-sector equivalence uses $P_k \cdot R \cdot P_k = Q$ and Lemma 11.71. $\square$

Lemma 11.75 (Orbit-sum construction, projection-step variant). *Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Assume the projection-step hypothesis of Lemma 11.74. Then the orbit-sum lift holds: for every orthogonal subprojection $Q \leq P_k$ whose corner is preserved by $(\mathcal{E}_A^\dagger)^m$, there is a global projection R invariant under $\mathcal{E}_A^\dagger$ with the same zero/full-sector tests.*

Proof. Combine the original orbit-sum construction with Lemma 11.73, which supplies the fixed-point upgrade from irreducibility and trace preservation. The irreducibility hypothesis needed there is obtained by transporting irreducibility from the adjoint transfer map to the tensor and then back to the primal transfer map. $\square$

Theorem 11.76 (Sector irreducibility from the orbit-sum construction). *Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Assume the projection-step and fixed-point-upgrade hypotheses of Lemma 11.74. Then $(\mathcal{E}_A^\dagger)^m$ is irreducible on each corner P_k .*

Proof. Apply Lemma 11.74 to obtain the orbit-sum lift, then apply Theorem 11.72. $\square$

Theorem 11.77 (Sector irreducibility from the projection-step hypothesis). *Let A be a left-canonical MPS tensor with irreducible adjoint transfer map, and let $(P_k)_{k=1}^m$ be cyclic orthogonal sector projections. Assume only the projection-step hypothesis from Lemma 11.74. Then $(\mathcal{E}_A^\dagger)^m$ is irreducible on each corner P_k .*

Proof. Apply Lemma 11.75 and then Theorem 11.72. $\square$

11.13 Reduction to primitive blocks

Theorem 11.78 (Arbitrary tensor to primitive block decomposition). *For any MPS tensor A , there exist a period $p \geq 1$, a zero-tail bond dimension D_0 , nonzero weights $(\mu_k)_{k=1}^r$, and left-canonical blocks $(B_k)_{k=1}^r$ with primitive transfer maps and positive bond dimensions, such that*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\mathbf{0}_{d[p], D_0})_\sigma + V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma.$$

Proof. Apply Theorem 11.48 to get the trivial block and left-canonical nontrivial blocks. Apply Theorem 11.53 to find a common blocking period p making all transfer maps primitive. Apply Theorem 11.49. $\square$

Theorem 11.79 (TP-primitive irreducible tensor is normal). *Let A be a left-canonical MPS tensor with $D > 0$, irreducible transfer map, and peripheral spectrum $\{1\}$. Then A is normal.*

Proof. Peripheral primitivity and irreducibility yield the spectral-gap condition. The PSD fixed point is positive definite by Theorem 8.3. Apply Theorem 10.53. $\square$

Theorem 11.80 (Blocking preserves normality). *If A is normal and $p \geq 1$, then $A^{[p]}$ is normal.*

Proof. Since A is normal, there exists N with $S_N(A) = M_D(\mathbb{C})$. Then $S_{Np}(A) = M_D(\mathbb{C})$ (word products at multiples still span), and $S_N(A^{[p]}) \supseteq S_{Np}(A) = M_D(\mathbb{C})$. $\square$

11.13.1 Basis of Normal Tensors

Theorem 11.81 (Sorted reindexing by strictly decreasing weight norms). *If nonzero weights $(\mu_k)_{k=1}^r$ have pairwise distinct norms, then there is a permutation of $\{1, \dots, r\}$ under which $\|\mu_k\|$ is strictly decreasing.*

Theorem 11.82 (Sorted decomposition preserves the represented MPS family). *Under the same distinct-norm hypothesis, the sorted block family is gauge equivalent to the original decomposition and has strictly decreasing norms.*

Theorem 11.83 (Normal canonical form after sorting). *If each block is irreducible, trace-preserving, primitive, and has positive dimension, then sorting by strictly decreasing weight norms produces a normal canonical form.*

Theorem 11.84 (Trivial sector decomposition for the sorted distinct-norm case). *If weights are already strictly decreasing in norm, one obtains a sector decomposition with one copy per sector that preserves the represented MPS family.*

Definition 11.85 (Norm-class partition). The grouping data consists of: norm classes, one representative value per class, class multiplicities, an enumeration of members in each class, and a regrouping identity recovering the full finite sum.

Definition 11.86 (Norm-class enumeration in decreasing order). From any finite weight family, construct norm classes ordered by strictly decreasing norm values.

Theorem 11.87 (BNT grouping theorem). *If equal-norm blocks represent the same MPS family, then one obtains a sector decomposition whose sector norms are strictly decreasing.*

11.13.2 Equal-Norm Blocks

Theorem 11.88 (Gauge equivalence from non-decaying cross-overlap). *For trace-preserving irreducible blocks, a non-decaying cross-overlap forces equal bond dimensions and gauge equivalence up to a unit-modulus phase.*

Theorem 11.89 (Gauge phases have unit modulus for primitive irreducible blocks). *For trace-preserving primitive irreducible blocks, every gauge-phase equivalence has a phase factor of modulus 1.*

Proof. Primitivity forces the peripheral spectrum to consist of $\{1\}$ alone, so the gauge phase must have modulus one. $\square$

Theorem 11.90 (BNT partition from gauge equivalence). *If equal-norm blocks are related by gauge equivalence with unit-modulus phase, the phases can be absorbed into sector weights to obtain a BNT grouping with strictly decreasing sector norms.*

Theorem 11.91 (Sector decomposition from TP primitive irreducible blocks). *From a trace-preserving primitive irreducible block decomposition with nonzero weights and non-decaying cross-overlaps for equal-norm blocks, one obtains a sector decomposition with strict norm ordering.*

11.13.3 Cyclic Sectors

Definition 11.92 (Left sector tensor). For a projection P , the left sector tensor is obtained by left multiplication of each Kraus operator by P .

Lemma 11.93 (Commuting letters imply commuting words). *If a projection commutes with every Kraus operator, it commutes with every finite word product.*

Lemma 11.94 (Left multiplication identity for sector words). *If a projection commutes with all Kraus operators, then word evaluation for the left sector tensor equals left multiplication by the projection.*

Lemma 11.95 (Sector support). *The left sector tensor is supported on the chosen projection.*

Theorem 11.96 (Adjoint-fixed projections commute with Kraus operators). *If a projection is fixed by the adjoint transfer map of a trace-preserving tensor, then it commutes with each Kraus operator.*

Theorem 11.97 (Left-canonical primitive blocking from irreducibility). *For a trace-preserving irreducible tensor, a positive blocking period yields a blocked tensor that is left-canonical and primitive.*

Theorem 11.98 (Normality from primitivity and a positive-definite fixed point). *A primitive tensor admitting a positive-definite fixed point of its transfer map is normal.*

Theorem 11.99 (Zero-word evaluation for the zero tensor). *Every nonempty finite word evaluates to zero when all Kraus operators are zero.*

Theorem 11.100 (Zero MPV for the zero tensor). *For a nonempty spin chain, the MPS value of every spin word is zero for the zero tensor.*

Theorem 11.101 (Nonzero Kraus operator in irreducible dimension at least two). *An irreducible tensor of bond dimension at least two has at least one nonzero Kraus operator.*

Theorem 11.102 (All-zero tensors are irreducible only in dimension at most one). *If every Kraus operator vanishes and the tensor is irreducible, then the bond dimension is at most one.*

Theorem 11.103 (Irreducible block decomposition with trace-preserving gauge). *After removing a trivial block, one obtains a block decomposition into irreducible blocks with a trace-preserving gauge.*

11.13.4 Reduction and structural after-blocking results

Theorem 11.104 (Blockwise TP-gauge normalization of irreducible blocks). *Suppose A generates the same MPV family as a block-diagonal tensor $\bigoplus_{k=1}^r B_k$ with each block B_k irreducible and containing at least one nonzero Kraus operator. Then, after rescaling the sector weights, one obtains blocks (C_k) such that $A \sim \bigoplus_k \mu_k C_k$, each C_k is irreducible, each C_k is left-canonical, all $\mu_k \neq 0$, and all bond dimensions are positive.*

Proof. The Perron–Frobenius TP-gauge construction is applied separately to each irreducible block. Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the new weights. $\square$

Theorem 11.105 (Primitive live blocks are normal). *A left-canonical irreducible tensor with primitive transfer map and positive bond dimension is normal.*

Proof. Apply Theorem 11.79. $\square$

Theorem 11.106 (Blocking preserves irreducibility for primitive irreducible blocks). *If A is left-canonical, irreducible, primitive, and has positive bond dimension, then every positive blocking length P preserves irreducibility: $A^{[P]}$ is again irreducible.*

Proof. A positive-definite fixed point of $\mathcal{E}_A$ remains fixed by the blocked transfer map. Primitivity gives uniqueness of positive semidefinite fixed points for the blocked map, hence the blocked transfer map is irreducible; this is exactly the irreducibility criterion for the blocked tensor. $\square$

Theorem 11.107 (Sorted TP-primitive irreducible data already forms a normal canonical form). *Let (μ_k, A_k) be a finite block family with strictly decreasing norms $|\mu_1| > \dots > |\mu_r|$. If every block A_k is left-canonical, primitive, irreducible, has positive bond dimension, and every $\mu_k \neq 0$, then (μ_k, A_k) is a normal canonical form.*

Proof. This is the defining constructor for normal canonical form with all required data supplied by the hypotheses. $\square$

Theorem 11.108 (Normal canonical form from already primitive input). *If A already admits a block decomposition into irreducible left-canonical primitive blocks with pairwise distinct weight norms, nonzero weights, and positive bond dimensions, then some blocked tensor $A^{[p]}$ is MPV-equivalent to a normal canonical form.*

Proof. Apply Theorem 11.44 to the given primitive block decomposition. $\square$

Theorem 11.109 (Conditional block matching after blocking). *Consider two block families in normal canonical form, with no two distinct blocks of the same dimension related by gauge-phase equivalence. If two tensors decompose into these families with coefficient sequences having nonzero limits, and the two tensors are asymptotically proportional coefficientwise, then the numbers of blocks agree and the blocks match up to permutation and gauge-phase equivalence.*

Proof. This is the block-matching statement from the multi-block fundamental theorem applied to two decompositions that already satisfy the normal-canonical-form and BNT-separation hypotheses. $\square$

Theorem 11.110 (Common blocking period for two primitive normal tensors). *Let A and B be left-canonical normal tensors of positive bond dimensions. Suppose $A^{[p_A]}$ and $B^{[p_B]}$ already have primitive transfer maps for some positive periods p_A, p_B . Then there exists a common positive period p such that both $A^{[p]}$ and $B^{[p]}$ are left-canonical, primitive, and normal.*

Proof. Take p to be the least common multiple of p_A and p_B . Primitivity follows from divisibility, and left-canonical normalization and normality are both preserved under blocking. $\square$

Theorem 11.111 (Structural after-blocking form of the fundamental theorem). *If two tensors generate the same MPV family, then each tensor admits some blocking after which it decomposes into nonzero left-canonical blocks with primitive transfer maps and positive bond dimensions.*

Proof. Apply Theorem 11.78 separately to the two tensors. $\square$

11.14 Open directions

Remark 11.112 (Remaining steps toward normal canonical form). The arbitrary-input reduction (Theorem 11.48) produces a trivial block together with left-canonical nontrivial blocks. After a common blocking step (Theorem 11.78), the blocked blocks have primitive transfer maps.

To conclude full normal canonical form, two further ingredients are still needed: tensor irreducibility for each blocked block (blocking does not in general preserve tensor irreducibility), and pairwise distinct weight norms. The cyclic-sector decomposition (Theorems 11.59 and 11.60) provides the infrastructure for the periodic-to-aperiodic splitting.

Chapter 12

Block Permutation and Separation

This chapter records the algebra of block permutations, the product-algebra maps built from per-block same-MPV data, the invariant-projection splitting step that reduces a tensor to smaller blocks, and the block-separation argument that recovers the individual blocks from a block-diagonal same-MPV hypothesis. The results correspond to the canonical-form uniqueness of [PGVWC07] and the block-separation results of [CPGSV21].

12.1 Block permutation algebra

Definition 12.1 (Block ideal). Let $\{R_k\}_{k=1}^r$ be simple rings. The k -th *block ideal* in $\prod_{j=1}^r R_j$ is

$$I_k := \{x \in \prod_{j=1}^r R_j : x_j = 0 \text{ for } j \neq k\}.$$

Theorem 12.2 (Each block ideal is an atom). *Each block ideal I_k is an atom in the lattice of two-sided ideals of $\prod_{j=1}^r R_j$.*

Proof. Under the order isomorphism between two-sided ideals of $\prod_{j=1}^r R_j$ and products of two-sided ideal lattices, I_k corresponds to the tuple with $\top$ in the k -th coordinate and $\perp$ elsewhere. This tuple is an atom, hence so is I_k . $\square$

Theorem 12.3 (Ring isomorphisms permute block ideals). *Let $\{R_k\}_{k=1}^r$ be simple rings. Every ring isomorphism $T : \prod_k R_k \rightarrow \prod_k R_k$ permutes the block ideals: there exists a permutation π such that T maps the k -th block ideal to the $\pi(k)$ -th block ideal.*

Proof. By Theorem 12.2, each block ideal is an atom in the lattice of two-sided ideals of $\prod_k R_k$. A ring isomorphism induces an order automorphism of this lattice, so it permutes the atoms. Since the atoms are exactly the block ideals, this gives the required permutation π . $\square$

Remark 12.4 (Ring level versus algebra level). Theorem 12.3 is a ring-level statement: it only uses the ideal structure of $\prod_k R_k$. Theorem 12.6 stays at this ring level: once the block ideals are permuted, one can extract the corresponding single-block maps without using scalar multiplication. Theorems 12.5 and 12.7 require an upgrade to $\mathbb{C}$ -algebra automorphisms. The reason

is threefold: dimension preservation compares the blocks as $\mathbb{C}$ -vector spaces; Skolem–Noether applies to the resulting central simple $\mathbb{C}$ -algebras; and the final per-block maps must commute with scalar multiplication. Thus the argument separates the ring-level permutation step from the later algebra-level decomposition.

Theorem 12.5 (Dimension preservation). *Assume $D_k \geq 1$ for all k . If T is an algebra automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and π is the induced permutation, then $D_{\pi(k)} = D_k$ for all k .*

Proof. The restriction of T to the k -th block ideal induces a $\mathbb{C}$ -algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$, which forces $D_k = D_{\pi(k)}$ by comparing dimensions as $\mathbb{C}$ -vector spaces. $\square$

Theorem 12.6 (Per-block component maps are bijections). *Assume $D_k \geq 1$ for all k . Let T be a ring automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and let π be the induced permutation from Theorem 12.3. For each block index k , the map*

$$M \mapsto (T(0, \dots, 0, M, 0, \dots, 0))_{\pi(k)}$$

from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ is bijective.

Proof. Theorem 12.3 forces a single-support element in the k -th block ideal to remain supported on the $\pi(k)$ -th block after applying T . Injectivity follows from injectivity of T . For surjectivity, apply the same argument to T^{-1} and the inverse permutation. $\square$

Theorem 12.7 (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$). *Assume $D_k \geq 1$ for all k . Every $\mathbb{C}$ -algebra automorphism T of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$ decomposes as a block permutation π followed by per-block inner automorphisms: there exist $\pi \in S_r$ with $D_{\pi(k)} = D_k$ and invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that, for every tuple $M = (M_1, \dots, M_r)$, $T(M)_k = X_k M_{\pi(k)} X_k^{-1}$, where $T(M)_k$ denotes the k -th block component of $T(M)$.*

Proof. Theorem 12.3 gives the permutation π . Theorem 12.5 identifies the matrix sizes $D_{\pi(k)} = D_k$. For each k , Theorem 12.6 gives a bijective multiplicative map from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ extracted from the single-support inputs of T ; after reindexing by the dimension equality, this is a $\mathbb{C}$ -algebra automorphism of $M_{D_k}(\mathbb{C})$. Skolem–Noether (Theorem 4.9) makes each such automorphism inner.

This is a standard result in the theory of semisimple algebras; see [CPGSV21, §IV.A] for the MPS context. $\square$

12.2 Product algebra linear extension

Definition 12.8 (Per-block linear extension). Given injective tensors A_k, B_k of bond dimension D_k with the same MPV for each block k , the *per-block linear extension* T_k is the unique linear map $M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$ satisfying $T_k(A_k^i) = B_k^i$ for all i . It exists (and is unique) because $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$ by injectivity.

Definition 12.9 (Product algebra automorphism). The *product algebra automorphism* is the blockwise map $T : \prod_k M_{D_k}(\mathbb{C}) \rightarrow \prod_k M_{D_k}(\mathbb{C})$ defined by $T(M)_k := T_k(M_k)$. Theorem 12.10 shows that each T_k is a $\mathbb{C}$ -algebra homomorphism and that this map is a $\mathbb{C}$ -algebra automorphism.

Theorem 12.10 (Per-block linear extensions define a product algebra automorphism). *Let A_k and B_k be injective tensors of bond dimension D_k for $k = 1, \dots, r$, and suppose the per-block same-MPV condition holds. Then the per-block linear extensions $T_k : A_k^i \mapsto B_k^i$ define an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$, which decomposes as a permutation π with $D_{\pi(k)} = D_k$ and per-block conjugation matrices X_k .*

Proof. Each T_k exists uniquely and is multiplicative by Theorems 4.6 and 4.7. Promotion to algebra homomorphisms (Lemma 4.10) and blockwise composition give an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$. Theorem 12.7 then decomposes this automorphism as a block permutation together with per-block inner automorphisms. $\square$

Theorem 12.11 (Per-block same MPV gives global gauge equivalence). *If all block tensors A_k are injective, and B_k are block tensors of the same bond dimensions such that per-block same-MPV holds, then each pair (A_k, B_k) is gauge equivalent, and the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are globally gauge equivalent.*

Proof. Per-block gauge equivalence follows from the single-block FT (Theorem 4.11) applied to each block. Global gauge equivalence follows by placing the per-block gauge matrices on the block diagonal (Theorem 11.4). $\square$

12.3 Burnside and invariant projections

Theorem 12.12 (The algebra span reaches a finite cumulative span). *If the generated unital algebra satisfies $\text{alg}(A) = M_D(\mathbb{C})$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.*

Proof. Every element of $\text{alg}(A)$ lies in some cumulative span $T_n(A)$: the generators lie in $T_1(A)$, scalar multiples of the identity lie in $T_0(A)$, and the family $\{T_n(A)\}_{n \geq 0}$ is closed under addition and under multiplication with lengths adding. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \subseteq T_1(A) \subseteq \dots$ stabilizes, so one can choose a single N with $T_N(A) = M_D(\mathbb{C})$. $\square$

Theorem 12.13 (Irreducible action gives an irreducible tensor). *If the letters of A act irreducibly on $\mathbb{C}^D$, then A has no nontrivial invariant orthogonal projection.*

Proof. If P were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of A . Irreducibility of the action forces this range to be either 0 or all of $\mathbb{C}^D$, so P must be 0 or $\mathbb{1}$, a contradiction. $\square$

Theorem 12.14 (Invariant projection gives a two-block decomposition). *Suppose P is an orthogonal projection satisfying*

$$(\mathbb{1} - P)A^i P = 0 \quad \text{for all } i.$$

Then there exist n, m with $n + m = D$ and MPS tensors A_1, A_2 of bond dimensions n and m such that A and the two-block tensor $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.

Proof. Diagonalize P by a unitary U . Since $P^2 = P$, the diagonal form of P has only 0- and 1-eigenvalues, so the basis splits as $n + m = D$. Conjugating A by U preserves the MPV family by Theorem 11.10. In this basis the relation $(\mathbb{1} - P)A^i P = 0$ makes each letter upper block triangular. Discarding the off-diagonal blocks does not change the trace of any word product, so the resulting block-diagonal tensor has the same MPV coefficients. Reindex the two diagonal blocks by $\{0, \dots, n-1\}$ and $\{0, \dots, m-1\}$. $\square$

Theorem 12.15 (Nontrivial invariant projection gives strict two-block reduction). *Under the hypotheses of Theorem 12.14, assume in addition that $P \neq 0$ and $P \neq \mathbb{1}$. Then there exist n, m with $n + m = D$, $n < D$, and $m < D$, together with block tensors A_1, A_2 , such that A and $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.*

Proof. Apply Theorem 12.14. In the diagonal form of P , the 1-eigenspace is nonempty because $P \neq 0$, and the 0-eigenspace is nonempty because $P \neq \mathbb{1}$. Hence both block sizes are positive. Since $n + m = D$, this implies $n < D$ and $m < D$. $\square$

12.4 Block separation

Recall from Definition 2.35 and Theorem 9.7 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \text{Tr}(F_{AB}^N).$$

We use this notation throughout the block-separation argument.

Definition 12.16 (Canonical form conditions). A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form conditions* if:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_k A_k}(N) \rightarrow 1$ for each k .

Only this one-sided normalization is assumed; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Thm. 6.7, item 3]), $T^n(\rho) \rightarrow \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Thm. 6.8] for the completely-positive characterisation and Chapter 9 for the spectral-gap proof.

Remark 12.17 (Normalization convention). The convention here differs from [PGVWC07, Thm. 4], which uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$. This chapter instead works in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides a gauge transformation between these two normalizations (Theorem 8.11).

Remark 12.18 (Overlap hypothesis). In the literature [CPGSV21], condition (5) is derived from primitivity of the transfer map (peripheral spectrum = $\{1\}$). Theorem 11.28 gives this implication: per-block primitivity implies overlap convergence, so condition (5) is redundant when primitivity holds. The canonical form conditions retain condition (5) explicitly because it can be verified independently of primitivity.

Remark 12.19 (Normal-canonical companion). Definition 12.16 gives the block-injective conditions. The normal-canonical analogue instead uses Definition 11.38: injectivity and the explicit self-overlap hypothesis are replaced by irreducibility, TP normalization, and primitive transfer maps. Theorem 11.39 then supplies the self-overlap limit automatically.

Theorem 12.20 (Vandermonde separation). *Let C be a canonical form with block weights $\mu_1, \dots, \mu_r$, and assume these weights are pairwise distinct. If $c_1, \dots, c_r \in \mathbb{C}$ satisfy*

$$\sum_{k=1}^r c_k \mu_k^N = 0 \quad \text{for } N = 0, 1, \dots, r-1,$$

then $c_k = 0$ for all k .

Proof. The coefficient vector c satisfies $Vc = 0$, where V is the Vandermonde matrix of the weights μ_k . Pairwise distinctness makes V invertible, so $c = 0$. $\square$

Lemma 12.21 (Repeated-word evaluation). *For any word w and any $L \geq 0$, the evaluation of A on the L -fold concatenation w^L is $(A^w)^L$.*

Proof. By induction on L , using the multiplicativity of word evaluation under concatenation. $\square$

Theorem 12.22 (Repeated-word trace identity). *If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then for every word w and every $L \geq 0$,*

$$\sum_k \mu_k^{|w|L} (\text{tr}((A_k^w)^L) - \text{tr}((B_k^w)^L)) = 0.$$

Proof. Apply the block-diagonal same-MPV identity to the word w^L obtained by repeating w exactly L times. The block-diagonal coefficient is a weighted sum over the blocks. Lemma 12.21 rewrites each block evaluation on w^L as a matrix power, giving the displayed trace identity. $\square$

Remark 12.23. Theorems 12.20 and 12.22 record an alternative Newton/Vandermonde route. The checked proof chain in this chapter instead continues with the mixed-transfer peeling argument of Lemma 12.24, and the later BNT weight-matching step uses the linear-independence results from Chapter 13 rather than Newton–Girard identities.

Lemma 12.24 (Block separation core). *Let $((\mu_k), (A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ for all N, σ , then each pair (A_k, B_k) generates the same MPV family.*

Proof. By induction on the number of blocks r . In the inductive step, take overlaps with the leading block A_0 ; after division by μ_0^N , the contributions from $k \geq 1$ decay because $|\mu_k/\mu_0| < 1$ (strict weight ordering) and the bilinear overlaps $O_{A_0 A_k}(N)$ and $O_{A_0 B_k}(N)$ are bounded (Theorems 9.23 and 9.25 supply the relevant bounds), leaving $O_{A_0 A_0}(N) - O_{A_0 B_0}(N) \rightarrow 0$. Condition (5) gives $O_{A_0 A_0}(N) \rightarrow 1$, hence $O_{A_0 B_0}(N) \rightarrow 1$. Since $O_{A_0 B_0}(N) \rightarrow 1 \neq 0$, the bond dimensions must match: $D_{A_0} = D_{B_0}$; otherwise Theorem 9.23 would give $O_{A_0 B_0}(N) \rightarrow 0$, a contradiction. With equal bond dimensions established, the contrapositive of Theorem 9.25 forces A_0 and B_0 to be gauge-phase equivalent; the limit $O_{A_0 B_0}(N) \rightarrow 1$ then makes the phase trivial, so A_0 and B_0 generate the same MPV family. Peeling off the leading block and dividing by μ_0^N , the induction hypothesis applies. $\square$

Theorem 12.25 (Block separation from canonical form). *Let $((\mu_k), (A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family.*

Proof. Apply Lemma 12.24 to the identity obtained from the equality of the two block-diagonal MPV families. The induction peels off the dominant block and treats the remainder as a decaying tail. Concretely, one isolates the leading term $\mu_1^N V(A_1)$ and bounds the remaining contribution by $O(|\mu_2/\mu_1|^N)$ after dividing by μ_1^N . The canonical form conditions supply injectivity, TP normalization, strict nonzero weights, and self-overlap convergence for (A_k) , while the hypotheses on (B_k) supply the comparison injectivity and TP normalization. $\square$

Theorem 12.26 (Block separation from normal canonical form). *Let $((\mu_k), (A_k))$ satisfy the normal canonical form conditions, and let (B_k) be irreducible tensors of the same bond dimensions, satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family.*

Proof. The proof follows the same peeling argument as Theorem 12.25, with Theorem 11.40 (modulus-one eigenvalue rigidity for irreducible tensors) replacing the contrapositive of Theorem 9.25 at the identification step, since the blocks are irreducible rather than injective. Theorem 11.39 replaces the explicit self-overlap hypothesis on the leading block. Since both block families already use the same dimension function, the conclusion is direct per-block same MPV. $\square$

Theorem 12.27 (Canonical form fundamental theorem, same-structure version). *Under the hypotheses of Theorem 12.25, each pair (A_k, B_k) is gauge equivalent. Hence the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. Theorem 12.25 gives per-block same MPV. Theorem 12.11 then converts this to per-block gauge equivalence and then to global gauge equivalence of the block-diagonal tensors. $\square$

Remark 12.28 (Relation to the literature). The block-permutation part follows the decomposition of automorphisms of semisimple matrix algebras used in [CPGSV21, §IV.A]. The separation part follows the induction on blocks from [PGVWC07, Appendix E]; the checked proof here uses overlap bounds and mixed-transfer decay rather than the alternative Newton/Vandermonde route recorded above.

Chapter 13

Basis of Normal Tensors

This chapter introduces the *basis of normal tensors* (BNT) notion, the canonical-form variants with BNT separation, the cross-overlap decay theorem, and the permutation-rigidity arguments used in the proof of the Fundamental Theorem. The chapter concludes with auxiliary results on coefficient convergence and Newton–Girard identities. The main references are [PGVWC07] and [CPGSV21].

13.1 Basis of normal tensors

Definition 13.1 (Basis of Normal Tensors (BNT)). A *basis of normal tensors* (BNT) for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that:

1. each A_j is normal (Definition 2.23),
2. at each system size N , the MPV of A_{tot} lies in the span of the block MPVs $\{|V^{(N)}(A_j)\}_{j=1}^g$,
3. for sufficiently large N , the block MPV vectors $\{|V^{(N)}(A_j)\}_{j=1}^g$ are linearly independent.

This is [CPGSV21, Definition 4.2].

Theorem 13.2 (Overlap-orthonormality implies eventual linear independence). *Let $(A_j)_{j=1}^g$ be a finite family of tensors such that $O_{A_j A_j}(N) \rightarrow 1$ for each j and $O_{A_i A_j}(N) \rightarrow 0$ for $i \neq j$. Then the MPV states $\{|V^{(N)}(A_j)\}_{j=1}^g$ are linearly independent for all sufficiently large N .*

Proof. The Gram matrix of the family $\{|V^{(N)}(A_j)\}_{j=1}^g$ converges entrywise to the identity matrix. Hence for all sufficiently large N it is invertible, so the MPV states are linearly independent. $\square$

Definition 13.3 (Canonical form with BNT separation). A family is in *canonical form with BNT separation* if it satisfies Definition 12.16 and, in addition:

1. the moduli are *strictly* decreasing: $|\mu_1| > |\mu_2| > \dots > |\mu_r|$ (strengthened from the non-increasing ordering of Definition 12.16),
2. distinct blocks of the same bond dimension are *not* gauge-phase equivalent.

The strictly decreasing condition is justified by the BNT grouping step, which merges gauge-phase-equivalent blocks with equal moduli.

Theorem 13.4 (Distinct bond dimensions force BNT separation). *If a canonical-form family has strictly decreasing weight moduli and pairwise distinct bond dimensions, then it is in canonical form with BNT separation.*

Proof. The only new clause in Definition 13.3 is the prohibition on gauge-phase equivalence between distinct blocks of the same bond dimension. When the bond dimensions are pairwise distinct, this clause is vacuous. $\square$

Definition 13.5 (Normal canonical form with BNT separation). A family is in *normal canonical form with BNT separation* if it satisfies Definition 11.38 and, in addition:

1. the moduli are strictly decreasing (strengthened from non-increasing),
2. distinct blocks of the same bond dimension are not gauge-phase equivalent.

Theorem 13.6 (Equal-norm grouping step). *Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family with $\mu_k \neq 0$ for every k . Assume that whenever $\|\mu_j\| = \|\mu_k\|$, the tensors A_j and A_k generate the same MPV family. Then one can regroup the blocks into a sector decomposition whose associated total tensor generates the same MPV family as $\bigoplus_k \mu_k A_k$, and whose sector-weight norms are strictly decreasing.*

Proof. Order the distinct norms of the weights in decreasing order, choose one representative block from each norm class, and retain the multiplicities inside each class as sector copies. The equal-MPV hypothesis lets every block in a class be replaced by the chosen representative, so the regrouped decomposition has the same MPV family. By construction, the sector-weight norms are strictly decreasing. $\square$

Theorem 13.7 (Cross-overlap decay from explicit BNT hypotheses). *Let $(A_j)_{j=1}^r$ be injective and trace-preserving normalized. Assume that distinct equal-dimension blocks are not gauge-phase equivalent. Then $O_{A_j A_k}(N) \rightarrow 0$ for every pair $j \neq k$.*

Proof. If the bond dimensions differ, Theorem 9.23 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 9.25 gives the decay. $\square$

Theorem 13.8 (Cross-overlap decay for distinct blocks in canonical form with BNT separation). *If $((\mu_k), (A_k))$ is in canonical form with BNT separation and $j \neq k$, then $O_{A_j A_k}(N) \rightarrow 0$.*

Proof. Definition 13.3 supplies injectivity, left-canonical normalization, and the BNT separation hypothesis, so Theorem 13.7 applies directly. $\square$

Theorem 13.9 (BNT from explicit blockwise hypotheses). *Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family. Assume that the blocks are injective and trace-preserving normalized, that $O_{A_k A_k}(N) \rightarrow 1$ for each k , and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of $\bigoplus_k \mu_k A_k$ in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 13.7 supplies the off-diagonal decay. Therefore Theorem 13.2 gives the eventual linear independence required in Definition 13.1. $\square$

Theorem 13.10 (Canonical form with BNT separation yields a basis of normal tensors). *If $((\mu_k), (A_k))$ is in canonical form with BNT separation, then the block tensors form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. Definition 13.3 provides injective blocks, left-canonical normalization, normalized self-overlaps, and BNT separation. Hence Theorem 13.9 applies. $\square$

Remark 13.11 (The normality gap for normal canonical form with BNT separation). Definition 13.5 references the normal canonical form conditions (Definition 11.38), which encode normality via the spectral characterization (primitive transfer map). Definition 13.1 requires normality via the algebraic characterization (eventual block injectivity, Definition 2.23). These are equivalent by Remark 2.20, but the equivalence must be invoked explicitly: the Wielandt theory of Chapter 10 provides the implication from primitivity to eventual block injectivity needed here.

Theorem 13.12 (Normal canonical form with BNT separation yields a BNT after algebraic normality is supplied). *If a family is in normal canonical form with BNT separation and each block is also known to be normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor.*

Proof. Normal-canonical BNT data already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 13.1. $\square$

13.2 Permutation rigidity for bases of normal tensors

Theorem 13.13 (Invertible change of basis). *Let $v_1, \dots, v_g$ and $w_1, \dots, w_g$ be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix $U \in \text{GL}_g(\mathbb{C})$ such that*

$$w_j = \sum_{i=1}^g U_{ij} v_i$$

for every j .

Proof. Because the two families span the same subspace, each w_j has a unique expansion in the basis $\{v_i\}_{i=1}^g$; these coefficients form the columns of U . If $Ua = 0$, then the corresponding linear relation among the w_j is trivial because the w_j are linearly independent. Thus U is injective, hence invertible. $\square$

Theorem 13.14 (Eventual change of basis from equal spans). *If two finite families of blocks have asymptotically orthonormal overlaps within each family and their MPV spans agree for every system size N , then for all sufficiently large N there exists an invertible matrix U_N expressing the MPV states of one family as linear combinations of the MPV states of the other.*

Proof. By Theorem 13.2, both families are linearly independent for all sufficiently large N . At such an N , the span equality lets us apply Theorem 13.13. $\square$

Theorem 13.15 (Permutation rigidity for equal block counts). *Let $(A_j)_{j=1}^g$ and $(B_j)_{j=1}^g$ be injective families satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for every system size N , then there is a permutation π of $\{1, \dots, g\}$ such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent.*

Proof. For large N , Theorem 13.14 gives an invertible matrix U_N relating the two MPV families. Fix j . If every mixed overlap $O_{A_i B_j}(N)$ tended to 0, then taking inner products of the relation with the A_i and using that the Gram matrix of the A -family tends to the identity would force the j th column of U_N to vanish asymptotically, contradicting the self-overlap limit of B_j . Thus each B_j has a matched block $A_{f(j)}$ with non-decaying mixed overlap. Theorems 9.23 and 9.25 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the B -family makes f injective, hence bijective, and its inverse is the required permutation. $\square$

Theorem 13.16 (Permutation rigidity for proportional MPV decompositions). *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be injective families satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. Assume moreover that there exist MPS tensors A_{tot} and B_{tot} , coefficient arrays $a_{N,j}$ and $b_{N,k}$, and a sequence c_N with nonzero limits a_j^∞ , b_k^∞ , c^∞ such that for every N ,*

$$|V^{(N)}(A_{\text{tot}})\rangle = \sum_{j=1}^{g_A} a_{N,j} |V^{(N)}(A_j)\rangle, \quad |V^{(N)}(B_{\text{tot}})\rangle = \sum_{k=1}^{g_B} b_{N,k} |V^{(N)}(B_k)\rangle,$$

with $a_{N,j} \rightarrow a_j^\infty \neq 0$ and $b_{N,k} \rightarrow b_k^\infty \neq 0$, and also

$$|V^{(N)}(A_{\text{tot}})\rangle = c_N |V^{(N)}(B_{\text{tot}})\rangle$$

with $c_N \rightarrow c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation π of the common block index set such that $A_{\pi(k)}$ and B_k have the same bond dimension and are gauge-phase equivalent for each k . The canonical-form application of these hypotheses is described in Remark 13.17.

Proof. The proportional decomposition implies that, for each block on the B -side, the mixed overlaps with A_{tot} are linear combinations of the overlaps with the A -family. If every mixed overlap with a fixed B_k decayed to 0, then the coefficient limits would force the limiting coefficient b_k^∞ to vanish, contrary to the hypotheses. Thus every B_k matches some A_j with non-decaying mixed overlap. Theorems 9.23 and 9.25 then force equal bond dimension and gauge-phase equivalence for each match. Repeating the same argument in the opposite direction gives an injective matching from the A -family to the B -family. Comparing the two injective matchings yields the permutation. $\square$

Remark 13.17. In the canonical-form application, the abstract data above come from block-diagonal tensors

$$A_{\text{tot}} = \bigoplus_{j=1}^{g_A} \mu_j A_j, \quad B_{\text{tot}} = \bigoplus_{k=1}^{g_B} \nu_k B_k,$$

and Theorem 2.28 gives the coefficient arrays $a_{N,j} = \mu_j^N$ and $b_{N,k} = \nu_k^N$. To fit the convergence hypotheses of Theorem 13.16, one first normalizes by the dominant weight on each side: the relevant arrays are $(\mu_j/\mu_0)^N$ and $(\nu_k/\nu_0)^N$, with $(\mu_j/\mu_0)^N \rightarrow 0$ for $j \geq 1$ and $(\nu_k/\nu_0)^N \rightarrow 0$ for $k \geq 1$ by Theorem 13.26, while the overall factors μ_0^N and ν_0^N are absorbed into the proportionality constant c_N . Definition 13.3 supplies injectivity, TP normalization, and the within-family overlap limits, so the theorem isolates the abstract permutation argument from this canonical-form setting.

Theorem 13.18 (Permutation rigidity with asymptotically orthonormal overlaps). *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$*

and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size N ,

$$\text{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g_A\} = \text{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \leq k \leq g_B\},$$

then $g_A = g_B$, and there is a permutation π of the common block index set such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each j .

Proof. By Theorem 13.2, both families are linearly independent for all sufficiently large N . At such an N , the common span has the same dimension when computed from the A -family or the B -family, so $g_A = g_B$. After identifying the two index sets, Theorem 13.15 yields the permutation conclusion. $\square$

Theorem 13.19 (Permutation rigidity for proportional decompositions, irreducible TP version). *Under the hypotheses of Theorem 13.16, replace injectivity of each block by irreducibility while keeping trace-preserving normalization and the same overlap limits. Then the same permutation conclusion holds.*

Proof. The same matching argument applies. The only change is at the two contradiction steps: the overlap-decay inputs for injective blocks are replaced by the corresponding irreducible trace-preserving decay theorems. $\square$

Theorem 13.20 (Canonical-form proportional decomposition theorem). *If two families in canonical form with BNT separation admit proportional MPV decompositions with convergent nonzero coefficients, then they have the same number of blocks and match blockwise up to permutation, bond-dimension equality, and gauge-phase equivalence.*

Proof. Project the two canonical-form hypotheses to blockwise injectivity, left-canonical normalization, self-overlap convergence, and BNT separation. Then Theorem 13.16 applies to the resulting proportional decomposition data. $\square$

Theorem 13.21 (Normal-canonical proportional decomposition theorem). *If two families in normal canonical form with BNT separation admit proportional MPV decompositions with convergent nonzero coefficients, then they have the same number of blocks and match blockwise up to permutation, bond-dimension equality, and gauge-phase equivalence.*

Proof. Project the two normal-canonical hypotheses to irreducibility, left-canonical normalization, self-overlap convergence, and BNT separation, and then apply Theorem 13.19. $\square$

Remark 13.22. Theorem 13.18 is retained for completeness but is not used in the proofs given here; the proportional version (Theorem 13.16) suffices for the Fundamental Theorem.

13.2.1 Fixed-size separation by Vandermonde inversion

Theorem 13.23 (Vandermonde separation). *Let C be a canonical form with r blocks and pairwise distinct weights μ_k . If coefficients c_k satisfy*

$$\sum_k c_k \mu_k^N = 0$$

for $N = 0, \dots, r-1$, then every c_k vanishes.

Proof. The Vandermonde matrix with entries $V_{Nk} = \mu_k^N$ is invertible when the weights are pairwise distinct. Therefore the displayed linear system has only the zero solution. $\square$

Theorem 13.24 (Fixed-size linear independence of block MPVs). *Let C be a canonical form with r blocks and pairwise distinct weights μ_k , and fix a system size N_0 . Assume that each block MPV $|V^{(N_0)}(A_k)\rangle$ is nonzero. If*

$$\sum_k c_k V^{(N_0)}(A_k)_\sigma \mu_k^i = 0$$

for every word σ of length N_0 and every $i = 0, \dots, r-1$, then $c_k = 0$ for all k .

Proof. Apply Theorem 13.23 pointwise in the word σ . This shows that every component of $c_k |V^{(N_0)}(A_k)\rangle$ vanishes. Since the MPV of each block is nonzero at size N_0 , some component survives, so $c_k = 0$. $\square$

Remark 13.25. These fixed-size Vandermonde theorems remain available for alternative finite-length arguments, but the present BNT proof chain uses the eventual linear independence from Theorem 13.2 instead.

13.3 Coefficient convergence

Theorem 13.26 (Coefficient ratio decay). *Under the canonical form with BNT separation of Definition 13.3, indexed so that μ_0 is the (strictly) dominant weight, one has $(\mu_k/\mu_0)^N \rightarrow 0$ for every $k \geq 1$ as $N \rightarrow \infty$.*

Proof. By the BNT separation condition in Definition 13.3, $|\mu_k| < |\mu_0|$ strictly, so $|\mu_k/\mu_0| < 1$ and the geometric sequence converges to 0. $\square$

Remark 13.27 (Strict-dominance regime). Theorem 13.26 applies when one modulus is strictly dominant: $|\mu_0| > |\mu_k|$ for $k \geq 1$. In the more general periodic setting of [CPGSV21, Eq. (72a)], the coefficient attached to block j can take the form $\sum_q \mu_{j,q}^N$, where the $\mu_{j,q}$ are the individual weights contributed by that block. The oscillatory case with $|\mu_{j,q}| = 1$ is treated separately in the full paper.

13.4 Newton–Girard identities

Theorem 13.28 (Newton–Girard trace recursion). *If $A, B \in M_n(\mathbb{C})$ satisfy $\text{tr}(A^k) = \text{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial.*

Proof. Let e_m denote the m th elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$me_m = \sum_{i=1}^m (-1)^{i-1} e_{m-i} \text{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\text{tr}(A^i)$. Equal power sums for A and B therefore give equal coefficients, hence equal characteristic polynomials. $\square$

Theorem 13.29 (Equal power sums imply equal multisets). *Let $\alpha, \beta : \{1, \dots, n\} \rightarrow \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \geq 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.*

Proof. Construct diagonal matrices $\text{diag}(\alpha)$ and $\text{diag}(\beta)$. The power-sum hypothesis gives $\text{tr}(\text{diag}(\alpha)^k) = \text{tr}(\text{diag}(\beta)^k)$. By Theorem 13.28, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. $\square$

Remark 13.30. Theorems 13.28 and 13.29 remain relevant for the general equal-MPV Theorem 14.5, where one expects a power-sum step to recover the multiplicity-weight data. The present explicit-coefficient special case, Theorem 14.6, instead uses linear independence of the block MPV family supplied by a basis of normal tensors.

Chapter 14

Proof of the Fundamental Theorem

This chapter proves the Fundamental Theorem of Matrix Product States. We state the Fundamental Theorem in two versions: one for block-injective tensors in canonical form with BNT separation, and one for the normal-canonical variant. We then treat the equal-MPV case, including the common-block-structure specialization that deduces gauge equivalence. The results combine the canonical form reduction (Chapter 11), block separation (Chapter 12), spectral gap (Chapter 9), and the BNT permutation arguments of Chapter 13.

The theorems in this chapter follow the approach of [CPGSV17a]: they first block the tensor to remove non-trivial periodicity, then apply the aperiodic fundamental theorem. Chapter 15 develops the stronger irreducible-form framework of [DCCSPG17], which gives a fundamental theorem valid for periodic tensors directly, with an extra Z -gauge degree of freedom.

14.1 Conditional Fundamental Theorem

The following theorem combines the reduction to canonical form (zero-block separation, TP gauge, blocking, primitivity) with the Fundamental Theorem for normal canonical form. It is conditional on the extra hypotheses (tensor irreducibility and distinct weight norms).

Theorem 14.1 (Conditional Fundamental Theorem). *Let $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$ be two families in normal canonical form, each satisfying the non-degeneracy condition that no two distinct blocks in the same family are gauge-phase equivalent. Let A_{tot} and B_{tot} be tensors with MPV decompositions*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_j a_{N,j} V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_k b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \rightarrow a_j \neq 0$ and $b_{N,k} \rightarrow b_k \neq 0$. Suppose moreover that $V^{(N)}(A_{\text{tot}})_\sigma = c_N V^{(N)}(B_{\text{tot}})_\sigma$ with $c_N \rightarrow c \neq 0$. Then $g_A = g_B$ and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for every j .

Proof. This is exactly the normal-canonical-form Fundamental Theorem (Theorem 14.4). The normal canonical form data provide the BNT separation and overlap convergence hypotheses; the non-degeneracy condition (no two distinct blocks in the same family are gauge-phase equivalent) ensures that the permutation π produced by Theorem 14.4 is uniquely determined, so that the per-block gauge equivalences can be assembled consistently. $\square$

14.2 Proportional Fundamental Theorem

Theorem 14.2 (Fundamental Theorem — proportional MPV case). *Let A_{tot} and B_{tot} be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞ , b_k^∞ , c^∞ such that*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \rightarrow a_j^\infty \neq 0$ and $b_{N,k} \rightarrow b_k^\infty \neq 0$, and suppose also that

$$V^{(N)}(A_{\text{tot}})_\sigma = c_N V^{(N)}(B_{\text{tot}})_\sigma$$

for a sequence $c_N \rightarrow c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for each j .

Proof. Apply Theorem 13.16 to the two canonical-form decompositions with BNT separation. This yields the permutation π that reindexes the separated sectors before comparing them one by one. These hypotheses provide injectivity, TP normalization, diagonal overlap limits, and off-diagonal decay via Theorem 13.8. With the supplied decomposition identities and convergence hypotheses, Theorem 13.16 yields the equality of block counts, the permutation, and the per-block gauge-phase equivalence. $\square$

Remark 14.3 (Coefficient arrays for canonical-form decompositions). In applications the coefficient arrays in Theorem 14.2 come from the block-diagonal decomposition itself. If

$$A_{\text{tot}} = \bigoplus_{j=1}^{g_A} \mu_j^A A_j, \quad B_{\text{tot}} = \bigoplus_{k=1}^{g_B} \mu_k^B B_k,$$

then Theorem 2.28 gives

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} (\mu_j^A)^N V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} (\mu_k^B)^N V^{(N)}(B_k)_\sigma.$$

Thus one takes $a_{N,j} = (\mu_j^A)^N$ and $b_{N,k} = (\mu_k^B)^N$, while the sequence c_N is the global proportionality factor from the hypothesis $V^{(N)}(A_{\text{tot}}) = c_N V^{(N)}(B_{\text{tot}})$. Theorem 14.2 keeps these arrays explicit because Theorem 13.16 is formulated in that abstract form.

Theorem 14.4 (Fundamental Theorem for normal canonical form in the proportional case). *Let A_{tot} and B_{tot} be MPS tensors in normal canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞ , b_k^∞ , c^∞ such that*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \rightarrow a_j^\infty \neq 0$ and $b_{N,k} \rightarrow b_k^\infty \neq 0$, and suppose also that

$$V^{(N)}(A_{\text{tot}})_\sigma = c_N V^{(N)}(B_{\text{tot}})_\sigma$$

for a sequence $c_N \rightarrow c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for each j .

Proof. Normal canonical form with BNT separation provides irreducibility, TP normalization, and primitive self-overlap convergence via Theorem 11.39. Theorem 11.40 supplies the leading-block identification step: a nondecaying mixed overlap forces the corresponding comparison block to be gauge-phase equivalent. The same peeling argument as in the block-injective case therefore yields the block permutation and gauge-phase equivalence. Remark 13.11 explains why the argument stops at this separated normal-canonical level. $\square$

14.3 Equal-MPV Fundamental Theorem

Theorem 14.5 (Fundamental Theorem — equal MPV case). *Let A_{tot} and B_{tot} be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. If*

$$V^{(N)}(A_{\text{tot}})_\sigma = V^{(N)}(B_{\text{tot}})_\sigma$$

for every N and σ , then the block-diagonal tensors $\bigoplus_{j=1}^{g_A} \mu_j^A A_j$ and $\bigoplus_{k=1}^{g_B} \mu_k^B B_k$ are gauge equivalent.

Proof. Begin by applying Theorem 14.2 with $c_N = 1$, which yields a permutation of the matched blocks and blockwise gauge-phase equivalences. One then compares the resulting MPV expansions from Theorem 2.28. The explicit-coefficient version (Theorem 14.6) proves gauge equivalence of the reindexed weighted block-diagonal tensors using BNT linear independence. The remaining step combines this with the coefficient-array construction of Remark 14.3 and the weight-recovery machinery of §14.3.1. After the weights are identified, the phase factors are absorbed into them and the blockwise conjugacies combine to a global gauge equivalence. $\square$

Theorem 14.6 (Fundamental Theorem — equal MPV case with explicit coefficients). *Let A_{tot} and B_{tot} be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞ , b_k^∞ such that*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \rightarrow a_j^\infty \neq 0$ and $b_{N,k} \rightarrow b_k^\infty \neq 0$. If

$$V^{(N)}(A_{\text{tot}})_\sigma = V^{(N)}(B_{\text{tot}})_\sigma$$

for every N and σ , then $g_A = g_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and, after reindexing the B -blocks by π , the weighted block-diagonal tensors $\bigoplus_{j=1}^{g_A} \mu_j^A A_j$ and $\bigoplus_{j=1}^{g_A} \mu_{\pi(j)}^B B_{\pi(j)}$ are gauge equivalent.

Proof. Apply Theorem 14.2 with $c_N = 1$. This yields $g_A = g_B$, a permutation π , equal block dimensions, and for each j a gauge-phase equivalence

$$B_{\pi(j)}^i = \zeta_j X_j A_j^i X_j^{-1}.$$

By Theorem 2.28,

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} (\mu_j^A)^N V^{(N)}(A_j)_\sigma,$$

and similarly for B_{tot} . Reindex the B -sum by π and use the gauge-phase identity to rewrite

$$V^{(N)}(B_{\pi(j)})_{\sigma} = \zeta_j^N V^{(N)}(A_j)_{\sigma}.$$

Then equality of MPVs implies

$$\sum_{j=1}^{g_A} \left((\mu_j^A)^N - (\mu_{\pi(j)}^B \zeta_j)^N \right) V^{(N)}(A_j)_{\sigma} = 0.$$

By Theorem 13.10 and Definition 13.1, the family $\{V^{(N)}(A_j)\}_{j=1}^{g_A}$ is linearly independent for all sufficiently large N , so

$$(\mu_j^A)^N = (\mu_{\pi(j)}^B \zeta_j)^N$$

for every j and all large N . Comparing two consecutive values of N and using that the weights are nonzero gives

$$\mu_j^A = \mu_{\pi(j)}^B \zeta_j.$$

Thus the phase factors can be absorbed into the weights, and the resulting blockwise conjugacies combine to give a gauge equivalence of the reindexed weighted block-diagonal tensors. $\square$

Remark 14.7. Theorem 14.6 recovers [CPGSV21, Corollary IV.5] for tensors in canonical form with BNT separation, without assuming common block structure and without Newton–Girard identities. The weight matching uses BNT linear independence (Theorem 13.10 and Definition 13.1) directly.

Theorem 14.8 (Fundamental Theorem — equal MPV case, common block structure). *Fix a common block structure: the same number of blocks g , the same bond dimensions $(D_k)_{k=1}^g$, and the same weights $(\mu_k)_{k=1}^g$. Let $(\mu_k, A_k)_{k=1}^g$ and $(\mu_k, B_k)_{k=1}^g$ be two families in canonical form with BNT separation on this common data. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) is gauge equivalent, and hence the corresponding block-diagonal tensors are gauge equivalent.*

Proof. Since both families share the canonical form data, the BNT separation condition is not needed. The canonical-form fundamental theorem (Theorem 12.27) then applies directly to this common (g, D_k, μ_k) structure, yielding per-block gauge equivalence and hence global gauge equivalence. $\square$

14.3.1 Multiplicity structure for repeated blocks

This subsection records the multiplicity data in which repeated copies of the same basis block are kept separate rather than merged into a single weighted block.

Definition 14.9 (Sector data and sector decomposition). A *sector datum* over g basis blocks specifies: multiplicities $r_j > 0$, sector weights $\mu_{j,q} \neq 0$ for $q \in \{1, \dots, r_j\}$, and the *sector coefficients*

$$c_{S,j}^{(N)} := \sum_{q=1}^{r_j} \mu_{j,q}^N$$

for a sector datum S . A *sector decomposition* consists of the basis blocks $(A_j)_{j=1}^g$ together with their sector data.

Theorem 14.10 (Sector decomposition formula). *For a sector decomposition S with associated tensor A_{tot} , the MPV satisfies*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^g c_{S,j}^{(N)} V^{(N)}(A_j)_\sigma.$$

Theorem 14.11 (Eventual coefficient comparison from BNT linear independence). *If two sector data S, T over the same basis, with coefficient arrays $c_{S,j}^{(N)}$ and $c_{T,j}^{(N)}$, produce equal total MPVs and the basis is eventually linearly independent (the BNT property), then $c_{S,j}^{(N)} = c_{T,j}^{(N)}$ for all sufficiently large N .*

Theorem 14.12 (Geometric sequence extrapolation). *If two finite families of nonzero complex numbers have equal power sums $\sum_i \alpha_i^k = \sum_j \beta_j^k$ for all $k > N_0$, then equality holds for all $k \geq 0$. The proof uses induction on the number of terms with a telescoping argument: subtract a geometric factor to reduce the number of terms, apply the induction hypothesis, and conclude via a geometric recurrence.*

Theorem 14.13 (Weight multiset recovery from multiplicity data). *Let S and T be two sector data over the same basis with the same multiplicities ($r_j^S = r_j^T$ for all j). If the basis is eventually linearly independent and the total MPVs of S and T agree, then for each basis block j the sector weight multisets $\{\mu_{j,q}^S\}$ and $\{\mu_{j,q}^T\}$ are equal.*

The argument proceeds as follows:

1. BNT LI + equal MPVs $\Rightarrow$ eventual coefficient equality (Theorem 14.11);
2. Geometric extrapolation $\Rightarrow$ full coefficient equality for all positive k (Theorem 14.12);
3. Newton–Girard $\Rightarrow$ equal weight multisets (Theorem 13.29).

At the level of block contributions, one compares the weighted sector sums $\sum_j c_{A,j}^{(N)} V^{(N)}(A_j)$ and $\sum_j c_{B,j}^{(N)} V^{(N)}(B_j)$, with the coefficients now aggregated over the multiplicity layers.

Remark 14.14 (Remaining step in the unconditional equal-case theorem). This chapter provides three complementary equal-case results:

1. Theorem 14.6: the explicit-coefficient equal-case theorem (upgrade of Theorem 14.2 with $c_N = 1$), concluding a global gauge equivalence of the reindexed weighted block-diagonal tensors.
2. Theorem 14.8: the common block-structure special case.
3. Theorem 14.13: the equal-case multiplicity corollary, which from the richer multiplicity data recovers the sector weight multisets without requiring explicit coefficient convergence hypotheses.

The unconditional equal-case theorem (Theorem 14.5, [CPGSV21, Corollary IV.5]) still requires one final global gauge-construction step: combining the weight-recovery result above with (a) the proportional FT for block matching and gauge-phase equivalences, (b) phase absorption into weights, and (c) global gauge construction via permutation and cast transport.

Chapter 15

Periodic MPS: irreducible form and the periodic Fundamental Theorem

This chapter develops the periodic equal-case Fundamental Theorem of MPS (Theorem 15.23) following [DICCSPG17], and records its two applications: the symmetry corollary lifting a physical on-site symmetry of a tensor in irreducible form to a virtual Z -gauge, and the equivalence between p -refinability of the matrix-product-vector family and p -divisibility of the transfer map.

15.1 Periodic irreducible form and physical-index rotation

Definition 15.1 (Periodic tensor). An irreducible MPS tensor A of bond dimension D is m -periodic (for some $m \geq 1$) if the peripheral spectrum of its transfer map $\mathcal{E}_A$ is exactly the m -th roots of unity $\{e^{2\pi i k/m} : 0 \leq k < m\}$. A 1-periodic tensor is exactly an irreducible tensor with primitive transfer map (Definition 5.49).

Definition 15.2 (Irreducible form). A tensor A is in *irreducible form* if it is block-diagonal

$$A^i = \bigoplus_{j \in J} \mu_j A_j^i,$$

where each block A_j is an m_j -periodic tensor, each weight $\mu_j \neq 0$, and the blocks are pairwise non-repeated (no two distinct j, j' have A_j gauge-equivalent to $A_{j'}$). This is the standard form introduced in [DICCSPG17, §II]. It extends the normal canonical form (Definition 11.38), which requires the stronger condition that every block is 1-periodic.

Remark 15.3 (Relation to normal canonical form). Every tensor in normal canonical form (Definition 11.38) is in irreducible form with all periods equal to 1. The irreducible form is strictly more general: it accommodates blocks such as the antiferromagnet, which are 2-periodic.

Definition 15.4 (Physical-index rotation). Given a matrix $u \in M_d(\mathbb{C})$ and an MPS tensor A , the *physical-index rotation* is the tensor $(A_u)^i := \sum_j u_{ij} A^j$.

Theorem 15.5 (Transfer map preserved by physical rotation). *If $uu^\dagger = I$, then the transfer map is invariant under physical-index rotation: $\mathcal{E}_{A_u} = \mathcal{E}_A$.*

Proof. Unitary freedom of the Kraus representation (Theorem 2.18, Wolf *Quantum Channels & Operations*). $\square$

Theorem 15.6 (Left-canonical preserved by physical rotation). *If $uu^\dagger = I$ and A is left-canonical, then A_u is left-canonical.*

Proof. Expand $\sum_i B_i^\dagger B_i$ where $B_i = \sum_j u_{ij} A_j$, swap sums, apply orthogonality $u^\dagger u = I$, and collapse to $\sum_j A_j^\dagger A_j = I$. $\square$

Theorem 15.7 (Irreducibility preserved by physical rotation). *If $uu^\dagger = I$ and A is irreducible, then A_u is irreducible.*

Proof. Contrapositive: any nontrivial invariant projection P for the rotated tensor yields $(1 - P)A_k P = 0$ for all k via the orthogonality of u , contradicting irreducibility of A . $\square$

Theorem 15.8 (Periodicity preserved by physical rotation). *If $uu^\dagger = I$ and A is periodic with period m , then A_u is periodic with the same period.*

Proof. Assembles transfer-map invariance, left-canonical preservation, and irreducibility preservation. $\square$

Theorem 15.9 (SameMPV preserved by physical rotation). *If $\mathcal{V}(A) = \mathcal{V}(B)$, then $\mathcal{V}(A_u) = \mathcal{V}(B_u)$.*

Proof. Induction on the word length with a strengthened hypothesis carrying an arbitrary prefix matrix, reducing each step to the hypothesis via word concatenation. $\square$

Theorem 15.10 (Physical rotation distributes over block assembly). *The rotated block-diagonal tensor satisfies $(\bigoplus_k \mu_k A_k)_u = \bigoplus_k \mu_k (A_k)_u$.*

Proof. Pointwise matrix-entry computation using linearity of the block-diagonal embedding and reindexing. $\square$

Theorem 15.11 (Irreducible form preserved by physical rotation). *If $uu^\dagger = I$ and A is in irreducible form II , then A_u is in irreducible form II with the same block structure and rotated blocks.*

Proof. Each block remains periodic by Theorem 15.8, the SameMPV condition transfers via Theorem 15.9, and block-assembly compatibility follows from Theorem 15.10. $\square$

15.1.1 The Z -gauge degree of freedom

Definition 15.12 (Entrywise periodic Z -gauge ratio). Given complex weights μ, ν , define the entrywise Z -gauge ratio by

$$z(\mu, \nu) := \mu/\nu.$$

Lemma 15.13 (Matched powers imply root-of-unity ratio). *If $\mu^m = \nu^m$ and $\nu \neq 0$, then*

$$z(\mu, \nu)^m = 1.$$

Lemma 15.14 (Ratio rescales denominator back to numerator). *If $\nu \neq 0$, then*

$$z(\mu, \nu) \nu = \mu.$$

Definition 15.15 (Diagonal periodic Z -gauge matrix). For matched weight families $(\mu_i)_i$ and $(\nu_i)_i$ on a finite index type, define the diagonal matrix

$$Z := \text{diag}(z(\mu_i, \nu_i))_i.$$

Lemma 15.16 (Diagonal Z -gauge has finite order under matched powers). If $\mu_i^m = \nu_i^m$ and $\nu_i \neq 0$ for all i , then

$$Z^m = \mathbb{1}.$$

Lemma 15.17 (Diagonal Z -gauge transports diagonal weights). Under $\nu_i \neq 0$ for all i ,

$$Z \text{ diag}(\nu) = \text{diag}(\mu).$$

Definition 15.18 (Z -gauge equivalence). Let A be an m -periodic irreducible tensor of bond dimension D . A Z -gauge transformation of A is a diagonal unitary matrix $Z \in M_D(\mathbb{C})$ satisfying:

1. Z commutes with every Kraus operator: $[A^i, Z] = 0$ for all physical indices i , and
2. $Z^m = \mathbb{1}_D$ (all diagonal entries are m -th roots of unity).

Replacing A by ZA (i.e. $A^i \mapsto ZA^i$) leaves the MPV family invariant for lengths N that are multiples of m , because the factor $\text{tr}(Z^N) = \text{tr}(\mathbb{1}) = D$ for $N \equiv 0 \pmod{m}$. Two tensors A, B are Z -gauge equivalent if there exists an invertible Y and a Z -gauge transformation Z for A such that $ZA^i = YB^iY^{-1}$ for all i .

Remark 15.19 (Z -gauge is trivial for period-1 blocks). When $m = 1$ the condition $Z^1 = \mathbb{1}$ forces $Z = \mathbb{1}$, so Z -gauge equivalence reduces to ordinary gauge equivalence (Definition 2.10). The Z -gauge is therefore a genuine new degree of freedom that appears only for $m > 1$.

15.2 Periodic Fundamental Theorem

The theorems in Chapter 14 all require that each block has a *primitive* transfer map, i.e. peripheral spectrum exactly $\{1\}$. This is achieved by blocking the original tensor by a common period. The paper [DICCSPG17] develops a framework that avoids this blocking step, giving a fundamental theorem that applies directly to periodic tensors.

15.2.1 Common-period blocking lemmas

Definition 15.20 (LCM period for a finite block family). Given a finite family of blocking periods (p_i) indexed by a finite set, their least common multiple serves as the shared blocking period.

Definition 15.21 (Common-period blocked family). Given a finite family of tensors (B_i) with designated periods (p_i) , block each B_i to the shared period $\text{lcm}(p_i)$, so every block lives in the same physical space.

Theorem 15.22 (Common-period blocking preserves transfer-map primitivity). If each tensor B_i has a primitive transfer map after blocking by its own positive period p_i , then blocking the entire family to the common LCM period preserves primitivity of each member's transfer map.

Proof. Each p_i divides $\text{lcm}(p_i)$. Apply the divisibility monotonicity theorem for primitive blocked transfer maps to each family member separately. $\square$

15.2.2 Statement of the periodic Fundamental Theorem

The following is Theorem 3.8 (“equal case”) of [DICCSPG17].

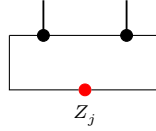
Theorem 15.23 (Periodic Fundamental Theorem). *Let A and B be MPS tensors in irreducible form, with bases of periodic blocks $\{A_j\}_{j \in J}$ and $\{B_k\}_{k \in K}$ respectively. Suppose that $\mathcal{V}(A) = \mathcal{V}(B)$ (equal MPV families for all lengths N).*

Then $|J| = |K|$ and there is a bijection $\pi : J \rightarrow K$ such that A_j and $B_{\pi(j)}$ are m_j -periodic for the same period m_j . Moreover, for each $j \in J$ there exists:

1. *an invertible matrix Y_j of size $D_j \times D_{\pi(j)}$, and*
 2. *a diagonal unitary matrix Z_j of size $D_j \times D_j$ satisfying $Z_j^{m_j} = \mathbb{1}$ and $[A_j^i, Z_j] = 0$ for all i ,*
- such that*

$$Z_j A_j^i = Y_j B_{\pi(j)}^i Y_j^{-1} \quad \text{for all physical indices } i.$$

For each periodic block, the extra freedom is encoded by



rather than by an ordinary aperiodic gauge alone: the operator Z_j sits on the closing virtual bond and satisfies $Z_j^{m_j} = \mathbb{1}$. Equivalently, setting $Y = \bigoplus_j Y_j \otimes \mathbb{1}_{r_j}$ and $Z = \bigoplus_j Z_j \otimes \mathbb{1}_{r_j}$, one has $ZA^i = YB^iY^{-1}$ for all i , with $[A^i, Z] = 0$ and $\mathcal{V}(A) = \mathcal{V}(ZA)$.

Remark. The proof is conditional on the periodic-overlap dichotomy (Proposition 3.3 of [DICCSPG17]), which remains to be proved.

15.2.3 Periodic overlap dichotomy

Definition 15.24 (Cyclic sector decomposition). A family of compressed sector tensors $(C_k)_{k=0}^{m-1}$ on corner bond spaces forms a *cyclic sector decomposition* of the blocked tensor $A^{(m)}$ if there exist orthogonal projections $P_0, \dots, P_{m-1}$ on the ambient bond space such that $\sum_k P_k = \mathbb{1}$, the projections satisfy the cyclic-shift relation $\mathcal{E}_A^\dagger(P_{k+1}) = P_k$ with the index $k+1$ taken cyclically modulo m , each P_k commutes with every blocked letter $P_k A_i^{(m)} = A_i^{(m)} P_k$, and $V^{(N)}(C_k)_\sigma = \text{tr}(P_k (A^{(m)})^\sigma)$ for every word σ , and each block carries a $*$ -algebra isomorphism $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k \cdot M_D(\mathbb{C}) \cdot P_k$ (multiplicative and adjoint-preserving) intertwining the compressed adjoint transfer map with the sector adjoint transfer map on the corner of P_k .

Definition 15.25 (One-site corner transition). For a cyclic projection family $(P_k)_{k \in \text{Fin } m}$ on the ambient bond algebra of an MPS tensor A , the *one-site corner transition* at sector k is the family of per-site matrices

$$\tau_k(A)^i := P_k \cdot A^i \cdot P_{k+1}, \quad i \in \text{Fin } d,$$

where the index $k+1$ is taken cyclically in $\text{Fin } m$.

Definition 15.26 (Staircase evaluation). For a word $w = (w_0, \dots, w_{L-1})$ on the physical index alphabet and a starting sector u , the *staircase evaluation* at u is

$$\text{cornerEvalWord}(A, P, u, w) := P_u \cdot A^{w_0} \cdot P_{u+1} \cdot A^{w_1} \cdot P_{u+2} \cdots A^{w_{L-1}} \cdot P_{u+L},$$

defined recursively by $\text{cornerEvalWord}(A, P, u, \varepsilon) = P_u$ and $\text{cornerEvalWord}(A, P, u, i \cdot w) = P_u \cdot A^i \cdot \text{cornerEvalWord}(A, P, u+1, w)$.

Definition 15.27 (Cyclic staircase product on blocked indices (Eq. A.8)). The *blocked corner transition tensor* at sector u is the MPS tensor on blocked physical indices defined by $i \mapsto \text{cornerEvalWord}(A, P, u, w_i)$, where w_i is the length- m physical word associated to the blocked index i . For a blocked index i with word $(i_0, \dots, i_{m-1})$ this returns the ambient-algebra matrix

$$P_u \cdot A^{i_0} \cdot P_{u+1} \cdot A^{i_1} \cdot P_{u+2} \cdots P_{u+m-1} \cdot A^{i_{m-1}} \cdot P_u.$$

This is the ambient matrix-algebra representative of the compressed blocked sector tensor C_u in a cyclic sector decomposition; the identification with C_u through the compression isometry φ_u is Equation A.8 of [DICCSPG17].

Theorem 15.28 (Self-overlap along period multiples). *If A is m -periodic, then the self-overlap along multiples of m converges to m (the period, viewed as a scalar).*

Theorem 15.29 (Different periods imply asymptotic orthogonality). *If A and B are periodic with different periods, then $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. Split on whether the bond dimensions agree. If not, the claim follows from Theorem 15.35. If $D_1 = D_2$, suppose for contradiction that A and B are gauge-phase equivalent; Perron–Frobenius eigenvalue uniqueness (Wolf Theorem 6.3) then forces equal periods, contradicting the hypothesis. Hence the tensors are not gauge-phase equivalent, and the overlap decays. $\square$

Lemma 15.30 (Sector-block normality for periodic tensors). *For a periodic tensor equipped with a cyclic sector decomposition, each nonzero blocked sector tensor is normal.*

Theorem 15.31 (No sector match implies asymptotic orthogonality). *If two periodic tensors with the same period admit no gauge-phase matching sector pair after blocking, then their overlap tends to zero.*

Lemma 15.32 (Sector-match propagation under translation). *A single gauge-phase match between blocked sectors propagates to all cyclic translates of that sector pair.*

Lemma 15.33 (Per-site proportionality from blocked sector match). *If all aligned blocked sectors match up to gauge and phase, then the original tensors are related by repeated-block proportionality.*

Theorem 15.34 (Sector match yields repeated-block equivalence). *For periodic tensors of the same period, existence of one matching sector pair implies repeated-block equivalence.*

Theorem 15.35 (Different bond dimensions imply asymptotic orthogonality). *If two periodic tensors have different bond dimensions, then $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$.*

Theorem 15.36 (Periodic overlap dichotomy). *Let A and B be periodic tensors. Then at least one of the following alternatives holds:*

1. $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$;
2. A and B are equivalent up to repeated blocks.

Theorem 15.37 (Eventual linear independence of a periodic basis). *For a finite family of pairwise non-equivalent periodic tensors whose periods divide a common integer p , there is N_0 such that for every $N \geq N_0$ the vectors $\{|V^{(pN)}(A_j)\rangle\}_j$ are linearly independent.*

Remark 15.38 (Comparison with the 1606.00608 approach). Theorem 15.23 and the equal-MPV Fundamental Theorem (Theorem 14.5) both characterize the gauge freedom between two tensors with equal MPV families, but they do so at different levels:

1. *Blocking approach* (Theorem 14.5, following [CPGSV17a]): Block both A and B by a common period p so that all blocks of $A^{[p]}$ and $B^{[p]}$ are primitive (1-periodic). Apply the aperiodic fundamental theorem to the blocked tensors. The conclusion is that $A^{[p]}$ and $B^{[p]}$ are gauge equivalent. The Z -gauge freedom does not appear because blocking absorbs the periodicity.
2. *Periodic approach* (Theorem 15.23, following [DICCSPG17]): Work directly with the periodic blocks of A and B . The conclusion includes an extra Z -gauge transformation reflecting the non-trivial phase ambiguity that arises for periodic blocks. The Z matrix satisfies $Z^m = \mathbb{1}$ and commutes with all Kraus operators; it is invisible to the MPV family because it only introduces m -th roots of unity in the trace.

Chapter 14 develops only the blocking approach; the periodic approach is the subject of the present chapter.

Remark 15.39 (Proportional periodic FT). There is also a proportional-case version of the periodic Fundamental Theorem (Theorem 3.4, “proportional case”, in [DICCSPG17]): if $\mathcal{V}(A) = \lambda_N \mathcal{V}(B)$ for scalars λ_N with $\lambda_N \rightarrow \lambda \neq 0$, then the bases of periodic tensors of A and B are equivalent (up to the same gauge and Z freedom as above). This extends Theorems 14.2 and 14.4 to the periodic setting.

15.3 Physical symmetries on periodic MPS

Theorem 15.40 (Cor. 4.1 — Physical symmetry yields a virtual Z -gauge). *Let A be an MPS tensor in irreducible form II and let $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$ be a representation of a group G on the physical leg, acting unitarily ($U(g)U(g)^\dagger = \mathbb{1}$). Assume moreover the periodic equal-case Fundamental Theorem of MPS (Theorem 15.23) as an explicit hypothesis on the pair (d, D) — formalised in Lean as the proposition dD . If A is on-site symmetric under U — that is, the twisted tensor $\tilde{A}_g$ has the same MPV family as A for every $g \in G$ — then for each $g \in G$ there exists a positive period m_g and a $\mathbb{Z}_{m_g}$ -gauge equivalence between A and $\tilde{A}_g$.*

Concretely there are matrices Z_g and Y_g , with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$, satisfying

$$Z_g A^i = Y_g \tilde{A}_g^i Y_g^{-1} \quad \text{for every physical index } i.$$

Proof. The twisted tensor $\tilde{A}_g$ coincides with the physical-index rotation $A_{U(g)}$ of Definition 15.4. By Theorem 15.11 the rotated tensor remains in irreducible form II, and the on-site symmetry hypothesis supplies the equality $\mathcal{V}(A) = \mathcal{V}(A_{U(g)})$. Apply the periodic equal-case Fundamental Theorem (Theorem 15.23) to obtain the asserted $\mathbb{Z}_{m_g}$ -gauge equivalence. $\square$

Definition 15.41 (Periodic projective-rigidity hypothesis). Let $A : \text{MPSTensor } d D$ be on-site symmetric under a group action $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$. The tensor A satisfies the *periodic projective-rigidity hypothesis* if one may choose a family of virtual gauges $Y : G \rightarrow \mathrm{GL}_D(\mathbb{C})$ and a scalar 2-cochain $u : G \times G \rightarrow \mathbb{C}^\times$ such that

1. every Y_g realises the Corollary 4.1 intertwining with the twisted tensor $\tilde{A}_g$ for some accompanying Z_g (with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$); and
2. the family obeys the projective multiplication law on the bond space: $Y_h Y_g = u(g, h) Y_{gh}$.

The injective case reduces the rigidity to scalar uniqueness of gauges (cf. §18.2); in the genuinely periodic setting the bond-space commutant of A is generated by the Z_g matrices, so coherence of the factor system u is an independent analytic input.

Theorem 15.42 (Cor. 4.1 projective-representation upgrade). *Under the hypotheses of Corollary 15.40 together with the periodic projective-rigidity hypothesis (Definition 15.41), the Y_g family assembles into a projective representation of G on the bond space: there exist a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$ and virtual gauges $X : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying*

$$X(g)X(h) = \omega(g, h)X(gh),$$

while for each g the matrix $X(g^{-1})$ still realises the Corollary 4.1 $\mathbb{Z}_{m_g}$ -intertwining between A and the twisted tensor $\tilde{A}_g$.

Proof. Unpack the rigidity hypothesis to extract the family (Y, u) together with the Corollary-4.1 intertwiners. Set $X(g) := Y(g^{-1})$ and $\omega(g, h) := u(h^{-1}, g^{-1})$. The projective multiplication law $X(g)X(h) = \omega(g, h)X(gh)$ is then exactly the (h^{-1}, g^{-1}) -instance of the factor-system identity, using $(gh)^{-1} = h^{-1}g^{-1}$. The intertwining relation between A and $\tilde{A}_g$ transports unchanged. $\square$

Theorem 15.43 (Cocycle identity for the upgraded factor system). *If the bond dimension D is positive, the factor system ω produced by Theorem 15.42 satisfies the multiplicative 2-cocycle identity $\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k)$.*

Proof. Associativity of matrix multiplication applied to $X(g)X(h)X(k)$, combined with the projective multiplication law, gives

$$\omega(g, h)\omega(gh, k)X((gh)k) = \omega(g, hk)\omega(h, k)X(g(hk)).$$

Since $(gh)k = g(hk)$, both sides are scalar multiples of the same matrix $X((gh)k)$. The matrix lies in $\text{GL}_D(\mathbb{C})$ and is therefore invertible, but the scalar cancellation leading from an equality of scalar multiples of a matrix to equality of the scalar coefficients additionally requires $D > 0$ so that the underlying index set is inhabited. Cancelling then yields $\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k)$ in $\mathbb{C}^\times$, hence in $\text{Units}(\mathbb{C})$. $\square$

Remark 15.44 (SPT classification). The cohomology class of ω in $H^2(G, U(1))$ classifies the SPT phase of the symmetric MPS (cf. Chen–Gu–Wen and Pérez-García et al.). This classification statement is not formalised at present; only the underlying 2-cocycle identity of Theorem 15.43 is available in the Lean development.

Theorem 15.45 (Bundled projective representation from symmetry). *Theorems 15.42 and 15.43 are packaged into a single high-level statement: from the periodic projective-rigidity hypothesis (Definition 15.41), one obtains a projective representation of G on the bond space whose factor system is a genuine 2-cocycle whenever $D > 0$.*

15.4 Refinement and divisibility

The next theorem characterises when the matrix-product-vector family of a periodic tensor admits a finer description in terms of a smaller-period tensor whose blocks have been merged. This was the original motivation for the irreducible-form framework in [DICCSPG17].

Definition 15.46 (p -divisible CP map). A linear endomorphism $\mathcal{E}$ of $M_D(\mathbb{C})$ is p -divisible if there exists a CPTP map $\mathcal{E}'$ of $M_D(\mathbb{C})$ such that $\mathcal{E} = (\mathcal{E}')^p$.

Definition 15.47 (*p-refinable MPS tensor*). An MPS tensor $B : \text{MPSTensor } d \ D$ is *p-refinable* if there exist another tensor $A : \text{MPSTensor } d \ D$ and an isometry $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$ (encoded as a matrix $W \in M_{d^p \times d}(\mathbb{C})$ with $W^\dagger W = \mathbb{1}$) such that the p -blocked tensor $A^{[p]}$ matches the W -image of B at the level of MPV coefficients:

$$\text{coeff}(A^{[p]})(\text{ofFn } \tau) = \sum_{\sigma : \text{Fin } N \rightarrow \text{Fin } d} \left(\prod_k W_{\tau(k), \sigma(k)} \right) \text{coeff } B(\text{ofFn } \sigma)$$

for every $N \in \mathbb{N}$ and every $\tau : \text{Fin } N \rightarrow \text{Fin } d^p$.

In paper notation, this encodes $|V_{pN}(A)\rangle = W^{\otimes N} |V_N(B)\rangle$ for all N .

Theorem 15.48 (Pullback of a refinement witness). *Let B be an MPS tensor in irreducible form II. If B is p-refinable, then there exist a tensor A , an isometry $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$, and the pullback tensor*

$$C^\alpha := \sum_{i=0}^{d-1} W_{\alpha, i} B^i \quad (\alpha \in \{0, \dots, d^p - 1\})$$

such that C is again in irreducible form II, $\mathcal{E}_C = \mathcal{E}_B$, and C has the same MPV family as $A^{[p]}$.

Proof. Choose (A, W) from the refinement data. The pullback construction gives $\mathcal{E}_C = \mathcal{E}_B$ and equality of MPV families between C and $A^{[p]}$. To see that C is still in irreducible form II, transport each periodic block B_j of B through the same physical isometry W . The left-canonical normalization is preserved because $W^\dagger W = \mathbb{1}$, while irreducibility and the peripheral spectrum are preserved because the transfer map of each transported block agrees with that of B_j . Reassembling these transported blocks with the same weights gives an irreducible-form decomposition of C . $\square$

Theorem 15.49 (Thm. 4.1 — *p-refinability* $\iff$ *p-divisibility*). *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume both canonicalization hypotheses used in the one-way statements: one for refinability $\Rightarrow$ divisibility and one for divisibility $\Rightarrow$ refinability. Then B is p-refinable iff its transfer map $\mathcal{E}_B$ is p-divisible.*

Proof. Bundle the forward implication (Theorem 15.50) with the reverse implication (Theorem 15.51) into a single biconditional. Both directions are formalised conditionally on their respective canonicalization hypotheses, whose combination yields the full equivalence. $\square$

Theorem 15.50 (Thm. 4.1 forward direction, conditional form). *Let B be an MPS tensor in irreducible form II. Assume a canonicalization hypothesis providing the remaining analytic assumptions (left-canonical witness with matching transfer map). Then p-refinability of B implies p-divisibility of $\mathcal{E}_B$.*

Proof. The canonicalization hypothesis supplies a witness A with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (left-canonical) and $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. The left-canonical property makes $\mathcal{E}_A$ a CPTP channel, and the blocking identity $\mathcal{E}_{A^{[p]}} = (\mathcal{E}_A)^p$ (formalised as) then identifies $\mathcal{E}_B$ with the p -th power of the channel $\mathcal{E}_A$. $\square$

Theorem 15.51 (Thm. 4.1 reverse direction, conditional form). *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume an inverse canonicalization hypothesis providing a witness $A : \text{MPSTensor } d \ D$ with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$ whenever $\mathcal{E}_B$ is p-divisible. Then p-divisibility of $\mathcal{E}_B$ implies p-refinability of B .*

Proof. From the inverse canonicalization hypothesis, extract A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. This matches two finite Kraus families of the same completely positive map: the blocked family $A^{[p]}$ with d^p operators and the original family B with d operators. The cardinality comparison $d \leq d^p$ holds whenever $p \geq 1$, so Wolf Theorem 2.18 (isometric freedom of Kraus representations, formalised as) supplies an isometry $V \in M_{d^p \times d}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $(A^{[p]})^\alpha = \sum_j V_{\alpha,j} B^j$. Expanding $\text{coeff}(A^{[p]})(\text{offn } \tau)$ by linearity of matrix multiplication and of the trace, the contributions reorganise into the V -weighted sum over length- N words of B required by Definition 15.47, exhibiting V as the isometry W in the statement of p -refinability. $\square$

Remark 15.52 (Continuum-limit application). The original motivation for Theorem 15.49 is a continuum-limit construction in the spirit of [DCCSPG17]: refinability along arbitrarily large p is equivalent to infinite divisibility of the transfer map, which selects out continuum limits of translationally invariant MPS. The further consequences are explored elsewhere.

Chapter 16

Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Ch. 7].

Section 16.1 establishes definitions and the basic exponential form (Proposition 16.14). Section 16.2 gives the Duhamel formula and perturbation bound. Section 16.3 develops the generator theory and proves the GKSL theorem.

16.1 Dynamical semigroups and the exponential form

Definition 16.1 (Dynamical semigroup). A family of linear maps $T : \mathbb{R} \rightarrow (M_D(\mathbb{C}) \rightarrow_\ell M_D(\mathbb{C}))$ is a *dynamical semigroup* if

1. (semigroup law) $T_{t+s} = T_t \circ T_s$ for all $t, s \geq 0$; and
2. (initial condition) $T_0 = \text{id}$.

This is [Wol12, Eq. (7.1)].

Definition 16.2 (Continuous dynamical semigroup). A dynamical semigroup T is *norm-continuous* if the map $t \mapsto T_t$ is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

Definition 16.3 (Exponential semigroup). Given a generator $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$, the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

Theorem 16.4 (Semigroup law for the exponential). *For all $t, s \in \mathbb{R}$, $e^{(t+s)L} = e^{tL} \cdot e^{sL}$.*

Proof. Since $(t \cdot L)$ commutes with $(s \cdot L)$, this follows from the exponential addition formula for commuting elements in a Banach algebra. $\square$

Theorem 16.5 (Exponential semigroup at time 0). *The continuous-linear-map exponential satisfies $e^{0L} = \mathbb{1}$.*

Proof. The exponential of the zero endomorphism is the identity. $\square$

Theorem 16.6 (Continuity of the exponential semigroup). *The map $t \mapsto e^{tL}$ is continuous in the operator norm topology.*

Proof. The exponential is analytic (convergence radius = ∞), hence continuous. $\square$

Theorem 16.7 (Derivative of the exponential semigroup). *For all $t \in \mathbb{R}$,*

$$\frac{d}{dt}e^{tL} = e^{tL} \cdot L.$$

This is [Wol12, Eq. (7.2)].

Proof. Chain rule applied to the composition $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \text{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \text{End}(M_D(\mathbb{C}))$. $\square$

Theorem 16.8 (Derivative at the origin). *One has*

$$\left. \frac{d}{dt}e^{tL} \right|_{t=0} = L.$$

Proof. This is Theorem 16.7 at $t = 0$ together with Theorem 16.5. $\square$

Theorem 16.9 (Generator uniqueness). *If $e^{tL} = e^{tL'}$ for all $t \geq 0$, then $L = L'$.*

Proof. Both semigroups agree on $[0, \infty)$, so their derivatives within $[0, \infty)$ at $t = 0$ agree. Since $[0, \infty)$ has unique differentials at 0, $L = L'$. $\square$

Theorem 16.10 (CLM and linear-map forms coincide). *The linear-map exponential semigroup and the continuous-linear-map exponential semigroup agree under the canonical identification of endomorphisms with continuous linear endomorphisms.*

Proof. This is a direct unpacking of the definition of the linear-map exponential semigroup. $\square$

Theorem 16.11 (Pointwise derivative formula). *For every $X \in M_D(\mathbb{C})$ and every $t \in \mathbb{R}$,*

$$\frac{d}{dt}(e^{tL}(X)) = e^{tL}(L(X)).$$

Proof. Evaluate the derivative of the continuous-linear-map semigroup at X and transport the statement through Theorem 16.10. $\square$

Theorem 16.12 (Linear-map exponential semigroup at time 0). *The linear-map exponential semigroup satisfies $e^{0L} = \mathbb{1}$.*

Proof. Transport Theorem 16.5 through Theorem 16.10. $\square$

Theorem 16.13 (Exponential semigroup as a dynamical semigroup). *The family $t \mapsto e^{tL}$ satisfies the semigroup law and the initial condition, hence is a dynamical semigroup.*

Proof. Transport the continuous-linear-map semigroup law from Theorem 16.4 through Theorem 16.10; the initial condition is Theorem 16.12. $\square$

Theorem 16.14 (Prop. 7.1: Every continuous semigroup is exponential). *Every norm-continuous dynamical semigroup T on $M_D(\mathbb{C})$ is of the form*

$$T_t = e^{tL}$$

for some generator $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ and every $t \geq 0$. This is [Wol12, Prop. 7.1].

Proof. Since $T_0 = \mathbb{1}$ and T is continuous, the integral $M_\varepsilon = \int_0^\varepsilon T_s ds$ is invertible for small $\varepsilon > 0$. Using the semigroup property, $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$ is differentiable for $t \geq 0$. ODE uniqueness then gives $T_t = e^{tL}$ for all $t \geq 0$. $\square$

16.2 Perturbation theory

Theorem 16.15 (Lem. 7.1: Duhamel formula). *Let $\Delta = L' - L$. For $t \geq 0$,*

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta e^{sL'} ds.$$

This is [Wol12, Lem. 7.1].

Proof. Define $f(s) = e^{(t-s)L} e^{sL'}$. Then $f'(s) = e^{(t-s)L} \Delta e^{sL'}$ (Theorem 16.7), and $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s) ds$. $\square$

Theorem 16.16 (Cor. 7.1: Perturbation bound). *For $t \geq 0$,*

$$\|e^{tL'} - e^{tL}\| \leq t \|\Delta\| \cdot \sup_{s \in [0, t]} \|e^{sL}\| \cdot \sup_{s \in [0, t]} \|e^{sL'}\|,$$

where $\Delta = L' - L$. This is [Wol12, Cor. 7.1].

Proof. Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 16.15). $\square$

Corollary 16.17 (Perturbation bound under unit-norm control). *For $t \geq 0$, if $\|e^{sL}\| \leq 1$ and $\|e^{sL'}\| \leq 1$ for every $s \in [0, t]$, then*

$$\|e^{tL'} - e^{tL}\| \leq t \|L' - L\|.$$

Proof. Apply Theorem 16.16 and bound both suprema by 1. $\square$

Lemma 16.18 (Dyson-series summability from factorial majorant). *Assume a factorial majorant of Dyson–Phillips iterates:*

$$\|D_n(t)\| \leq M \frac{(t \|L' - L\| M)^n}{n!}.$$

Then the Dyson series is summable in operator norm.

Proof. This is the Weierstrass M-test step, using the scalar convergence of $\sum_n a^n/n!$ and lifting via norm-bounded summability. $\square$

Theorem 16.19 (Factorial norm bound for Dyson iterates). *For $s \in [0, t]$ and $M = \sup_{u \in [0, t]} \|e^{uL}\|$,*

$$\|\tilde{T}_s^{(n)}\| \leq M \frac{(s \|\Delta\| M)^n}{n!}.$$

Proof. Induction on n . The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by $M \|\Delta\| \cdot M(u \|\Delta\| M)^n/n!$ and integrates $\int_0^s u^n du = s^{n+1}/(n+1)$, recovering the factorial. $\square$

Corollary 16.20 (Dyson series summability). *The Dyson–Phillips series $\sum_n \tilde{T}_t^{(n)}$ converges in operator norm for every $t \geq 0$.*

Proof. Compose the factorial norm bound (Theorem 16.19) with the Weierstrass M-test step (Lemma 16.18). $\square$

Corollary 16.21 (Factorial norm bound at the final time). *For $t \geq 0$, setting $s = t$ in Theorem 16.19 gives*

$$\|\tilde{T}_t^{(n)}\| \leq M \frac{(t \|\Delta\| M)^n}{n!},$$

where $M = \sup_{u \in [0, t]} \|e^{uL}\|$.

Proof. This is the special case $s = t$ of Theorem 16.19. $\square$

Theorem 16.22 (Continuity of Dyson iterates). *Each Dyson iterate $t \mapsto \tilde{T}_t^{(n)}$ is continuous in the time parameter.*

Proof. By induction on n . The base case is continuity of the matrix exponential. For the inductive step, the parametric integral $t \mapsto \int_0^t e^{(t-s)L} \Delta \tilde{T}_s^{(n)} ds$ is continuous by the continuous-parameter primitive theorem; the pasting lemma extends this to all of $\mathbb{R}$ since $\tilde{T}_t^{(n+1)} = 0$ for $t \leq 0$. $\square$

Theorem 16.23 (Factorial bound on the Dyson remainder). *For $s \in [0, t]$, $M = \sup_{u \in [0, t]} \|e^{uL}\|$, and $M' = \sup_{u \in [0, t]} \|e^{uL'}\|$:*

$$\|T'_s - \sum_{n < N} \tilde{T}_s^{(n)}\| \leq M' \frac{(s \|\Delta\| M)^N}{N!}.$$

Proof. Induction on N . The base case is the semigroup supremum bound. The inductive step uses the telescoping remainder identity $R_{N+1}(s) = \int_0^s e^{(s-u)L} \Delta R_N(u) du$ (derived from the Duhamel formula), bounds the integrand pointwise, and integrates to recover the factorial. $\square$

Theorem 16.24 (Dyson series identity [Wol12, Eq. (7.13)]). *For $t \geq 0$, the Dyson–Phillips series equals the perturbed semigroup:*

$$\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}.$$

Proof. The partial-sum remainder $\|e^{tL'} - \sum_{n < N} \tilde{T}_t^{(n)}\| \leq M' (t \|\Delta\| M)^N / N!$ tends to zero as $N \rightarrow \infty$, since $c^N / N! \rightarrow 0$ for any constant c . Together with the summability of the series (Corollary 16.20), this identifies the sum via uniqueness of limits. $\square$

Corollary 16.25 (Tsum form of the Dyson series). $\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}$ (*tsum form*).

Proof. Immediate from Theorem 16.24. $\square$

16.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 16.54), which shows that such generators have the standard Lindblad form.

16.3.1 Generator decomposition and conditional complete positivity

Definition 16.26 (Generator decomposition). A *generator decomposition* consists of a completely positive map $\phi : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ and a matrix $\kappa \in M_d(\mathbb{C})$, defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Eq. (7.14)].

Definition 16.27 (Conditional complete positivity). A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map ϕ and a matrix κ such that $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. This is condition 1 of [Wol12, Prop. 7.2].

Definition 16.28 (Trace-annihilating map). A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is *trace-annihilating* if $\text{tr}(L(\rho)) = 0$ for all $\rho \in M_d(\mathbb{C})$. This is the infinitesimal version of trace preservation.

Definition 16.29 (Trace constraint). A generator decomposition (ϕ, κ) satisfies the *trace constraint* if $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, where ϕ^* is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Eq. (7.20)].

Theorem 16.30 (Trace constraint implies trace-annihilating). *If (ϕ, κ) satisfies $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ is trace-annihilating.*

Proof. Let $\phi(\rho) = \sum_i K_i \rho K_i^\dagger$. By the cyclic property of trace, $\text{tr}(L(\rho)) = \text{tr}((\sum_i K_i^\dagger K_i)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = \text{tr}((\kappa + \kappa^\dagger)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = 0$. $\square$

16.3.2 The Lindblad form

Definition 16.31 (Lindblad form). A *Lindblad form* consists of a Hermitian matrix $H = H^\dagger$ (the Hamiltonian) and a family of matrices $\{L_j\}_{j=0}^{r-1}$ (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\}_+ \right),$$

where $[A, B] = AB - BA$ and $\{A, B\}_+ = AB + BA$. This is [Wol12, Eq. (7.21)].

Theorem 16.32 (Lindblad form is trace-annihilating). *Every Lindblad form defines a trace-annihilating linear map.*

Proof. The Hamiltonian part satisfies $\text{tr}(i[\rho, H]) = 0$ by the cyclic property. Each dissipator term satisfies $\text{tr}(L_j \rho L_j^\dagger) = \text{tr}(L_j^\dagger L_j \rho) = \text{tr}(\rho L_j^\dagger L_j)$, so the three terms sum to zero. $\square$

Theorem 16.33 (Lindblad form equals generator decomposition). *A Lindblad form with Hamiltonian H and operators $\{L_j\}$ defines the same linear map as the generator decomposition (ϕ, κ) with $\phi(\rho) = \sum_j L_j \rho L_j^\dagger$ and $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$. This is [Wol12, Eq. (7.24)].*

Proof. Compute $\kappa^\dagger = -iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ using $H^\dagger = H$ and $(A^\dagger A)^\dagger = A^\dagger A$. Setting $S = \sum_j L_j^\dagger L_j$, expand $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$, which is $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j \rho - \frac{1}{2}\rho L_j^\dagger L_j)$. $\square$

Theorem 16.34 (Lindblad form is CCP). *Every Lindblad form defines a conditionally completely positive map.*

Proof. By Theorem 16.33, the Lindblad form equals a generator decomposition with ϕ CP. $\square$

Definition 16.35 (Commutator form of Lindblad equation). The *commutator form* of the Lindblad equation writes the generator as

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_j ([L_j, \rho L_j^\dagger] + [L_j \rho, L_j^\dagger]),$$

where $[A, B] = AB - BA$. This is [Wol12, Eq. (7.22)].

Theorem 16.36 (Commutator form equals standard Lindblad form). *The commutator form (Definition 16.35) defines the same linear map as the standard Lindblad form (Definition 16.31).*

Proof. Expanding the commutator brackets $[L_j, \rho L_j^\dagger]$ and $[L_j \rho, L_j^\dagger]$ yields the standard dissipator terms. $\square$

16.3.3 Characterization of conditional complete positivity

Theorem 16.37 (Prop. 7.2, direction 1 $\rightarrow$ 2). *If L is CCP and $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then*

$$P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

This is [Wol12, Prop. 7.2].

Proof. Write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. The left- and right-multiplication terms are annihilated by P , so the projected Choi matrix reduces to the projected Choi matrix of ϕ , which is positive. $\square$

Theorem 16.38 (Prop. 7.2, direction 2 $\rightarrow$ 1). *If L is Hermiticity-preserving and $P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0$ where $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then L is CCP. This is [Wol12, Prop. 7.2].*

Proof. Decompose the Hermitian Choi matrix as $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$ with $Q \geq 0$ supported on $|\Omega\rangle^\perp$. The Choi–Jamiołkowski isomorphism applied to Q gives the CP map ϕ , and $(\kappa \otimes \mathbb{1})|\Omega\rangle = |\psi\rangle$ defines κ . $\square$

16.3.4 CP semigroups and CCP generators

Theorem 16.39 (Prop. 7.3: CP semigroup implies CCP generator). *If e^{tL} is CP for all $t \geq 0$, then L is CCP. This is [Wol12, Prop. 7.3].*

Proof. From $(e^{tL} \otimes \text{id})(|\Omega\rangle\langle\Omega|) \geq 0$, expand at infinitesimal t and project onto P to get $P(L \otimes \text{id})(|\Omega\rangle\langle\Omega|)P \geq 0$. Then Theorem 16.38 gives CCP. $\square$

Theorem 16.40 (Prop. 7.3: CCP generator implies CP semigroup). *If L is CCP, then e^{tL} is CP for all $t \geq 0$. This is [Wol12, Prop. 7.3].*

Proof. Approximate e^{tL} by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \quad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to e^{tL} , and the cone of CP maps is closed. $\square$

Theorem 16.41 (Prop. 7.3: CP semigroup iff CCP generator). *e^{tL} is CP for all $t \geq 0$ if and only if L is CCP.*

Proof. Combine Theorems 16.39 and 16.40. $\square$

16.3.5 Freedom in generator representation

Theorem 16.42 (Traceless Kraus operators exist). *Assume $d > 0$. Given any Kraus operators $\{K_j\}$, there exist $c_j \in \mathbb{C}$ such that $K'_j = K_j + c_j \mathbb{1}$ is traceless. This is part of [Wol12, Prop. 7.4].*

Proof. Set $c_j = -\text{tr}(K_j)/d$. □

Theorem 16.43 (Shift invariance of generators). *If $K'_j = K_j + c_j \mathbb{1}$ and κ' is adjusted per [Wol12, Eq. (7.19)], then (ϕ', κ') and (ϕ, κ) define the same generator.*

Proof. Direct algebraic expansion. □

16.3.6 Uniqueness of the traceless Lindblad form

Definition 16.44 (Traceless Kraus operators). A Lindblad form has *traceless Kraus operators* if $\text{tr}(L_j) = 0$ for every j .

Theorem 16.45 (Prop. 7.4, item 2: CP-part uniqueness). *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their CP parts agree: $\phi = \phi'$. This is part of [Wol12, Prop. 7.4, item 2].*

Proof. Compare projected Choi matrices and use Choi injectivity. □

Theorem 16.46 (Prop. 7.4, item 2: drift uniqueness modulo phase). *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar: $\kappa' = \kappa + i\lambda \mathbb{1}$ for some $\lambda \in \mathbb{R}$. This is part of [Wol12, Prop. 7.4, item 2].*

Proof. After identifying $\phi = \phi'$ (Theorem 16.45), the generator equality forces $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$ for all ρ . Setting $\rho = \mathbb{1}$ gives $\Delta\kappa + (\Delta\kappa)^\dagger = 0$ (skew-Hermiticity), which converts the equation to $[\Delta\kappa, \rho] = 0$ for all ρ . The scalar commutant lemma gives $\Delta\kappa = c \cdot \mathbb{1}$, and skew-Hermiticity forces $c = i\lambda$ for some $\lambda \in \mathbb{R}$. □

Theorem 16.47 (Prop. 7.4, item 2: combined uniqueness). *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then $\phi = \phi'$ and $\kappa' = \kappa + i\lambda \mathbb{1}$ for some $\lambda \in \mathbb{R}$.*

Proof. Combine Theorems 16.45 and 16.46. □

16.3.7 Trace-annihilation and trace preservation

Theorem 16.48 (Trace-annihilating generators yield trace-preserving semigroups). *If L is trace-annihilating, then e^{tL} is trace-preserving for every $t \in \mathbb{R}$.*

Proof. For fixed ρ , the function $f(t) = \text{tr}(e^{tL}(\rho))$ has derivative $f'(t) = \text{tr}(L(e^{tL}(\rho))) = 0$, so f is constant and $\text{tr}(e^{tL}(\rho)) = f(t) = f(0) = \text{tr}(\rho)$. □

Theorem 16.49 (Trace-preserving semigroups have trace-annihilating generators). *If e^{tL} is trace-preserving for every $t \geq 0$, then L is trace-annihilating.*

Proof. Differentiate the identity $\text{tr}(e^{tL}(\rho)) = \text{tr}(\rho)$ at $t = 0$ from the right. □

16.3.8 GKSL/Lindblad theorem

Definition 16.50 (GKSL generator). A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is a *GKSL generator* if e^{tL} is a quantum channel (CPTP) for all $t \geq 0$.

Theorem 16.51 (Thm. 7.1, Form (i)). *If $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with ϕ CP and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then L is a GKSL generator. This is [Wol12, Thm. 7.1, Eq. (7.20)].*

Proof. CP follows from Theorem 16.40. The trace constraint implies trace-annihilation by Theorem 16.30, so Theorem 16.48 gives trace preservation. $\square$

Theorem 16.52 (Thm. 7.1, Form (i) conversely). *If L is a GKSL generator, then there exist a CP map ϕ and a matrix κ such that*

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

and $\phi^(\mathbb{1}) = \kappa + \kappa^\dagger$.*

Proof. Apply Theorem 16.39 to obtain a CCP decomposition. Then apply Theorem 16.49 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$. $\square$

Theorem 16.53 (Thm. 7.1: GKSL iff CCP + trace-annihilating). *L is a GKSL generator if and only if L is CCP and trace-annihilating.*

Proof. Forward: apply Theorem 16.39 to the CP part and Theorem 16.49 to the TP part. Reverse: combine Theorem 16.40 with Theorem 16.48. $\square$

Theorem 16.54 (Thm. 7.1: GKSL iff Lindblad form). *L is a GKSL generator if and only if*

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho \}_+ \right)$$

with $H = H^\dagger$. This is [Wol12, Thm. 7.1].

Proof. Forward: apply Theorem 16.52 to write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with trace constraint. Then choose traceless Kraus operators by Theorem 16.42 and rewrite κ as $iH + \frac{1}{2}\phi^*(\mathbb{1})$; Theorem 16.33 gives the Lindblad formula. Reverse: Theorems 16.34 and 16.32 show that every Lindblad form satisfies the right-hand side of Theorem 16.53. $\square$

16.3.9 Kossakowski matrix form

Definition 16.55 (Kossakowski form). A *Kossakowski form* consists of $H = H^\dagger$, a finite family of matrices $\{F_k\}_{k=0}^{n-1}$, and a positive semidefinite matrix $C \in M_n(\mathbb{C})$, defining

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_{k,l} C_{lk} ([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]).$$

In Wolf's formula one takes $\{F_k\}$ to be a basis of traceless matrices; here we keep only the algebraic data needed for the conversion to Lindblad form.

Theorem 16.56 (Kossakowski form iff Lindblad form). *A linear map admits a Kossakowski form iff it admits a Lindblad form.*

Proof. Forward: diagonalize $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Reverse: keep the same Hamiltonian and family $F_k = L_k$, and take $C = \mathbb{1}$. $\square$

16.4 Primitivity and irreducibility of QDS

16.4.1 Auxiliary spectral and semigroup lemmas

Definition 16.57 (Quantum dynamical semigroup). A quantum dynamical semigroup is a norm-continuous dynamical semigroup whose time slices are quantum channels for every $t \geq 0$.

Theorem 16.58 (Semigroup power identity). *For $t \geq 0$ and $n \in \mathbb{N}$, one has $T_{nt} = (T_t)^n$.*

Theorem 16.59 (Eigenvector propagation under powers). *If $E(v) = \mu v$, then $E^n(v) = \mu^n v$ for all $n \in \mathbb{N}$.*

Lemma 16.60 (Density matrices are nonzero). *Every density matrix is nonzero.*

Theorem 16.61 (Compression invariance is stable under powers). *If a corner compression is invariant under E , it remains invariant under every power E^n .*

Theorem 16.62 (Generator eigenvectors along the exponential semigroup). *If $L(X) = \mu X$, then $e^{tL}(X) = e^{t\mu} X$ for all $t \in \mathbb{R}$.*

Theorem 16.63 (Spectral mapping for semigroup generators). *If μ is in the spectrum of L , then $e^{t\mu}$ is in the spectrum of e^{tL} .*

Theorem 16.64 (Root-of-unity rigidity for phases). *If $e^{it\theta}$ is a root of unity for every $t > 0$, then $\theta = 0$.*

Theorem 16.65 (Peripheral generator eigenvalues are purely imaginary). *Peripheral spectral points of the generator have vanishing real part.*

Theorem 16.66 (Peripheral cardinality bound). *The number of peripheral eigenvalues is bounded by $\dim(M_D(\mathbb{C}))$.*

Theorem 16.67 (Bounded order under peripheral power-closure). *If all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding $\dim(M_D(\mathbb{C}))$ equals 1.*

Theorem 16.68 (Peripheral powers stay peripheral under irreducibility). *For an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.*

Theorem 16.69 (Continuous-linear-map power evaluation). *Evaluating powers of a linear map agrees with powers in its continuous linear realization.*

Theorem 16.70 (Spectral-radius criterion from eigenvalue bounds). *If all eigenvalues satisfy $|\lambda| < 1$, then the spectral radius is < 1 .*

Theorem 16.71 (Primitive powers annihilate trace-zero matrices). *If E is an irreducible primitive channel with faithful fixed density σ , then $E^n(X) \rightarrow 0$ for every matrix X with $\text{tr}(X) = 0$.*

Proof. Decompose E^n into the fixed-point projection and its complementary part. On the complement, every eigenvalue has modulus < 1 , so the complementary powers tend to 0; the trace-zero hypothesis removes the fixed-point component. $\square$

Theorem 16.72 (Fixed points of an irreducible channel are trace multiples). *If E is an irreducible channel with faithful fixed density σ , then every nonzero fixed point V of E satisfies $V = \text{tr}(V)\sigma$.*

Proof. Subtract the trace multiple $\text{tr}(V)\sigma$ and apply the vanishing theorem for trace-zero fixed points of an irreducible channel. $\square$

Corollary 16.73 (Nonzero fixed points have nonzero trace). *Every nonzero fixed point of an irreducible channel has nonzero trace.*

Proof. Otherwise the preceding theorem would force the fixed point to vanish. $\square$

Theorem 16.74 (Peripheral eigenvectors become periodic fixed points). *For every positive time t , a quantum dynamical semigroup irreducible at every positive time has the following property: every peripheral eigenvalue μ of T_t admits a nonzero eigenvector V and a positive integer p such that $T_t(V) = \mu V$ and $T_{pt}(V) = V$.*

Proof. Peripheral powers remain peripheral for an irreducible channel with faithful fixed point. The bounded-order lemma then gives $\mu^p = 1$, and the semigroup power identity turns the eigenvector equation for T_t into a fixed-point equation for T_{pt} . $\square$

Theorem 16.75 (Peripheral eigenvectors with nonzero trace). *For every positive time t , under the same hypotheses every peripheral eigenvalue of T_t has a nonzero eigenvector with nonzero trace.*

Proof. Apply Theorem 16.74 and then use the previous corollary at the irreducible time pt . $\square$

Theorem 16.76 (Trace-preserving eigenvectors force eigenvalue 1). *If E is trace-preserving, $E(V) = \mu V$, and $\text{tr}(V) \neq 0$, then $\mu = 1$.*

Proof. Take traces in $E(V) = \mu V$ and use trace preservation. $\square$

Theorem 16.77 (Peripheral collapse under irreducibility at all times). *If a quantum dynamical semigroup is irreducible at every positive time, then every peripheral eigenvalue of every positive-time slice equals 1.*

Proof. Combine Theorem 16.75 with Theorem 16.76. $\square$

Theorem 16.78 (Primitivity from irreducibility at all times). *If a quantum dynamical semigroup is irreducible at every positive time, then every positive-time slice is primitive.*

Proof. The unique faithful fixed density gives the one-dimensional peripheral spectrum criterion, and Theorem 16.77 collapses the peripheral set to $\{1\}$. $\square$

Remark 16.79 (Semigroup irreducible/primitive equivalence). The semigroup analogue of the irreducible/primitive equivalence starts from one irreducible time slice, constructs a primitive smaller time step, and lifts the conclusion to the full semigroup (Theorems 16.95 and 16.96).

Remark 16.80 (Reduction to a smaller time step in Proposition 7.5). Starting from one irreducible time t_0 , we construct a positive time step $u = t_0/N$ with $T_{t_0} = (T_u)^N$, prove peripheral eigenvalue collapse at time u , and deduce primitivity of T_u .

Theorem 16.81 (The stationary density is fixed for all times). *If $t_0 > 0$, T_{t_0} is irreducible, and σ is its unique fixed density matrix, then $T_u(\sigma) = \sigma$ for every $u \geq 0$.*

Proof. By semigroup commutativity, $T_{t_0}(T_u(\sigma)) = T_u(T_{t_0}(\sigma)) = T_u(\sigma)$, and uniqueness of the fixed density at time t_0 identifies $T_u(\sigma)$ with σ . $\square$

Theorem 16.82 (Fixed points at the irreducible time are trace multiples). *If T_{t_0} is irreducible with faithful fixed density σ , then every fixed point V of T_{t_0} satisfies $V = \text{tr}(V)\sigma$.*

Proof. This is the fixed-point trace-scalar theorem for the irreducible channel T_{t_0} . $\square$

Definition 16.83 (Residual slice index). For $u \geq 0$ and $s > 0$, the residual slice index is

$$m_n = \left\lfloor \frac{nu}{s} \right\rfloor.$$

Definition 16.84 (Residual slice time). For $u \geq 0$ and $s > 0$, the residual slice time is the remainder

$$r_n = s \operatorname{fract}\left(\frac{nu}{s}\right).$$

Theorem 16.85 (Residual slice decomposition). For $u \geq 0$ and $s > 0$, one has $r_n \in [0, s]$ and

$$nu = m_n s + r_n.$$

Proof. This is the floor-plus-fraction decomposition of nu/s , multiplied by s . $\square$

Theorem 16.86 (A residual slice can vanish). Suppose $u \geq 0$ and $s > 0$, and that $T_s(\delta) = \delta$ and $T_{nu}(\delta) \rightarrow 0$ as $n \rightarrow \infty$. Then there exists $a \in [0, s]$ such that $T_a(\delta) = 0$.

Proof. The residual slice times lie in the compact interval $[0, s]$, so a subsequence converges to some $a \in [0, s]$. The residual decomposition rewrites $T_{nu}(\delta)$ in terms of $T_{r_n}(\delta)$, and continuity of the semigroup identifies the limit with $T_a(\delta)$. $\square$

Theorem 16.87 (Reduced-time eigenvector has nonzero trace). Under the reduced-time hypotheses $T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!}$, if μ is a peripheral eigenvalue of T_u , then T_u admits a μ -eigenvector with nonzero trace.

Theorem 16.88 (Peripheral collapse at the reduced time step). Under the same hypotheses, every peripheral eigenvalue of T_u equals 1.

Theorem 16.89 (Primitivity at the reduced time step). Under the same hypotheses, T_u is primitive.

Theorem 16.90 (Irreducibility descends to a reduced time step). If $T_{t_0} = (T_u)^N$ and T_{t_0} is irreducible, then T_u is irreducible.

Proof. Any invariant compression for T_u remains invariant under powers, hence also for T_{t_0} ; irreducibility of T_{t_0} rules this out. $\square$

Theorem 16.91 (Existence of an irreducible reduced time step). If $t_0 > 0$, T_{t_0} is irreducible, and σ is fixed for all times, then there exists a positive time step u such that

$$T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!},$$

the slice T_u is a channel, T_u is irreducible, and $T_u(\sigma) = \sigma$.

Proof. Take $u = t_0/N$ with $N = (\dim M_D(\mathbb{C}))!$. The semigroup power identity gives the factorization of T_{t_0} , channelity comes from the semigroup hypotheses, and Theorem 16.90 gives irreducibility of T_u . $\square$

Theorem 16.92 (Fixed points of the primitive reduced time step are trace-scalars). If T_u is primitive and σ is the common fixed density, then every fixed point of any positive-time slice T_s has the form $\tau = \operatorname{tr}(\tau) \sigma$.

Theorem 16.93 (Existence of a primitive reduced time step). If T_{t_0} is irreducible for some $t_0 > 0$, then there exists a positive reduced time step u such that T_u is primitive.

Theorem 16.94 (Irreducibility propagates to all positive times). *If T_{t_0} is irreducible for some $t_0 > 0$, then every positive-time slice T_s is irreducible.*

Proof. Let σ be the unique faithful fixed density of T_{t_0} . By Theorem 16.81, it is fixed for all times, and Theorem 16.82 makes the fixed-point space of T_{t_0} one-dimensional. Choose a primitive reduced time step by Theorem 16.93; then the trace-scalar description of fixed points from Theorem 16.92 gives the uniqueness criterion for irreducibility at every positive time. $\square$

Theorem 16.95 (Prop. 7.5: Irreducible semigroup implies primitive). *Let $T_t = e^{tL}$ be a quantum dynamical semigroup. If T_{t_0} is irreducible for some $t_0 > 0$, then T_t is primitive (and irreducible) for every $t > 0$. This is [Wol12, Prop. 7.5].*

Proof. First apply Theorem 16.94 to propagate irreducibility from the single time t_0 to every positive time. Then apply Theorem 16.78 to conclude primitivity of every positive-time slice. $\square$

Theorem 16.96 (Prop. 7.5: Irreducibility iff primitivity for QDS). *For a quantum dynamical semigroup $T_t = e^{tL}$:*

$$(\exists t_0 > 0, T_{t_0} \text{ irreducible}) \iff (\forall t > 0, T_t \text{ primitive and irreducible}).$$

This is [Wol12, Prop. 7.5].

Proof. Forward: apply Theorem 16.95. Reverse: take $t_0 = 1$ (or any positive time); the right-hand side already includes irreducibility of T_t for every $t > 0$. $\square$

16.5 Kernel of the adjoint Liouvillian

Definition 16.97 (Faithful stationary state). A generator L has a faithful stationary state if there exists a positive definite density matrix ρ_0 with $L(\rho_0) = 0$.

Definition 16.98 (Adjoint dissipator). For a Lindblad operator L_j , the adjoint dissipator on observables is

$$\mathcal{D}_{L_j}^*(A) = L_j^\dagger A L_j - \frac{1}{2} L_j^\dagger L_j A - \frac{1}{2} A L_j^\dagger L_j.$$

Definition 16.99 (Adjoint generator). The adjoint generator of a Lindblad form is

$$L^*(A) = i[H, A] + \sum_j \mathcal{D}_{L_j}^*(A).$$

Theorem 16.100 (Trace-pairing characterization of the adjoint generator). *For every density matrix ρ and every observable A ,*

$$\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A).$$

Proof. Expand the Schrödinger and Heisenberg formulas and apply cyclicity of the trace term by term. $\square$

Definition 16.101 (Commutant of the Lindblad data). The commutant of a Lindblad form is the set of matrices commuting with H , with every Lindblad operator L_j , and with every adjoint $L_j^\dagger$.

Definition 16.102 (Adjoint kernel). The adjoint kernel is the set of observables A satisfying $L^*(A) = 0$.

Theorem 16.103 (Commutant lies in the adjoint kernel). *If an observable commutes with H , with every L_j , and with every $L_j^\dagger$, then it lies in $\ker(L^*)$.*

Proof. The Hamiltonian commutator term vanishes, and each adjoint dissipator $\mathcal{D}_{L_j}^*(A)$ is zero under the commutation hypotheses. $\square$

Theorem 16.104 (Thm. 7.2: Adjoint kernel equals commutant (faithful state)). *Let L be a GKSL generator in Lindblad form with Hamiltonian H and Lindblad operators $\{L_j\}$. Define the adjoint generator L^* by $\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A)$. If L has a faithful stationary state $\rho_0 > 0$ with $L(\rho_0) = 0$, then for any A :*

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0 \quad \forall j.$$

That is, $\ker(L^) \subseteq \{H, L_j, L_j^\dagger\}'$. This is [Wol12, Thm. 7.2].*

Proof. From $L^*(A) = 0$ and $L^*(A^\dagger) = 0$, compute $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$. Faithfulness of ρ_0 and $\text{tr}(\rho_0 L^*(A^\dagger A)) = 0$ force $[A, L_j] = 0$ for all j . Similarly $[A, L_j^\dagger] = 0$, and then $L^*(A) = 0$ directly gives $[A, H] = 0$. $\square$

Theorem 16.105 (Thm. 7.2: Adjoint kernel equals commutant (full equivalence)). *Under the same hypotheses as Theorem 16.104,*

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

i.e. the adjoint kernel equals the commutant of $\{H, L_j, L_j^\dagger\}$. This is [Wol12, Thm. 7.2].

Proof. The inclusion $\supseteq$ is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse $\subseteq$ is Theorem 16.104. $\square$

16.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Prop. 7.6] are as follows.

Definition 16.106 (Nontrivial projection). A projection is nontrivial if it is orthogonal and neither 0 nor $\mathbb{1}$.

Definition 16.107 (Rank-deficient fixed density). Condition (1): there exists a density matrix ρ_0 with nontrivial kernel such that $e^{tL}(\rho_0) = \rho_0$ for all $t \geq 0$.

Definition 16.108 (Rank-deficient kernel element). Condition (2): there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$.

Definition 16.109 (Invariant compression). Condition (3): there exists a nontrivial orthogonal projector P such that $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ for all $t \geq 0$.

Definition 16.110 (Block-upper-triangular Lindblad form). Condition (4): there exists a non-trivial projector P and a Lindblad form $(H, \{L_j\})$ for L such that $(1 - P)L_j P = 0$ and $(1 - P)\kappa P = 0$ for all j , where $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$.

Definition 16.111 (Generator-preserved compression). For a projection P , the generator preserves the compression $PM_d(\mathbb{C})P$ if

$$P L(PXP) P = L(PXP)$$

for every $X \in M_d(\mathbb{C})$.

Definition 16.112 (Reducible quantum dynamical semigroup). A quantum dynamical semigroup is reducible if it has a nontrivial invariant compression.

Theorem 16.113 (Semigroup invariance implies generator invariance). *If the semigroup preserves $PM_d(\mathbb{C})P$ for all nonnegative times, then the generator preserves the same compression.*

Proof. Differentiate the identity $Pe^{tL}(PXP)P = e^{tL}(PXP)$ at $t = 0$. $\square$

Theorem 16.114 (Generator invariance implies semigroup invariance). *If the generator preserves $PM_d(\mathbb{C})P$, then so does the exponential semigroup e^{tL} for every $t \geq 0$.*

Proof. Apply the compression map termwise to the exponential series. Every iterate of L preserves the compression, so the whole series does as well. $\square$

Theorem 16.115 (Reducibility as a generator-level criterion). *A quantum dynamical semigroup is reducible if and only if its generator preserves some nontrivial compression.*

Proof. Combine Theorems 16.113 and 16.114. $\square$

Theorem 16.116 (Prop. 7.6, (1) $\leftrightarrow$ (2)). *A rank-deficient density matrix is a fixed point of e^{tL} for all $t \geq 0$ if and only if it lies in $\ker L$. This is [Wol12, Prop. 7.6, (1) $\leftrightarrow$ (2)].*

Proof. The equivalence $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$ for all $t \geq 0$ follows from the generator-semigroup correspondence: differentiate at $t = 0$ for the forward direction, and exponentiate for the reverse. $\square$

Theorem 16.117 (Prop. 7.6, (1) $\rightarrow$ (3)). *If L is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Prop. 7.6, (1) $\rightarrow$ (3)].*

Proof. Let ρ_0 be a rank-deficient fixed density. Let P be the support projection of ρ_0 . Since ρ_0 is rank-deficient, $P \neq \mathbb{1}$; since $\rho_0 \neq 0$, $P \neq 0$. Channel positivity and trace preservation force e^{tL} to preserve $PM_D(\mathbb{C})P$. $\square$

Theorem 16.118 (Invariant compression yields block-upper-triangular Lindblad data). *If a GKSL generator preserves a nontrivial compression, then its Lindblad data can be chosen block-upper-triangular with respect to that compression.*

Proof. First pass from semigroup invariance to generator invariance by Theorem 16.113. The generator-level vanishing condition forces $(1 - P)L_jP = 0$ and $(1 - P)\kappa P = 0$ through the sum-of-squares identity. $\square$

Theorem 16.119 (Block-upper-triangular Lindblad data yield invariant compression). *If the Lindblad data are block-upper-triangular with respect to a nontrivial projection, then the associated semigroup preserves that compression.*

Proof. The block-upper-triangular relations imply generator-level compression invariance; then apply Theorem 16.114. $\square$

Theorem 16.120 (Prop. 7.6, (3) $\leftrightarrow$ (4)). *For a GKSL generator L , the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Prop. 7.6, (3) $\leftrightarrow$ (4)].*

Proof. Combine Theorems 16.119 and 16.118. $\square$

Theorem 16.121 (Prop. 7.6, (4)→(2)). *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector P , then there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$. This is [Wol12, Prop. 7.6, (4)→(2)].*

Proof. The block-upper-triangular structure implies L preserves the compression $PM_d(\mathbb{C})P$. Therefore the semigroup e^{tL} also preserves $PM_d(\mathbb{C})P$ for all $t \geq 0$. By the Brouwer fixed-point theorem, $e^{(1/m)L}$ has a fixed density matrix ρ_m supported in $PM_d(\mathbb{C})P$ for each m . Passing to a subsequential limit gives a density matrix ρ_0 with $P\rho_0P = \rho_0$ and $L(\rho_0) = 0$. Since $P \neq \mathbb{1}$, the matrix ρ_0 has nontrivial kernel. $\square$

Theorem 16.122 (Prop. 7.6: Full equivalence). *For a GKSL generator $L : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the four conditions*

1. *rank-deficient fixed density,*
2. *rank-deficient kernel element,*
3. *invariant compression, and*
4. *block-upper-triangular Lindblad form*

are equivalent. This is [Wol12, Prop. 7.6].

Proof. The cycle (1)↔(2), (1)→(3), (3)↔(4), and (4)→(2) closes the equivalence chain. $\square$

16.6.1 Sufficient conditions for non-reducibility

Definition 16.123 (Lindblad span). The $\mathbb{C}$ -linear span of the Lindblad operators $\{L_j\}$.

Definition 16.124 (Hermitian-closed Lindblad span). The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a $*$ -subspace of $M_d(\mathbb{C})$.

Definition 16.125 (Trivial Lindblad commutant). The commutant of the Lindblad family $\{L_j\}'$ equals $\mathbb{C} \cdot \mathbb{1}$, i.e. the Lindblad operators act irreducibly on $M_d(\mathbb{C})$.

Definition 16.126 (Kossakowski rank). The minimal number of Lindblad operators across all GKSL representations of L , which equals the rank of the Kossakowski matrix in the orthonormal-basis representation.

Theorem 16.127 (κ block vanishing from generator compression). *Let $(H, \{L_j\})$ be a Lindblad form with $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$, and let P be an orthogonal projection such that the generator preserves the compression $PM_d(\mathbb{C})P$. If $(1 - P)L_jP = 0$ for every j , then $(1 - P)\kappa P = 0$.*

Proof. From $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$ and $(1 - P)L(P) = 0$: the $(1 - P)\varphi(P)$ term vanishes because each $(1 - P)L_jP = 0$, and the $(1 - P)(P\kappa^\dagger)$ term vanishes because $(1 - P)P = 0$. What remains is $(1 - P)\kappa P = 0$. $\square$

Theorem 16.128 (Full algebra generation forbids block-upper-triangular form). *Let $(H, \{L_j\})$ be a Lindblad form. If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. Assume for contradiction that a block-upper-triangular decomposition exists with non-trivial projection P . By Theorem 16.127, $(1 - P)\kappa P = 0$. Induction on elements of the subalgebra shows $(1 - P)AP = 0$ for every generator. Since the generated algebra is all of $M_d(\mathbb{C})$, every matrix satisfies $(1 - P)AP = 0$, forcing $P = 0$ or $P = \mathbb{1}$ — a contradiction. $\square$

Theorem 16.129 (Hermitian span + trivial commutant forbids block-upper-triangular form). *If the Lindblad span is closed under Hermitian conjugation and its commutant is $\mathbb{C} \cdot \mathbb{1}$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P . The block vanishing $(1-P)L_jP = 0$ and Hermitian closure give $PL_j(1-P) = 0$ as well, so $[P, L_j] = 0$ for all j . But then P lies in the commutant of the Lindblad span, contradicting triviality. $\square$

Theorem 16.130 (Large Kossakowski rank forbids block-upper-triangular form). *If the Kossakowski rank satisfies*

$$\text{rank}(C) + d \geq d^2 + 1,$$

then no block-upper-triangular Lindblad form exists.

Proof. A block-upper-triangular traceless Lindblad family lies in a proper subspace whose dimension is at most $d^2 - d$. The formal rank minimization argument then contradicts the large-rank bound. $\square$

Theorem 16.131 (No block-upper-triangular Lindblad form implies non-reducibility). *If a GKSL generator admits no block-upper-triangular Lindblad form, then the associated quantum dynamical semigroup is not reducible.*

Proof. A reducible semigroup would have an invariant compression, and Theorem 16.118 would then produce a block-upper-triangular Lindblad form. $\square$

Theorem 16.132 (Cor. 7.2(1): full algebra generation implies non-reducibility). *If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the QDS is not reducible. This is [Wol12, Cor. 7.2, condition 1].*

Proof. By Theorem 16.128, no block-upper-triangular decomposition exists. Then apply Theorem 16.131. $\square$

Theorem 16.133 (Cor. 7.2(2): Hermitian span + trivial commutant implies non-reducibility). *If the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Cor. 7.2, condition 2].*

Proof. By Theorem 16.129, no block-upper-triangular decomposition exists. Then apply Theorem 16.131. $\square$

Theorem 16.134 (Cor. 7.2(3): large Kossakowski rank implies non-reducibility). *If $\text{rank}(C) > d^2 - d$, then the QDS is not reducible. This is [Wol12, Cor. 7.2, condition 3].*

Proof. The rank hypothesis rules out block-upper-triangular Lindblad data by Theorem 16.130. Then apply Theorem 16.131. $\square$

Chapter 17

The Algebraic Fundamental Theorem

This chapter develops the one-dimensional MPS results of [MGRPG⁺18]: the Fundamental Theorem for injective and normal MPS on a finite periodic chain of fixed length. The argument is purely algebraic, working directly with $M_D(\mathbb{C})$. Two preparatory lemmas—the virtual bond gauge (Lemma 17.11) and the proportionality lemma (Lemma 17.12)—carry the entire argument, yielding the main theorem and its corollaries for translation-invariant chains, non-translation-invariant descriptions of translation-invariant states, and blocked chains.

Throughout, D is the bond dimension and d is the physical dimension with $d \geq D^2$. The extensions to injective and normal PEPS on arbitrary graphs and the improved system-size bound ($2L + 1$ in place of $3L$) are left for future work. The chain data (Definition 4.12), including same chain state, injectivity, and cyclic gauge equivalence, was introduced in Chapter 4.

17.1 One-sided inverse and decomposition

If A is injective, the linear combination map $\Phi_A(c) := \sum_i c_i A^i : \mathbb{C}^d \rightarrow M_D(\mathbb{C})$ is surjective and admits a right inverse.

Lemma 17.1 (Existence of a right inverse). *If A is injective, then there exists a $\mathbb{C}$ -linear map $\Psi : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ satisfying $\Phi_A \circ \Psi = \text{id}_{M_D(\mathbb{C})}$.*

Proof. Surjectivity of Φ_A yields a linear section by choice. □

Definition 17.2 (Decomposition map). Let A be an injective MPS tensor. The *decomposition map* $\Psi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ is a fixed choice of right inverse of Φ_A .

Lemma 17.3 (Decomposition property). *If A is injective, then for every $X \in M_D(\mathbb{C})$ there exist coefficients $c \in \mathbb{C}^d$ such that*

$$X = \sum_{i=0}^{d-1} c_i A^i.$$

Proof. Take $c = \Psi_A(X)$. Then $\Phi_A(c) = \Phi_A(\Psi_A(X)) = X$. □

Remark 17.4 (Non-uniqueness of the decomposition). When $d > D^2$, the map Φ_A has a nontrivial kernel, so the section Ψ_A is not unique. The statements that follow depend only on $\Phi_A \circ \Psi_A = \text{id}$, so the choice does not affect the conclusions.

17.2 Physical realisation of virtual insertions

For an injective tensor A , the physical realisation map encodes the effect of inserting a virtual matrix X on the bond as a $d \times d$ operation on the physical indices.

Definition 17.5 (Physical realisation map). Let A be an injective tensor. For $X \in M_D(\mathbb{C})$, define the $d \times d$ matrix $O_A(X) \in M_d(\mathbb{C})$ by

$$O_A(X)_{ij} := (\Psi_A(A^i X))_j,$$

so that row i of $O_A(X)$ contains the decomposition coefficients of $A^i X$ in the spanning set $\{A^j\}$. Diagrammatically, the inserted virtual matrix is re-expressed as a physical operation on the same local tensor:

The map $X \mapsto O_A(X)$ is $\mathbb{C}$ -linear.

Lemma 17.6 (Defining property of the physical realisation map). For every $X \in M_D(\mathbb{C})$ and every physical index i ,

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Proof. By definition, $O_A(X)_{ij} = (\Psi_A(A^i X))_j$. Applying Φ_A gives $\sum_j O_A(X)_{ij} A^j = \Phi_A(\Psi_A(A^i X)) = A^i X$, where the last step uses $\Phi_A \circ \Psi_A = \text{id}$. $\square$

Theorem 17.7 (Physical realisation is multiplicative). Let A be an injective MPS tensor. For all $X, Y \in M_D(\mathbb{C})$,

$$O_A(XY) = O_A(X) O_A(Y).$$

Proof. Applying Lemma 17.6 to X yields

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Multiply by Y on the right and apply the linear map Ψ_A :

$$\Psi_A(A^i XY) = \sum_{j=0}^{d-1} O_A(X)_{ij} \Psi_A(A^j Y).$$

Taking the k th component of both sides gives

$$O_A(XY)_{ik} = \sum_{j=0}^{d-1} O_A(X)_{ij} O_A(Y)_{jk} = (O_A(X) O_A(Y))_{ik}.$$

Since this holds for all i and k , the matrices are equal. $\square$

Remark 17.8 (General injective case). When $d > D^2$, the decomposition map Ψ_A need not be unique. Nevertheless, Theorem 17.7 is already the general injective statement: for the fixed choice of Ψ_A in Definition 17.2, the associated physical realisation map O_A is multiplicative. This supersedes the earlier basis-only formulation.

17.3 Intermediate lemmas

Lemma 17.9 (Blocking preserves injectivity). *Let $A_0, \dots, A_{m-1}$ be injective MPS tensors with compatible bond dimensions. Then the blocked tensor $(A_0 \otimes \dots \otimes A_{m-1})^{(i_0, \dots, i_{m-1})} := A_0^{i_0} A_1^{i_1} \dots A_{m-1}^{i_{m-1}}$ is again injective.*

Proof. The span of the blocked matrices contains all products $A_0^{i_0} \dots A_{m-1}^{i_{m-1}}$. Since each $\{A_k^i\}_i$ spans $M_D(\mathbb{C})$, these products span all of $M_D(\mathbb{C})$. $\square$

Lemma 17.10 (Centre of $M_D(\mathbb{C})$). *If $Z \in M_D(\mathbb{C})$ commutes with every element of $M_D(\mathbb{C})$, then $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}$.*

Proof. Standard: Z commutes with every matrix unit E_{ij} , which forces all off-diagonal entries to vanish and all diagonal entries to agree. $\square$

17.4 Same state forces a virtual bond gauge

This section records the 3-site virtual-bond statement used later in the paper proof. In the current Lean development the hypothesis is already the all-length MPV equality of the combined tensors, and the conclusion is the conjugacy of the middle-bond insertion data.

Lemma 17.11 (Virtual bond gauge – 3-site case). *Let $A = (A_0, A_1, A_2)$ and $B = (B_0, B_1, B_2)$ be two 3-site injective periodic chains with common bond dimension $D \geq 1$ and $\mathcal{V}(A) = \mathcal{V}(B)$ (same MPV family). Then at the middle bond there exists an invertible matrix $Z \in \text{GL}_D(\mathbb{C})$ such that the virtual-insertion trace coefficients agree after conjugation:*

$$\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Z^{-1} X Z B_1^{i_1} B_2^{i_2})$$

for all $X \in M_D(\mathbb{C})$ and all physical indices i_0, i_1, i_2 .

In the paper [MGRPG⁺18], Lemma 1 is stated for arbitrary chain length $n \geq 3$ and any bond position, with uniqueness of Z up to scalar. The chain-level generalization to arbitrary $n \geq 3$ and any bond position is future work.

Proof. The $\mathcal{V}(A) = \mathcal{V}(B)$ hypothesis gives, for every $X \in M_D(\mathbb{C})$, equality of the virtual-insertion trace coefficients: $\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Y B_1^{i_1} B_2^{i_2})$ for some Y depending on X . Since all three site tensors on each side are injective, the physical realisation maps O_A, O_B are algebra homomorphisms by Theorem 17.7. Setting $T(X) = Y$ defines a $\mathbb{C}$ -linear multiplicative map, hence a nonzero $\mathbb{C}$ -algebra endomorphism of $M_D(\mathbb{C})$, which is bijective by Theorem 4.8. Skolem-Noether (Theorem 4.9) gives $Z \in \text{GL}_D(\mathbb{C})$ with $T(X) = Z^{-1} X Z$. $\square$

17.5 Aligned gauges force proportionality

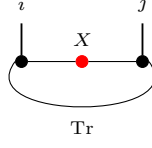
After applying Lemma 17.11 to each bond and absorbing the gauge matrices, one obtains two chains whose virtual-insertion operations agree on every bond. Lemma 2 of [MGRPG⁺18] shows that this agreement forces the local tensors to be proportional.

Lemma 17.12 (Aligned gauges force proportionality – Lemma 2 of [MGRPG⁺18]). *Let A_1, A_2 and B_1, B_2 be injective MPS tensors with the same bond dimension D . Suppose that for all $X \in M_D(\mathbb{C})$ (on the internal bond) and all $Y \in M_D(\mathbb{C})$ (on the external bond),*

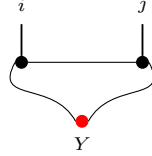
$$\text{tr}(A_1^i X A_2^j) = \text{tr}(B_1^i X B_2^j) \quad \text{for all } i, j, \quad (17.1)$$

$$\text{tr}(A_2^j Y A_1^i) = \text{tr}(B_2^j Y B_1^i) \quad \text{for all } i, j. \quad (17.2)$$

Diagrammatically, Eq. (17.1) is the equality obtained by inserting X on the bond between the two sites,



while Eq. (17.2) is the corresponding equality with the insertion Y on the outer trace bond,



so the two hypotheses distinguish the internal and external virtual loops of the two-site periodic chain.

Then there exists a nonzero scalar $\lambda \in \mathbb{C}^\times$ such that

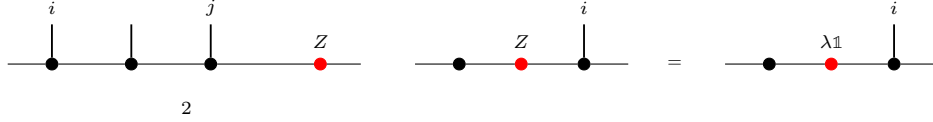
$$A_1^i = \lambda B_1^i \quad \text{and} \quad A_2^j = \lambda^{-1} B_2^j$$

for all physical indices i, j .

Proof. By cyclicity and non-degeneracy of the trace pairing, the conditions in Eqs. (17.1) and (17.2) give

$$A_2^j A_1^i = B_2^j B_1^i \quad \text{and} \quad A_1^i A_2^j = B_1^i B_2^j \quad \text{for all } i, j.$$

Write $\mathbb{1}_D = \sum_j c_j A_2^j$ using the decomposition map Ψ_{A_2} , and define $Z := \sum_j c_j B_2^j \in M_D(\mathbb{C})$. This is the point where the two-site comparison collapses to a single boundary matrix:



The first product identity gives $A_1^i = Z B_1^i$, and the second gives $A_1^i = B_1^i Z$. Hence Z commutes with every B_1^i ; since $\{B_1^i\}$ spans $M_D(\mathbb{C})$, Z is central. By Lemma 17.10, $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}^\times$ (nonzero since the tensors are nonzero). Thus $A_1^i = \lambda B_1^i$, and substituting into the first product identity forces $A_2^j = \lambda^{-1} B_2^j$. $\square$

17.6 The Fundamental Theorem for injective chains

The current Lean endpoint for chains is obtained by bundling the site index into the physical index and applying the single-block Fundamental Theorem to the resulting tensor. The paper-style fixed-length same-state statement is recovered only after adding a separate same-state reduction hypothesis.

Theorem 17.13 (Fundamental Theorem for injective chains (combined-tensor MPV version)). *Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two n -site chains with common bond dimension D , and assume that $\mathbf{A}$ is injective. Suppose that, after bundling the site index and physical index into a single MPS tensor, the resulting two tensors generate the same MPV family. Then $\mathbf{A}$ and $\mathbf{B}$ are cyclically gauge equivalent.*

Proof. Choose one site of $\mathbf{A}$; its injectivity implies injectivity of the bundled tensor by Lemma 17.9. Apply the single-block Fundamental Theorem (Theorem 4.11) to the two bundled tensors. This yields one invertible matrix X such that every bundled matrix of $\mathbf{B}$ equals the corresponding bundled matrix of $\mathbf{A}$ conjugated by X . Reading this identity back at each site gives $B_k^i = X A_k^i X^{-1}$ for all k and i , so the constant choice $Z_k = X$ is a cyclic gauge. $\square$

17.6.1 From fixed-length chain states to bundled-tensor MPV agreement

Definition 17.14 (Same-state reduction hypothesis). A *same-state reduction hypothesis* for dimensions (d, D) is the assertion that whenever $\mathbf{A}$ is an injective n -site chain with $n \geq 3$ and $\mathbf{A}, \mathbf{B}$ generate the same chain state, then the bundled tensors of $\mathbf{A}$ and $\mathbf{B}$ generate the same MPV family.

Lemma 17.15 (Same state gives bundled-tensor MPV agreement). *Assume a same-state reduction hypothesis for (d, D) . Let $\mathbf{A}$ and $\mathbf{B}$ be n -site chains with common bond dimension D , with $n \geq 3$, and assume that $\mathbf{A}$ is injective. If $\mathbf{A}$ and $\mathbf{B}$ generate the same chain state, then the bundled tensors of $\mathbf{A}$ and $\mathbf{B}$ generate the same MPV family.*

Proof. This is exactly the defining property of the same-state reduction hypothesis. $\square$

Theorem 17.16 (Fundamental Theorem for injective chains from same state). *Assume a same-state reduction hypothesis for (d, D) . Let $\mathbf{A}$ and $\mathbf{B}$ be n -site chains with common bond dimension D , with $n \geq 3$, and assume that $\mathbf{A}$ is injective. If $\mathbf{A}$ and $\mathbf{B}$ generate the same chain state, then $\mathbf{A}$ and $\mathbf{B}$ are cyclically gauge equivalent.*

Proof. Apply Lemma 17.15 to convert the fixed-length chain-state hypothesis into equality of the bundled-tensor MPV families, then invoke Theorem 17.13. $\square$

Remark 17.17 (Relationship to the translation-invariant all-sizes theorem). Theorem 17.13 treats a fixed chain length and assumes equality of the bundled-tensor MPV families. The single-block Fundamental Theorem (Theorem 4.11) instead studies one tensor at all system sizes and concludes exact gauge equivalence $B^i = X A^i X^{-1}$. The chain theorem is recovered by converting the chain into one bundled tensor; the price is that the hypothesis is all-length MPV equality for that bundled tensor rather than equality of the chain state at one fixed length.

17.6.2 Finite-length MPV agreement

Definition 17.18 (Finite-length MPV agreement). Two tensors A and B of the same bond dimension satisfy *finite-length MPV agreement from N_0 onward* if they generate the same MPV family for all system sizes $N \geq N_0$:

$$V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma \quad \text{for all } N \geq N_0, \sigma \in [d]^N.$$

Theorem 17.19 (Finite-length agreement implies full agreement). *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B generate the same MPV family for every system size $N \geq N_0$, then they generate the same MPV family for every system size.*

Proof. Write $1 = \sum_i c_i A^i$ (by injectivity) and set $S = \sum_i c_i B^i$.

1. Show that the length- N_0 word span of B is all of $M_D(\mathbb{C})$ via the linear relation between the length- N_0 word-evaluation maps of A and B induced by trace agreement at length $N_0 + 1$, together with a finrank transfer argument.
2. Derive $S = 1$ by trace nondegeneracy: for every $M \in M_D(\mathbb{C})$, $\text{tr}(M(S - 1)) = 0$.
3. Downward induction on $k = N_0 - |w|$: decompose $\text{tr}(w_A)$ and $\text{tr}(w_B)$ using $1 = \sum c_i A^i$ and $S = 1$ respectively, then apply the inductive hypothesis at length $|w| + 1$.

□

Theorem 17.20 (Strengthened single-block FT (finite-length)). *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B agree on all MPV coefficients for system sizes $N \geq N_0$ (for any threshold N_0), then A and B are gauge equivalent.*

Proof. Apply Theorem 17.19 to obtain equality of the MPV families for all system sizes, then invoke the single-block Fundamental Theorem (Theorem 4.11). □

17.7 Corollaries

17.7.1 Translation-invariant chains

Theorem 17.21 (Single gauge for translation-invariant chains). *Let A and B be MPS tensors of bond dimension D , and fix a chain length $n \geq 1$. Assume that A is injective and that the bundled tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that*

$$B^i = X A^i X^{-1}$$

for every physical index i .

Proof. Bundle the constant chains into one tensor. Injectivity of A implies injectivity of the bundled tensor by Lemma 17.9, and the bundled MPV equality is exactly the hypothesis needed for the single-block Fundamental Theorem. The resulting gauge matrix is independent of the site because every site carries the same local tensor. □

Corollary 17.22 (Translation-invariant collapse to one gauge). *Let A and B be MPS tensors of bond dimension D , and fix a chain length $n \geq 1$. Assume that A is injective and that the bundled tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exist an invertible matrix $Z \in \text{GL}_D(\mathbb{C})$ and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$ such that*

$$B^i = \lambda Z^{-1} A^i Z$$

for every physical index i . In the current Lean theorem one can take $\lambda = 1$.

Proof. Apply Theorem 17.21 to obtain $B^i = X A^i X^{-1}$. Set $Z = X^{-1}$ and $\lambda = 1$. Then $\lambda^n = 1$ and the displayed identity is exactly the same gauge relation rewritten in the form used by the corollary. □

17.7.2 Non-translation-invariant descriptions of translation-invariant states

Corollary 17.23 (Non-translation-invariant realisation of a translation-invariant state). *Assume a same-state reduction hypothesis for (d, D) . Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an injective non-translation-invariant chain of length $n \geq 3$. Suppose the state generated by $\mathbf{A}$ is translation-invariant, i.e. it equals the state generated by the cyclically shifted chain $(A_1, \dots, A_{n-1}, A_0)$. Then $\mathbf{A}$ admits a translation-invariant description with the same bond dimension: there exists an injective tensor B such that the constant chain $(B, \dots, B)$ generates the same state as $\mathbf{A}$. Explicitly, $B^i = A_0^i Z_0^{-1}$ (right multiplication, not conjugation), where Z_0 is extracted from the cyclic shift.*

Proof. Apply Theorem 17.16 to $\mathbf{A}$ and its cyclic shift. This gives invertible gauges $Z_0, \dots, Z_{n-1}$ with $A_{k+1}^i = Z_k^{-1} A_k^i Z_{k+1}$. Iterating these identities yields $A_k^i = L_k^{-1} A_0^i R_k$, where the products L_k, R_k satisfy $R_k L_{k+1}^{-1} = Z_0^{-1}$. Substituting into the chain trace shows that the same state is generated by the constant tensor $B^i := A_0^i Z_0^{-1}$. Since A_0 is injective and Z_0 is invertible, B is injective. $\square$

Symmetry-theoretic consequences of the Fundamental Theorem, including virtual representations and string order, are developed in Chapter 18.

17.7.3 Blocked chains

Lemma 17.24 (Block injectivity and blocked-tensor injectivity). *A tensor A is L -block injective if and only if its L -blocked tensor $A^{[L]}$ is injective.*

Proof. The blocked physical alphabet is just a reindexing of words of length L . Decoding blocked indices therefore identifies the two spanning families, so the two injectivity conditions are equivalent. $\square$

Definition 17.25 (Constant blocked chain). For an MPS tensor A , a blocking length L , and a chain length n , the *blocked chain* is the constant n -site chain whose local tensor is the blocked tensor $A^{[L]}$.

Lemma 17.26 (Blocked chains are injective). *If A is L -block injective, then the constant blocked chain built from $A^{[L]}$ is injective at every site.*

Proof. By Lemma 17.24, the local blocked tensor $A^{[L]}$ is injective, and therefore every site of the constant blocked chain is injective. $\square$

Theorem 17.27 (Fundamental Theorem for blocked chains). *Let A and B be MPS tensors of bond dimension D , let L be a blocking length, and let n be a chain length. Assume that A is L -block injective and that the bundled tensors of the two constant blocked chains built from $A^{[L]}$ and $B^{[L]}$ generate the same MPV family. Then the two blocked chains are gauge equivalent.*

Proof. By Lemma 17.26, the blocked chain built from $A^{[L]}$ is injective. Apply Theorem 17.13 to the two constant blocked chains. $\square$

Remark 17.28. The theorem above is the exact blocked-chain endpoint presently formalized in Lean. Recovering sitewise gauges for the original unblocked tensors, and hence the paper's normal-MPS corollary, remains future work.

17.8 Product algebra construction and per-block gauge

The chain Fundamental Theorem (Theorem 17.13) handles non-translation-invariant periodic chains. A parallel line of argument addresses block-diagonal tensors: given a family of injective block tensors $\{A_k^i\}_{k=0}^{r-1}$ with block dimensions D_k and a second family $\{B_k^i\}$ with per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$, one seeks per-block gauge equivalence $B_k^i = X_k A_k^i X_k^{-1}$ for each k independently, and a global block-permutation plus inner-automorphism decomposition.

17.8.1 Per-block linear extension

Definition 17.29 (Per-block linear extension). Let A_k, B_k be MPS tensors of bond dimension D_k with A_k injective and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block $k \in \{0, \dots, r-1\}$. The *per-block linear extension* $T_k : M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$ is the unique $\mathbb{C}$ -linear map satisfying $T_k(A_k^i) = B_k^i$ for every physical index i .

Lemma 17.30 (Per-block multiplicativity). *For each block k , the per-block linear extension T_k is multiplicative: $T_k(MN) = T_k(M)T_k(N)$ for all $M, N \in M_{D_k}(\mathbb{C})$.*

Proof. Since $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$, it suffices to check on spanning elements, where the result follows from the MPV identity. $\square$

Lemma 17.31 (Per-block bijectivity). *Each T_k is bijective.*

Proof. A nonzero multiplicative linear endomorphism of $M_{D_k}(\mathbb{C})$ is automatically bijective: injectivity follows from the simplicity of $M_{D_k}(\mathbb{C})$ (the kernel is a proper ideal, hence zero), and surjectivity from finite dimension. Non-vanishing of T_k is established via the trace pairing: if $T_k = 0$, then all $B_k^i = 0$, contradicting the non-vanishing of the trace pairing for injective tensors. $\square$

17.8.2 Product algebra automorphism

Definition 17.32 (Product algebra automorphism). The *product algebra automorphism* is the $\mathbb{C}$ -algebra equivalence

$$T : \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$$

defined by $T(M)_k = T_k(M_k)$, where each T_k is the per-block linear extension. It is an algebra automorphism since each T_k is a multiplicative, unital, bijective linear map. Diagrammatically, this is the blockwise map on the product algebra obtained by applying the corresponding per-block extension to each summand.

Theorem 17.33 (Block-permutation decomposition). *The product algebra automorphism T decomposes as a block permutation composed with per-block inner automorphisms: there exist a permutation $\sigma \in S_r$ (with $D_{\sigma(k)} = D_k$ for all k) and invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that*

$$(T(M))_{\sigma(k)} = X_k M X_k^{-1}$$

for all $M \in M_{D_k}(\mathbb{C})$ (viewed as supported in block k) and all k . Equivalently, the product-algebra automorphism first permutes the simple matrix blocks and then acts inside each block by an inner conjugation.

Proof. The automorphism T permutes the block ideals of $\prod_k M_{D_k}(\mathbb{C})$ (each block ideal is simple, and an automorphism sends simple ideals to simple ideals of the same dimension). This gives the permutation σ . On each block, the restricted map is a $\mathbb{C}$ -algebra automorphism of $M_{D_k}(\mathbb{C})$; by Skolem–Noether (Theorem 4.9), it is inner. Thus T splits algebraically into the permutation of block ideals followed by inner conjugation on each surviving block. $\square$

17.8.3 Nondegeneracy of the product trace pairing

Lemma 17.34 (Nondegeneracy of the product trace pairing). *Let $M = (M_0, \dots, M_{r-1}) \in \prod_k M_{D_k}(\mathbb{C})$. If $\sum_k \text{tr}(M_k N_k) = 0$ for all $N = (N_0, \dots, N_{r-1})$, then $M = 0$.*

Proof. Choose N supported on a single block k (all other components zero). The hypothesis reduces to $\text{tr}(M_k N_k) = 0$ for all N_k , so $M_k = 0$ by nondegeneracy of the matrix trace pairing on $M_{D_k}(\mathbb{C})$. Since k was arbitrary, $M = 0$. $\square$

Lemma 17.35 (Injectivity of the product Gram map). *Let $\{A_k\}$ be a family of injective block tensors. The product Gram map $M \mapsto (k, i \mapsto \text{tr}(M_k \cdot A_k^i))$ is injective.*

Proof. Reduces to per-block injectivity of the Gram map $M_k \mapsto (i \mapsto \text{tr}(M_k \cdot A_k^i))$, which holds by injectivity of A_k (the matrices $\{A_k^i\}$ span $M_{D_k}(\mathbb{C})$, so the trace pairing against them separates points). $\square$

17.9 The multi-block Fundamental Theorem

Theorem 17.36 (Multi-block Fundamental Theorem). *Let $\{A_k\}_{k=0}^{r-1}$ and $\{B_k\}_{k=0}^{r-1}$ be two families of MPS tensors with block dimensions D_k , each A_k injective, and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all k . Then:*

1. *Per-block gauge equivalence: $B_k^i = X_k A_k^i X_k^{-1}$ for some $X_k \in \text{GL}_{D_k}(\mathbb{C})$, for every k and i .*
2. *Global gauge equivalence: the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

The second clause is obtained from the first by assembling the per-block gauges into one block-diagonal gauge.

Proof. Per-block gauge equivalence follows from applying the single-block Fundamental Theorem (existing) to each block, using injectivity of A_k and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. Global gauge equivalence is assembled from the per-block gauges via the block-diagonal structure. $\square$

Theorem 17.37 (Explicit per-block gauges). *Under the hypotheses of Theorem 17.36, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that*

$$B_k^i = X_k A_k^i X_k^{-1}$$

for every block k and every physical index i .

Proof. Extract the gauge matrix from the per-block part of Theorem 17.36 for each block. $\square$

Theorem 17.38 (Product-algebra decomposition of the multi-block theorem). *Assume that every block dimension is nonzero. Under the hypotheses of Theorem 17.36, the product-algebra automorphism determined by the per-block linear extensions decomposes into a block permutation together with inner conjugation on each block: there exist a permutation σ of the blocks, compatible block-dimension equalities, and invertible matrices X_k such that, after reindexing the target block $\sigma(k)$ back to block k , the action on the k th matrix block is $M \mapsto X_k M X_k^{-1}$.*

Proof. Apply Theorem 17.33 to the product-algebra automorphism built from the per-block linear extensions. $\square$

Theorem 17.39 (Per-block equality of MPV families and per-block gauge equivalence). *Let $\{A_k\}$ be a family of injective block tensors. Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all k if and only if A_k and B_k are gauge equivalent for all k .*

Proof. The forward direction is the single-block Fundamental Theorem applied blockwise. The reverse direction holds because gauge equivalence preserves all traces of word evaluations. $\square$

17.10 Block separation from canonical form

The multi-block Fundamental Theorem (Theorem 17.36) assumes per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. In practice, one starts from the weaker global hypothesis that the block-diagonal tensors satisfy the same MPV family: $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$. Block separation extracts the per-block identities from the global one under canonical-form normalization. The first step is the weighted block-sum identity

$$\sum_k \mu_k^N V^{(N)}(A_k)_\sigma = \sum_k \mu_k^N V^{(N)}(B_k)_\sigma,$$

which is then combined with overlap estimates and the leading-block peel.

Definition 17.40 (Canonical form). A family of block tensors $\{A_k\}_{k=0}^{r-1}$ with scaling weights $\mu_0, \dots, \mu_{r-1} \in \mathbb{C}^\times$ is in *canonical form* if:

1. each A_k is injective,
2. each A_k is left-canonical: $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}_{D_k}$,
3. the weights are strictly ordered by norm: $|\mu_0| > |\mu_1| > \dots > |\mu_{r-1}| > 0$,
4. the self-overlaps are normalised: $O_{A_k A_k}(N) \rightarrow 1$ as $N \rightarrow \infty$.

Definition 17.41 (Normal canonical form). A family is in *normal canonical form* if it satisfies the canonical-form conditions with each block additionally *irreducible* (no nontrivial invariant projection) and *primitive* (the transfer map has spectral gap).

Theorem 17.42 (Block separation from weighted word identities). *Let $\{A_k\}$ and $\{B_k\}$ be block families with strictly ordered, nonzero weights μ_k . Assume that every A_k and B_k is injective and left-canonical, that each self-overlap $O_{A_k A_k}(N)$ tends to 1, and that for every system size N and every word σ one has the weighted identity*

$$\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0.$$

Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k .

Proof. The proof is the peeling argument described below for canonical form. The strict weight ordering isolates the leading block, the overlap bounds show that the tail decays geometrically after dividing by the dominant weight, and one then iterates on the remaining blocks. $\square$

Theorem 17.43 (Block separation from canonical form). *Let $\{A_k\}$ be in canonical form with weights $\{\mu_k\}$, and let $\{B_k\}$ be another family with each B_k injective and left-canonical. If $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$, then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block k individually. The proof isolates the leading block and treats the remaining blocks as an exponentially decaying tail with $\rho = |\mu_1|/|\mu_0| < 1$.*

Proof. The proof proceeds by strong induction on the number of blocks r . For $r = 1$, the weighted sum degenerates and the result is immediate after cancelling $\mu_0^N \neq 0$.

For $r \geq 2$, the global identity $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ holds for all N and σ .

$$\mu_0^N (V^{(N)}(A_0)_\sigma - V^{(N)}(B_0)_\sigma) = - \sum_{k \geq 1} \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma).$$

Dividing by μ_0^N shows that the right-hand side is $O(\rho^N)$ with $\rho = |\mu_1|/|\mu_0| < 1$. Multiplying by the MPV components of A_0 and summing over σ converts this to an overlap identity. Dividing by $|\mu_0|^{2N}$ and using the strict norm ordering $|\mu_1|/|\mu_0| < 1$, the tail terms ($k \geq 1$) decay geometrically with ratio $\rho = |\mu_1|/|\mu_0|$. The leading term ($k = 0$) therefore converges: $O_{A_0 B_0}(N) \rightarrow 1$ as $N \rightarrow \infty$. Combined with the normalised self-overlap $O_{A_0 A_0}(N) \rightarrow 1$, this gives $\mathcal{V}(A_0) = \mathcal{V}(B_0)$ via the gauge-phase equivalence criterion. Subtracting the leading block and applying the inductive hypothesis yields $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all remaining k . $\square$

Theorem 17.44 (Block separation from separated canonical data). *Suppose the weights are nonzero with strictly decreasing norms, the A -blocks are injective and left-canonical with normalized self-overlaps, the B -blocks are injective and left-canonical, and the assembled block-diagonal tensors generate the same MPV family. Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k .*

Proof. Equality of the assembled MPV families gives the weighted word identity required by Theorem 17.42. The separated hypotheses supply exactly the injectivity, left-canonicity, strict-weight, and self-overlap assumptions needed there. $\square$

Theorem 17.45 (Fundamental Theorem from separated canonical data). *Under the hypotheses of Theorem 17.44, every block pair A_k, B_k is gauge equivalent, and the assembled block-diagonal tensors are globally gauge equivalent.*

Proof. First apply Theorem 17.44 to get $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ blockwise. Then apply Theorem 17.36. $\square$

Theorem 17.46 (Explicit gauge from separated canonical data). *Under the same hypotheses, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that*

$$B_k^i = X_k A_k^i X_k^{-1}$$

for every block k and physical index i .

Proof. Combine blockwise MPV equality from Theorem 17.44 with the explicit-gauge form of the multi-block Fundamental Theorem. $\square$

Theorem 17.47 (Fundamental Theorem from canonical form). *Under the hypotheses of Theorem 17.43, every block pair A_k, B_k is gauge equivalent, and the block-diagonal tensors are globally gauge equivalent.*

Proof. Theorem 17.43 gives per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. Theorem 17.36 converts this to per-block and global gauge equivalence. $\square$

Theorem 17.48 (Explicit gauge from canonical form). *Under the same hypotheses, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that $B_k^i = X_k A_k^i X_k^{-1}$ for every block k and physical index i .*

Proof. Extract the gauge matrices from Theorem 17.47 by choice. $\square$

17.11 Heterogeneous equal-case endpoint

Theorem 17.49 (Heterogeneous equal-case Fundamental Theorem). *Let $\{\mu_j A_j\}_{j=0}^{r_A-1}$ and $\{\nu_k B_k\}_{k=0}^{r_B-1}$ be two canonical-form BNT families, possibly with different block counts and different bond dimensions in corresponding blocks. If the assembled block-diagonal tensors generate the same MPV family, then the two families have the same number of blocks, the blocks can be matched by a permutation preserving the bond dimensions, and each matched pair is gauge-phase equivalent.*

Proof. The proof is the block-matching argument carried out in the supporting modules for the heterogeneous equal case: one first produces a non-decaying cross-overlap for each block, applies the overlap dichotomy to show that the matched blocks must have equal dimension and differ only by a phase gauge, and finally upgrades the resulting matching to a permutation of all blocks. $\square$

17.12 Translation-invariant reduction

Theorem 17.50 (Translation-invariant reduction corollary). *Let A and B be MPS tensors of bond dimension D with A injective. Consider the two constant translation-invariant chains $(A, \dots, A)$ and $(B, \dots, B)$ of length $n \geq 1$. If the bundled tensors of the two chains generate the same MPV family, then:*

1. *B is gauge equivalent to A : there exists $X \in \text{GL}_D(\mathbb{C})$ with $B^i = X A^i X^{-1}$ for all i .*
2. *B is injective.*

Proof. Theorem 17.21 gives a single gauge matrix X with $B^i = X A^i X^{-1}$ for all i . Since gauge equivalence preserves injectivity, B is injective as well. $\square$

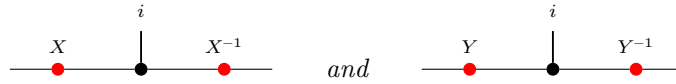
Theorem 17.51 (Translation-invariant reduction from same state). *Assume a same-state reduction hypothesis for (d, D) . Let A and B be MPS tensors of bond dimension D , and let $n \geq 3$. If the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same chain state, with A injective, then the same conclusion as in Theorem 17.50 holds: B is gauge equivalent to A , and B is injective.*

Proof. The same-state reduction hypothesis upgrades equality of the two constant chain states to equality of their bundled-tensor MPV families. Then Theorem 17.50 applies. $\square$

17.12.1 Global symmetry via gauge composition

The virtual representation theorem (Theorem 18.14) relies on the fact that composing two gauge transformations yields another gauge up to a scalar factor.

Lemma 17.52 (Gauge ratio commutes with tensor matrices). *Diagrammatically, the hypothesis is that the two local gauges*



produce the same target tensor B^i from the same source tensor A^i . If $X, Y \in \text{GL}_D(\mathbb{C})$ both conjugate an MPS tensor A to the same tensor B (i.e. $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i), then $Y^{-1}X$ commutes with every A^i .

Proof. Direct matrix calculation: conjugating both gauge relations and composing gives $(Y^{-1}X)A^i = A^i(Y^{-1}X)$. $\square$

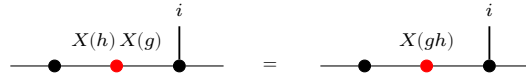
Lemma 17.53 (Gauge ratio is scalar for injective tensors). *Under the same hypotheses, if A is additionally injective, then $Y^{-1}X = \lambda \mathbb{1}_D$ for some scalar $\lambda \in \mathbb{C}$.*

Proof. The same two-gauge picture shows that the ratio $Y^{-1}X$ acts invisibly on the local tensor. By Lemma 17.52, $Y^{-1}X$ commutes with all A^i . Since $\{A^i\}$ spans $M_D(\mathbb{C})$, the matrix $Y^{-1}X$ commutes with all of $M_D(\mathbb{C})$. By Lemma 17.10, it is a scalar matrix. $\square$

Theorem 17.54 (Projective multiplication law for gauge matrices). *Let A be an injective MPS tensor with on-site symmetry under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. For each $g \in G$, let $X(g) \in \text{GL}_D(\mathbb{C})$ satisfy $\tilde{A}_g^i = X(g) A^i X(g)^{-1}$ for all i . Then for all $g, h \in G$, there exists $c(g, h) \in \mathbb{C}$ such that*

$$X(h) X(g) = c(g, h) X(g \cdot h).$$

In fact $c(g, h) \neq 0$ since both sides are invertible, so $c(g, h) \in \mathbb{C}^\times$. Diagrammatically, the two candidate gauges are competing local replacements for the same twisted tensor:



and both sides conjugate A^i to the same twisted tensor $\tilde{A}_{gh}^i$.

Proof. By the composition law $\tilde{A}_{gh} = \widetilde{(\tilde{A}_h)_g}$ (Lemma 18.5), both $X(h) X(g)$ and $X(gh)$ conjugate A to $\tilde{A}_{gh}$. Lemma 17.53 then gives the scalar factor. $\square$

17.13 Fundamental Theorem for injective PEPS

The following definitions and results extend the Fundamental Theorem to injective PEPS on simple graphs, following Theorem 2 of [MGRPG⁺18].

Definition 17.55 (Graph edge for PEPS). For a finite simple graph G , an *edge* is an ordered pair (u, v) with $u < v$ and u adjacent to v . The ordering is only used to avoid double counting.

Definition 17.56 (Incident edges at a vertex). For a vertex v , an *incident edge* is an edge of G having v as one of its endpoints.

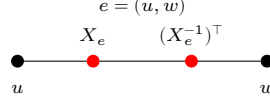
Definition 17.57 (PEPS tensor on a graph). A PEPS tensor on G with physical dimension d consists of a bond dimension D_e for every edge e , together with a local tensor at each vertex v whose virtual indices are indexed by the edges incident to v and whose physical index lies in $\{0, \dots, d-1\}$.

Definition 17.58 (Virtual configuration). A *virtual configuration* assigns to every edge e one basis index in the corresponding bond space of dimension D_e .

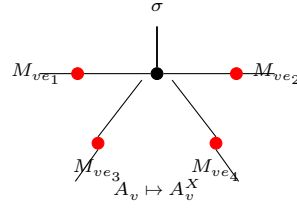
Definition 17.59 (PEPS state coefficient). For a physical configuration σ on the vertices, the PEPS state coefficient is obtained by summing over all virtual configurations and, for each such configuration, multiplying the local tensor entries over all vertices.

Definition 17.60 (Equality of PEPS states). Two PEPS tensors represent the same state if all their state coefficients agree for every physical configuration.

Definition 17.61 (Edge gauge at a vertex). For an edge $e = (u, w)$ with $u < w$ and gauge $X_e \in \text{GL}(D_e)$, the gauge matrix applied at vertex v is X_e if $v = u$ and $(X_e^{-1})^\top$ if $v = w$. Diagrammatically, the ordered edge carries opposite endpoint actions:



Definition 17.62 (Gauge vertex action). The gauged tensor at vertex v is A_v^X , where $A_v^X(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) A_v(\eta', \sigma)$. In tensor-network notation, one inserts the appropriate gauge matrix on each incident virtual leg of the local tensor:



Definition 17.63 (Apply gauge). The PEPS tensor obtained by applying edge-gauge matrices to A , with vertex tensors A_v^X .

Definition 17.64 (Gauge equivalence for PEPS). Two PEPS tensors A, B are gauge-equivalent if they have the same bond dimensions and B is obtained from A by applying invertible gauge matrices on each edge.

Theorem 17.65 (Gauge invariance of PEPS state). *Applying a gauge to a PEPS tensor preserves the state coefficients. Along each internal edge, the two endpoint insertions cancel:*

$$\begin{array}{ccc} \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet & \sum_j X_e(j, i) ((X_e^{-1})^\top)(j, k) = \delta(i, k) & \text{---} \bullet \text{---} \bullet \end{array}$$

Theorem 17.66 (Gauge equivalence implies same state). *Gauge equivalence implies that the two PEPS tensors represent the same state.*

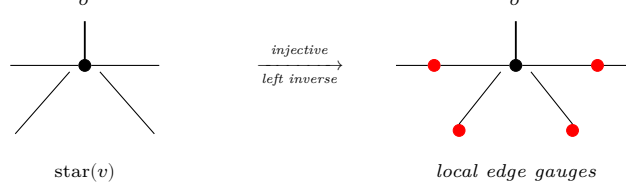
Definition 17.67 (Local tensor evaluation). The local tensor evaluated at vertex v with virtual-index weighting f , computing $\sum_{\eta} (\prod_{ie} f(ie)(\eta(ie))) \cdot A_v(\eta, \sigma)$. This is multilinear in the components of f , not linear in the full tuple.

Definition 17.68 (Vertex injectivity). A PEPS tensor A is *vertex-injective* if, at every vertex v , the family of physical vectors

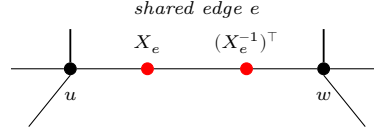
$$(A_v(\eta, \cdot) \in \mathbb{C}^d)_\eta$$

indexed by virtual configurations η on the edges incident to v is linearly independent in $\mathbb{C}^d$. Equivalently, extending A_v by linearity to a map from the free vector space on virtual configurations into the physical space $\mathbb{C}^d$ gives a linear map with trivial kernel. This matches the paper's formulation that "all tensors interpreted as maps from the virtual space to the physical one are injective" ([MGRPG⁺18], §3).

Theorem 17.69 (Local gauge extraction). *At a single vertex, vertex-injectivity of both tensors and the same-state condition force a local gauge relation. Schematically, injectivity on the star neighbourhood provides a left inverse that isolates the local edge gauges:*



Theorem 17.70 (Gauge consistency). *Local gauges extracted at adjacent vertices are consistent across shared edges, yielding a global gauge family. On a shared edge, the two neighbouring local extractions match as one edge gauge:*



Theorem 17.71 (Fundamental Theorem for injective PEPS). *If two PEPS tensors on a simple graph are both vertex-injective and generate the same state, then they are gauge-equivalent (Theorem 2 of [MGRPG⁺18]).*

Remark 17.72 (Current scalar-uniqueness placeholder). The present Lean declaration records a global-scalar uniqueness claim, but this conclusion is known to be too strong. On a triangle with all bond dimensions equal to 1, edge scalars (a, a^{-1}, a) preserve every local tensor yet are not all multiples of $(1, 1, 1)$ when $a \neq \pm 1$. The expected uniqueness statement must therefore quotient the residual edge-gauge freedom by a cocycle-type normalization on the graph.

Remark 17.73. The first three PEPS results above still wait for the graph-blocking / virtual-insertion argument of [MGRPG⁺18], §3. For the uniqueness endpoint, the current global-scalar conclusion is too strong: a connected triangle with bond dimension 1 admits nontrivial edge rescalings whose vertex products are all 1. A corrected statement will need an explicit normalization of the edge gauges.

17.14 Closing remarks

Remark 17.74. Appendix A of [MGRPG⁺18] strengthens the normal-MPS bound from $n \geq 3L$ to $n \geq 2L + 1$ by a more refined blocking argument. The injective PEPS theorem and the normal PEPS theorem on arbitrary graphs both rest on the same two core lemmas (Lemma 17.11 and Lemma 17.12); the graph geometry enters only through the blocking step that reduces to a three-site chain.

Chapter 18

Symmetries and String Order

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem. For injective tensors, physical on-site symmetries give virtual gauges, those gauges define projective representations on the bond space, and the same virtual data controls string-order phenomena. The algebraic theorem supplies the gauge mechanism, and this chapter develops its symmetry consequences.

18.1 Physical Symmetries Give Virtual Gauges

Corollary 18.1 (Physical symmetry implies virtual gauge). *Assume a same-state reduction hypothesis for (d, D) . Let A be an injective MPS tensor and let $U \in \text{GL}_d(\mathbb{C})$ be a unitary acting on the physical index. Fix $n \geq 3$ and suppose the n -site translation-invariant state satisfies*

$$U^{\otimes n} |\psi_n(A)\rangle = |\psi_n(A)\rangle.$$

Then there exist $V \in \text{GL}_D(\mathbb{C})$ and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$ such that

$$\sum_{j=0}^{d-1} U_{ij} A^j = \lambda V^{-1} A^i V$$

for every physical index i .

Proof. Define $B^i := \sum_j U_{ij} A^j$. Since U is invertible, $\{B^i\}$ spans $M_D(\mathbb{C})$, so B is injective. The symmetry condition means the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ of length n generate the same chain state. Theorem 17.51 then gives the required virtual gauge relation. $\square$

For periodic tensors, the physical-index rotation track is developed in Chapter 15: Definition 15.4 and Theorems 15.5, 15.6, 15.7, 15.8, 15.9, 15.10, and 15.11. We use these results here when deriving symmetry consequences, without restating their proofs.

Corollary 18.2 (Periodic symmetry yields $\mathbb{Z}$ -gauge equivalence). *Let A be in irreducible form II and let u be a matrix such that $B := A_u$ is also in irreducible form II and $\mathcal{V}(A) = \mathcal{V}(B)$. Then there exists $m > 0$ such that A and B are $\mathbb{Z}_m$ -gauge equivalent.*

Proof. Direct application of the periodic equal-case fundamental theorem (Theorem 15.23) to A and B . $\square$

18.2 Virtual Representation Theorem

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

Definition 18.3 (Twisted tensor). Given a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$ on the physical index, the g -twisted tensor is

$$\tilde{A}_g^i := \sum_{j=0}^{d-1} U(g)_{ij} A^j \quad \text{for each } i \in \{0, \dots, d-1\}.$$

Definition 18.4 (On-site symmetry). A tensor A is *on-site symmetric* under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$ if $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$ for every $g \in G$.

Lemma 18.5 (Functoriality of twisting). *The twisted tensor satisfies the composition law*

$$\tilde{A}_{gh} = \widetilde{(\tilde{A}_h)_g},$$

i.e. twisting by gh is the same as first twisting by h and then by g .

Proof. Expanding both sides using $U(gh) = U(g)U(h)$, then swapping the order of summation. $\square$

Lemma 18.6 (Identity twist). *Twisting by the identity group element is trivial: $\tilde{A}_1 = A$.*

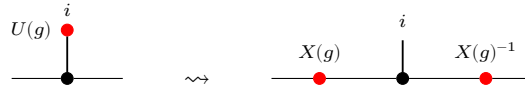
Proof. Since $U(1) = I_d$, each component reduces to $\sum_j \delta_{ij} A^j = A^i$. $\square$

Lemma 18.7 (Symmetric twists are gauge equivalent). *If A is injective and on-site symmetric under U , then for every $g \in G$ the twisted tensor $\tilde{A}_g$ is gauge equivalent to A .*

Proof. On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$, and equal MPV for injective tensors implies gauge equivalence. $\square$

Theorem 18.8 (Virtual symmetry equation). *If A is injective and on-site symmetric under U , then for every $g \in G$ there exist an invertible matrix $X(g) \in \text{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that $\tilde{A}_g^i = \varphi(g) X(g) A^i X(g)^{-1}$ for all i .*

Proof. The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 18.7 to obtain $X(g)$, then set $\varphi(g) = 1$. $\square$

Lemma 18.9 (Gauge uniqueness up to scalar). *Let A be an injective tensor. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. The matrix $Z = Y^{-1}X$ commutes with every A^i . Since A is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so Z lies in the centre of $M_D(\mathbb{C})$. By Lemma 17.10, Z is a scalar matrix. Since Z is invertible, that scalar is nonzero, and $Y = u \cdot X$. $\square$

Theorem 18.10 (Gauge uniqueness from equality of MPV families). *Let A be an injective tensor and assume that A and B generate the same MPV family. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there is a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. This is exactly Lemma 18.9; the extra MPV hypothesis records the setting in which the two gauges are usually produced. $\square$

Definition 18.11 (Scalar 2-cochain). A scalar 2-cochain on a group G is a function $\omega : G \times G \rightarrow \mathbb{C}^\times$.

Definition 18.12 (Projective representation). Given a scalar 2-cochain ω , a projective representation of G on $\mathbb{C}^D$ is a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh) \quad \text{for all } g, h \in G.$$

Lemma 18.13 (Associativity forces the cocycle condition). *If ρ is a projective representation with factor system ω and $D > 0$, then ω satisfies the 2-cocycle identity:*

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k).$$

Proof. Expanding $\rho(g) (\rho(h) \rho(k))$ and $(\rho(g) \rho(h)) \rho(k)$ using the projective multiplication law and equating via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor. $\square$

Theorem 18.14 (Virtual representation theorem for injective MPS). *Let A be an injective MPS tensor that is on-site symmetric under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. Then there exist:*

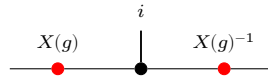
1. a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$, and
2. a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh) \quad \text{for all } g, h \in G,$$

such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has

$$\tilde{A}_g^i = X(g) A^i X(g)^{-1} \quad \text{for every } g \in G, i \in \{0, \dots, d-1\}.$$

In tensor-network notation, for each fixed g this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular, ρ is a projective representation of G on the bond space $\mathbb{C}^D$.

Proof. Since A is injective and on-site symmetric, each twisted tensor $\tilde{A}_g$ is gauge equivalent to A (Lemma 18.7). Choose $X(g) \in \text{GL}_D(\mathbb{C})$ with $\tilde{A}_g^i = X(g) A^i X(g)^{-1}$ for all i .

By functoriality (Lemma 18.5), the product $X(h) X(g)$ also intertwines A with $\tilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 18.9) gives a nonzero scalar $\omega'(h, g)$ with $X(h) X(g) = \omega'(h, g) X(gh)$.

Defining $\rho(g) := X(g^{-1})$ converts this to the standard law: $\rho(g) \rho(h) = \omega(g, h) \rho(gh)$ where $\omega(g, h) := \omega'(h^{-1}, g^{-1})$. $\square$

Corollary 18.15 (2-cocycle identity for the factor system). *If $D > 0$, the factor system ω from Theorem 18.14 satisfies the 2-cocycle identity:*

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k) \quad \text{for all } g, h, k \in G.$$

Proof. Theorem 18.14 gives ρ and ω . By Lemma 18.13, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices. $\square$

18.3 Cohomology Classes of Virtual Symmetry Data

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient $H^2(G, \mathbb{C}^\times)$.

Definition 18.16 (Coboundary). A scalar 2-cocycle ω is a *coboundary* if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ such that

$$\omega(g, h) = \varphi(g) \varphi(h) \varphi(gh)^{-1} \quad \text{for all } g, h \in G.$$

Definition 18.17 (Cohomologous cocycles). Two cocycles ω_1, ω_2 are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ with

$$\omega_1(g, h) = \varphi(g) \varphi(h) \varphi(gh)^{-1} \omega_2(g, h) \quad \text{for all } g, h \in G.$$

Theorem 18.18 (Cohomologous-to is an equivalence relation). *The relation "cohomologous to" is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.*

Proof. Reflexivity: take $\varphi = 1$. Symmetry: replace φ by φ^{-1} . Transitivity: compose the witnesses φ and ψ pointwise. $\square$

Lemma 18.19 (Coboundary iff cohomologous to the trivial cocycle). *A cocycle ω is a coboundary if and only if $\omega \sim \mathbf{1}$, where $\mathbf{1}(g, h) = 1$ for all g, h .*

Proof. Both directions follow by absorbing or introducing the factor $\mathbf{1}(g, h) = 1$ via $x \cdot 1 = x$. $\square$

Theorem 18.20 (Gauge independence of the cocycle class). *If the bond dimension satisfies $D \geq 1$ and two projective representations ρ_1, ρ_2 (with cocycles ω_1, ω_2) both arise as virtual representations of the same injective MPS tensor A under an on-site symmetry U , then $\omega_1 \sim \omega_2$.*

Proof. For each g , gauge uniqueness gives a nonzero scalar $f(g)$ with $X_2(g^{-1}) = f(g) X_1(g^{-1})$. Equivalently, for each fixed g the two local gauges

$$\begin{array}{c} i \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} X_1(g^{-1}) \\ \bullet \\ X_1(g^{-1})^{-1} \end{array} \quad \text{and} \quad \begin{array}{c} i \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} X_2(g^{-1}) \\ \bullet \\ X_2(g^{-1})^{-1} \end{array}$$

describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling $X_1(g \cdot h)$ yields $\omega_2(g, h) = f(g) f(h) f(gh)^{-1} \omega_1(g, h)$. $\square$

Definition 18.21 (Multiplicative 2-cocycle condition). A scalar 2-cochain $\omega: G \times G \rightarrow \mathbb{C}^\times$ is a *2-cocycle* if it satisfies

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k) \quad \text{for all } g, h, k \in G.$$

Definition 18.22 (Second cohomology quotient). The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times) / \sim,$$

where $\sim$ is the coboundary-equivalence relation.

Definition 18.23 (Projective equivalence). Two projective representations are *projectively equivalent* when their factor systems are cohomologous.

Theorem 18.24 (Projective equivalence and cohomologous cocycles). *Let ρ_1, ρ_2 be projective representations with factor systems ω_1, ω_2 . Then*

$$\rho_1 \sim_{\text{proj}} \rho_2 \iff \omega_1 \sim \omega_2.$$

Proof. This is immediate from the definition of projective equivalence as cohomology-class equality for factor systems. $\square$

18.4 Permutation-Based Physical Symmetry

The linear-combination twist $\tilde{A}_g^i = \sum_j U(g)_{ij} A^j$ is the standard form used in the virtual representation theorem. An alternative formulation replaces the matrix action by a permutation of the physical index; this is useful when the symmetry acts by reordering basis states rather than by a general linear map.

Definition 18.25 (Permutation symmetry data). An *on-site symmetry datum* for a group G on a physical space of dimension d consists of a matrix-valued map $u : G \rightarrow M_d(\mathbb{C})$ and a physical-index action $\sigma : G \rightarrow \text{End}(\{0, \dots, d-1\})$.

Definition 18.26 (Permutation-twisted tensor). Given on-site symmetry data (u, σ) , the *permutation-twisted tensor* at group element g is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}.$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:

Lemma 18.27 (Identity permutation twist). *If $\sigma(1) = \text{id}$, then $(A^{\sigma_1})^i = A^i$ for all i .*

Proof. Immediate from $\sigma(1)(i) = i$ for all i . $\square$

Composition order. The linear-combination twist (Lemma 18.5) composes as $\widetilde{(\tilde{A}_h)_g} = \tilde{A}_{gh}$, applying h first then g . The permutation twist below uses the same left-action convention $\sigma(gh) = \sigma(g) \circ \sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner permutation $\sigma(h)$ acts first on the index.

Lemma 18.28 (Composition of permutation twists). *If $\sigma(gh) = \sigma(g) \circ \sigma(h)$ for all $g, h \in G$, then*

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

Proof. Expanding both sides: $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. Diagrammatically this is just two consecutive relabellings of the same local tensor:



□

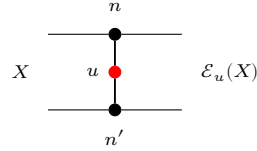
18.5 String Order and Local Symmetry Equivalence

The string order parameter [PGWS⁺08] detects whether an on-site unitary u acts as a local symmetry on the state generated by an injective MPS tensor.

Definition 18.29 (Twisted transfer map). Given an MPS tensor A and a physical-index matrix u , the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n' | u | n \rangle A_n X A_{n'}^\dagger.$$

Diagrammatically one inserts the physical operator u on the doubled local transfer cell:

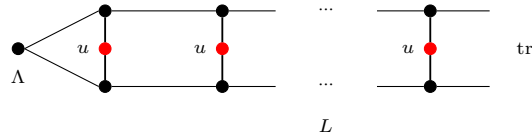


The upper and lower black nodes denote A_n and $A_{n'}^\dagger$, and the red node labelled u couples the physical legs.

Definition 18.30 (String order parameter). The *string order parameter* for twist u and stationary state Λ at length L is

$$R_L(u) := \text{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

In tensor-network notation this is the length- L doubled chain with a copy of u inserted on each physical rung:



The boundary matrix Λ is contracted into the virtual boundary, the red nodes labelled u sit on the physical rungs, and the right boundary is traced.

Definition 18.31 (Boundary string order parameter). For a boundary state Λ and boundary matrices $X, Y \in M_D(\mathbb{C})$, the *boundary string order parameter* is

$$R_L(u; X, Y) := \text{tr}(\Lambda X \mathcal{E}_u^L(Y)).$$

The ordinary string order parameter is the special case $X = Y = \mathbb{1}$.

Definition 18.32 (Local symmetry). An MPS tensor A has *local symmetry* under a unitary u with respect to a boundary state Λ if there exist a unitary matrix V and a phase μ with $|\mu| = 1$ such that $V^\dagger \Lambda V = \Lambda$ and

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger \quad \text{for all } i.$$

Informally, this means $\|R_L(u)\| = \|R_L(\mathbb{1})\|$ for all L .

Definition 18.33 (String order). String order *exists* for u with respect to a boundary state Λ if there exist boundary matrices X, Y and a positive constant $c > 0$ such that $c \leq \|R_L(u; X, Y)\|$ for all L , where $R_L(u; X, Y)$ is the boundary-refined string-order parameter. Informally, this means the scalar string-order parameter $R_L(u)$ does not decay to zero.

Lemma 18.34 (String order prevents decay to zero). *If A has string order for u with respect to a boundary state Λ , then for some boundary matrices X, Y the sequence $R_L(u; X, Y)$ does not tend to 0 as $L \rightarrow \infty$.*

Proof. A uniform lower bound $c \leq \|R_L(u; X, Y)\|$ is incompatible with convergence to 0. $\square$

Definition 18.35 (Local physical intertwining). The local physical intertwining relation is $\sum_j u_{ij} A^j = V A^i V^\dagger$. Locally, the physical action of u can therefore be pushed to the virtual legs:

Definition 18.36 (Transfer-map covariance). Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation: $\mathcal{E}(V X V^\dagger) = V \mathcal{E}(X) V^\dagger$. In doubled-layer notation, the virtual operators slide through the local transfer cell:

Definition 18.37 (Commutation in the doubled transfer picture). In the doubled transfer picture, one asks that the doubled transfer matrix commutes with $V \otimes \bar{V}$, expressed as $V \mathcal{E}_1(X) V^\dagger = \mathcal{E}_1(V X V^\dagger)$.

Theorem 18.38 (Transfer-map covariance and doubled commutation are equivalent). *Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix V .*

Proof. Both sides express the same local picture with opposite orientation:

The two formulas are literally the same identity read from opposite sides. $\square$

Lemma 18.39 (Unitary Kraus mixing). *If u is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel: $\sum_i (\sum_j u_{ij} A_j) Y (\sum_j u_{ij} A_j)^\dagger = \sum_i A_i Y A_i^\dagger$.*

Proof. This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Section 5.8. $\square$

Theorem 18.40 (Local physical intertwining implies transfer-map covariance). *If V and u are unitary and the local physical intertwining relation holds, then the transfer map is covariant.*

Proof. Starting from

one inserts $V^\dagger V = \mathbb{1}$ so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical u -boxes disappear by the unitary Kraus-mixing identity. $\square$

Theorem 18.41 (Transfer-map covariance implies local physical intertwining). *For an injective MPS, if V is unitary and the transfer map is covariant, then there exists a unitary u satisfying the local physical intertwining relation.*

Proof. Set $B^i := V A^i V^\dagger$. The transfer-map covariance relation

implies that the two Kraus families B and A define the same transfer map. The missing step — supplied by Kraus freedom for equal channels — converts equality of transfer maps $\mathcal{E}_B = \mathcal{E}_A$ into a unitary Kraus mixing matrix u with $B^i = \sum_j u_{ij} A^j$. $\square$

Definition 18.42 (Twisted companion tensor). For an MPS tensor A and a physical-index matrix u , define the companion tensor B_u by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}} A^{n'}.$$

Lemma 18.43 (Twisted transfer as a mixed transfer operator). *The twisted transfer map of A is the mixed transfer operator of the pair (A, B_u) :*

$$\mathcal{E}_u = F_{A, B_u}.$$

Proof. Expand both definitions and rearrange the double sum. $\square$

Theorem 18.44 (Twisted-transfer eigenvalues have modulus at most one). *Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If*

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$, then $|\lambda| \leq 1$.

Proof. By Lemma 18.43, the twisted transfer map is the mixed transfer operator F_{A,B_u} . Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common TP gauge. Theorem 9.14 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue λ . $\square$

Theorem 18.45 (Modulus-one twisted eigenvalue implies gauge-phase equivalence). *Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If*

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$ with $|\lambda| = 1$, then A is gauge-phase equivalent to the companion tensor B_u .

Proof. By Lemma 18.43, the twisted transfer map is the mixed transfer operator F_{A,B_u} . Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common TP gauge. In that gauge the eigenvalue equation becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 9.28 yields gauge-phase equivalence. Undoing the common TP gauge gives the claim for A and B_u . $\square$

Lemma 18.46 (Spectral radius < 1 forces boundary-string decay). *If $\rho(\mathcal{E}_u) < 1$, then for all boundary matrices $X, Y, \Lambda \in M_D(\mathbb{C})$ one has*

$$R_L(u; X, Y) \rightarrow 0 \quad \text{as } L \rightarrow \infty.$$

Proof. In finite dimension, $\rho(\mathcal{E}_u) < 1$ implies $\mathcal{E}_u^L \rightarrow 0$ in operator norm. Composing with the fixed trace pairing $Z \mapsto \text{tr}(\Lambda X Z)$ gives the stated scalar convergence. $\square$

Theorem 18.47 (String order implies gauge-phase equivalence with the companion tensor). *Let A be injective, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ , then A is gauge-phase equivalent to the companion tensor B_u .*

Proof. If $\rho(\mathcal{E}_u) < 1$, Lemma 18.46 forces every boundary string-order sequence to decay to zero, contradicting Definition 18.33. Hence $\rho(\mathcal{E}_u) \geq 1$. Theorem 18.44 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 18.45 then gives the gauge-phase equivalence. $\square$

Theorem 18.48 (Gauge-phase equivalence with the companion tensor yields a virtual unitary). *Let A be injective, assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A is gauge-phase equivalent to the companion tensor B_u , then there exist a unitary V and a phase μ with $|\mu| = 1$ such that*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger \quad \text{for all } i.$$

Proof. Start from an invertible intertwiner X between A and B_u . The positive matrix $Q := XX^\dagger$ is a positive fixed point of $\mathcal{E}_A$. By uniqueness of positive fixed points for an injective normalized tensor, Q is a positive scalar multiple of the identity. Rescaling X by the square root of that scalar produces a unitary V , and the original intertwining relation becomes the phased covariance equation above, with $|\mu| = 1$. $\square$

Theorem 18.49 (Virtual unitary gives a twisted-transfer eigenmatrix). *Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ and*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger \quad \text{for all } i,$$

with V unitary. Then V is an eigenmatrix of the twisted transfer map:

$$\mathcal{E}_u(V) = \mu V.$$

Proof. Expand $\mathcal{E}_u(V)$, insert the intertwining relation into each local term, and use $V^\dagger V = \mathbb{1}$ together with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. $\square$

Theorem 18.50 (Boundary-state invariance of the virtual unitary). *Let A be injective, assume $uu^\dagger = \mathbb{1}$, and let Λ be a positive definite boundary state with $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. If V is a unitary and μ a phase satisfying*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger \quad \text{for all } i,$$

then

$$V^\dagger \Lambda V = \Lambda.$$

Proof. The covariance relation shows that $V^\dagger \Lambda V$ is another positive fixed point of the adjoint transfer map. Since Λ is the normalized positive fixed point, uniqueness forces $V^\dagger \Lambda V = \Lambda$. $\square$

Lemma 18.51 (Local symmetry implies string order). *Assume $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If u is a local symmetry of A with respect to Λ , then string order exists for u .*

Proof. Choose the boundary matrices $X = V^\dagger$ and $Y = V$, where V is the unitary from the local-symmetry datum. By Theorem 18.49, one has $\mathcal{E}_u^L(V) = \mu^L V$. Therefore

$$R_L(u; V^\dagger, V) = \mu^L \text{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1. $\square$

Theorem 18.52 (Local symmetry iff a unitary peripheral eigenmatrix exists). *Let A be injective, and assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then u is a local symmetry if and only if there exist a unitary $V \in M_D(\mathbb{C})$ and a phase μ with $|\mu| = 1$ such that*

$$\mathcal{E}_u(V) = \mu V.$$

By Theorem 18.44, this witness form is equivalent to the peripheral-spectrum condition $\rho(\mathcal{E}_u) = 1$.

Proof. The forward direction is Theorem 18.49. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 18.45 turns it into a gauge-phase equivalence with the companion tensor; then Theorem 18.48 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 18.50 gives the boundary-state invariance needed to assemble the local-symmetry data. $\square$

Theorem 18.53 (String order $\Leftrightarrow$ local symmetry). *For an injective MPS, assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order for u exists iff u is a local symmetry.*

Proof. If string order exists, Theorem 18.47 gives a gauge-phase equivalence with the companion tensor. Theorem 18.48 extracts a virtual unitary and phase from that equivalence, and Theorem 18.50 shows that the same unitary preserves the boundary state, so one obtains local symmetry. Conversely, Lemma 18.51 turns local symmetry into a non-decaying boundary string-order sequence. $\square$

Theorem 18.54 (Virtual unitary from string order). *For an injective MPS, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ , then there exists a virtual unitary V and a unit-modulus scalar μ such that $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ for all i . Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:*

Proof. String order first gives gauge-phase equivalence with the companion tensor by Theorem 18.47. Theorem 18.48 then normalizes the resulting intertwiner to a unitary and produces the phase μ . $\square$

18.6 SPT Phase Classification

Two injective symmetric tensors belong to the same symmetry-protected topological phase when their virtual representation cocycles are cohomologous [CGW11].

Definition 18.55 (Same SPT phase). Injective tensors A and B , both symmetric under an on-site representation $U: G \rightarrow \text{GL}_d(\mathbb{C})$, are in the *same SPT phase* if there exist virtual cocycles ω_A, ω_B with corresponding projective representations that intertwine the respective twisted tensors and satisfy $\omega_A \sim \omega_B$ (cohomologous).

Theorem 18.56 (String order holds for injective symmetric MPS). *Let A be an injective tensor, symmetric under a unitary representation U . Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order exists for every group element g .*

Proof. The virtual representation theorem produces $\rho(g^{-1}) \in \text{GL}_D(\mathbb{C})$ satisfying $\sum_j U(g)_{ij} A^j = \rho(g^{-1}) A^i \rho(g^{-1})^{-1}$. A direct computation using $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ shows that $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$, so $\rho(g^{-1})$ is a nonzero eigenvector with eigenvalue 1. Theorem 18.45 upgrades this to gauge-phase equivalence with the companion tensor, and Theorem 18.48 extracts a virtual unitary and phase. By Theorem 18.50, this unitary preserves the boundary state, so Lemma 18.51 gives string order. $\square$

Theorem 18.57 (String order is an SPT invariant). *If A and B are injective, both on-site symmetric under a unitary representation U , and if $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$, Λ_A and Λ_B are positive definite, $\text{tr}(\Lambda_A) = 1$, $\text{tr}(\Lambda_B) = 1$, $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$, and $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$, then for every $g \in G$, string order exists for A with twist $U(g)$ if and only if it exists for B . In particular, since Definition 18.55 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase.*

Proof. Both directions follow from Theorem 18.56, which shows that string order holds universally for injective symmetric tensors with canonical normalisations. $\square$

Chapter 19

Parent Hamiltonians

This chapter introduces the local ground-space construction and the parent interaction for translation-invariant periodic MPS. We follow the standard setup from [CPGSV21, PGVWC07], with the N -site Hilbert space modelled as coefficient functions

$$\{0, \dots, d-1\}^N \rightarrow \mathbb{C},$$

rather than as an explicit tensor product.

Definition 19.1 (N -site coefficient space). For local dimension d and chain length N , the ambient coefficient space is

$$\mathcal{H}_N^{(d)} := \{0, \dots, d-1\}^N \rightarrow \mathbb{C}.$$

This is the concrete model used throughout the chapter for periodic-chain states.

19.1 Local ground space $G_L(A)$

Fix an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ with $A^i \in M_D(\mathbb{C})$ and a block length $L \geq 1$. For a word $w = (i_1, \dots, i_L)$, write $A^w := A^{i_1} \dots A^{i_L}$. The local ground-space map is

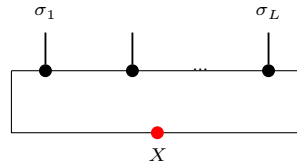
$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}), \quad \Gamma_L(X)(\sigma) := \text{tr}(A^\sigma X),$$

where A^σ abbreviates $A^{\sigma(0)} \dots A^{\sigma(L-1)}$.

Definition 19.2 (Ground-space map). The map Γ_L above is a $\mathbb{C}$ -linear map

$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

In tensor-network notation,



the upper chain denotes the word tensor A^σ with open physical indices $\sigma_1, \dots, \sigma_L$, and the lower red node denotes the boundary matrix X inserted on the closing virtual bond before taking the trace. The chain length L is fixed by the surrounding formula, not by an extra label inside the diagram.

Lemma 19.3 (Evaluation formula). *For every boundary matrix $X \in M_D(\mathbb{C})$ and every length- L word σ ,*

$$\Gamma_L(X)(\sigma) = \text{tr}(A^\sigma X).$$

Definition 19.4 (Local ground space). The *local ground space* is the image

$$G_L(A) := \text{im}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

Lemma 19.5 (Dimension bound). *For every L ,*

$$\dim G_L(A) \leq D^2.$$

Proof. Since $G_L(A)$ is the range of a linear map from $M_D(\mathbb{C})$,

$$\dim G_L(A) \leq \dim M_D(\mathbb{C}) = D^2.$$

□

Lemma 19.6 (Ambient-space dimension). *The coefficient-function model satisfies*

$$\dim(\{0, \dots, d-1\}^L \rightarrow \mathbb{C}) = d^L.$$

Lemma 19.7 (Nontriviality criterion). *If $d^L > D^2$, then $G_L(A)$ is a proper subspace of the local Hilbert space:*

$$G_L(A) \neq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

Proof. If $G_L(A)$ were the whole space, its dimension would be d^L by Lemma 19.6; Lemma 19.5 gives the contradiction $d^L \leq D^2$. □

19.2 Parent interaction and chain Hamiltonian

The parent interaction at range L is the orthogonal projector onto $G_L(A)^\perp$ in the L -site Euclidean space.

Definition 19.8 (Euclidean transport of the local ground space). Via the canonical identification between coefficient functions and the Euclidean space $\ell^2(\{0, \dots, d-1\}^L)$, one transports the local ground space to a Hilbert-space subspace

$$G_L^{\text{ES}}(A) \subseteq \ell^2(\{0, \dots, d-1\}^L).$$

Definition 19.9 (Parent interaction). The parent interaction at range L is the orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp},$$

on the local L -site space.

Lemma 19.10 (Parent interaction kills ground-space elements). *For any $v \in G_L(A)$, we have $h_L(A)v = 0$.*

Proof. Since $h_L(A)$ is the orthogonal projector onto $G_L(A)^\perp$, it annihilates every element of $G_L(A)$. □

Definition 19.11 (Periodic window extraction). For a configuration $\sigma \in \{0, \dots, d-1\}^N$ on the periodic chain and a starting site i , the extracted word

$$\text{extractWindow}_{L,i}(\sigma) \in \{0, \dots, d-1\}^L$$

records the entries of σ on the cyclic interval $i, i+1, \dots, i+L-1$ modulo N .

Definition 19.12 (Periodic window replacement). If $L \leq N$, the configuration

$$\text{replaceWindow}_{L,i}(\sigma, \tau)$$

is obtained from σ by replacing the cyclic window starting at i by the length- L word τ .

Lemma 19.13 (Extracting a replaced window). *If $L \leq N$, then extracting the same cyclic window after replacement recovers the inserted word:*

$$\text{extractWindow}_{L,i}(\text{replaceWindow}_{L,i}(\sigma, \tau)) = \tau.$$

For a periodic chain of length N , the parent Hamiltonian is the sum of translated copies of the local interaction.

Definition 19.14 (Translated local term). For each site index $i \in \{0, \dots, N-1\}$, the translated local term h_i embeds the parent interaction into the full N -site space at position i (with periodic boundary conditions). Thus h_i acts on the cyclic interval $[i, i+L-1]$ and as the identity outside that interval.

Definition 19.15 (Parent Hamiltonian). The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

Definition 19.16 (Frustration-free condition). A state ψ is frustration-free if each local term annihilates it:

$$\forall i \in \{0, \dots, N-1\}, \quad h_i \psi = 0.$$

Lemma 19.17 (MPV window membership). *For any window of L consecutive sites starting at position i in an N -site periodic chain ($L \leq N$), the restriction of the MPV state to that window lies in $G_L(A)$.*

Proof. The witness is the product of A -matrices on the complement sites. Trace cyclicity rotates the full N -site product so that window indices come first, matching the ground-space map definition. $\square$

Lemma 19.18 (Local term annihilates the MPV). *For each site i , $h_i V^{(N)}(A) = 0$.*

Proof. Combines MPV window membership with the fact that the parent interaction kills ground-space elements. $\square$

Lemma 19.19 (Parent Hamiltonian annihilates the MPV). *For every N , the MPV state satisfies*

$$H_N(A, L) V^{(N)}(A) = 0.$$

Proof. Sum of zeros, since each local term annihilates the MPV. $\square$

Lemma 19.20 (Frustration-freeness of the MPV). *The MPV state is frustration-free for the parent model.*

Proof. Immediate from Lemma 19.18. $\square$

19.3 Intersection property and unique ground state

This section develops the intersection property of local ground spaces for injective MPS and the resulting uniqueness of the ground state on the periodic chain.

19.3.1 Restriction maps

Given a state ψ on $L + 1$ sites, we define two restriction maps.

Remark 19.21 (Word and restriction helper formulas). Appending a last letter to a word corresponds to right multiplication by the corresponding tensor letter, while prepending a first letter corresponds to left multiplication. The restriction operators are the associated linear maps on coefficient spaces, together with their pointwise formulas.

Definition 19.22 (Left restriction). Fix the last physical index j :

$$(\text{restrictLast}\psi j)(\sigma) := \psi(\sigma_1, \dots, \sigma_L, j).$$

Definition 19.23 (Right restriction). Fix the first physical index i :

$$(\text{restrictFirst}\psi i)(\sigma) := \psi(i, \sigma_1, \dots, \sigma_L).$$

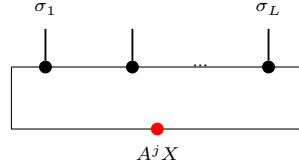
Definition 19.24 (Left ground membership). A state ψ on $L + 1$ sites satisfies the left ground condition if $\text{restrictLast}\psi j \in G_L(A)$ for every j .

Definition 19.25 (Right ground membership). A state ψ on $L + 1$ sites satisfies the right ground condition if $\text{restrictFirst}\psi i \in G_L(A)$ for every i .

19.3.2 Forward direction: ground space restricts

Lemma 19.26 (Left restriction). *If $\psi \in G_{L+1}(A)$, i.e. $\psi(\sigma) = \text{tr}(A^\sigma X)$ for some X , then $\text{restrictLast}\psi j \in G_L(A)$ with witness $A^j X$.*

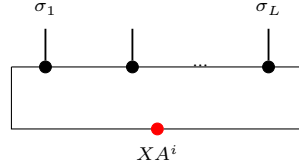
Proof. The restriction simply absorbs the last tensor into the boundary matrix:



Direct computation: $\psi(\sigma_1, \dots, \sigma_L, j) = \text{tr}(A^{\sigma_1} \dots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma)$. $\square$

Lemma 19.27 (Right restriction). *If $\psi \in G_{L+1}(A)$ with $\psi(\sigma) = \text{tr}(A^\sigma X)$, then $\text{restrictFirst}\psi i \in G_L(A)$ with witness XA^i .*

Proof. On the right restriction one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity: $\psi(i, \sigma_1, \dots, \sigma_L) = \text{tr}(A^i A^\sigma X) = \text{tr}(A^\sigma \cdot XA^i) = \Gamma_L(XA^i)(\sigma)$. $\square$

19.3.3 Injectivity of the ground-space map

Theorem 19.28 (Ground-space map injectivity). *If A is injective and $L \geq 1$, then Γ_L is injective.*

Proof. If $\Gamma_L(X) = 0$ then $\text{tr}(A^\sigma X) = 0$ for every length- L word σ . Since A is injective, the products $\{A^\sigma\}_\sigma$ span $M_D(\mathbb{C})$, so the trace pairing gives $X = 0$. $\square$

Theorem 19.29 (Ground-space map injectivity from word span). *If the length- L products $\{A^\sigma : |\sigma| = L\}$ span $M_D(\mathbb{C})$, then Γ_L is injective.*

Proof. If $\Gamma_L(X) = 0$, then $\text{tr}(A^\sigma X) = 0$ for every word σ of length L . The spanning hypothesis turns this into $\text{tr}(MX) = 0$ for every $M \in M_D(\mathbb{C})$, and the trace pairing gives $X = 0$. $\square$

Lemma 19.30 (Ground-space map injectivity for block-injective tensors). *If A is L_0 -block-injective, then Γ_{L_0} is injective.*

Proof. By definition, L_0 -block injectivity says precisely that the length- L_0 products span $M_D(\mathbb{C})$. Apply Theorem 19.29. $\square$

Theorem 19.31 (Ground-space dimension). *If A is injective and $L \geq 1$, then $\dim G_L(A) = D^2$.*

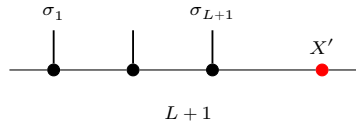
Proof. The upper bound $\dim G_L(A) \leq D^2$ is Lemma 19.5. Injectivity of Γ_L gives $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$. $\square$

19.3.4 The intersection property

Theorem 19.32 (Intersection property). *If A is injective and $L \geq 2$, then a state ψ on $L+1$ sites satisfying both the left and right ground conditions lies in $G_{L+1}(A)$:*

$$\psi \in G_L^{\text{left}} \cap G_L^{\text{right}} \implies \psi \in G_{L+1}(A).$$

Proof. The “invert-and-regrow” argument compares the left and right restrictions on their common $(L-1)$ -site overlap and then reconstructs the full $(L+1)$ -site word from a single boundary matrix:



1. From the right ground condition, for each physical index i there exists a unique $Y_i \in M_D(\mathbb{C})$ (by injectivity of Γ_L) such that $\psi(i, \sigma) = \text{tr}(A^\sigma Y_i)$.
2. Similarly, from the left ground condition, for each j there exists a unique Z_j with $\psi(\sigma, j) = \text{tr}(A^\sigma Z_j)$.
3. Matching on the $(L-1)$ -site overlap and using nondegeneracy of the trace pairing: $A^j Y_i = Z_j A^i$ for all i, j .
4. Since A is injective, the matrices $\{A^j\}$ span $M_D(\mathbb{C})$ (Definition 2.21), so $\mathbb{1} = \sum_j c_j A^j$ for some coefficients $c_j \in \mathbb{C}$. Then $Y_i = \mathbb{1} \cdot Y_i = \sum_j c_j A^j Y_i = \sum_j c_j Z_j A^i = X' A^i$, where $X' := \sum_j c_j Z_j$.
5. By trace cyclicity, $\psi(\sigma) = \text{tr}(A^\sigma X')$, so $\psi \in G_{L+1}(A)$.

□

Theorem 19.33 (Intersection characterization). *For injective A and $L \geq 2$: $\psi \in G_{L+1}(A) \iff \psi \in G_L^{\text{left}} \cap G_L^{\text{right}}$.*

Proof. Immediate from the two forward-direction lemmas and Theorem 19.32. □

19.3.5 Unique ground state

Contiguous and cyclic window operators

Remark 19.34 (Contiguous window operators). For a non-wrapping interval $[s, s + M - 1] \subseteq \{0, \dots, N - 1\}$, one can insert a length- M word into that interval and keep the complementary sites fixed. The associated restriction map evaluates an N -site state on those configurations. The auxiliary lemmas record the full-window case $s = 0$, $M = N$, and the compatibility of this restriction with deleting the last or first site.

Remark 19.35 (Cyclic window operators). On the periodic chain, a cyclic window starting at i reads the sites $i, i + 1, \dots, i + L - 1$ modulo N . When this window does not wrap around the ring, the cyclic construction agrees with the contiguous one, and so do the two restriction maps.

Lemma 19.36 (Iterated contiguous-window intersection). *If an N -site state satisfies the L -site ground condition on every non-wrapping contiguous window (with $L \geq 2$), then it lies in $G_N(A)$.*

Proof. The induction repeatedly merges two neighbouring admissible windows $[i, i + L - 1]$ and $[i + 1, i + L]$ into the larger window $[i, i + L]$ by applying Theorem 19.32 to their common overlap $[i + 1, i + L - 1]$. Strong induction on window size, combining adjacent windows via Theorem 19.32 at each step. □

Definition 19.37 (Suffix restriction). Fix a prefix $u : \{0, \dots, K - 1\} \rightarrow \{0, \dots, d - 1\}$. The suffix restriction of an $(K + L)$ -site state ψ is the L -site state given by

$$(\text{tailRestrict}_u \psi)(\sigma) = \psi(u, \sigma).$$

Definition 19.38 (Suffix ground membership). A state ψ on $K + L$ sites lies in the suffix ground condition if every fixed prefix u gives a suffix state in $G_L(A)$.

Lemma 19.39 (Ground-space suffix restriction). *If $\psi \in G_{K+L}(A)$, then every fixed-prefix suffix restriction of ψ lies in $G_L(A)$.*

Proof. Write $\psi(\rho) = \text{tr}(A^\rho X)$ for a boundary matrix X . Fixing a prefix u gives

$$\psi(u, \sigma) = \text{tr}(A^u A^\sigma X) = \text{tr}(A^\sigma X A^u),$$

so the suffix restriction again has the form of a ground-space vector. □

Lemma 19.40 (Left-tail compatibility). *Assume ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every fixed prefix of length K . Then there exist matrices $(Z_j)_j$ and $(Y_u)_u$ such that*

$$\psi(\sigma, j) = \text{tr}(A^\sigma Z_j), \quad \psi(u, \tau) = \text{tr}(A^\tau Y_u),$$

and for all j and u ,

$$Z_j A^u = A_j Y_u.$$

Proof. Choose Z_j from the long left-window condition and Y_u from the suffix ground condition. Restricting the suffix state at its last site gives the boundary matrix $A_j Y_u$, while first fixing the last site and then taking the suffix gives $Z_j A^u$. The two restrictions are the same because appending the fixed prefix and then the last letter gives the same configuration as appending the last letter to the suffix and then placing the prefix in front. Injectivity of Γ_{L_0} , obtained from L_0 -block injectivity, yields $Z_j A^u = A_j Y_u$. $\square$

Theorem 19.41 (Common right factor from word compatibility). *Assume A is L_0 -block injective with $L_0 > 0$. If matrices $(Z_j)_j$ satisfy*

$$Z_j A^\sigma = A_j Y_\sigma$$

for every word σ of length K , then there exists a matrix X such that $Z_j = A_j X$ for all j .

Proof. For $K = 1$, the hypothesis gives $Z_j A_i = A_j Y_i$ for every letter i . Every blocked word of length L_0 is nonempty, so this identity extends to the coefficients of the blocked tensor $A^{[L_0]}$. Since $A^{[L_0]}$ is injective, the identity matrix is a linear combination of those blocked coefficients, and substituting this decomposition yields $Z_j = A_j X$ for a common matrix X . For larger K , strip the first letter: for each i , the matrices $Z_j A_i$ satisfy the same compatibility hypothesis with words of length $K - 1$, so the induction hypothesis reduces the problem to the case $K = 1$. $\square$

Theorem 19.42 (Open-chain right extension at the injectivity length). *Assume A is L_0 -block injective with $L_0 > 0$. If an element ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every prefix of length K , then $\psi \in G_{K+L_0+1}(A)$.*

Proof. Lemma 19.40 produces matrices $(Z_j)_j$ and $(Y_u)_u$ with $Z_j A^u = A_j Y_u$ for every prefix word u of length K . Theorem 19.41 gives a common right factor X such that $Z_j = A_j X$ for all j . Therefore each last-site restriction of ψ agrees with the corresponding last-site restriction of the ground-space vector determined by X . Since every configuration is obtained by adjoining its final letter to its initial $(K + L_0)$ -tuple, the two states coincide, so ψ lies in $G_{K+L_0+1}(A)$. $\square$

Theorem 19.43 (Boundary matrix commutation from the wrapping window). *Let A be injective with $D \geq 1$, and let $\psi = \Gamma_N(X)$ for some boundary matrix $X \in M_D(\mathbb{C})$. If every cyclic restriction of length L of ψ lies in $G_L(A)$, with $N \geq 2$ and $1 < L \leq N$, then*

$$X A^j = A^j X \quad \text{for every } j = 0, \dots, d - 1.$$

Proof. The wrapping window across the boundary produces matrix identities of the form $X M W = M Y$ for arbitrary test matrices M and complementary words W . Spanning first over the length- $(L - 1)$ tails and then over the complementary words, injectivity forces $(X M - M X) W = 0$ for every W , hence $X M = M X$ for every matrix M . In particular, X commutes with each letter A^j . $\square$

Definition 19.44 (MPV submodule). The subspace spanned by the MPV state $V^{(N)}(A)(\sigma) = \text{tr}(A^{\sigma_0} \dots A^{\sigma_{N-1}})$.

Lemma 19.45 (The MPV is the ground-space map at the identity). *For every chain length N ,*

$$V^{(N)}(A) = \Gamma_N(\mathbb{1}).$$

Lemma 19.46 (MPV lies in the ground space). *$V^{(N)}(A) \in G_N(A)$ for every N , with boundary matrix $X = I$.*

Proof. $V^{(N)}(A)(\sigma) = \text{tr}(A^\sigma) = \text{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$. $\square$

Definition 19.47 (Unique ground state). A submodule S has a unique ground state if $\dim S = 1$.

Theorem 19.48 (One-dimensionality criterion). *For finite-dimensional S , $\dim S = 1$ iff there exists a nonzero $\psi_0 \in S$ such that every $\psi \in S$ is of the form $c\psi_0$.*

Proof. Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector. $\square$

Definition 19.49 (Periodic chain ground space). For $N > 0$ and $L \leq N$, define the *periodic chain ground space*

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0, \dots, d-1\}^N \rightarrow \mathbb{C}) : \text{for every cyclic window } [i, i+L-1], \psi|_{[i, i+L-1]} \in G_L(A)\},$$

where the condition $\psi|_{[i, i+L-1]} \in G_L(A)$ means that for every choice of physical indices outside the window, the restriction of ψ to that window lies in $G_L(A)$. When $N = 0$ or $L > N$, set $\mathcal{G}_{N,L}(A) := \mathbb{T}$ by convention.

Lemma 19.50 (The MPV satisfies every cyclic window constraint). *If $0 < N$ and $L \leq N$, then*

$$V^{(N)}(A) \in \mathcal{G}_{N,L}(A).$$

Theorem 19.51 (MPV nonvanishing for block-injective tensors). *If A is L_0 -block-injective with $L_0 > 0$, $D \geq 1$, and $N \geq L_0 + 1$, then $V^{(N)}(A) \neq 0$ in the N -site space.*

Proof. Suppose for contradiction that $V^{(N)}(A) = 0$, i.e. $\text{tr}(A^\sigma) = 0$ for every word σ of length N . For any words w and u with $|w| = N - L_0$ and $|u| = L_0$, we have $\text{tr}(A^w A^u) = 0$. Since A is L_0 -block-injective, $\{A^u : |u| = L_0\}$ spans $M_D(\mathbb{C})$. By nondegeneracy of the trace pairing, $A^w = 0$ for every $|w| = N - L_0$. Repeating the argument with words of length $N - 2L_0$ (since $|w'| + L_0 = N - L_0$, the product $A^{w'} A^{u_1}$ has length $N - L_0$, so $A^{w'} A^{u_1} = 0$ by the previous step) and iterating, we eventually reach $\text{tr}(A^u) = 0$ for $|u| = L_0$, which gives $I = 0$, a contradiction. $\square$

19.3.6 Block-word stripping

Theorem 19.52 (Common right factor from block-word compatibility). *Assume A is L_0 -block-injective with $L_0 > 0$. Let $\{Z_u\}_{|u|=L_0}$ be a family indexed by words of length L_0 . If for some $K \geq 0$ and every suffix word w of length K there exists $Y_w \in M_D(\mathbb{C})$ such that*

$$Z_u A^w = A^u Y_w \quad \text{for every word } u \text{ of length } L_0,$$

then there exists $X \in M_D(\mathbb{C})$ with $Z_u = A^u X$ for every word u of length L_0 .

Proof. Induct on K . For $K = 0$ the empty suffix gives $Z_u = A^u Y_\emptyset$. For the successor step, fix the first physical index and apply the induction hypothesis to the shortened suffix. This reduces the statement to letter compatibility on the right. Extending that letter compatibility to all length- L_0 words and decomposing $\mathbb{1}$ in the spanning family $\{A^u : |u| = L_0\}$ yields a common factor X . $\square$

Theorem 19.53 (Block-word commutation from long-word commutation). *If A is L_0 -block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length- m product A^w , then X already commutes with every length- L_0 product A^u .*

Proof. Write each length- m word as a concatenation uw with $|u| = L_0$ and $|w| = m - L_0$. The hypothesis gives

$$XA^u A^w = A^u A^w X = A^u (A^w X).$$

Thus the family $Z_u := XA^u$ satisfies the hypothesis of Theorem 19.52. One obtains $XA^u = A^u R$ for all $|u| = L_0$. Decomposing $\mathbb{1}$ in the span of the length- L_0 words shows $R = X$, hence $XA^u = A^u X$. $\square$

Theorem 19.54 (Centrality from long-word commutation). *If A is L_0 -block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length- m word product, then X commutes with every matrix in $M_D(\mathbb{C})$.*

Proof. By Theorem 19.53, X commutes with every length- L_0 word product. These products span $M_D(\mathbb{C})$ by block injectivity, so linearity gives $XM = MX$ for every $M \in M_D(\mathbb{C})$. $\square$

Lemma 19.55 (Generator commutation from long-word commutation). *Under the hypotheses of Theorem 19.54, one has $XA^i = A^i X$ for every physical index i .*

Proof. Apply Theorem 19.54 with $M = A^i$. $\square$

Theorem 19.56 (Chain ground space equals the MPV line in the injective case). *If A is injective, $D \geq 1$, $N \geq 2$, and $1 < L \leq N$, then*

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}.$$

Proof. Let $\psi \in \mathcal{G}_{N,L}(A)$. The non-wrapping window conditions and Lemma 19.36 imply $\psi \in G_N(A)$, so $\psi = \Gamma_N(X)$ for some boundary matrix X . The wrapping window condition now applies Theorem 19.43 and forces X to commute with every letter A^j . Since A is injective, the letters generate the full matrix algebra, hence X is scalar. Therefore ψ is proportional to $V^{(N)}(A)$, while the reverse inclusion is Lemma 19.50. $\square$

Theorem 19.57 (Normal-range reduction: containment in the MPV line). *If A is normal, $D \geq 1$, L_0 -block-injective, with $N \geq 2$ and $L_0 < L \leq N$, then*

$$\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}.$$

Proof. The remaining step is the periodic range-reduction theorem from [CPGSV21, § IV.C]: one must show that the reduced cyclic window constraints still force the same boundary-matrix commutation relation as in Theorem 19.43. $\square$

Theorem 19.58 (Chain ground space equals the MPV line in the normal case). *If A is normal, $D \geq 1$, L_0 -block-injective, with $N \geq 2$ and $L_0 < L \leq N$, then*

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}.$$

Theorem 19.59 (Unique ground state on the periodic chain). *For an injective tensor A with $D \geq 1$, if $N \geq 2$ and $1 < L \leq N$, then the periodic chain ground space is one-dimensional.*

Proof. By Theorem 19.56, the ground space is the line $\text{span}\{V^{(N)}(A)\}$. It remains to show that the MPV is nonzero. If $V^{(N)}(A) = 0$, then Lemma 19.45 gives

$$\Gamma_N(\mathbb{1}) = \Gamma_N(0).$$

Since A is injective and $N > 0$, Theorem 19.28 forces $\mathbb{1} = 0$, a contradiction. Therefore the spanning line is one-dimensional. $\square$

Theorem 19.60 (Unique ground state for block-injective tensors). *If A is L_0 -block-injective with $L_0 > 0$ and $D \geq 1$, the parent Hamiltonian with interaction range $2L_0$ has a unique ground state on the periodic chain.*

Theorem 19.61 (Optimal unique ground state for normal tensors). *If A is normal (hence L_0 -block-injective for some $L_0 > 0$), $D \geq 1$, and in normal form, the interaction range can be reduced to $L_0 + 1$.*

19.4 Commuting parent Hamiltonians

A parent Hamiltonian has *commuting* local terms when all translated interaction projectors commute with each other.

Definition 19.62 (Commuting parent Hamiltonian). The parent Hamiltonian with block length L on N sites has *commuting* local terms if $h_i h_j = h_j h_i$ for all $i, j \in \{0, \dots, N-1\}$. See Ref. [CPGSV21, Definition 3.9].

Definition 19.63 (Nearest-neighbor commuting parent Hamiltonian). A *nearest-neighbor commuting parent Hamiltonian* (NNCPH) is a commuting parent Hamiltonian with block length $L = 2$. See Ref. [CPGSV21, Definition 3.9].

Remark 19.64 (Elementary consequences of the commuting definition). The nearest-neighbor condition is simply the length-2 specialization of the general commuting condition; the latter is symmetric in the two sites, and it implies that the full parent Hamiltonian commutes with each local interaction term.

Theorem 19.65 (RFP implies NNCPH). *If A is a normal renormalization fixed point, then for every chain length $N \geq 2$ the parent Hamiltonian with $L = 2$ has commuting local terms. In the current formal development, this implication is supplied by the sanctioned axiom recording the product-of-entangled-pairs description from Appendix B of Theorem 3.10 in [CPGSV21].*

Remark 19.66 (Product-pair data for future NNCPH proofs). On an even chain, one can extract the p -th adjacent two-site word of a configuration and compare the MPS coefficients with a repeated two-site pair state. The corresponding data record both that coefficient identity and a family of idempotent chain-level projectors whose values are the nearest-neighbor local terms. The final theorem in these data shows that these local terms commute. Consequently the same data yields the nearest-neighbor commuting condition on every finite chain. This isolates the remaining proof of Theorem 19.65 to the extraction of such data from the Appendix B structural form.

Theorem 19.67 (NNCPH implies RFP). *If A is normal, in left-canonical form, and its parent Hamiltonian with $L = 2$ has commuting local terms for every chain length $N \geq 2$, then A is a renormalization fixed point. In the current formal development, this implication is supplied by the sanctioned axiom recording Beigi’s ground-space characterization of one-dimensional commuting nearest-neighbor Hamiltonians; the left-canonical hypothesis fixes the normalization of the transfer map.*

19.5 Decorrelation and commuting parent Hamiltonians

This section develops the abstract operator-algebraic formulation of decorrelation and its connection to commuting parent Hamiltonians, following Appendix D, §D.2 of [CPGSV21].

Definition 19.68 (Decorrelation). Given an idempotent endomorphism P_K and sets of observables $\mathcal{O}_A, \mathcal{O}_B$, regions A and B are *decorrelated* w.r.t. K when $P_K \circ O_A \circ (1 - P_K) \circ O_B \circ P_K = 0$ for all $O_A \in \mathcal{O}_A, O_B \in \mathcal{O}_B$.

Definition 19.69 (Commuting parent Hamiltonian structure). A subspace K (given by its idempotent endomorphism P_K) has a *commuting parent Hamiltonian structure* if there exist idempotent endomorphisms P_{AX} and P_{XB} that commute and satisfy $P_{AX} \circ P_{XB} = P_K$.

Theorem 19.70 (Idempotence gives a trivial commuting parent Hamiltonian). *If*

$$P_K \circ P_K = P_K,$$

then P_K has a commuting parent Hamiltonian structure.

Proof. Take $P_{AX} = P_{XB} = P_K$. Their commutativity is immediate, and the product identity is exactly the assumed idempotence of P_K . $\square$

Lemma 19.71 (Cancellation for commuting idempotents). *If P and Q are commuting idempotents, then*

$$Q \circ (1 - P \circ Q) \circ P = 0.$$

Proof. Expand the middle factor, use commutativity to rewrite $QPQP$ as $QPPQ$, and then apply idempotence of P and Q . $\square$

Theorem 19.72 (Idempotence of the product projector). *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_K = P_K.$$

Proof. Since P_{AX} and P_{XB} are idempotent and commute, their product $P_{AX} \circ P_{XB}$ is idempotent. Substituting $P_K = P_{AX} \circ P_{XB}$ gives the claim. $\square$

Theorem 19.73 (Absorption by the left projector). *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_{AX} \circ P_K = P_K.$$

Proof. Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{AX} ,

$$P_{AX} \circ P_K = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

$\square$

Theorem 19.74 (Absorption by the right projector). *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_{XB} = P_K.$$

Proof. Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{XB} ,

$$P_K \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

$\square$

Theorem 19.75 (Cross-absorption by the right projector). *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_{XB} \circ P_K = P_K.$$

Proof. By commutativity, $P_{XB} \circ P_K = P_{XB} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB}$, and idempotence of P_{XB} yields $P_{XB} \circ P_K = P_K$. $\square$

Theorem 19.76 (Cross-absorption by the left projector). *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_{AX} = P_K.$$

Proof. By commutativity, $P_K \circ P_{AX} = P_{AX} \circ P_{XB} \circ P_{AX} = P_{AX} \circ P_{AX} \circ P_{XB}$, and idempotence of P_{AX} yields $P_K \circ P_{AX} = P_K$. $\square$

Theorem 19.77 (Reverse product identity). *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_{XB} \circ P_{AX} = P_K.$$

Proof. This is immediate from commutativity of P_{AX} and P_{XB} together with the identity $P_{AX} \circ P_{XB} = P_K$. $\square$

Theorem 19.78 (Commuting complementary projectors). *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition and*

$$Q_{AX} = 1 - P_{AX}, \quad Q_{XB} = 1 - P_{XB},$$

then

$$Q_{AX} \circ Q_{XB} = Q_{XB} \circ Q_{AX}.$$

Proof. Expanding both products and using $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$ shows that the complementary projectors commute. $\square$

Theorem 19.79 (Commuting parent Hamiltonian implies decorrelation). *If a subspace K has a commuting parent Hamiltonian decomposition $P_K = P_{AX} \circ P_{XB}$ with $[P_{AX}, P_{XB}] = 0$, and A -observables commute with P_{XB} while B -observables commute with P_{AX} , then regions A and B are decorrelated w.r.t. K . This is the backward direction of Proposition D.3 from [CPGSV21], Appendix D, §D.2.*

Proof. The key identity is $P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$. Using the commutation hypotheses to slide O_A past P_{XB} and O_B past P_{AX} , the five-fold composition factors through this zero term. $\square$

Theorem 19.80 (Commuting parent Hamiltonian implies decorrelation). *Corollary of Theorem 19.79: if P_K has a commuting parent Hamiltonian decomposition and observables respect locality, then regions A and B are decorrelated w.r.t. K .*

Lemma 19.81 (Frustration-free identity). *For any endomorphisms P and Q ,*

$$(1 - P) + (1 - Q) - (1 - P) \circ (1 - Q) = 1 - P \circ Q.$$

Proof. Expanding the left-hand side and cancelling gives the result. $\square$

Theorem 19.82 (Commuting parent Hamiltonian identity). *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = 1 - P_K,$$

where $Q_{AX} = 1 - P_{AX}$ and $Q_{XB} = 1 - P_{XB}$ are the complementary projectors. This is the frustration-free form of the parent Hamiltonian from [CPGSV21], Appendix D, §D.2.

Proof. Applying Lemma 19.81 to P_{AX} and P_{XB} , and substituting $P_{AX} \circ P_{XB} = P_K$. $\square$

Theorem 19.83 (Ground-space membership). *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K v = v \iff P_{AX} v = v \text{ and } P_{XB} v = v.$$

This is the algebraic form of equation (D.2) from [CPGSV21]: $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$.

Proof. ($\Rightarrow$) If $P_K v = v$, then $P_{AX}(P_K v) = (P_{AX} \circ P_K)v = P_K v = v$ by left absorption, and similarly for P_{XB} .

($\Leftarrow$) If $P_{AX} v = v$ and $P_{XB} v = v$, then $P_K v = (P_{AX} \circ P_{XB})v = P_{AX}(P_{XB} v) = P_{AX} v = v$. $\square$

19.6 Spectral gap

A frustration-free Hamiltonian is *gapped* if the first excited eigenvalue is bounded away from zero uniformly in the chain length N . For MPS parent Hamiltonians, the spectral gap follows from a martingale criterion due to Kastoryano and Lucia [KL18].

Definition 19.84 (Transported parent ground space). Denote by $\mathcal{G}_{N,L}^{\text{ES}}(A)$ the parent ground space after identifying the coefficient space with its Euclidean ℓ^2 -model.

Definition 19.85 (Transported parent Hamiltonian). Denote by $H_{N,L}^{\text{ES}}(A)$ the conjugate of $H_N(A, L)$ by the canonical identification between the coefficient space and its Euclidean avatar.

Theorem 19.86 (Euclidean transport of the parent ground space). *Under the canonical ℓ^2 transport of the N -site state space, the transported parent ground space is exactly the kernel of the transported parent Hamiltonian.*

Proof. This is an immediate unraveling of the definitions: both constructions are obtained from the same parent Hamiltonian by conjugating with the canonical identification between the site space and its Euclidean-space avatar, so the transported ground space is precisely the transported kernel. $\square$

Lemma 19.87 (Membership in the transported parent ground space). *A vector belongs to the transported parent ground space exactly when the transported parent Hamiltonian annihilates it.*

Lemma 19.88 (Quadratic form to norm bound). *Let H be a positive semidefinite self-adjoint operator on a finite-dimensional complex Hilbert space, and suppose that*

$$\gamma \langle v, Hv \rangle \leq \langle Hv, Hv \rangle \quad \text{for all } v,$$

i.e. $H^2 \geq \gamma H$ as a quadratic form, for some $\gamma > 0$. Then for every v in the orthogonal complement of $\ker H$,

$$\gamma \|v\| \leq \|Hv\|.$$

In particular, every positive eigenvalue of H is at least γ . The positive-semidefiniteness hypothesis is essential: without it, $H = -\text{Id}$ with $\gamma = 2$ satisfies the quadratic-form inequality vacuously but violates the norm bound.

Proof. By the finite-dimensional spectral theorem, H diagonalises in an orthonormal eigenbasis with real eigenvalues λ_k . For a unit eigenvector ψ_k with eigenvalue λ_k the hypothesis gives $\gamma \lambda_k \leq \lambda_k^2$, whence $\lambda_k \leq 0$ or $\lambda_k \geq \gamma$. Since H is positive semidefinite (its quadratic form is ≥ 0), the negative case is excluded and every nonzero eigenvalue satisfies $\lambda_k \geq \gamma$. Writing $v = \sum_k c_k \psi_k$ and projecting onto $(\ker H)^\perp$ keeps only terms with $\lambda_k \geq \gamma$, so $\|Hv\|^2 = \sum_k |c_k|^2 \lambda_k^2 \geq \gamma^2 \sum_k |c_k|^2 = \gamma^2 \|v\|^2$. $\square$

Theorem 19.89 (Martingale criterion for spectral gap). *Consider a frustration-free Hamiltonian*

$$H = \sum_{i=1}^n h_i$$

with projector interaction terms ($h_i^2 = h_i$). Suppose that for every pair of overlapping terms h_i, h_j there exist constants $c_{ij} \geq 0$ and $\gamma > 0$ satisfying

$$h_i h_j + h_j h_i \geq -c_{ij}(1 - \gamma)(h_i + h_j),$$

and suppose further that $c_{ij} = c_{ji}$ for every overlapping pair (i, j) and that $\sum_{j: j \neq i} c_{ij} \leq 1$ for every i . Then the smallest positive eigenvalue of H satisfies $\lambda_{\min}^+(H) \geq \gamma$.

Proof. The relevant geometry is a family of translated length- L projector windows on the ring; two terms h_i and h_j interact only when their supports overlap, equivalently when the cyclic distance $d_N(i, j)$ is smaller than L . From $h_i^2 = h_i$,

$$H^2 = \sum_i h_i + \sum_{\substack{i < j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i < j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ($h_i h_j = h_j h_i$), so their product $h_i h_j$ is itself a projector and hence PSD; thus the last sum is ≥ 0 . For the overlapping sum, the martingale condition gives $h_i h_j + h_j h_i \geq -c_{ij}(1 - \gamma)(h_i + h_j)$. Summing over unordered pairs $\{i, j\}$:

$$\sum_{\{i, j\}, \text{overlap}} (h_i h_j + h_j h_i) \geq -(1 - \gamma) \sum_{\{i, j\}, \text{overlap}} c_{ij} (h_i + h_j).$$

Collecting terms for each h_i and using the symmetry assumption $c_{ij} = c_{ji}$:

$$\sum_{\{i, j\}, \text{overlap}} c_{ij} (h_i + h_j) = \sum_i \left(\sum_{\substack{j \neq i \\ \text{overlap}}} c_{ij} \right) h_i \leq H,$$

where the last inequality uses the row-sum hypothesis $\sum_{j \neq i} c_{ij} \leq 1$. Combining, $H^2 \geq H - (1 - \gamma)H = \gamma H$. For any eigenvector with $H\psi = \lambda\psi$ and $\lambda > 0$, this gives $\lambda^2 \geq \gamma\lambda$, hence $\lambda \geq \gamma$ [KL18]. $\square$

Lemma 19.90 (Martingale condition for MPS). *Let A be injective and fix an interaction range $L \geq 2$ as in Definition 19.9. Let $h_L(A) := \Pi_{G_L(A)^\perp}$ be the L -site parent interaction, and for each chain length and site i let h_i denote its translate as in Definition 19.14. Then for every pair of overlapping terms h_i and h_j with $0 < d_N(i, j) < L$, where $d_N(i, j) := \min(|i - j|, N - |i - j|)$ is the cyclic distance on the ring (i.e. whose L -site supports share at least one site under periodic boundary conditions), the martingale condition of Theorem 19.89 holds with constants c_{ij} and $\gamma > 0$ depending only on A (not on N). Note: The proof below fully establishes the martingale*

constants only for $L = 2$; for $L > 2$ the existence of suitable constants is asserted without complete proof (see the remark at the end of the proof). Only the $L = 2$ case is used in this chapter (Theorem 19.92). (When $L = 1$ the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale machinery.)

Proof. Two terms h_i, h_j with $0 < d_N(i, j) < L$ have supports overlapping in $L - d_N(i, j)$ sites (where d_N is the cyclic distance on the N -site ring). Since A is injective, the intersection property (Theorem 19.32) gives the qualitative identity $\ker(h_i) \cap \ker(h_j) = G_{[i, j+L-1]}(A)$. Because the local Hilbert space is finite-dimensional, the subspaces $\ker(h_i)$ and $\ker(h_j)$ live in a fixed finite-dimensional space. The intersection identity shows that $\ker(h_i) \cap \ker(h_j)$ is exactly $G_{[i, j+L-1]}(A)$; in particular there is no non-trivial vector in one kernel that is orthogonal to the other yet lies in both. Hence the Friedrichs angle $\theta_{d_N(i, j)}$ between $\ker(h_i)$ and $\ker(h_j)$ is strictly positive (for the fixed injective tensor A). Because the angle depends only on the overlap pattern $d_N(i, j) \in \{1, \dots, L-1\}$ and not on the absolute position, the number of distinct angles is at most $L-1$. Let $\alpha := \max_{1 \leq s < L} \cos \theta_s < 1$ be the worst-case cosine of the Friedrichs angles.

Row-sum bound. Since $L \geq 2$, each site i overlaps with at most $2(L-1)$ neighbours (those j with $0 < d_N(i, j) < L$). Set $c_{ij} := \frac{1}{2(L-1)}$ for every overlapping pair and $c_{ij} := 0$ otherwise. These constants depend only on the cyclic distance $d_N(i, j)$ and therefore satisfy $c_{ij} = c_{ji}$. Then $\sum_{j \neq i} c_{ij} \leq 2(L-1) \cdot \frac{1}{2(L-1)} = 1$, satisfying the row-sum hypothesis of Theorem 19.89.

Positivity of γ . The Friedrichs angle bound gives, for each overlapping pair, $h_i h_j + h_j h_i \geq -\alpha(h_i + h_j)$. The martingale inequality holds with $\gamma = 1 - 2(L-1)\alpha$, which requires $c_{ij}(1-\gamma) = \frac{1}{2(L-1)} \cdot 2(L-1)\alpha = \alpha$ (matching the Friedrichs bound exactly). Since $\alpha < 1$ (the Friedrichs angle is strictly positive), it remains to show $\gamma > 0$, i.e. $\alpha < \frac{1}{2(L-1)}$. In the blocked picture of Theorem 19.92, $L = 2$ and each site overlaps with at most 2 neighbours, so the condition reduces to $\alpha < \frac{1}{2}$. For $L = 2$ with an injective tensor, the two projectors h_i and h_{i+1} act on a finite-dimensional space, and the intersection property (Theorem 19.32) guarantees that $\ker(h_i) \cap \ker(h_{i+1})$ is exactly $G_{[i, i+2]}(A)$; hence $\|P_i P_{i+1}\| < 1$ (the Friedrichs cosine is strictly less than 1). On a finite-dimensional space with two projectors, the Friedrichs cosine satisfies the quantitative bound $\alpha \leq 1 - c_{\min}$ where $c_{\min} > 0$ depends on the minimal principal angle between the two kernels. The bound $\alpha < \frac{1}{2}$ then follows from the Nachtergaele finite-size criterion [Nac96], which establishes that the overlap $\|P_i P_{i+1}\|$ for injective MPS is bounded away from 1 by a constant depending only on the tensor A . For general $L > 2$, the uniform constants $c_{ij} = \frac{1}{2(L-1)}$ make the condition $\alpha < \frac{1}{2(L-1)}$ increasingly restrictive as L grows. A more refined choice of separation-dependent constants $c_{ij} = c_{d_N(i, j)}$ that weight small-separation (large-overlap) pairs more heavily can exploit the exponential decay of the Friedrichs cosine α_s in s to satisfy the row-sum constraint. However, this refinement is not needed for the main result: in Theorem 19.92 below, the argument is applied only with $L = 2$ in the blocked picture. In all cases, the constants depend only on A and are independent of N [FNW92, KL18]. $\square$

Theorem 19.91 (Explicit Friedrichs-angle gap bound). *For an injective tensor A and a range $L > 1$, the current Lean statement gives the missing analytic estimate with the explicit constant*

$$\gamma_L := \frac{1}{4L} > 0.$$

Concretely, it asserts that if $N \geq 2L$ and $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$, then

$$\gamma_L \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

Proof. This is exactly the missing MPS-specific Friedrichs-angle estimate together with the row-sum bound. Once those analytic inputs are formalized, the result feeds directly into the abstract martingale criterion. $\square$

Theorem 19.92 (Spectral gap for MPS parent Hamiltonians). *For an injective tensor A and a range $L > 1$, there exists $\gamma > 0$ such that for every $N \geq 2L$ and every vector $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ one has*

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

By Lemma 19.88, this implies the usual uniform lower bound on every nonzero eigenvalue of the parent Hamiltonian.

Proof. This follows immediately from Theorem 19.91 by taking as γ the explicit constant produced there, which yields the required existential spectral-gap constant. $\square$

19.7 Ground space of non-injective MPS

For a general (non-injective) MPS tensor in block-diagonal canonical form, the ground space is no longer one-dimensional. Instead, it is spanned by the MPVs of the individual normal blocks — the basis of normal tensors (BNT) from Chapter 13 [FNW92, PGVWC07].

Definition 19.93 (Assembled parent ground space). For a family of blocks $A = (A_j)_{j=1}^r$ and weights μ , write $\text{ParentGround}_{N,L}(A)$ for the periodic chain ground space of the block-diagonal tensor assembled from the blocks.

Lemma 19.94 (Definition by assembled chain ground space). *By definition, this subspace is exactly the chain ground space of the assembled tensor.*

Definition 19.95 (Span of BNT block states). The BNT span is the linear span of the periodic MPV states of the individual blocks:

$$\text{BNTSpan}_N(A) := \text{span}\{V^{(N)}(A_j) : j = 1, \dots, r\}.$$

Lemma 19.96 (Each BNT MPV lies in the assembled parent ground space). *If $1 < L$ and $N \geq L + 1$, then for each block A_j of a canonical-form/BNT decomposition, the corresponding MPV state lies in the assembled parent ground space.*

Proof. One first applies Lemma 19.50 to the individual block. The proof then inserts the resulting boundary matrix into the j -th diagonal block of the assembled tensor, with the compensating scalar factor coming from the nonzero weight μ_j . $\square$

Theorem 19.97 (Containment in the BNT span by block decomposition). *If A is in canonical-form/BNT and $1 < L$ and $N \geq L + 1$, every vector in the assembled parent ground space should decompose into blockwise chain ground states, and therefore lie in the BNT span.*

Proof. The missing step is the periodic block decomposition of the assembled chain ground space. Once such a decomposition is available, one applies the normal uniqueness theorem to each block separately and recovers the BNT span. $\square$

Theorem 19.98 (Ground space equals span of BNT). *If $1 < L$ and $N \geq L + 1$, then for a canonical-form/BNT family the assembled parent ground space equals the BNT span:*

$$\text{ParentGround}_{N,L}(A) = \text{BNTSpan}_N(A).$$

Proof. We prove the two inclusions separately. The inclusion

$$\text{ParentGround}_{N,L}(A) \subseteq \text{BNTSpan}_N(A)$$

is exactly Theorem 19.97. For the reverse inclusion, Lemma 19.96 shows that each BNT MPV lies in $\text{ParentGround}_{N,L}(A)$. Since the parent ground space is a linear subspace, it therefore contains every linear combination of such vectors, and hence contains $\text{BNTSpan}_N(A)$. Thus

$$\text{BNTSpan}_N(A) \subseteq \text{ParentGround}_{N,L}(A),$$

and the two spaces are equal. □

Remark. A separate kernel-identification statement for these assembled block families is still to be recorded; at present, the formal results are stated in the chain-ground-space model rather than directly as $\ker(H_N)$.

Chapter 20

Exponential Decay of Correlations

This chapter records the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 19), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound $\rho(F_{AB}) \leq 1$, the strict spectral gap $\rho(F_{AB}) < 1$ for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 9. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

20.1 Connected correlators from the transfer map

Definition 20.1 (One-point expectation). Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as A and the fixed point as ρ^R . Under the hypotheses of Theorem 8.9, that fixed point exists and is unique up to scaling; we normalize it so that $\text{tr}(\rho^R) = 1$. For a local observable $X \in M_D(\mathbb{C})$ define

$$\langle X_0 \rangle := \text{tr}(X \rho^R).$$

Definition 20.2 (Two-point expectation at distance n). For observables $X, Y \in M_D(\mathbb{C})$ and $n \geq 0$,

$$\langle X_0 Y_n \rangle := \text{tr}(Y \mathcal{E}_A^n(X \rho^R)).$$

Equivalently: insert X at site 0, propagate by $\mathcal{E}_A^n$, then insert Y and take the trace.

Definition 20.3 (Connected correlator). The connected two-point function is

$$C(X, Y; n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle \langle Y_0 \rangle.$$

20.2 Spectral expansion and exponential decay

Theorem 20.4 (Sum of exponentials). *For a normal MPS, the connected correlator admits a spectral expansion of the form*

$$C(X, Y; n) = \sum_{j=1}^{D^2-1} c_j \lambda_j^n,$$

where λ_j are subleading transfer-matrix eigenvalues.

Proof. The leading eigenvalue contribution ($\lambda_1 = 1$) equals $\langle X_0 \rangle \langle Y_0 \rangle$ and cancels in the connected part. Apply spectral decomposition to $\mathcal{E}_A^n$. $\square$

Theorem 20.5 (Exponential decay bound). *If $|\lambda_j| \leq |\lambda_2| < 1$ for all subleading eigenvalues, then there exists $C_{XY} \geq 0$ such that*

$$|C(X, Y; n)| \leq C_{XY} |\lambda_2|^n.$$

Proof. Combine the sum-of-exponentials expansion with the triangle inequality. $\square$

Remark 20.6 (Spectral hypothesis and the spectral gap). The hypothesis “ $|\lambda_2| < 1$ ” in Theorem 20.5 is precisely the *spectral gap* condition for the transfer map $\mathcal{E}_A$. For primitive channels, the complementary spectral gap $\rho(\mathcal{E}_A - P) < 1$ is established in Theorem 5.51 (Chapter 5); for irreducible trace-preserving tensors, the analogous result is Theorem 9.29. Thus the exponential decay of connected correlations follows directly from the block separation theory once the spectral decomposition is in hand.

Definition 20.7 (Correlation length). The correlation length associated with λ_2 is

$$\xi := -\frac{1}{\log |\lambda_2|}.$$

Lemma 20.8 (Correlation length is positive). *If $|\lambda_2| < 1$ and $\lambda_2 \neq 0$, then $\xi > 0$.*

Proof. Since $0 < |\lambda_2| < 1$, we have $\log |\lambda_2| < 0$, so $\xi = -1/\log |\lambda_2| > 0$. $\square$

20.3 Quantitative spectral gap bounds

The spectral gap theorems in Chapter 5 establish $\rho(\mathcal{E}_A - P) < 1$ for primitive channels qualitatively. This section records the *quantitative* refinements that give explicit constants and rates for the single-tensor transfer map $\mathcal{E}_A$.

Theorem 20.9 (Exponential convergence of injective primitive channels). *Fix an injective, trace-preserving MPS tensor A with positive definite fixed point ρ . Let*

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the corresponding fixed-point projection. Then there exist $C > 0$ and $0 < \delta \leq 1$ such that for all $n \geq 0$ and $X \in M_D(\mathbb{C})$,

$$\|\mathcal{E}_A^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|.$$

The spectral gap δ exists by primitivity (Theorem 5.51, the complementary spectral gap).

Proof. Injectivity implies primitivity of the transfer map. The complementary spectral radius $\rho(\mathcal{E}_A - P) < 1$ then follows from the primitive spectral gap theorem. The Gelfand formula yields a geometric bound $\|(\mathcal{E}_A - P)^n\| \leq C(1 - \delta)^n$ for suitable constants, giving the claimed estimate. $\square$

Theorem 20.10 (Correlation length bound). *For an injective trace-preserving MPS tensor, there exist $C > 0$ and $\xi > 0$ such that for all $n \geq 0$ and all traceless $X \in M_D(\mathbb{C})$ (i.e. $\text{tr}(X) = 0$),*

$$\|\mathcal{E}_A^n(X)\| \leq C e^{-n/\xi} \|X\|.$$

Traceless matrices lie in $\ker P$, where P is the fixed-point projection. Since $\mathcal{E}_A - P$ has spectral radius strictly less than 1, the Gelfand formula gives exponential decay at rate $\xi = -1/\log \rho(\mathcal{E}_A - P)$.

Proof. For traceless X we have $P(X) = 0$, so $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$. The exponential convergence theorem gives a geometric bound $C(1 - \delta)^n \|X\|$. Rewriting $(1 - \delta)^n = e^{-n/\xi}$ with $\xi = -1/\log(1 - \delta)$ yields the claimed estimate. $\square$

Theorem 20.11 (Spectral gap from injectivity). *For an injective trace-preserving MPS tensor, there exists $\delta > 0$ such that every eigenvalue $\mu \neq 1$ of the transfer map satisfies $|\mu| \leq 1 - \delta$.*

Proof. Injectivity implies eventual full Kraus rank. This yields primitivity, hence a spectral gap for the complementary map. Since the transfer map has only finitely many eigenvalues, one may choose a uniform $\delta > 0$ separating 1 from the remaining spectrum. $\square$

20.4 Relation to parent Hamiltonians

Remark 20.12. When the MPS tensor A generates a parent Hamiltonian $H_N(A, L)$ (Definition 19.15) whose ground state is the matrix product vector (Lemma 19.19), the exponential decay bound (Theorem 20.5) shows that ground-state correlations decay with a finite correlation length ξ .

20.5 Renormalization fixed points and zero correlation length

Definition 20.13 (Renormalization fixed point). An MPS tensor A is a *renormalization fixed point* (RFP) when its transfer map is idempotent, $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$. See [CPGSV21, Definition 3.2].

Theorem 20.14 (Kraus-isometry criterion for RFP). *Suppose there exists an isometry $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$, written as coefficients $V_{(i_1, i_2), j}$, such that*

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j$$

for all $i_1, i_2 \in \{0, \dots, d-1\}$. Then A is a renormalization fixed point. This is the backward direction of [CPGSV21, Theorem 3.1].

Proof. Expanding $\mathcal{E}_A^2$ gives a double Kraus sum indexed by pairs (i_1, i_2) . Substituting the decomposition above and using the isometry relation $V^\dagger V = 1$ collapses the double sum to the single Kraus family $\{A^j\}$, hence $\mathcal{E}_A^2 = \mathcal{E}_A$. $\square$

Definition 20.15 (Local orthogonality). A single BNT block is *locally orthogonal* when its self-transfer map is idempotent. For a single tensor this coincides with the RFP condition (Definition 20.13); in the full multi-block BNT setting, local orthogonality additionally requires that mixed transfer operators vanish. See [CPGSV21, Definition 3.5].

Theorem 20.16 (RFP normal tensor is injective). *If A is a nonzero normal RFP tensor, then A is injective.*

Proof. The RFP condition $\mathcal{E}_A^2 = \mathcal{E}_A$ implies that the cumulative span $\text{span}\{A^w : |w| \leq n\}$ stabilises after one step. Since A is normal (hence has an injective blocked form), the one-step span already equals $M_D(\mathbb{C})$, so A is injective. $\square$

Theorem 20.17 (Left-canonical normal RFP tensor is injective). *Assume $D > 0$. Let A be a normal renormalization fixed point such that*

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then A is injective. This is the left-canonical injectivity step in [CPGSV21, Appendix B].

Proof. The normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ forces at least one matrix A^i to be nonzero, since otherwise the left-hand side would vanish. Hence A is a nonzero normal renormalization fixed point, and Theorem 20.16 applies. $\square$

Theorem 20.18 (Unitary diagonal fixed point for a left-canonical normal RFP). *Assume $D > 0$. Let A be a normal renormalization fixed point such that*

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \quad \text{and} \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

This is the diagonal fixed-point reduction in [CPGSV21, Appendix B].

Proof. By Theorem 20.17, the tensor A is injective. Therefore its transfer map is irreducible, and hence A is irreducible as a tensor. Applying Theorem 11.17 gives a unitary conjugate B for which the trace-preserving normalization remains valid and a diagonal positive definite matrix Λ with $\mathcal{E}_B(\Lambda) = \Lambda$. $\square$

Theorem 20.19 (Rank-one classification for RFP transfer maps). *Let A be an injective left-canonical RFP tensor with positive-definite fixed point ρ of the transfer map $\mathcal{E}_A$. Then $\mathcal{E}_A = P_\rho$, where $P_\rho(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$ is the rank-one fixed-point projection.*

Proof. The exponential convergence bound gives $\|\mathcal{E}_A^n(X) - P_\rho(X)\| \leq C(1-\delta)^n \|X\|$. Idempotence of $\mathcal{E}_A$ implies $\mathcal{E}_A^{n+1} = \mathcal{E}_A$ for all $n \geq 0$, so $\|\mathcal{E}_A(X) - P_\rho(X)\| \leq C(1-\delta)^{n+1} \|X\|$ for all n . Taking $n \rightarrow \infty$ gives $\mathcal{E}_A = P_\rho$. $\square$

Theorem 20.20 (Full structural form for RFP tensors). *A normal tensor A in canonical form II that is a renormalization fixed point admits a decomposition $A^i = X \Lambda U^i X^{-1}$, where Λ is diagonal positive and the family $U = (U^i)$ is an isometry in the physical indices.*

Proof. Combine the diagonal fixed-point reduction with the rank-one classification of injective left-canonical RFP tensors. One then writes the resulting rank-one projector in a canonical Kraus form, applies Kraus freedom to identify the physical-index family with an isometry U , and transports the decomposition back through the conjugation. $\square$

Theorem 20.21 (RG flow convergence for canonical-form tensors). *For a tensor in canonical form with blocks (A_k) and weights (μ_k) , the iterated blocking $\mathcal{E}_{A_k}^{2^n}$ converges pointwise to an idempotent linear map $\mathcal{E}_\infty$ as $n \rightarrow \infty$. That is, there exists $\mathcal{E}_\infty$ with $\mathcal{E}_\infty^2 = \mathcal{E}_\infty$ such that $\mathcal{E}_{A_k}^{2^n}(\rho) \rightarrow \mathcal{E}_\infty(\rho)$ for all density matrices ρ . This is [CPGSV21, Appendix B].*

Proof. Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point ρ_0 . The exponential convergence bound $\|\mathcal{E}^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|$ (Theorem 20.9) then gives pointwise convergence $\mathcal{E}^n(X) \rightarrow P(X)$, and composing with the subsequence $2^n \rightarrow \infty$ yields the result. The witness $\mathcal{E}_\infty = P$ is the rank-one fixed-point projection, which is idempotent. $\square$

Remark 20.22. When the transfer map is idempotent ($\mathcal{E}^2 = \mathcal{E}$), connected correlations become separation-independent for all separations $n \geq 1$. Informally, this corresponds to zero correlation length ($\xi = 0$). The full spectral equivalence (idempotence iff vanishing subleading spectrum) is deferred here.

Definition 20.23 (Correlations independent of distance). For a positive definite right fixed point $\rho^R > 0$, an MPS tensor has *correlations independent of distance* (CID) when for all local observables X and Y and all separations $n, m \geq 1$,

$$C(X, Y; n) = C(X, Y; m).$$

Definition 20.24 (Zero correlation length). An MPS tensor has *zero correlation length* when it is both locally orthogonal and correlations are independent of distance. See [CPGSV21, Definition 3.6].

Theorem 20.25 (CID implies RFP). *If an MPS tensor admits a positive definite right fixed point of its transfer map and has correlations independent of distance, then its transfer map is idempotent. This is the reverse direction of [CPGSV21, Theorem 3.8].*

Proof. Trace nondegeneracy and invertibility of the fixed point reduce idempotence to equality of the connected correlator at separations $n = 2$ and $n = 1$. $\square$

Theorem 20.26 (Zero correlation length iff idempotent transfer). *An MPS tensor has zero correlation length (i.e. is both locally orthogonal and CID) if and only if its transfer map is idempotent. This is [CPGSV21, Theorem 3.8].*

Proof. The forward direction extracts local orthogonality (which equals idempotence by definition). The reverse uses $\mathcal{E}^n = \mathcal{E}$ for $n \geq 1$ to show the connected correlator is separation-independent. $\square$

Theorem 20.27 (Canonical-form blocks satisfy RFP iff ZCL). *For a tensor in canonical form, each block is a renormalization fixed point if and only if it has zero correlation length. This is the RFP-ZCL part of [CPGSV21, Theorem 3.10].*

Proof. Apply Theorem 20.26 to the chosen canonical-form block. $\square$

Chapter 21

Concrete Examples

This chapter collects concrete MPS tensors that illustrate the general theory developed in earlier chapters. Each example is specified by its physical dimension d , bond dimension D , and explicit matrix family $\{A^i\}$. Key structural properties (injectivity, RFP status, symmetry) are stated as propositions linking back to the general definitions.

References follow [CPGSV21], Section III.

21.1 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state is the prototypical non-injective, renormalization-fixed-point MPS.

Definition 21.1 (GHZ tensor). Set $d = D = 2$. The GHZ tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |0\rangle\langle 0| = \text{diag}(1, 0), \quad A^1 = |1\rangle\langle 1| = \text{diag}(0, 1).$$

For a system of N sites the resulting MPV is the GHZ state

$$|\text{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

Theorem 21.2 (GHZ transfer map is idempotent). *The transfer map of the GHZ tensor satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$.*

Proof. The transfer map acts by $\mathcal{E}_A(X)_{ij} = \delta_{ij} X_{ij}$ (extraction of the diagonal). This operation is manifestly idempotent. $\square$

Theorem 21.3 (GHZ tensor is an RFP). *The GHZ tensor is a renormalization fixed point.*

Proof. Immediate from Theorem 21.2. $\square$

Theorem 21.4 (GHZ tensor is not injective). *The GHZ tensor is not injective: $\text{span}_{\mathbb{C}}\{A^0, A^1\}$ is the two-dimensional space of diagonal matrices in $M_2(\mathbb{C})$, which does not equal $M_2(\mathbb{C})$.*

Proof. Every element of $\text{span}_{\mathbb{C}}\{A^0, A^1\}$ has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span. $\square$

Theorem 21.5 ($\mathbb{Z}_2$ on-site symmetry of the GHZ tensor). *Let $\mathbb{Z}_2 = \{1, \sigma_x\}$ act on $\mathbb{C}^2$ by the Pauli- X matrix σ_x . The GHZ tensor is on-site symmetric under this action: for each $g \in \mathbb{Z}_2$, the twisted tensor $\tilde{A}_g$ defines the same MPV family as A , i.e. $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$.*

Proof. The identity element is trivial. For the generator $g = \sigma_x$, the twisted tensor satisfies $\widetilde{A}_g^i = A^{1-i}$, i.e. the two matrices are swapped. Conjugation by σ_x implements the same swap, giving gauge equivalence and hence the same MPV family. $\square$

21.2 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order.

Definition 21.6 (AKLT tensor). Set $d = 3$ (spin-1) and $D = 2$. The AKLT tensor is the family $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$ defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}} \sigma_z, \quad A^1 = \frac{\sqrt{2}}{\sqrt{3}} |0\rangle\langle 1|, \quad A^2 = -\frac{\sqrt{2}}{\sqrt{3}} |1\rangle\langle 0|,$$

where $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, so equivalently $A^1 = \frac{1}{\sqrt{3}} \sigma_+$ and $A^2 = -\frac{1}{\sqrt{3}} \sigma_-$ for $\sigma_+ = \sqrt{2} |0\rangle\langle 1|$ and $\sigma_- = \sqrt{2} |1\rangle\langle 0|$. Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin- $\frac{1}{2}$ to spin- $\frac{1}{2}$.

Theorem 21.7 (AKLT tensor is not injective). *The AKLT tensor is not injective at a single site: $\text{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$ is the three-dimensional space of traceless matrices $\mathfrak{sl}(2, \mathbb{C})$, which is a proper subspace of $M_2(\mathbb{C})$.*

Theorem 21.8 (AKLT tensor is normal). *The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-2 blocked tensor is injective.*

Theorem 21.9 (SO(3) on-site symmetry of the AKLT tensor). *The AKLT tensor is on-site symmetric under the spin-1 representation of SO(3) (the spin-1 representation of SU(2) factors through SO(3)), with the virtual representation being the projective spin- $\frac{1}{2}$ representation. The projective phase is the non-trivial element of $H^2(\text{SO}(3), \text{U}(1)) \cong \mathbb{Z}_2$, witnessing a non-trivial SPT phase. The $\mathbb{Z}_2$ subgroup symmetry is proved below; the full SO(3) projective representation is deferred.*

21.3 The cluster state

The cluster state is a universal resource for measurement-based quantum computation and provides another example of a non-trivial SPT phase.

Definition 21.10 (Cluster-state tensor). Set $d = D = 2$. The cluster-state tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.

Theorem 21.11 (Cluster-state tensor is not injective). *The cluster-state tensor is not injective: $\text{span}_{\mathbb{C}}\{A^0, A^1\} = \text{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$ has dimension 2, not $\dim M_2(\mathbb{C}) = 4$.*

Proof. Both A^0 and A^1 are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than $\dim M_2(\mathbb{C}) = 4$. $\square$

Remark 21.12. Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$, $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$, and $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$, the four products $A^{ij} = A^i A^j$ are

$$A^{00} = \frac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \frac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \frac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\frac{1}{\sqrt{2}}|-\rangle\langle 1|.$$

Since $\{|+\rangle, |-\rangle\}$ and $\{|0\rangle, |1\rangle\}$ are bases, these four rank-one matrices are linearly independent and span $M_2(\mathbb{C})$.

Theorem 21.13 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked cluster-state tensor). *The length-2 blocked cluster-state tensor is on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$. The two generators act on the 4-dimensional blocked physical space $(\mathbb{C}^2)^{\otimes 2}$ by $\sigma_x \otimes I$ and $I \otimes \sigma_x$, which commute and each square to the identity, defining a linear representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$. The corresponding virtual matrices are $V_1 = \sigma_z$ and $V_2 = \sigma_x$, which anticommute ($V_1 V_2 = -V_2 V_1$). This projective phase is the non-trivial element of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$, witnessing a non-trivial SPT phase.*

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